SUPPLEMENTARY MATERIAL TO "EXPLICIT HAMILTONIAN FULL PACKINGS OF THE FOUR-BALL BY EQUAL BALLS"

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Abstract. This supplement contains the complete construction and every proof that the main text states without proof or proves only in outline: the coordinate rearrangement and the identities of the linear comparison datum, the explicit planar rearrangements, the cubic model on the cover of the Fermat torus, the four-dimensional handle, the strip and layer maps, the return to the four-ball, the common isotopy and the global Hamiltonian, and a worked instance for ten balls. The appendix describes the Lean explicitness check and the formal statements that differ in form from the text.

Authorship and the role of AI. See the corresponding section at the end of the main text. The source tags, indexed in `verification/PAPER-INDEX.md`, locate the formal proofs.

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1. A checking route

This section lists, in construction order, the statements to check for Theorem A and Corollary C. For each step it gives input, output, where it is proved, and what is new. Prefix M and the lettered results refer to the main text. Interface conventions are in Table 1.

Organisation. After §2, the sections are grouped as follows: the coordinate rearrangement and the planar maps of the linear comparison datum (§§3–5); the four-dimensional model, handle, strip and layer maps and return to the ball (§§6–10); and the global construction and the worked

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instance (§§11–13). The general comparator, the scalar comparison datum for unequal capacities and $k = 9$, the higher-dimensional models and the finite-output algorithm are in the companion paper [Son26].

1.1. **The four-dimensional chain.** The data are $n = 2$, $k \geq 10$, equal capacities c and C with $\frac{3}{4} < \rho\sqrt{k} < 1$, $\rho = c/C$; this covers the data of Theorem A and Corollary C. Steps F5–F12 apply verbatim to any valid cubic data (Definition 8.1); the companion paper supplies such data for all admissible $(2, k, C, c)$.

F1 *Normalisation.* Admissible $(2, k, C, c) \mapsto \rho = c/C < \frac{1}{3}$ and the linear branch (Definition M7.1, Construction 11.1).

F2 *Auxiliary capacities and profile.* $\rho \mapsto t_0 = \frac{1}{2}(1 + \rho\sqrt{k})$, $a = t_0/\sqrt{k}$, $h = t_0/3$, and $P_{\mathrm{b}} = H_{\mathrm{b}} - k(a - p)_+$ on $[0, h]$ with $\int_0^h P_{\mathrm{b}} = t_0(1 - t_0) > 0$ (Remark 10.25).

F3 *Comparison function.* $K(p) = (3\sqrt{k} - 9)p$; $K - P_{\mathrm{b}}$ is a constant plus a tent whose increasing rearrangement is linear (Proposition M4.4, §3.3). New: no fixed point of the rearrangement contraction of [Ope26] is needed.

F4 *Planar realisation.* A Hamiltonian isotopy q_t of $(\mathbb{C}, \omega_{\mathbb{C}})$ with generator supported in $\{\pi|z|^2 \leq P_{\mathrm{supp}}\}$, $P_{\mathrm{supp}} < h$, satisfying (16) (Theorems 5.34 and 4.21). New: built from Morse charts of an explicit convex potential, with a radial source primitive as Hamiltonian.

F5 *Model.* $h < d_n < \frac{1}{3} \mapsto$ the cubic cover model $(\mathbb{R}^2 \times \mathbb{C}, \Omega_0)$ with deck maps, marked fibres, annulus and dual one-form (Definition 6.1, Propositions 6.3, 7.16). New: an explicit theta lift Z of the Fermat torus, giving Biran's disc bundle [Bir01] by formulas on the cover.

F6 *Handle.* Cubic data and model $\mapsto$ maps f_i^N on $B^\circ(\hat{\rho}_i)$ into $\{p < h\}$, with $(f_i^N)^*\Omega_0 = \omega_0$, injective, disjoint images, no two image points deck equivalent (§8.1: Propositions 8.9, 8.13, 8.16, 8.17; strip map Proposition 8.20; outline Remark M5.3). New: the three Moser paths have explicit deck-invariant primitives vanishing on the marked fibres.

F7 *Transfer.* Compact subsets of $\{p < h\}$ avoiding marked fibres $\mapsto \mathcal{T}$ into $B^\circ(1)$, $\mathcal{T}^*\omega_0 = \Omega_0$, identifying exactly deck-equivalent points, with compact margin (Theorem 10.21). New: explicit injection, transport and Moser return flows; a cube-root symmetry keeps the image off the line removed by the ball chart.

F8 *Embeddings.* F6, F7 $\mapsto$ normalised $f_i^\natural$ on $B^\circ(\rho_i^{\mathrm{dom}})$, injective on $B(\rho_i^+)$, disjoint images in $B(1 - m)$ (Theorem 10.23; Theorem M6.1).

F9 *Common isotopy.* Rescaled datum (Proposition M7.3) and centres with disjoint closed balls $\mapsto$ symplectic $F_{i,t}$, disjoint at all times, from $u \mapsto s_i + u$ to f_i (Proposition M7.6, Proposition 12.14). New: explicit route, path in $\mathrm{Sp}(2n)$ and common Alexander tail, the same in every dimension.

F10 *Hamiltonian.* $F_{i,t} \mapsto$ smooth $H_{\mathfrak{E}}$ vanishing for $t \notin (0, 1)$ and outside $\mathbb{R} \times \mathbb{B}(R_F)$, $X_H = \dot{F}_{i,t}$ on the moving balls, satisfying Pack (Definition M7.10, Lemma M7.12, Proposition M7.15, Theorem M7.14). New: pushforward of a cut-off radial primitive in the source balls.

F11 *Statements.* With $H[2, k, C, c, s] = H_{\mathfrak{E}}$ (Definition M7.16), Theorem A and Corollary C follow after checking admissibility (§M7.4).

F12 *Robustness.* Proposition B follows from Theorem A and Grönwall's inequality.

1.2. **What the formal proof covers.** Scope, trusted base and the unverified metaprogram are in §M8; the explicitness check and the formal statements that differ in form from the text are in Appendix A. The Lean declarations for each statement are named in the source tags and indexed in `verification/PAPER-INDEX.md`.

2. Conventions and elementary tools

2.1. **Conventions.** This expands §M1.4. Manifolds are finite dimensional, Hausdorff and second countable; boundaries are stated explicitly.

Phase space. $\mathbb{R}^{2n} = \mathbb{C}^n$, $z_j = x_j + iy_j$, $\omega_0 = \sum_j dx_j \wedge dy_j = \langle \cdot, J_0 \cdot \rangle$ with $J_0 = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$ (multiplication by $-i$, so the J_0 of §M7 is conjugate to that of $\mathbb{C}^n$). For smooth $H(t,z)$, $\iota_{X_H}\omega_0 = dH_t$, whence $X_H = J_0\nabla H$, $\dot{x}_j = \partial_{y_j}H$, $\dot{y}_j = -\partial_{x_j}H$, $\dot{z}_j = -2i\,\partial_{\bar{z}_j}H$. φ_H^t is the time-t flow; $H = \pi|z_1|^2$ generates $z_1 \mapsto e^{-2\pi i t}z_1$. The letter p is reserved for moment maps.

Capacities and balls. For $c > 0$, $s \in \mathbb{C}^n$: $B(c) = \{\pi|z|^2 \le c\}$ (Euclidean radius $\sqrt{c/\pi}$), $B^\circ(c) = \{\pi|z|^2 < c\}$, $B(s,c) = s + B(c)$; $\mathbb{B}(r)$, $\mathbb{B}^\circ(r)$ are the closed and open Euclidean balls of radius r about 0. Volumes use $\omega^n/n!$, so $\operatorname{vol} B(c) = c^n/n!$.

Lemma 2.1 (Dilations)**.** *Let $\lambda > 0$, $\delta_\lambda(z) = \lambda z$, and let $H(t,z)$ be smooth. Then $\delta_\lambda^*\omega_0 = \lambda^2\omega_0$, $\delta_\lambda(B(s,c)) = B(\lambda s, \lambda^2 c)$, and the function $\tilde{H}(t,z) = \lambda^2 H(t, z/\lambda)$ satisfies $X_{\tilde{H}}(t,z) = \lambda X_H(t, z/\lambda)$; hence an integral curve γ of X_H gives the integral curve $\lambda\gamma$ of $X_{\tilde{H}}$, and $\varphi_{\tilde{H}}^t = \delta_\lambda \circ \varphi_H^t \circ \delta_\lambda^{-1}$ wherever either side is defined.*

Proof. For $X(t,z) = \lambda X_H(t, z/\lambda)$, $\omega_0(X, w) = \lambda\, dH_{(t,z/\lambda)}(w) = d\tilde{H}_{(t,z)}(w)$; the rest is immediate. □

With $\lambda = 1/\sqrt{C}$ this normalises the target capacity to 1 (§M7).

Moments and angles. On a fibre $\mathbb{C}^m$, $w_j = \xi_j + i\eta_j$, put $p_j = \pi|w_j|^2$, $\theta_j = \arg w_j/2\pi \in \mathbb{R}/\mathbb{Z}$, $|p| = \sum_j p_j$, and $\omega_V = \sum_j dp_j \wedge d\theta_j = \sum_j d\xi_j \wedge d\eta_j$ (both equal $r\,dr \wedge d\phi$ where $w_j \neq 0$). The form $p_j\,d\theta_j = \frac{1}{2}(\xi_j\,d\eta_j - \eta_j\,d\xi_j)$ is smooth on $\mathbb{C}$ with differential $d\xi_j \wedge d\eta_j$. Functions $f(p)$, f smooth on an open subset of $\mathbb{R}^m$, are smooth in w and called *toric*.

Fibre Hamiltonians. Following [Ope26], fibre Hamiltonians Q, Hamiltonians on toric models and Moser fields use $\iota_{X_Q}\omega_V = -dQ$ (so X_Q is X_{-Q} in the ω_0 convention, and $X_{p_j} = \partial_{\theta_j}$ generates $w_j \mapsto e^{2\pi i t}w_j$); the global Hamiltonian on $\mathbb{R}^{2n}$ uses $\iota_{X_H}\omega_0 = dH$ (§M7, §12). F, G *Poisson-commute* if $\omega_V(X_F, X_G) = 0$; this is independent of the convention and means that F is constant along the flow of X_G.

Complex conventions. Put $d^c = \frac{i}{4\pi}(\bar{\partial} - \partial)$, so $dd^c = \frac{i}{2\pi}\partial\bar{\partial}$, $d^c u$ is real for real u, and $d^c F(u) = F'(u)\,d^c u$. For $u = F(|z|^2)$ on $\mathbb{C}$, $\bar{\partial}u - \partial u = -2i\,F'(|z|^2)|z|^2\,d\phi$, hence

$$d^c F(|z|^2) = F'(|z|^2)\,|z|^2\,d\theta, \qquad \text{in particular} \qquad d^c \log|z|^2 = d\theta, \tag{1}$$

with $\theta = \arg z/2\pi$; this holds on $\mathbb{C}$ for smooth F (as $|z|^2 d\theta = (x\,dy - y\,dx)/2\pi$) and on $\mathbb{C}\setminus 0$ for $F = \log$. Since $\int_{|z|=\varepsilon} d\theta = 1$, $dd^c\log|z|^2$ has unit mass at 0. On $\mathbb{CP}^n$, $\omega_{\mathrm{FS}} = dd^c \log\|Z\|^2$.

Lemma 2.2 (The affine Fubini–Study chart)**.** *On $\mathbb{C}^N$, put $D = 1 + |z|^2$ and*

$$\lambda_{\mathrm{FS}} = \frac{1}{2\pi D}\sum_{j=1}^{N}(x_j\,dy_j - y_j\,dx_j), \qquad \omega_{\mathrm{FS}} = d\lambda_{\mathrm{FS}}.$$

Then:

- *(a) $\lambda_{\mathrm{FS}} = d^c \log D$ is smooth. Where every coordinate is nonzero it equals $\sum_j p_j^{\mathrm{FS}}\,d\theta_j$, with $p_j^{\mathrm{FS}} = |z_j|^2/D$ and $\theta_j = \arg z_j/(2\pi)$; its differential is $\sum_j dp_j^{\mathrm{FS}} \wedge d\theta_j$ there.*
- *(b) The map $\mathrm{ch}(z) = z/\sqrt{\pi D}$ is a smooth bijection of $\mathbb{C}^N$ onto $B^\circ(1)$ with $\mathrm{ch}^*\omega_0 = \omega_{\mathrm{FS}}$, with smooth inverse $w \mapsto \sqrt{\pi}\,w/\sqrt{1-\pi|w|^2}$.*
- *(c) $\mathrm{ch}^*\lambda_{\mathrm{std}} = \lambda_{\mathrm{FS}}$, where $\lambda_{\mathrm{std}} = \frac{1}{2}\sum_j(x_j\,dy_j - y_j\,dx_j)$, and $\pi|\mathrm{ch}(z)|^2 = 1 - D^{-1} < 1$.*

Proof. (a) follows from $d^c|z|^2 = \sum_j (x_j\,dy_j - y_j\,dx_j)/(2\pi)$ and the chain rule; the inverse in (b) comes from solving $\pi|w|^2 = |z|^2/(1+|z|^2)$. For (c), a radial rescaling $z \mapsto r(z)z$ pulls λ_{std} back to $r^2\lambda_{\mathrm{std}}$ (the dr term contributes $\omega_0(z,z) = 0$); take $r = (\pi D)^{-1/2}$ and apply d. □

The cubic model form (§6) uses this normalisation.

Bidegree and orientation. Over an open $U \subset \Sigma$ with oriented coordinates (x, y) and a unitary frame $E|_U = U \times \mathbb{C}^m$ of $E = L_1 \oplus \cdots \oplus L_m$, forms split uniquely into bidegrees (r, s), r counting dx, dy and s counting $d\xi_j, d\eta_j$ (coefficients in all variables; components vanish if $r > 2$ or $s > 2m$). Bidegree $(r, 0)$ forms are *basic*. A top form is *positive* if it is a positive multiple of $\omega_V^m \wedge dx \wedge dy$, i.e. of $\omega_V^m \wedge \kappa$ for a positive area form κ. A one-form is *horizontal* if it annihilates vertical vectors.

2.2. **The flat step.**

Definition 2.3 (Flat step)**.** Let $a_\flat(x) = 0$ for $x \le 0$ and $a_\flat(x) = e^{-1/x}$ for $x > 0$, and

$$S(x) = \frac{a_\flat(x)}{a_\flat(x) + a_\flat(1-x)} \qquad (x \in \mathbb{R}).$$

The denominator is positive; $S = 0$ on $(-\infty, 0]$ and $S = 1$ on $[1, \infty)$.

Lemma 2.4 (Properties of S)**.** *S is C^∞ on $\mathbb{R}$, takes values in $[0,1]$, is nondecreasing on $\mathbb{R}$ and strictly increasing on $[0,1]$, and $S(1-x) = 1 - S(x)$. For every $j \ge 1$, $S^{(j)} = 0$ on $(-\infty, 0] \cup [1, \infty)$; in particular $S^{(j)}(0) = S^{(j)}(1) = 0$.*

Proof. For $x > 0$, $a_\flat^{(j)} = Q_j(1/x)e^{-1/x}$ with Q_j polynomial, tending to 0 as $x \downarrow 0$, so $a_\flat$ and S are C^∞. Symmetry is immediate. On $(0,1)$, $S = 1/(1 + \exp(1/x - 1/(1-x)))$ with strictly decreasing exponent, giving strict increase, monotonicity and $0 \le S \le 1$. S is constant outside $(0,1)$, so $S^{(j)}$ vanishes there and, by continuity, at $0, 1$. □

Lemma 2.5 (Flat concatenation)**.** *Let $0 = t_0 < t_1 < \cdots < t_N = 1$ and let $\sigma_j(t) = S\big((t - t_{j-1})/(t_j - t_{j-1})\big)$, $1 \le j \le N$, be the clocks.*

(a) Let $\mathcal{P}$ and M be smooth manifolds and $\gamma_1, \dots, \gamma_N \colon [0,1] \times \mathcal{P} \to M$ smooth maps (smooth up to the boundary, i.e. restrictions of smooth maps on $(-\delta, 1+\delta) \times \mathcal{P}$ for some $\delta > 0$) such that $\gamma_j(1, y) = \gamma_{j+1}(0, y)$ for all y. Define $\bar\gamma \colon \mathbb{R} \times \mathcal{P} \to M$ by $\bar\gamma(t, y) = \gamma_j(\sigma_j(t), y)$ for $t \in [t_{j-1}, t_j]$, $\bar\gamma(t, y) = \gamma_1(0, y)$ for $t \le 0$ and $\bar\gamma(t, y) = \gamma_N(1, y)$ for $t \ge 1$. Then $\bar\gamma$ is well defined and C^∞, and at every point (t_j, y), $0 \le j \le N$, every partial derivative of $\bar\gamma$ (in charts) that contains at least one derivative in t vanishes.

(b) Let $\psi^{(1)}, \dots, \psi^{(N)} \colon [0,1] \times M \to M$ be smooth isotopies of M (smooth up to the boundary as in (a)) with $\psi_0^{(j)} = \mathrm{id}$, generated by smooth time-dependent vector fields $Y_s^{(j)}$, so that $\partial_s \psi_s^{(j)} = Y_s^{(j)} \circ \psi_s^{(j)}$. Put $\Phi_0 = \mathrm{id}$, $\Phi_j = \psi_1^{(j)} \circ \Phi_{j-1}$, and

$$\varphi_t = \psi^{(j)}_{\sigma_j(t)} \circ \Phi_{j-1} \qquad (t \in [t_{j-1}, t_j]),$$

$\varphi_t = \mathrm{id}$ for $t \le 0$ and $\varphi_t = \Phi_N$ for $t \ge 1$. Then φ is a smooth isotopy with $\varphi_0 = \mathrm{id}$ and $\varphi_1 = \Phi_N = \psi_1^{(N)} \circ \cdots \circ \psi_1^{(1)}$, generated by the field $Y_t = \sigma_j'(t)\, Y^{(j)}_{\sigma_j(t)}$ on $[t_{j-1}, t_j]$ and $Y_t = 0$ for $t \notin (0,1)$. The field Y is smooth on $\mathbb{R} \times M$, and Y and all its partial derivatives vanish at $t = t_j$. If M is symplectic and $Y_s^{(j)}$ is the Hamiltonian field of $Q_s^{(j)}$ (in either sign convention), with $Q^{(j)}$ smooth on $[0,1] \times M$ (up to the boundary), then Y_t is the Hamiltonian field of $Q_t = \sigma_j'(t)\, Q^{(j)}_{\sigma_j(t)}$ on $[t_{j-1}, t_j]$, $Q_t = 0$ for $t \notin (0,1)$, and Q is smooth on $\mathbb{R} \times M$.

Proof. (a) The definitions agree at the t_j, so $\bar\gamma$ is well defined and continuous. In charts, on $[t_{j-1},t_j]\times\mathcal{P}$ we have $\bar\gamma=\gamma_j\circ(\sigma_j\times\mathrm{id})$, so by Faà di Bruno every partial derivative containing a t-derivative has a factor $\sigma_j^{(i)}$, $i\ge1$, vanishing at t_{j-1},t_j by Lemma 2.4; pure y-derivatives on both sides are those of $\gamma_j(1,\cdot)=\gamma_{j+1}(0,\cdot)$, and likewise at $t=0,1$ against the constant extensions. Thus one-sided jets agree along each seam and $\bar\gamma$ is C^∞.

(b) Apply (a) to $\gamma_j(s,x)=\psi_s^{(j)}(\Phi_{j-1}(x))$, using $\gamma_j(1,\cdot)=\Phi_j=\gamma_{j+1}(0,\cdot)$; each φ_t is a diffeomorphism. On $[t_{j-1},t_j]$, $\partial_t\varphi_t=\sigma_j'(t)\,Y^{(j)}_{\sigma_j(t)}\circ\varphi_t$, and as in (a) all partial derivatives of $\sigma_j'(t)Y^{(j)}(\sigma_j(t),x)$ vanish at the seams, while the field is 0 for $t\notin(0,1)$; so Y is smooth. The Hamiltonian statement follows from $X_{cQ}=cX_Q$ (either convention), and Q is smooth by the same argument. □

2.3. **Cutoffs.**

Definition 2.6 (Radial and interval cutoffs)**.** For $0\le r<r'$ and u in a Euclidean space $\mathbb{R}^d$ put

$$\chi_{r,r'}(u)=1-S\Big(\frac{|u|^2-r^2}{r'^2-r^2}\Big).$$

For $\ell_0<\ell_1$ and $0<w<(\ell_1-\ell_0)/2$ put $\chi^{\mathrm{I}}_{\ell_0,\ell_1,w}(x)=S\big((x-\ell_0)/w\big)\,S\big((\ell_1-x)/w\big)$ for $x\in\mathbb{R}$.

Put $s_1=\max_{\mathbb{R}}S'$, which is finite since S' is continuous and vanishes outside $[0,1]$.

Lemma 2.7 (Cutoffs)**.** *(a) $\chi_{r,r'}$ is C^∞, $O(d)$-invariant and nonincreasing in $|u|$, takes values in $[0,1]$, equals 1 on $\mathbb{B}(r)$ and 0 outside $\mathbb{B}^\circ(r')$, and all its derivatives of order ≥1 vanish on $\mathbb{B}(r)\cup(\mathbb{R}^d\setminus\mathbb{B}^\circ(r'))$, in particular on the spheres $|u|=r$ and $|u|=r'$. Its gradient is*

$$\nabla\chi_{r,r'}(u)=-\frac{2u}{r'^2-r^2}\,S'\Big(\frac{|u|^2-r^2}{r'^2-r^2}\Big),\qquad |\nabla\chi_{r,r'}|\le\frac{2r's_1}{r'^2-r^2}.$$

(b) $\chi^{\mathrm{I}}_{\ell_0,\ell_1,w}$ is C^∞ with values in $[0,1]$, equals 1 on $[\ell_0+w,\ell_1-w]$ and 0 outside (ℓ_0,ℓ_1), and all its derivatives of order ≥1 vanish outside $(\ell_0,\ell_0+w)\cup(\ell_1-w,\ell_1)$.

Proof. (a) Write $\chi_{r,r'}=1-S\circ\ell$ with $\ell(u)=(|u|^2-r^2)/(r'^2-r^2)$. Smoothness, invariance, monotonicity and the range follow from Lemma 2.4. Moreover $\ell\le0$ for $|u|\le r$ and $\ell\ge1$ for $|u|\ge r'$. By Faà di Bruno, every derivative of order ≥1 of $S\circ\ell$ contains a factor $S^{(i)}(\ell(u))$ with $i\ge1$, and these factors vanish there. The gradient formula is the chain rule, and $|u|\le r'$ wherever $S'(\ell(u))\ne0$. (b) One factor vanishes for $x\le\ell_0$ or $x\ge\ell_1$, and both arguments are ≥1 on $[\ell_0+w,\ell_1-w]$. Since $2w<\ell_1-\ell_0$, the second factor is 1 on (ℓ_0,ℓ_0+w), and symmetrically on (ℓ_1-w,ℓ_1). Derivatives are treated as in (a). □

A cutoff around a coordinate disc means the squared-distance cutoff $\chi_{r,r'}$ in that coordinate.

2.4. **Toric forms.**

Lemma 2.8 (Top power)**.** *Let $m\ge1$, $n=m+1$, and let U be an oriented surface chart with coordinates (x,y). On an open subset of $U\times\mathbb{C}^m$, write $w_j=\xi_j+i\eta_j$, $\omega_V=\sum_j d\xi_j\wedge d\eta_j$, and $\kappa'=\varkappa\,dx\wedge dy$. The mixed term ϑ is of one of two types:*

(a) $\vartheta=\sum_j dp_j\wedge a_j'$, where $p_j=\pi|w_j|^2$ and the a_j' are basic one-forms whose coefficients may depend on the base and fibre variables.

(b) $\vartheta=\zeta\wedge\beta'$, where ζ is vertical and β' is basic.

In both cases $\omega_V^{m-1} \wedge \vartheta^2 = 0$, *and*

$$(\omega_V + \kappa' + \vartheta)^n = n\,\omega_V^m \wedge \kappa' = n!\,\varkappa\, d\xi_1 \wedge d\eta_1 \wedge \cdots \wedge d\xi_m \wedge d\eta_m \wedge dx \wedge dy.$$

Thus the two-form is nondegenerate exactly where $\varkappa \neq 0$, *and positive where* $\varkappa > 0$. *On the handle annulus take* $\beta = b(v)\,dv$ *closed,* $\nu = c_\nu\, ds \wedge dv$, $e > 0$, $c_\nu > 0$, $b \geq 0$, *and a smooth* $G(s, w)$ *independent of* v. *Then*

$$\big(\omega_V + e\nu + d(G\beta)\big)^n = n\big(ec_\nu + b(v)\partial_s G\big)\omega_V^m \wedge ds \wedge dv.$$

It is symplectic and positive where $\partial_s G \geq 0$. *This contains Lemma M5.1.*

Proof. Even forms commute, so $(\omega_V + \kappa' + \vartheta)^n$ is the multinomial sum of the terms $\frac{n!}{i_1!\,i_2!\,i_3!}\,\omega_V^{i_1} \wedge \kappa'^{i_2} \wedge \vartheta^{i_3}$. Such a term has bidegree $(2i_2 + i_3,\, 2i_1 + i_3)$, so it can be nonzero only if $(i_2, i_3) \in \{(0,0), (0,1), (0,2), (1,0)\}$. For $(0,0)$ and $(0,1)$ the fibre degree exceeds $2m$; $(1,0)$ gives $n\,\omega_V^m \wedge \kappa'$; $(0,2)$ gives a multiple of $\omega_V^{m-1} \wedge \vartheta^2$, which vanishes as shown below. The formula follows from $\omega_V^m = m!\, d\xi_1 \wedge d\eta_1 \wedge \cdots \wedge d\xi_m \wedge d\eta_m$, and nondegeneracy is equivalent to the n-th power being nonzero.

(a) $\vartheta^2 = -\sum_{j,l} dp_j \wedge dp_l \wedge a_j' \wedge a_l'$, and $\omega_V^{m-1} = (m-1)! \sum_i \bigwedge_{r \neq i} d\xi_r \wedge d\eta_r$. Since $dp_j = 2\pi(\xi_j d\xi_j + \eta_j d\eta_j)$, we have $d\xi_j \wedge d\eta_j \wedge dp_j = 0$ and $dp_j \wedge dp_j = 0$. For $j \neq l$, one of the indices, say j, differs from i, so $\bigwedge_{r \neq i} d\xi_r \wedge d\eta_r$ kills dp_j.

(b) $(\zeta \wedge \beta')^2 = -\zeta \wedge \zeta \wedge \beta' \wedge \beta' = 0$. On the annulus, $d\beta = 0$ and $\partial_v G = 0$ give $d(G\beta) = dG \wedge \beta = \partial_s G\, b(v)\, ds \wedge dv + d_w G \wedge \beta$, where $d_w G$ is the fibre differential. This is case (b) with $\kappa' = (ec_\nu + b\,\partial_s G)\, ds \wedge dv$, whose coefficient is positive where $\partial_s G \geq 0$ (as $e, c_\nu > 0$, $b \geq 0$), and $\vartheta = d_w G \wedge \beta$. □

The two types cannot be mixed: for $m = 1$, $(dp \wedge a' + \zeta \wedge \beta)^2 = 2\, dp \wedge a' \wedge \zeta \wedge \beta$ is in general nonzero. The handle construction uses (a) off the annulus $\mathcal{A}$ and (b) on it.

Local model. On an oriented base chart U, let a_j be smooth real one-forms, $F_j = -da_j$, and let σ be a smooth two-form. On $U \times \mathbb{C}^m$ put $p_j = \pi|w_j|^2$, $\lambda_j = \frac{1}{2}(\xi_j\, d\eta_j - \eta_j\, d\xi_j)$ and $p_j\alpha_j = \lambda_j + p_j a_j$. Where $w_j \neq 0$, $\alpha_j = d\theta_j + a_j$; $p_j\alpha_j$ is smooth everywhere. No global frame is needed: in a cover presentation, descent is checked by the deck transformations.

Lemma 2.9 (Local form of the model form)**.** *On the above product chart put*

$$\Omega_0 = \sigma + \sum_{j=1}^m d(p_j\alpha_j).$$

(a) The one-forms $p_j\alpha_j$ *are smooth, and* Ω_0 *is closed and invariant under fibre rotations.*
(b)

$$\Omega_0 = \omega_V + \kappa(p) + \sum_j dp_j \wedge a_j, \qquad \kappa(p) = \sigma - \sum_j p_j F_j.$$

(c) For the circle generator $X_{p_j} = \partial_{\theta_j}$, $\iota_{X_{p_j}}\Omega_0 = -dp_j$, *including the fibre axis.*
(d) With $n = m + 1$, $\Omega_0^n = n\,\omega_V^m \wedge \kappa(p)$, *so* Ω_0 *is nondegenerate where* $\kappa(p) \neq 0$ *and positive where* $\kappa(p) > 0$. *If* $F_j = d_j\sigma$ *for constants* d_j, *then* $\kappa(p) = (1 - \sum_j d_j p_j)\sigma$.
(e) On a fixed compact base patch, integration at fixed moment gives $\int \kappa(p) = \int \sigma - \sum_j p_j \int F_j$.

Proof. The form $\lambda_j + p_j a_j$ is smooth, and $d(\lambda_j + p_j a_j) = d\xi_j \wedge d\eta_j + dp_j \wedge a_j - p_j F_j$. Since two-forms on the base are closed, this gives (a) and (b). Contracting with a fibre rotation gives (c); on the axis both sides vanish. Part (d) is Lemma 2.8 with toric mixed terms, and (e) is linearity of integration. □

For the periodic cubic input (§6), $m = 1$, $F_1 = 3\sigma$ and $\kappa = (1-3p)\sigma$. For the quadric curves of [Son26], $F_j = 2\sigma$ and $\kappa = (1-2|p|)\sigma$.

2.5. **The Moser lemma.**

Lemma 2.10 (Moser lemma)**.** *Let M be a smooth manifold, $U \subset M$ open and $I \subset \mathbb{R}$ an open interval containing $[0,1]$. Let Ω_t and $\mathfrak{v}_t$, $t \in I$, be two-forms and one-forms on U depending smoothly on $(t,x) \in I \times U$, such that each Ω_t is closed and nondegenerate at every point of U and*

$$\partial_t \Omega_t = d\mathfrak{v}_t.$$

Let Y_t be the vector field on U defined by $\iota_{Y_t}\Omega_t = -\mathfrak{v}_t$, and let Ψ_t be its flow from time 0 ($\Psi_0 = \mathrm{id}$), where defined.

- *(a) Y is smooth on $I \times U$, and $Y_t(x) = 0$ whenever $\mathfrak{v}_t(x) = 0$ as an element of T_x^*M.*
- *(b) If $\Psi_t(x)$ is defined for $t \in [0,1]$, then Ψ_t is defined and smooth on a neighbourhood of x and $(\Psi_t^*\Omega_t)_x = (\Omega_0)_x$ for $t \in [0,1]$.*
- *(c) If $Y_t(x) = 0$ for all $t \in [0,1]$, then $\Psi_t(x) = x$ for $t \in [0,1]$. In particular Ψ_t fixes pointwise every set at whose points $\mathfrak{v}_t$ is the zero covector for all $t \in [0,1]$.*
- *(d) (Tangency.) Let $U' \subset U$ be open, $\mu \in C^\infty(U')$ and V a vector field on U' with $\iota_V\Omega_t = -d\mu$ and $\mathfrak{v}_t(V) = 0$ on U' for $t \in [0,1]$. Then $d\mu(Y_t) = 0$ on U', and μ is constant along every integral curve of Y on a subinterval of $[0,1]$ as long as the curve stays in U'.*
- *(e) (Completeness.) Let $\mathcal{K} \subset U$ be compact, and suppose that every integral curve $\gamma\colon J \to U$ of Y, $J \subset [0,1]$ an interval, which meets $\mathcal{K}$ satisfies $\gamma(J) \subset \mathcal{K}$. Then Ψ_t is defined on $\mathcal{K}$ for every $t \in [0,1]$, $\Psi_t(\mathcal{K}) = \mathcal{K}$, $\Psi_t|_{\mathcal{K}}$ is the restriction of a diffeomorphism between open neighbourhoods of $\mathcal{K}$, and $\Psi_t^*\Omega_t = \Omega_0$ at every point of $\mathcal{K}$.*
- *(f) (Reparametrisation and concatenation.) If $\varsigma\colon I \to I$ is smooth with $\varsigma(0) = 0$, the pair $(\Omega_{\varsigma(t)}, \varsigma'(t)\mathfrak{v}_{\varsigma(t)})$ satisfies the hypotheses, its field is $\varsigma'(t)Y_{\varsigma(t)}$, and its flow is $\Psi_{\varsigma(t)}$. Consequently, let $(\Omega_s^{(j)}, \mathfrak{v}_s^{(j)})$, $1 \le j \le N$, be pairs on U, smooth on $[0,1] \times U$ (up to the boundary), with every $\Omega_s^{(j)}$ closed and nondegenerate on U, $\partial_s\Omega_s^{(j)} = d\mathfrak{v}_s^{(j)}$, and $\Omega_1^{(j)} = \Omega_0^{(j+1)}$. With the clocks σ_j of Lemma 2.5 put $(\Omega_t, \mathfrak{v}_t) = (\Omega^{(j)}_{\sigma_j(t)}, \sigma_j'(t)\mathfrak{v}^{(j)}_{\sigma_j(t)})$ on $[t_{j-1}, t_j]$, $(\Omega_t, \mathfrak{v}_t) = (\Omega_0^{(1)}, 0)$ for $t \le 0$ and $(\Omega_t, \mathfrak{v}_t) = (\Omega_1^{(N)}, 0)$ for $t \ge 1$. Then $(\Omega_t, \mathfrak{v}_t)_{t\in\mathbb{R}}$ satisfies the hypotheses with $I = \mathbb{R}$, and its time-one map is $\Psi_1 = \Psi_1^{(N)} \circ \cdots \circ \Psi_1^{(1)}$ wherever the right side is defined.*
- *(g) (Equivariance.) If a diffeomorphism g of U satisfies $g^*\Omega_t = \Omega_t$ and $g^*\mathfrak{v}_t = \mathfrak{v}_t$ for all t, then $g_*Y_t = Y_t$ and $g \circ \Psi_t = \Psi_t \circ g$ wherever defined.*

Proof. (a) Locally $Y_t = -\Omega_t^{-1}\mathfrak{v}_t$, which is smooth by Cramer's rule.

(b) The flow domain is open. By Cartan's formula and $d\Omega_t = 0$, $\mathcal{L}_{Y_t}\Omega_t = d\iota_{Y_t}\Omega_t = -d\mathfrak{v}_t$. Hence $\frac{d}{dt}\Psi_t^*\Omega_t = \Psi_t^*(\partial_t\Omega_t + \mathcal{L}_{Y_t}\Omega_t) = 0$.

(c) By uniqueness the constant curve is the integral curve; then apply (a).

(d) $d\mu(Y_t) = -\Omega_t(V, Y_t) = (\iota_{Y_t}\Omega_t)(V) = -\mathfrak{v}_t(V) = 0$.

(e) Multiply Y by a compactly supported cutoff in $I \times U$ equal to 1 near $[0,1] \times \mathcal{K}$, and extend by zero. The resulting flow Φ_t is global and jointly smooth. For a trajectory meeting $\mathcal{K}$, the set of times it spends in $\mathcal{K}$ is relatively closed in $[0,1]$. It is also relatively open, since near $[0,1] \times \mathcal{K}$ the trajectory solves the original equation and trapping applies. By connectedness the trajectory stays in $\mathcal{K}$. Applying this forwards and backwards gives $\Phi_t(\mathcal{K}) = \mathcal{K}$. By the tube lemma, $\Phi = \Psi$ (with smooth inverse) near $\mathcal{K}$, and (b) gives the pullback identity.

(f) For reparametrisation, $\iota_{\varsigma' Y_\varsigma}\Omega_\varsigma = -\varsigma'\mathfrak{v}_\varsigma$ and $\partial_t\Omega_{\varsigma(t)} = \varsigma'\, d\mathfrak{v}_\varsigma$. The curve $\Psi_{\varsigma(t)}(x)$ has derivative $\varsigma' Y_\varsigma$, so it is the flow by uniqueness.

For concatenation, each Ω_t is some $\Omega_s^{(j)}$, hence closed and nondegenerate. Ω is smooth by Lemma 2.5(a), applied to $(s,x)\mapsto\Omega_s^{(j)}(x)$. $\mathfrak{v}$ is smooth by the argument of Lemma 2.5(b): every partial derivative of $\sigma_j'\mathfrak{v}_{\sigma_j}^{(j)}$ contains a factor $\sigma_j^{(i)}$ with $i\geq 1$, so all one-sided t-jets vanish at each t_j. On $[t_{j-1},t_j]$ we have $\partial_t\Omega_t=\sigma_j'(\partial_s\Omega^{(j)})_{\sigma_j}=d\mathfrak{v}_t$, and both sides vanish off $(0,1)$. By the reparametrisation statement, the flow from t_{j-1} to t is $\Psi^{(j)}_{\sigma_j(t)}$. Composing the stages gives Ψ_1.

(g) $\iota_{g_*Y_t}\Omega_t=(g^{-1})^*(\iota_{Y_t}g^*\Omega_t)=-(g^{-1})^*\mathfrak{v}_t=-\mathfrak{v}_t$, so $g_*Y_t=Y_t$, and flows of g-related fields are conjugate. □

Signs. This matches the fibre convention: for constant $\Omega_t=\omega$ and $\mathfrak{v}_t=dQ_t$, $Y_t=X_{Q_t}$ with $\iota_{X_Q}\omega=-dQ$. Y_t depends on the chosen primitive $\mathfrak{v}_t$, and (c), (d) are properties of that primitive. Part (c) requires $\mathfrak{v}_t$ to vanish as a covector on M; if only its pullback to a submanifold vanishes, Y_t is merely Ω_t-orthogonal to that submanifold.

2.6. **Conventions at the interfaces.** Table 1 lists the conventions at the interfaces between the steps of the construction.

Table 1: Conventions at the interfaces and where they are converted.

interface	conventions and conversion	where
comparison → planar	R_j acts on p_j; planar map in the rectangle chart, fibre sign; for $n=2$ it acts on the fibre $\mathbb{C}$	§3, §5
fibre → bundle	$\mathcal{D}_\lambda(s,v,w)=(s,v,q_{\lambda f(s)}(w))$, $Q_s^\lambda=\lambda f'(s)Q_{\lambda f(s)}$, $\mathcal{D}_\lambda^*\omega_V=\omega_V+d[(Q_s^\lambda\circ q_{\lambda f(s)})\,ds]$	Lemma M5.2, §8.1
model → $B^\circ(1)$	$\psi_i^*\Omega=\omega_0$; $\mathcal{T}^*\omega_0=\Omega$ (resp. Ω_0); cubic: $b_{\mathrm{ball}}^*\omega_0=\omega_{\mathrm{FS}}$ on unit representatives	Thm. 10.21, §10.5
embeddings → isotopy	$f_i^\natural$ for $(n,k,1,\rho)$, $\rho_i<\rho_i^+<\rho_i^{\mathrm{dom}}$ (enlarged source, domain capacities); $f_i(u)=\sqrt{C}\,f_i^\natural(u/\sqrt{C})$, $c_i'=C\rho_i^+$, $\bar c_i=C\rho_i^{\mathrm{dom}}$	Def. M7.1, Prop. M7.3, Def. M7.4

2.7. **Notation.** Table 2 lists the notation used in several sections; a statement reusing a letter redefines it locally.

Table 2: Notation used in more than one section. The letters s, y, b, J have several meanings.

symbol	meaning	defined in
$n\geq 2$, $m=n-1$	complex dimension; fibre rank	§M1
k, c_i, s_i, C	number, capacities, centres of balls; target capacity	§M2
a_i; $\bar\rho_i$, $\hat\rho_i$	auxiliary capacities; enlarged source and domain capacities, $\rho_i<\bar\rho_i<\hat\rho_i<a_i$; $\delta_i=\frac12(a_i-\rho_i)$, $(\bar\rho_i,\hat\rho_i)=(a_i-\delta_i,a_i-\frac34\delta_i)$ for $n=2$ ($n\geq 3$: [Son26])	§M5, Def. 9.8
$\eta_\star=1-\rho\sqrt{k}$	volume defect, equal capacities ($n=2$); $t_0=1-\eta_\star/2$	§3.3, Rem. 10.25
Δ, $\langle b_\nu,p\rangle\leq\ell_\nu$, τ_Δ	moment polytope, outer faces, core fraction	§§M3, M4
h; h_1,h_2	fibre height(s) of Δ	§M3
V_Δ, $V_\Delta(E)$	$\{p(w)\in\Delta\}$; fibrewise version	§M3
Γ, Z; E, α_j	base, marked set; model space, angular forms	§6, [Son26]
Σ; σ	base surface; base area form	§M3, §2
a_j, $a_j(p)$, a_j'; F_j	connection forms, mixed terms (not a_i); curvatures	§M3, §2

Continued on the next page

Table 2 (continued)

symbol	meaning	defined in
$\kappa(p)$; $H_{\mathrm{b}}(p)=\int_\Sigma \kappa(p)$; $\mathcal{B}_\Delta$	horizontal area form; area profile; its moment budget	§M3
$P_{\mathrm{b}}(p)$; P	$H_{\mathrm{b}}(p)-\sum_i (a_i-\lvert p\rvert)_+$; its minorant	§§M3, M4
K, $F=K-P$, γ, γ_*	comparison function, defect, gap, reserve	§§M3, M4
R_j, $\mathcal{R}=R_m\circ\cdots\circ R_1$	coordinatewise increasing rearrangement	§M4
q_t, $q=q_1$, Q_t	fibre isotopy, endpoint, generator	§§M3, M4
$\mathcal{A}\subset\mathcal{A}^+$, $\mathcal{B}$, (s,v), β	handle annulus, band, coordinates, closed form of period 1	§M5
ν, c_ν; Λ	periodic unit-area form, density; affine form on bordered base	§M5
$e_\Lambda\Lambda$, H^*, J	background form, $H_{\mathrm{b}}-\epsilon_{\mathrm{sm}}$, $K-H^*$	§M5
$h_i(p)$, R^*	caps $h_i(p)=\frac{1}{2}\big(a_i-p+\sqrt{(a_i-p)^2+\mu_h^2}\big)$, smoothing the tents from above: $(a_i-p)_+<h_i(p)\le(a_i-p)_++\mu_h/2$ (Lemma 8.3(a)); residual	§M5, Def. 8.2
$\mathcal{D}$, Φ; Λ_i, E_i	twist, Moser map; layers, layer maps ($\mathcal{E}_i$ in main text)	§M5
$\mathcal{T}$, $\mathcal{C}_\mathcal{K}$	transfer, compact set containing image of $\mathcal{K}$	§M3, §6
S, s_1, σ_j; $\chi_{r,r'}$, χ^{I}	flat step, $\max S'$, clocks; cutoffs	§2
the(h)	definite description	Def. M2.4
$r_{1,i}<\cdots<r_{4,i}$	enlarged source radii	§M7
$y_i=f_i(0)$, $Y_i=y_i/\tau$, L, δ_c, τ	centres, far centres, Lipschitz bound, separation, tail scale	§M7
$F_{i,t}$; $\mathfrak{a}_{i,t}$, $\mathfrak{h}_{i,t}$	common isotopy; source form, primitive	§M7
$H=H[n,k,C,c,s]$	the global Hamiltonian	§M7
L1, L2, L3	levels of explicitness	Def. M2.5

3. Coordinate rearrangement and the equal-ball identities

We prove the properties of the coordinate rearrangement used in §M4 and record the scalar identities behind Proposition M4.4. General comparison data are constructed in the companion paper [Son26].

3.1. **Setting.** Fix $m\ge 1$. Let $\Delta=\{p\in\mathbb{R}^m_{\ge0}:\langle b_\nu,p\rangle\le\ell_\nu,\ \nu=1,\dots,N_\Delta\}$, with $b_\nu\in\mathbb{Q}^m_{\ge0}$ and $\ell_\nu\in\mathbb{Q}_{>0}$, be a compact full-dimensional polytope. Since $b_\nu\ge0$, Δ is *downward closed*: $p\in\Delta$ and $0\le p'\le p$ imply $p'\in\Delta$; so $0\in\Delta$ and $\tau\Delta\subset\Delta$ for $0\le\tau\le1$.

We use p, $|p|=\sum_j p_j$ and ω_V of §2, and put $V_\Delta=p^{-1}(\Delta)\subset\mathbb{C}^m$. We write $\mathbf{e}_1,\dots,\mathbf{e}_m$ for the standard basis and $\|\cdot\|$ for the uniform norm on Δ. As in §M4, P is an arbitrary concave function on Δ (applied with $P=P_{\mathrm{b}}$ or a concave minorant of P_{b}). The letters y, L, and in §3.3 s,a,c_s,g,A,B,Λ, are local to this section and §M4, with the same meaning there.

Fix j and write $p=(y,x)$ with $x=p_j$ and $y=(p_i)_{i\ne j}$ (y absent if $m=1$). By compactness some $b_{\nu j}>0$. Put

$$L(y)=\min_{\nu:\,b_{\nu j}>0}\big(\ell_\nu-\textstyle\sum_{i\ne j}b_{\nu i}y_i\big)/b_{\nu j};$$

it is concave, continuous and piecewise affine. Let $\Delta_j=\{y:(y,0)\in\Delta\}$, a compact convex polytope. On Δ_j we have $L\ge0$ and

$$\Delta=\{(y,x):y\in\Delta_j,\ 0\le x\le L(y)\}. \tag{2}$$

The *slice* through $y\in\Delta_j$ is $\{y\}\times[0,L(y)]$, possibly of length zero.

3.2. **Coordinate rearrangement.** For $L\ge0$ and continuous f on $[0,L]$, let $\mathcal{L}_f(b)=\big|\{v\in[0,L]:f(v)\le b\}\big|$ (Lebesgue measure, $b\in\mathbb{R}$).

Lemma 3.1 (Increasing rearrangement on an interval)**.** *Let $L \geq 0$, let $f\colon [0,L] \to \mathbb{R}$ be continuous and convex, and for $0 \leq x \leq L$ put*

$$f^\sharp(x) = \min_{0\leq a\leq L-x} \max\bigl(f(a), f(a+x)\bigr).$$

(a) For $b \in \mathbb{R}$ and $0 \leq x \leq L$: $f^\sharp(x) \leq b$ if and only if $\{f \leq b\} \neq \emptyset$ and $\mathcal{L}_f(b) \geq x$. Hence $f^\sharp(x) = \min\{b \geq \min f : \mathcal{L}_f(b) \geq x\}$, the increasing rearrangement of f.
(b) $\mathcal{L}_{f^\sharp} = \mathcal{L}_f$; in particular $\int_0^L f^\sharp = \int_0^L f$.
(c) $f^\sharp$ is nondecreasing, $f^\sharp(0) = \min f$ and $f^\sharp(L) = \max f$.

Proof. By convexity, each $\{f \leq b\}$ is a closed interval and $f \leq \max(f(a), f(a+x))$ on $[a, a+x]$.

(a) A minimiser a gives $[a, a+x] \subset \{f \leq b\}$. Conversely, if $\{f \leq b\} = [u, v]$ with $v - u \geq x$, take $a = u$.

(b) By (a), $\{f^\sharp \leq b\} = [0, \mathcal{L}_f(b)]$ for $b \geq \min f$, and both sets are empty for $b < \min f$. Equal distribution functions give equal integrals via $\int_0^L \varphi = c_0 L + \int_{c_0}^\infty (L - \mathcal{L}_\varphi(b))\, db$.

(c) A minimiser a for $x' \geq x$ is admissible for x, and $\max(f(a), f(a+x)) \leq f^\sharp(x')$. At $x = 0$ every $a \in [0, L]$ is admissible. At $x = L$ only $a = 0$ is admissible, and $\max(f(0), f(L)) = \max f$ by convexity. □

For a continuous convex function F on Δ and a coordinate j put

$$(R_j F)(y, x) = \min_{0\leq a,\ a+x\leq L(y)} \max\bigl(F(y, a), F(y, a+x)\bigr) \qquad ((y, x) \in \Delta),$$

and $\mathcal{R} = R_m \circ \cdots \circ R_1$; for $m = 1$, $\Delta = [0, h]$ this is Definition M4.1. For continuous F on Δ, put $E = \{(y, x, a) : (y, x) \in \Delta,\ a \geq 0,\ a + x \leq L(y)\}$ and $\Phi_F(y, x, a) = \max\bigl(F(y, a), F(y, a+x)\bigr)$. Then $(R_j F)(y, x) = \min_a \Phi_F(y, x, a)$ over $(y, x, a) \in E$; E is compact, Φ_F is continuous, and $(y, a), (y, a+x) \in \Delta$ on E by (2).

Lemma 3.2 (Properties of R_j)**.** *Let F, G be continuous convex functions on Δ and $j \in \{1, \ldots, m\}$.*

(i) For every $y \in \Delta_j$, $x \mapsto (R_j F)(y, x)$ is the increasing rearrangement $f^\sharp$ (Lemma 3.1) of $f = F(y, \cdot)$ on $[0, L(y)]$. It is nondecreasing, $(R_j F)(y, 0) = \min_{[0,L(y)]} F(y, \cdot)$ and $(R_j F)(y, L(y)) = \max_{[0,L(y)]} F(y, \cdot)$.
(ii) $R_j F$ is continuous and convex on Δ.
(iii) $\int_\Delta R_j F = \int_\Delta F$.
(iv) If $F \leq G$ then $R_j F \leq R_j G$. For $c \in \mathbb{R}$ and $\alpha \geq 0$, $R_j(F + c) = R_j F + c$ and $R_j(\alpha F) = \alpha R_j F$; in particular $R_j 0 = 0$. Moreover $\|R_j F - R_j G\| \leq \|F - G\|$, $\max_\Delta R_j F = \max_\Delta F$ and $\min_\Delta R_j F = \min_\Delta F$.
(v) If $i \neq j$ and F is nondecreasing in p_i (that is, $F(p) \leq F(p')$ whenever $p, p' \in \Delta$ differ only in the i-th coordinate and $p_i \leq p_i'$), then so is $R_j F$. Consequently $\mathcal{R}F = R_m \circ \cdots \circ R_1 F$ is nondecreasing in every coordinate, and $\mathcal{R}F(0) = \min_\Delta \mathcal{R}F$.

Proof. (i) By (2), this is Lemma 3.1(a),(c) applied to $F(y, \cdot)$ on $[0, L(y)]$.

(ii) Let $(y_n, x_n) \to (y, x)$. Upper semicontinuity: if a is a minimiser at (y, x), then $a_n = \min(a, L(y_n) - x_n) \to a$ is admissible. Lower semicontinuity: along a subsequence, minimisers $a_n \to a^*$ with $(y, x, a^*) \in E$, since E is closed. Convexity: convex combinations of minimisers are admissible by concavity of L, and convexity of F gives the inequality.

(iii) Apply Fubini over (2), with (i) and Lemma 3.1(b) on each slice.

(iv) Φ is monotone in F, and $\Phi_{F+c} = \Phi_F + c$, $\Phi_{\alpha F} = \alpha \Phi_F$. Apply these to $G - \|F - G\| \leq F \leq G + \|F - G\|$. The max/min statements follow from (i).

(v) Since $b_{\nu i} \geq 0$, we have $L(y + t\mathbf{e}_i) \leq L(y)$ for $t \geq 0$. So a minimiser at $(y + t\mathbf{e}_i, x)$ is admissible at (y, x), and monotonicity of F gives the claim. Inductively, R_j makes the output nondecreasing in p_j (by (i)) and preserves monotonicity in $p_1, \dots, p_{j-1}$. Downward closure then gives $\mathcal{R}F(0) \leq \mathcal{R}F(p)$. □

3.3. **Equal balls in dimension four.** We record the identities and margins used in Proposition M4.4 and §5.

Let $k \geq 10$, $\eta_\star \in (0, \frac{1}{4})$ and $t_0 = 1 - \eta_\star/2$. Let $s = \sqrt{k}$, $a = t_0/s$, $h = t_0/3$, $c_s = 3s - 9$, $g = 3(1 - t_0) = \frac{3}{2}\eta_\star$, $A = h/a$, $B = h/(h - a)$, and let the tent Λ be as in §M4.2. In the equal-capacity branch, $\eta_\star = 1 - \rho\sqrt{k}$ is the volume defect of the normalised capacity ρ (Remark 10.25). Put $K_{\mathrm{lin}}(p) = c_s p$, $P_{\mathrm{b}}(p) = 3(1 - 3p) - k(a - p)_+$, $F = K_{\mathrm{lin}} - P_{\mathrm{b}}$ on $\mathbb{R}$, and $\bar{D}(h) = \{z \in \mathbb{C} : \pi|z|^2 \leq h\}$. By §M4.2: $0 < a < h$, $c_s > 0$, $A, B > 0$, $1/A + 1/B = 1$ (M5), and $F = -g + c_s\Lambda$ (M6).

Lemma 3.3 (The slopes of the tent)**.** *$A + B = AB$, $h/A = a$, $h/B = h - a$, and*

$$c_s A = k - 3s, \qquad c_s B = 3s, \qquad 3sa = 3 - g.$$

Proof. Multiplying $1/A + 1/B = 1$ by AB gives $A + B = AB$; $h/A = a$ and $h/B = h - a$ are the definitions. Using $h - a = t_0(s - 3)/(3s)$,

$$c_s A = (3s - 9)\frac{s}{3} = s^2 - 3s = k - 3s, \qquad c_s B = (3s - 9)\frac{s}{s - 3} = 3s, \qquad 3sa = 3t_0 = 3 - g. \ \square$$

We have $\Lambda(0) = h$, $\Lambda(a) = 0$, $\Lambda(h) = h$, and by Corollary M4.3 the increasing rearrangement of Λ on $[0, h]$ is $x \mapsto x$.

No area-preserving homeomorphism q of $\bar{D}(h)$ satisfies $\Lambda(p(q(z))) \leq \pi|z|^2$ exactly: for $0 < b < h$ it would map $\bar{D}(b)$ into $\{\Lambda(\pi|z|^2) \leq b\}$, which by Lemma M4.2 is the closed annulus $\{a - b/A \leq \pi|z|^2 \leq a + b/B\}$ of area b with $a - b/A > 0$; a compact subset of area b is the whole annulus (nonempty relatively open subsets have positive measure), and a disc is not homeomorphic to an annulus. The inner map of §5 achieves the inequality up to an error $4e_q$; the bound $e_q \leq h/100$ keeps its errors small relative to h.

Definition 3.4 (Inner smoothing parameter)**.** Put

$$e_q = \min\Big(\frac{h}{100},\ \frac{g}{128\, c_s}\Big).$$

Then $0 < e_q \leq h/100$.

Lemma 3.5 (Margin)**.** *$g - 4c_s e_q \geq \frac{31}{32}g$. Consequently, for all real u and x,*

$$\Lambda(x) \leq u + 4e_q \quad \Longrightarrow \quad K_{\mathrm{lin}}(u) - F(x) \geq \tfrac{31}{32}g.$$

In particular, if q satisfies $\Lambda(p(q(z))) \leq \pi|z|^2 + 4e_q$ on $\bar{D}(h)$, then $(K_{\mathrm{lin}} - P_{\mathrm{b}})(p(q(z))) \leq K_{\mathrm{lin}}(\pi|z|^2) - \frac{31}{32}g$ there.

Proof. $4c_s e_q \leq g/32$. By (M6) and $c_s > 0$, $K_{\mathrm{lin}}(u) - F(x) = c_s u + g - c_s\Lambda(x) \geq g - 4c_s e_q \geq \frac{31}{32}g$. The last statement is the case $u = \pi|z|^2$, $x = p(q(z))$. □

4. Area-preserving disc maps from Morse charts

The fibre map of the handle construction (§M5) must be the endpoint of a compactly supported Hamiltonian isotopy of a disc, the identity near the boundary and satisfying a comparison inequality. In [Ope26] such an isotopy comes from a rearrangement lemma by nested discs (credited there to [Sch99, Lemma 3.2]) and Moser's method. Here it is constructed explicitly, for the inner map for equal balls (§5); [Son26] reuses it. From potentials $U(p)$, $V(\theta)$, each with a single

nondegenerate minimum near which it is exactly quadratic, we build a compactly supported Hamiltonian isotopy of the plane preserving a disc of prescribed area, the identity on an outer collar, whose endpoint on a prescribed core disc is an explicit area-preserving map carrying round circles onto level sets of $U + V$. §§4.2–4.4 produce the endpoint: a Morse chart from signed square roots, made area preserving by rotation and area normalisations and placed in the disc by the polar chart (Theorem 4.7). §§4.5–4.10 join it to the identity, unchanged on the core: interpolation of the Morse chart with its linearisation, a five-segment path of embeddings, the cut-off ambient flow, one area normalisation of the whole path, and its radial Hamiltonian (Theorems 4.21 and 4.23).

Conventions. For $P > 0$, $D(P) = \{z \in \mathbb{C} : \pi|z|^2 < P\}$ is the open disc of *area* P, $\bar{D}(P)$ its closure, $D(\infty) = \mathbb{C}$. Area forms: $\omega_{\mathbb{C}}$ on source and target discs, $du \wedge dv$ on the Morse plane $\mathbb{R}^2 \ni w = (u, v) = u + iv$, $dx \wedge d\theta$ on rectangle coordinates (x, θ). A map between open subsets of planes is *symplectic* if its Jacobian determinant is 1. In polar area coordinates $p = \pi|z|^2$, $\vartheta = \arg(z)/2\pi \in \mathbb{R}/\mathbb{Z}$ (the fibre angle θ_j of the main text), $\omega_{\mathbb{C}} = dp \wedge d\vartheta$ on $\mathbb{C} \setminus \{0\}$; under the polar chart C_y of §4.4, $\theta = \vartheta - \frac{1}{2}$. Rot_α is rotation by α radians. S is the flat step of Definition 2.3: by Lemma 2.4 it is C^∞, nondecreasing, 0 on $(-\infty, 0]$, 1 on $[1, \infty)$, with all derivatives vanishing at 0 and 1. E denotes an energy level, J a positive density on a plane, $h > 0$ the area of the preserved disc $D(h)$. A function on an arbitrary $\Omega \subset \mathbb{R}^N$ is *smooth* if near each point of Ω it is the restriction of a C^∞ function on an open subset of $\mathbb{R}^N$ (one-sided at endpoints of parameter intervals). Fibre Hamiltonians follow the convention $\iota_X \omega_{\mathbb{C}} = -dQ$ of §M4.

4.1. **Admissible potential pairs.**

Definition 4.1 (Admissible potential pair)**.** An *admissible potential pair* is the following family: the modified equal-ball wells of §5 ($k \geq 10$, $\frac{7}{8} < t_0 < 1$; Lemma 5.15); [Son26] adds the modified slice wells. Let $\mathcal{Y} \subset \mathbb{R}^d$ be its open parameter domain ($d = 0$ meaning a single point), $h > 0$ its disc area, and $U_y(p)$, $V_y(\theta)$ its potentials on open intervals I_U, I_V. By that lemma:

- (P1) The potentials are jointly smooth. There is a unique $p^*(y) \in I_U \cap (0, h)$ with $U_y'(p)(p - p^*(y)) > 0$ away from the minimum, $U_y''(p^*(y)) > 0$, and $V_y(0) = 0$, $V_y'(\theta)\theta > 0$ away from zero, $V_y''(0) > 0$.
- (P2) Both potentials tend to $+\infty$ at their interval endpoints. Their centred wells are exactly quadratic on neighbourhoods of zero; the widths are positive uniformly on the parameter domain used in a construction. The quadratic coefficients are smooth positive functions of the parameters.

Write $U_{\min}(y) = U_y(p^*(y))$. The pair is *exactly quadratic near the minimum* if there are $\varepsilon_U, \varepsilon_V > 0$ with $U_y(p) - U_{\min}(y) = \frac{1}{2}U_y''(p^*(y))(p - p^*(y))^2$ for $|p - p^*(y)| \leq \varepsilon_U$ and $V_y(\theta) = \frac{1}{2}V_y''(0)\theta^2$ for $|\theta| \leq \varepsilon_V$.

The modified wells need not be convex; only the signs of their derivatives and their quadratic germs are used.

4.2. **Morse inverse charts.**

Lemma 4.2 (Morse inverse charts)**.** *Let (U, V) be an admissible potential pair over $\mathcal{Y}$.*

(i) The minimum point $y \mapsto p^*(y)$ and $y \mapsto U_{\min}(y)$ are C^∞ on $\mathcal{Y}$.

(ii) On the open set $\mathcal{D}_U = \{(y, x) : y \in \mathcal{Y},\ p^*(y) + x \in I_U\}$ put

$$\Sigma_U(y, x) = \mathrm{sign}(x)\sqrt{U_y(p^*(y) + x) - U_{\min}(y)}.$$

(removable singularity at $x = 0$). Σ_U is C^∞ with $\partial_x \Sigma_U > 0$. For each y, $\Sigma_U(y,\cdot)$ is an increasing diffeomorphism of $I_U - p^*(y)$ onto $\mathbb{R}$; its inverse X_y is C^∞ jointly in $(y,u) \in \mathcal{Y} \times \mathbb{R}$, and

$$U_y\big(p^*(y) + X_y(u)\big) - U_{\min}(y) = u^2,$$
$$\operatorname{sign} X_y(u) = \operatorname{sign} u, \qquad X_y'(0) = a_0(y) := \sqrt{2/U_y''(p^*(y))}.$$

(iii) In the same way $\Sigma_V(y,\theta) = \operatorname{sign}(\theta)\sqrt{V_y(\theta)}$ is an increasing diffeomorphism of I_V onto $\mathbb{R}$ for each y, and its inverse Y_y is C^∞ in (y,v), with $V_y(Y_y(v)) = v^2$, $\operatorname{sign} Y_y(v) = \operatorname{sign} v$ and $Y_y'(0) = b_0(y) := \sqrt{2/V_y''(0)}$.

(iv) The Morse chart $F_y(u,v) = (X_y(u), Y_y(v))$ is a C^∞ diffeomorphism of $\mathbb{R}^2$ onto the Morse rectangle $\mathcal{R}_y = (I_U - p^*(y)) \times I_V$, jointly smooth in (y,w), with

$$F_y^*(dx \wedge d\theta) = J_y\, du \wedge dv, \qquad J_y(u,v) = X_y'(u)\, Y_y'(v) > 0, \qquad J_y(0) = j_0(y) := a_0(y) b_0(y),$$

and $U_y(p^*(y)+x) - U_{\min}(y) + V_y(\theta) = u^2 + v^2$ for $(x,\theta) = F_y(u,v)$. Consequently, for $E \geq 0$,

$$\int_{|w|^2 \leq E} J_y(w)\, dw = \operatorname{area}\big\{(p,\theta) \in I_U \times I_V :\ U_y(p) - U_{\min}(y) + V_y(\theta) \leq E\big\},$$

and these areas increase to $|I_U|\,|I_V| \in (0,\infty]$ as $E \to \infty$.

(v) If the pair is exactly quadratic near the minimum, then $X_y(u) = a_0(y)u$ for $|u| \leq \varepsilon_U/a_0(y)$ and $Y_y(v) = b_0(y)v$ for $|v| \leq \varepsilon_V/b_0(y)$; hence $J_y = j_0(y)$ on the disc $|w| \leq \rho_T(y)$, $\rho_T(y) = \min(\varepsilon_U/a_0(y), \varepsilon_V/b_0(y))$.

Proof. (i) As $U_y''(p^*(y)) > 0$, the implicit function theorem and uniqueness in (P1) make p^*, hence $U_{\min}$, smooth.

(ii) Near $x = 0$ the quadratic germ gives $\Sigma_U = \sqrt{U_y''(p^*)/2}\, x$; elsewhere the radicand is positive. For $x \neq 0$, $2\Sigma_U \partial_x \Sigma_U = U_y'(p^* + x)$, both factors having the sign of x, so $\partial_x \Sigma_U > 0$. By (P2), $\Sigma_U(y,\cdot) \to \mp\infty$ at the endpoints, so it is onto $\mathbb{R}$. The map $(y,x) \mapsto (y, \Sigma_U)$ has invertible triangular derivative, so by the inverse function theorem and injectivity $(y,u) \mapsto (y, X_y(u))$ is smooth; the identities follow from $\Sigma_U(y, X_y(u)) = u$.

(iii) is (ii) for V. (iv) F_y is a product of diffeomorphisms with Jacobian $X_y' Y_y' > 0$; summing (ii), (iii) gives the level identity, change of variables the area formula, and exhaustion the limit. (v) On $|x| \leq \varepsilon_U$, $\Sigma_U = x/a_0$, so $X_y(u) = a_0 u$ for $|u| \leq \varepsilon_U/a_0$; likewise for V, and $J_y = a_0 b_0$ for $|w| \leq \rho_T$. □

4.3. **Rotation and area normalisation.** *Setting (N)*: $\mathcal{P} \subset \mathbb{R}^N$ is an arbitrary parameter set and $\mathcal{U} \subset \mathbb{R}^N \times \mathbb{R}^2$ an open set containing $\mathcal{P} \times \mathbb{R}^2$. The density $J\colon \mathcal{U} \to \mathbb{R}$, $(\sigma, w) \mapsto J_\sigma(w)$, is C^∞ and positive on $\mathcal{P} \times \mathbb{R}^2$, with constant germ at $w = 0$: locally in σ there are $\rho_0 > 0$ and a smooth positive $j(\sigma)$ with $J_\sigma(w) = j(\sigma)$ for $|w| \leq \rho_0$. (Applied in §§4.5, 4.8, §9 and in the scalar comparison datum of [Son26].) Points of $\mathbb{R}^2$ are complex numbers, $w\zeta$ the complex product.

Definition 4.3 (Rotation and area normalisation)**.** In setting (N), for $\sigma \in \mathcal{P}$, $w \in \mathbb{R}^2$ and $E \geq 0$ put

$$\bar{J}_\sigma(|w|^2) = \frac{1}{2\pi}\int_0^{2\pi} J_\sigma(\mathrm{Rot}_\alpha w)\, d\alpha, \qquad \mathcal{M}_\sigma(w) = \frac{1}{2\pi}\int_0^{2\pi} (\alpha-\pi)\, J_\sigma(\mathrm{Rot}_\alpha w)\, d\alpha,$$
$$T_\sigma(w) = \mathrm{Rot}_{\tau_\sigma(w)}\, w, \qquad \tau_\sigma(w) = \mathcal{M}_\sigma(w)/\bar{J}_\sigma(|w|^2),$$
$$\mathcal{A}_\sigma(E) = \int_{|w|^2\leq E} J_\sigma(w)\, dw, \qquad \mathcal{A}_\sigma(\infty) = \lim_{E\to\infty} \mathcal{A}_\sigma(E) \in (0,\infty],$$
$$\mathcal{G}_\sigma(w) = \frac{1}{\pi}\int_{|\zeta|\leq 1} J_\sigma(w\zeta)\, d\zeta.$$

Let $E_\sigma\colon [0,\mathcal{A}_\sigma(\infty)) \to [0,\infty)$ be the inverse of $\mathcal{A}_\sigma$ (Lemma 4.5) and, for $z \in D(\mathcal{A}_\sigma(\infty))$ with $I = \pi|z|^2$,

$$D_\sigma(z) = \sqrt{\pi E_\sigma(I)/I}\; z \quad (z \neq 0), \qquad D_\sigma(0) = 0.$$

T_σ is the *rotation normalisation* and D_σ the *area normalisation* of J_σ. For a single density J we write $\bar{J}$, $\mathcal{M}_J$, T_J, $\mathcal{A}_J$, E_J, $\mathcal{G}_J$, D_J.

At $|w| \leq R$, or z with $E_\sigma(\pi|z|^2) \leq R^2$, the normalisations depend only on $J_\sigma|_{\{|w|\leq R\}}$, so there they coincide with those of the smooth positive density

$$J^{\mathrm{ext}}_\sigma(w) = \vartheta_R(w) J_\sigma(w) + (1-\vartheta_R(w)) j(\sigma), \qquad \vartheta_R(w) = 1 - S\left(\frac{|w|^2-R^2}{3R^2}\right),$$

which has the same germ, equals J_σ for $|w| \leq R$, and has $\mathcal{A}_{J^{\mathrm{ext}}_\sigma}(\infty) = \infty$. T_σ carries the measure of J_σ on each circle to the uniform one, the rotation fixed by the base-point-free mean $\mathcal{M}_\sigma$; $\mathcal{G}_\sigma$ makes D_σ smooth at the origin.

Lemma 4.4 (Rotation normalisation)**.** *In setting (N), let $\sigma \in \mathcal{P}$.*

(i) $\partial_{\arg}\mathcal{M}_\sigma = J_\sigma - \bar{J}_\sigma(|w|^2)$*, where $\partial_{\arg}$ is the derivative along ψ in $w = \rho e^{i\psi}$.*

(ii) T_σ *maps each circle $\{|w| = \rho\}$ onto itself; for $\rho > 0$, writing $T_\sigma(\rho e^{i\psi}) = \rho e^{i\psi^\sharp(\psi)}$, one has $(\psi^\sharp)' = J_\sigma/\bar{J}_\sigma > 0$ and $\psi^\sharp(\psi+2\pi) = \psi^\sharp(\psi) + 2\pi$. T_σ is a bijection of $\mathbb{R}^2$ and $T_\sigma(0) = 0$.*

(iii) The maps $(\sigma, w) \mapsto T_\sigma(w)$ and $(\sigma, w) \mapsto T_\sigma^{-1}(w)$ are smooth on $\mathcal{P}\times\mathbb{R}^2$, each T_σ is a diffeomorphism of $\mathbb{R}^2$ with $DT_\sigma(0) = \mathrm{Id}$, and $T_\sigma^*\big(\bar{J}_\sigma(|w|^2)\, du\wedge dv\big) = J_\sigma\, du\wedge dv$ on all of $\mathbb{R}^2$.

(iv) If $J_\sigma(w) = j$ for $|w| \leq \rho_0$, then $\mathcal{M}_\sigma = 0$ and $T_\sigma = T_\sigma^{-1} = \mathrm{id}$ on $\{|w| \leq \rho_0\}$.

Proof. For $\sigma_0 \in \mathcal{P}$, $\rho_1 > 0$ there is an open ball $N_0 \ni \sigma_0$ with $N_0 \times B(0,\rho_1) \subset \mathcal{U} \cap \{J > 0\}$; differentiating under the integral, $\bar{J} > 0$ and $\mathcal{M}$ are C^∞ there, giving a local C^∞ extension $\tilde{T}$ of T.

(i) Integrate by parts in $\mathcal{M}_\sigma(\rho e^{i\psi}) = \frac{1}{2\pi}\int_0^{2\pi}(\alpha-\pi) J_\sigma(\rho e^{i(\psi+\alpha)})\, d\alpha$. (ii) By (i), $\psi^\sharp = \psi + \mathcal{M}_\sigma/\bar{J}_\sigma$ has $(\psi^\sharp)' = J_\sigma/\bar{J}_\sigma > 0$, and is periodic since $\mathcal{M}_\sigma$ is single-valued.

(iii) $\mathcal{M}_\sigma(0) = 0$ gives $DT_\sigma(0) = \mathrm{Id}$. In polar coordinates (ii) gives the pull-back identity, so $\det DT_\sigma = J_\sigma/\bar{J}_\sigma > 0$ (at 0 by continuity) and T_σ is a diffeomorphism. For joint smoothness of T_σ^{-1}, the local C^∞ inverse of $(\sigma, w) \mapsto (\sigma, \tilde{T}(\sigma, w))$ near $(\sigma_0, T_{\sigma_0}^{-1}(x_0))$ equals $(\sigma, T_\sigma^{-1}(x))$ for $\sigma \in \mathcal{P}$ by injectivity.

(iv) For $|w| \leq \rho_0$ all $\mathrm{Rot}_\alpha w$ lie in the disc, so $\mathcal{M}_\sigma = 0$, $T_\sigma = \mathrm{id}$, and $T_\sigma^{-1} = \mathrm{id}$ as T_σ preserves circles. □

Lemma 4.5 (Area normalisation)**.** *In setting (N), let $\sigma \in \mathcal{P}$ and put $\Omega = \{(\sigma, z) : \sigma \in \mathcal{P},\ \pi|z|^2 < \mathcal{A}_\sigma(\infty)\}$.*

(i) $\mathcal{A}_\sigma(E) = \pi\int_0^E \bar J_\sigma(t)\,dt$; *$\mathcal{A}_\sigma$ is a continuous strictly increasing bijection of $[0,\infty)$ onto $[0,\mathcal{A}_\sigma(\infty))$, so E_σ is well defined.*

(ii) *$(\sigma,w)\mapsto \mathcal{G}_\sigma(w)$ is smooth and positive on $\mathcal{P}\times\mathbb{R}^2$, $\mathcal{G}_\sigma(0)=J_\sigma(0)$, and $\mathcal{A}_\sigma(|w|^2)=\pi|w|^2\mathcal{G}_\sigma(w)$.*

(iii) *$\Psi_\sigma(w)=\sqrt{\mathcal{G}_\sigma(w)}\,w$ is a diffeomorphism of $\mathbb{R}^2$ onto $D(\mathcal{A}_\sigma(\infty))$ with $\pi|\Psi_\sigma(w)|^2=\mathcal{A}_\sigma(|w|^2)$ and $\Psi_\sigma^*\omega_{\mathbb{C}}=\bar J_\sigma(|w|^2)\,du\wedge dv$; its inverse is D_σ. The map $(\sigma,z)\mapsto D_\sigma(z)$ is smooth on Ω, including $z=0$; $|D_\sigma(z)|^2=E_\sigma(\pi|z|^2)$, $D_\sigma^*(\bar J_\sigma(|w|^2)\,du\wedge dv)=\omega_{\mathbb{C}}$, $DD_\sigma(0)=J_\sigma(0)^{-1/2}\,\mathrm{Id}$, and the quotient in the definition of D_σ is*

$$\frac{\pi E_\sigma(\pi|z|^2)}{\pi|z|^2}=\frac{1}{\mathcal{G}_\sigma(D_\sigma(z))}\qquad(z\neq 0),$$

a smooth positive function of $(\sigma,z)\in\Omega$ with value $1/J_\sigma(0)$ at $z=0$.

(iv) *(Composite.) Suppose $(\sigma,w)\mapsto F_\sigma(w)$ is C^∞ on an open neighbourhood of $\mathcal{P}\times\mathbb{R}^2$ and each F_σ is a diffeomorphism of $\mathbb{R}^2$ onto an open set $\mathcal{V}_\sigma$ with $F_\sigma^*(dx\wedge d\theta)=J_\sigma\,du\wedge dv$. Then $f_\sigma=F_\sigma\circ T_\sigma^{-1}\circ D_\sigma$ is a symplectic diffeomorphism of $D(\mathcal{A}_\sigma(\infty))$ onto $\mathcal{V}_\sigma$, $(\sigma,z)\mapsto f_\sigma(z)$ is smooth on Ω, f_σ maps the circle $\{\pi|z|^2=I\}$ onto $F_\sigma(\{|w|^2=E_\sigma(I)\})$, $f_\sigma(0)=F_\sigma(0)$ and $Df_\sigma(0)=J_\sigma(0)^{-1/2}DF_\sigma(0)$.*

(v) *(Constant near the origin.) If $J_\sigma(w)=j$ for $|w|\le\rho_0$, then $\mathcal{G}_\sigma=j$ there, $\mathcal{A}_\sigma(E)=\pi jE$ for $E\le\rho_0^2$, $D_\sigma(z)=z/\sqrt{j}$ for $|z|\le\sqrt{j}\,\rho_0$, and in (iv) $f_\sigma(z)=F_\sigma(z/\sqrt{j})$ for $|z|\le\sqrt{j}\,\rho_0$.*

Proof. (i) is polar coordinates and $J_\sigma>0$. (ii) $\zeta\mapsto w\zeta$ maps the unit disc onto $\{|x|\le|w|\}$ with determinant $|w|^2$; with N_0,ρ_1 as above, the integrand is C^∞ and positive on $N_0\times B(0,\rho_1)\times\bar B(0,1)$, giving a positive C^∞ extension $\tilde{\mathcal{G}}$.

(iii) $\Psi_\sigma(\rho e^{i\psi})=r(\rho)e^{i\psi}$ with $\pi r^2=\mathcal{A}_\sigma(\rho^2)$ strictly increasing from 0 to $\mathcal{A}_\sigma(\infty)$, so Ψ_σ is a bijection onto $D(\mathcal{A}_\sigma(\infty))$. By (i), $\Psi_\sigma^*\omega_{\mathbb{C}}=\frac{1}{2\pi}d(\pi r^2)\wedge d\psi=\bar J_\sigma\,du\wedge dv$, so $\det D\Psi_\sigma=\bar J_\sigma>0$ (at 0 by continuity, $D\Psi_\sigma(0)=\sqrt{\mathcal{G}_\sigma(0)}\,\mathrm{Id}$) and Ψ_σ is a diffeomorphism; its inverse is radial, mapping $\{\pi|z|^2=I\}$ onto $\{|w|^2=E_\sigma(I)\}$, i.e. it is D_σ, and the stated identities follow. Joint smoothness: the local C^∞ inverse of $(\sigma,w)\mapsto(\sigma,\sqrt{\tilde{\mathcal{G}}(\sigma,w)}\,w)$ (Jacobian $\bar J_{\sigma_0}(|w_0|^2)>0$) equals $(\sigma,D_\sigma(z))$ on Ω by injectivity; and $\Psi_\sigma(D_\sigma(z))=z$ gives the quotient formula.

(iv) By Lemma 4.4(iii) and (iii), $f_\sigma^*(dx\wedge d\theta)=D_\sigma^*(T_\sigma^{-1})^*(J_\sigma\,du\wedge dv)=D_\sigma^*(\bar J_\sigma\,du\wedge dv)=\omega_{\mathbb{C}}$; circles are mapped as stated since T_σ^{-1} preserves them, and the values at 0 follow from Lemma 4.4(iii).

(v) For $|w|\le\rho_0$, $|\zeta|\le1$, $|w\zeta|\le\rho_0$, so $\mathcal{G}_\sigma(w)=j$ and $\Psi_\sigma(w)=\sqrt{j}\,w$; the last claim is Lemma 4.4(iv). □

4.4. **The inner map.** For $y\in\mathcal{Y}$ let

$$C_y(x,\theta)=-\sqrt{\frac{p^*(y)+x}{\pi}}\,e^{2\pi i\theta},\qquad(x,\theta)\in\mathcal{C}_y=\{p^*(y)+x>0,\ |\theta|<\tfrac12\}.\tag{3}$$

Its polar angle is $2\pi\theta+\pi\in(0,2\pi)$, so C_y is injective with image $\mathbb{C}\setminus[0,\infty)$, $\pi|C_y(x,\theta)|^2=p^*(y)+x$ and $C_y^*(dp\wedge d\vartheta)=dx\wedge d\theta$; the centre line $\theta=0$ maps to the negative real axis, and $C_y(0)=b_y:=(-r_b(y),0)$, $r_b(y)=\sqrt{p^*(y)/\pi}$.

Definition 4.6 (The inner map)**.** Let (U,V) be an admissible potential pair over $\mathcal{Y}$ and $y\in\mathcal{Y}$. Apply the formulas of Definition 4.3 to the density $J_y=X_y'Y_y'$ of Lemma 4.2 (parameters $\sigma=y\in\mathcal{P}=\mathcal{Y}$), and write T_y, D_y, $\mathcal{A}_y$, E_y for its normalisations. Put

$$f_y=F_y\circ T_y^{-1}\circ D_y\colon D(\mathcal{A}_y(\infty))\to\mathcal{R}_y.$$

Let $E_{\mathrm{rect}}(y) \in (0,\infty]$ be the infimum of $U_y(p)-U_{\min}(y)+V_y(\theta)$ over $(I_U\times I_V)\setminus\big((0,h)\times(-\frac12,\frac12)\big)$ (and $+\infty$ if this set is empty), let $I_{\mathrm{rect}}(y)=\mathcal{A}_y(E_{\mathrm{rect}}(y))$ (with $\mathcal{A}_y(+\infty)=\mathcal{A}_y(\infty)$), and define the *inner map*

$$q_{\mathrm{in},y} = C_y \circ f_y \colon D\big(I_{\mathrm{rect}}(y)\big) \to \mathbb{C}.$$

We also put $c_x(y)=\sqrt{a_0(y)/b_0(y)}$ and $\mathbf{L}_y=\mathrm{diag}(c_x(y),c_x(y)^{-1})$.

By (P1), $E_{\mathrm{rect}}(y)$ is the least of $U_y(0)-U_{\min}$, $U_y(h)-U_{\min}$, $V_y(\pm\frac12)$ over those of $0,h$ in I_U and $\pm\frac12$ in I_V; it is positive since $p^*(y)\in(0,h)$. By Lemma 4.2(iv), $\mathcal{A}_y(E)$ is the area of $\{U_y-U_{\min}+V_y\le E\}$ and $\mathcal{A}_y(\infty)=|I_U|\,|I_V|$; for $E<E_{\mathrm{rect}}(y)$ this sublevel lies in $(0,h)\times(-\frac12,\frac12)$, so $I_{\mathrm{rect}}(y)\le h$.

Theorem 4.7 (The inner map)**.** *Let (U,V) be an admissible potential pair. Put $\mathcal{D}_y^{\mathrm{src}}=\mathbb{C}$ (the equal-ball family).*

(a) For each y, f_y is a symplectic embedding of $\mathcal{D}_y^{\mathrm{src}}$ into the Morse rectangle $\mathcal{R}_y$, and $(y,z)\mapsto f_y(z)$ is smooth on the corresponding source set. In the equal-ball case it is a diffeomorphism onto $\mathcal{R}_y=\mathbb{R}^2$. Moreover $f_y(0)=0$ and $Df_y(0)=\mathbf{L}_y$, a diagonal matrix of determinant 1.

(b) (Energy identity.) If $f_y(z)=(x,\theta)$, then

$$U_y\big(p^*(y)+x\big)+V_y(\theta)=U_{\min}(y)+E_y(\pi|z|^2),$$

so f_y maps each circle $\{\pi|z|^2=I\}$ onto the level curve of $U_y(p^*+\cdot)-U_{\min}+V_y$ which encloses area I.

(c) The inner map $q_{\mathrm{in},y}$ is a symplectic embedding of $D(I_{\mathrm{rect}}(y))\cap\mathcal{D}_y^{\mathrm{src}}$ into $D(h)\setminus[0,\infty)$, jointly smooth in (y,z), with $q_{\mathrm{in},y}(0)=b_y$, $Dq_{\mathrm{in},y}(0)=\mathbf{A}_y:=DC_y(0)\,\mathbf{L}_y$, and, for $f_y(z)=(x,\theta)$,

$$\pi|q_{\mathrm{in},y}(z)|^2=p^*(y)+x,\qquad U_y\big(\pi|q_{\mathrm{in},y}(z)|^2\big)+V_y(\theta)=U_{\min}(y)+E_y(\pi|z|^2).$$

In particular $U_y(\pi|q_{\mathrm{in},y}(z)|^2)\le U_{\min}(y)+E_y(\pi|z|^2)$, with equality on the centre line $\theta=0$.

(d) If the pair is exactly quadratic near the minimum, then $f_y(z)=\mathbf{L}_y z$ for $|z|\le\sqrt{j_0(y)}\,\rho_T(y)$.

Proof. (a) Apply Lemma 4.5(iv) with $\mathcal{P}=\mathcal{Y},\mathcal{U}=\mathcal{Y}\times\mathbb{R}^2$, the density J_y and F_y (Lemma 4.2(iv)), whose image is $\mathcal{R}_y$; also $f_y(0)=0$ and $Df_y(0)=j_0^{-1/2}\,\mathrm{diag}(a_0,b_0)=\mathbf{L}_y$.

(b) $(u,v)=T_y^{-1}D_y(z)$ satisfies $u^2+v^2=E_y(\pi|z|^2)$, since T_y^{-1} preserves $|w|$ (Lemmas 4.4(ii), 4.5(iii)); apply the level identity of Lemma 4.2(iv).

(c) If $\pi|z|^2<I_{\mathrm{rect}}(y)$ then $E_y(\pi|z|^2)<E_{\mathrm{rect}}(y)$, so by (b) $(p^*+x,\theta)\in(0,h)\times(-\frac12,\frac12)$; hence $(x,\theta)\in\mathcal{C}_y$, $0<\pi|q_{\mathrm{in},y}(z)|^2<h$, and $q_{\mathrm{in},y}$ is a composite of symplectic embeddings, smooth in (y,z). The value and derivative at 0 come from (a), the identities from the moment formula for C_y and (b), and equality on $\theta=0$ from $V_y(0)=0$.

(d) By Lemmas 4.2(v) and 4.5(v), $f_y(z)=F_y(z/\sqrt{j_0})$ with $F_y(w)=(a_0u,b_0v)$ for $|z|\le\sqrt{j_0}\rho_T$, so $f_y(z)=\mathbf{L}_yz$. □

From (3), $DC_y(0)=\mathrm{diag}\big(-1/(2\pi r_b(y)),\,-2\pi r_b(y)\big)$, so

$$\mathbf{A}_y=\mathrm{diag}\Big(-\frac{c_x(y)}{2\pi r_b(y)},\,-\frac{2\pi r_b(y)}{c_x(y)}\Big),\qquad \det\mathbf{A}_y=1. \tag{4}$$

4.5. **Density interpolation and the charts C_λ.**

Definition 4.8 (Density family, rescaled charts and schedule)**.** Let (U,V) be an admissible potential pair over $\mathcal{Y}$.

(a) For $y\in\mathcal{Y}$ and $s\in\mathbb{R}$ put

$$X_{y,s}(u)=(1-s)a_0(y)u+sX_y(u),\quad Y_{y,s}(v)=(1-s)b_0(y)v+sY_y(v),\quad J_{y,s}(u,v)=X'_{y,s}(u)\,Y'_{y,s}(v),$$

$F_{y,s}=(X_{y,s},Y_{y,s})$, and for $s\in[0,1]$ let $T_{y,s}$, $D_{y,s}$, $\mathcal{A}_{y,s}$, $E_{y,s}$ be the normalisations of $J_{y,s}$ (Definition 4.3 with parameters $(y,s)\in\mathcal{Y}\times[0,1]$), and $f_{y,s}=F_{y,s}\circ T_{y,s}^{-1}\circ D_{y,s}$ on $D(\mathcal{A}_{y,s}(\infty))$.

(b) For (y,λ,w) in the open set $\mathcal{W}=\{(y,\lambda,w)\in\mathcal{Y}\times\mathbb{R}\times\mathbb{R}^2:\ \lambda w\in\mathcal{C}_y\}$ put

$$C_{y,\lambda}(w)=b_y+\int_0^1 DC_y(t\lambda w)\,w\,dt.$$

(c) For numbers $\lambda_0\in(0,\frac12]$ and $\delta_s\in(0,\frac14]$, the *schedule* is

$$\lambda(s)=\lambda_0\,S(2s)+(1-\lambda_0)\,S\Big(\frac{s-1+\delta_s}{\delta_s}\Big),\qquad s\in[0,1].$$

As $\mathcal{C}_y$ is convex and contains 0, the integrand in (b) is defined and $C_{y,\lambda}$ is C^∞ on $\mathcal{W}$. As $\delta_s\le\frac14$, the schedule is smooth with $\lambda(0)=0$, $\lambda(1)=1$, $\lambda(s)=\lambda_0 S(2s)\le\lambda_0$ for $s\le1-\delta_s$ (the *early range*), $\lambda=\lambda_0$ on $[\frac12,1-\delta_s]$, and $\lambda(s)\in[\lambda_0,1]$ for $s\ge1-\delta_s$ (the *late range*).

Lemma 4.9 (The rescaled charts)**.** *For $(y,\lambda,w)\in\mathcal{W}$: if $\lambda\neq0$ then $C_{y,\lambda}(w)=b_y+[C_y(\lambda w)-b_y]/\lambda$; $C_{y,0}(w)=b_y+DC_y(0)w$; $DC_{y,\lambda}(w)=DC_y(\lambda w)$, so $\det DC_{y,\lambda}=1$; and $C_{y,\lambda}$ is injective on $\{w:\lambda w\in\mathcal{C}_y\}$.*

Proof. For $\lambda\neq0$, divide $C_y(\lambda w)-C_y(0)=\int_0^1 DC_y(t\lambda w)\lambda w\,dt$ by λ; the w-derivative is $DC_y(\lambda w)$, and at $\lambda=0$ the map is affine with linear part $DC_y(0)$. Injectivity follows from that of C_y on $\mathcal{C}_y$, and from $\det DC_y(0)=1$ when $\lambda=0$. □

Equivalently, with $g(\zeta)=(e^\zeta-1)/\zeta$ ($g(0)=1$) and $w=(u,v)$,

$$C_{y,\lambda}(u,v)=b_y-2\pi r_b(y)\,i\,v\,g(2\pi i\lambda v)-\frac{u\,e^{2\pi i\lambda v}}{\pi\big(r_b(y)+\sqrt{(p^*(y)+\lambda u)/\pi}\big)}\qquad\text{on }\mathcal{W},$$

obtained for $\lambda\neq0$ by rationalising the second bracket in $C_y(\lambda w)-b_y=r_b(1-e^{2\pi i\lambda v})+\big(r_b-\sqrt{(p^*+\lambda u)/\pi}\big)e^{2\pi i\lambda v}$, and equal to $b_y+DC_y(0)w$ for $\lambda=0$.

Lemma 4.10 (The density family and the uniform energy bound)**.** *Let (U,V) be an admissible potential pair over $\mathcal{Y}$, $y\in\mathcal{Y}$, $s\in[0,1]$, and abbreviate $a_0=a_0(y)$, $b_0=b_0(y)$, $j_0=a_0b_0$.*

(a) $X_{y,s},Y_{y,s}$ are C^∞ and strictly increasing, and

$$J_{y,s}=(1-s)^2j_0+s(1-s)\big(a_0Y'_y+b_0X'_y\big)+s^2J_y\ \ge\ (1-s)^2j_0+s^2J_y>0,\qquad J_{y,s}(0)=j_0.$$

(b) For $E\ge0$,

$$\mathcal{A}_{y,s}(E)\ \ge\ \pi j_0(1-s)^2E+s^2\mathcal{A}_y(E)\ \ge\ \mathcal{A}_y^-(E):=\frac{\pi j_0E\,\mathcal{A}_y(E)}{\pi j_0E+\mathcal{A}_y(E)}\quad(E>0).$$

In particular $\mathcal{A}_{y,s}(\infty)=\infty$ for $s<1$, and $\mathcal{A}_{y,1}(\infty)=\mathcal{A}_y(\infty)$.

(c) (Uniform energy bound.) $\mathcal{A}_y^-$ is continuous and strictly increasing. If $\bar E>0$ and $\mathcal{A}_{y,s}(\bar E)\ge p^\dagger$ for every $s\in[0,1]$ (this holds for the explicit $\bar E$ of Lemma 5.29(d)), then $E_{y,s}(I)\le\bar E$ for $0\le I\le p^\dagger$ and $s\in[0,1]$.

(d) $f_{y,s}$ is a symplectic embedding on the source domain $\mathcal{D}_y^{\mathrm{src}}$ of Theorem 4.7; the map $(y,s,z)\mapsto f_{y,s}(z)$ is smooth for $s\in[0,1]$ and $z\in\mathcal{D}_y^{\mathrm{src}}$. In the equal-ball case it is a diffeomorphism onto $\mathbb{R}^2$; $f_{y,s}(0)=0$ and $Df_{y,s}(0)=\mathbf{L}_y$ for every s; $f_{y,0}=\mathbf{L}_y$ and $f_{y,1}=f_y$. If $(u,v)=T_{y,s}^{-1}D_{y,s}(z)$, then $f_{y,s}(z)=(X_{y,s}(u),Y_{y,s}(v))$ and $u^2+v^2=E_{y,s}(\pi|z|^2)$.

(e) If the pair is exactly quadratic near the minimum, then $J_{y,s}=j_0$ on $|w|\le\rho_T(y)$ and $f_{y,s}(z)=\mathbf{L}_y z$ for $|z|\le\sqrt{j_0}\,\rho_T(y)$, for every $s\in[0,1]$.

Proof. (a) $X'_{y,s}=(1-s)a_0+sX'_y>0$, $Y'_{y,s}=(1-s)b_0+sY'_y>0$; expanding the product gives the formula, with nonnegative middle term, and $X'_y(0)=a_0$, $Y'_y(0)=b_0$.

(b) Integrate (a) over $\{|w|^2\le E\}$ and use $\min_s\bigl(\alpha(1-s)^2+\beta s^2\bigr)=\alpha\beta/(\alpha+\beta)$ for $\alpha,\beta>0$. For $s<1$ the first term is unbounded in E; $J_{y,1}=J_y$.

(c) $1/\mathcal{A}_y^-(E)=1/(\pi j_0E)+1/\mathcal{A}_y(E)$. Since $\mathcal{A}_{y,s}$ is strictly increasing, $\mathcal{A}_{y,s}(\bar E)\ge p^\dagger\ge I$ gives $E_{y,s}(I)\le\bar E$.

(d) Setting (N) applies: $J_{y,s}$ is C^∞ on $\mathcal{U}=\mathcal{Y}\times\mathbb{R}\times\mathbb{R}^2$, positive on $\mathcal{P}\times\mathbb{R}^2$ with $\mathcal{P}=\mathcal{Y}\times[0,1]$, with the common constant germ. $F_{y,s}$ is a diffeomorphism of $\mathbb{R}^2$ onto $X_{y,s}(\mathbb{R})\times Y_{y,s}(\mathbb{R})$ with Jacobian $J_{y,s}$, both factors being onto $\mathbb{R}$ for $s<1$ since $(1-s)a_0u$, $(1-s)b_0v$ are unbounded and X_y,Y_y increasing. Lemma 4.5(iv) gives all but the last three assertions, with $Df_{y,s}(0)=\mathbf{L}_y$. For $s=0$ the density is constant, so $T_{y,0}=\mathrm{id}$, $D_{y,0}=j_0^{-1/2}\mathrm{id}$, $f_{y,0}=\mathbf{L}_y$; clearly $f_{y,1}=f_y$; the last assertion is the computation in the proof of Theorem 4.7(b).

(e) For $|w|\le\rho_T$, Lemma 4.2(v) gives $X'_{y,s}=a_0$, $Y'_{y,s}=b_0$, so $J_{y,s}=j_0$; conclude as in Theorem 4.7(d). □

Definition 4.11 (Isotopy data)**.** Let (U,V) be an admissible potential pair over $\mathcal{Y}$ with disc area h, and consider positive numbers, independent of y,

$$p_{\mathrm{on}},\ p_{\mathrm{zero}},\ p^\dagger,\ p^*_{\mathrm{lo}},\ p^*_{\mathrm{hi}},\ E_c,\ \Delta P,\ \Delta\theta,\ R_q,\ \delta_x,\ \delta_\theta,\ \delta_s,\ \bar E,\ \bar X,\ \bar Y,\ \lambda_0,\ P_{\min},\ P_{\max},\ B_{\mathrm{lin}},\ B^*,\ c_{\mathrm{sh}},$$

where the radial range bounds satisfy

$$P_{\min}\le\tfrac12\min(p^*_{\mathrm{lo}},\Delta P),\qquad P_{\max}\ge\max(\tfrac32 p^*_{\mathrm{hi}},h).$$

From these numbers we form the *derived constants*

$$M_C=\frac{1}{2\sqrt{\pi P_{\min}}}+2\sqrt{\pi P_{\max}},$$
$$R^\dagger=\sqrt{p^\dagger/\pi},\quad R_h=\sqrt{h/\pi},\quad r_b^+=\sqrt{p^*_{\mathrm{hi}}/\pi},$$
$$R_{\mathrm{supp}}=\max\bigl(R^\dagger,\ r_b^+ + c_{\mathrm{sh}}B_{\mathrm{lin}},\ r_b^+ + c_{\mathrm{sh}}B^*,\ R_q\bigr),\quad P_{\mathrm{supp}}=\pi R_{\mathrm{supp}}^2,\quad p_{\mathrm{bd}}=\tfrac12(h+P_{\mathrm{supp}}).$$

The numbers are *isotopy data* for the pair if, for every $y\in\mathcal{Y}$:

(D1) $p_{\mathrm{on}}<p_{\mathrm{zero}}<p^\dagger$ and $p^*_{\mathrm{lo}}\le p^*(y)\le p^*_{\mathrm{hi}}<h$;

(D2) (endpoint margins) for all u,v with $u^2<E_c$, $v^2<E_c$:

$$\Delta P\le p^*(y)+X_y(u)\le h-\Delta P,\qquad p^*(y)+X_y(u)\le\pi R_q^2<h,\qquad |Y_y(v)|\le\tfrac12-\Delta\theta;$$

(D3) (interpolation error) $|a_0(y)u-X_y(u)|\le\delta_x$ and $|b_0(y)v-Y_y(v)|\le\delta_\theta$ for $u^2<E_c$, $v^2<E_c$;

(D4) (late range) $\delta_s\le\frac14$, $\delta_s\delta_x\le\frac12\Delta P$, $\delta_s\delta_\theta\le\frac12\Delta\theta$ and either $(1-\delta_s)^2\mathcal{A}_y(E_c)>p^\dagger$, or the uniform-energy alternative $\mathcal{A}_{y,s}(E_c)>p^\dagger$ for all $s\in[0,1]$;

(D5) (uniform energy) $\mathcal{A}_{y,s}(\bar E)\ge p^\dagger$ for all $s\in[0,1]$, and $\max(a_0(y)|u|,|X_y(u)|)\le\bar X$, $\max(b_0(y)|v|,|Y_y(v)|)\le\bar Y$ for $u^2\le\bar E$, $v^2\le\bar E$;

(D6) (early range) $\lambda_0\le\frac12$, $\lambda_0\bar X\le\frac12 p^*_{\mathrm{lo}}$ and $\lambda_0\bar Y\le\frac14$;

(D7) (ranges of the path)

$$R^\dagger \max\bigl(1, |(\mathbf{A}_y)_{11}|, |(\mathbf{A}_y)_{22}|\bigr) \le B_{\mathrm{lin}}, \qquad B^* \ge \min\Bigl(M_C\sqrt{\bar X^2+\bar Y^2},\ \frac{\bar X}{\sqrt{\pi p^*_{\mathrm{lo}}}} + 2\pi r_b^+ \bar Y\Bigr),$$

$$c_{\mathrm{sh}} \le \min\Bigl(\frac14,\ \frac{R_h - r_b^+}{8\max(R^\dagger, B_{\mathrm{lin}}, B^*)}\Bigr).$$

The schedule $\lambda(s)$ of Definition 4.8(c) is formed with these λ_0 and δ_s.

Lemma 4.12 (Existence and order of the data)**.** *(a) The equal-ball potential family has the explicit isotopy data given in §5.6.*

(b) For any isotopy data: $0 < \delta_s \le \frac14$, $p^\dagger < \mathcal{A}_y(E_c) < h$, $E_c \le E_{\mathrm{rect}}(y)$ and $p^\dagger < I_{\mathrm{rect}}(y)$ for every y, and

$$R_{\mathrm{supp}} < R_h, \qquad \textit{i.e.} \qquad p_{\mathrm{on}} < p^\dagger \le P_{\mathrm{supp}} < p_{\mathrm{bd}} < h.$$

Proof. (a) The positivity and source order are Lemma 5.28, the pointwise and area gates are Lemma 5.29, and the range and shrinking constants are Definition 5.32, assembled in Lemma 5.33.

(b) $\delta_s > 0$ is part of the data. $\mathcal{A}_y(E_c) > p^\dagger$ follows from either alternative of (D4), taking $s = 1$ in the uniform-energy case. If $|w|^2 \le E_c'$ for some $E_c' < E_c$, then $u^2, v^2 < E_c$, so by (D2) $F_y(w)$ lies in $[\Delta P - p^*, h - \Delta P - p^*] \times [-\frac12 + \Delta\theta, \frac12 - \Delta\theta]$; letting $E_c' \uparrow E_c$, $\mathcal{A}_y(E_c) \le (h - 2\Delta P)(1 - 2\Delta\theta) < h$. The same inclusion shows that every point of $\{U_y - U_{\min} + V_y < E_c\}$ lies in $(0,h) \times (-\frac12, \frac12)$, so $E_c \le E_{\mathrm{rect}}(y)$, and then $I_{\mathrm{rect}}(y) \ge \mathcal{A}_y(E_c) > p^\dagger$. Consequently $R^\dagger < R_h$. By (D1), $r_b^+ < R_h$, and (D7) gives $r_b^+ + c_{\mathrm{sh}}\max(B_{\mathrm{lin}}, B^*) \le r_b^+ + \frac18(R_h - r_b^+) < R_h$. By (D2), $\pi R_q^2 < h$. Hence $R_{\mathrm{supp}} < R_h$, $P_{\mathrm{supp}} < h$, and $P_{\mathrm{supp}} \ge \pi(R^\dagger)^2 = p^\dagger$. □

Proposition 4.13 (The common inner family)**.** *Let (U, V) be an admissible potential pair over $\mathcal{Y}$ with isotopy data. For $y \in \mathcal{Y}$, $s \in [0,1]$ and $z \in \bar D(p^\dagger)$ write $(u, v) = T_{y,s}^{-1} D_{y,s}(z)$.*

(a) (A priori bounds.) $u^2 + v^2 = E_{y,s}(\pi|z|^2) \le \bar E$, $|X_{y,s}(u)| \le \bar X$ and $|Y_{y,s}(v)| \le \bar Y$.

(b) (Late margins.) If $s \ge 1 - \delta_s$, then $u^2 + v^2 < E_c$ and

$$\tfrac12\Delta P \le p^*(y) + X_{y,s}(u) \le h - \tfrac12\Delta P, \qquad |Y_{y,s}(v)| \le \tfrac12 - \tfrac12\Delta\theta.$$

(c) (The family.) For $t \in [0,1]$ the point $t\lambda(s) f_{y,s}(z)$ lies in $\mathcal{C}_y$ and has radial coordinate $p^*(y) + t\lambda(s) X_{y,s}(u) \in [P_{\min}, P_{\max}]$. Hence

$$q^{\mathrm{in}}_{y,s} = C_{y,\lambda(s)} \circ f_{y,s} \colon D(p^\dagger) \to \mathbb{C}$$

is defined, $(y, s, z) \mapsto q^{\mathrm{in}}_{y,s}(z)$ is smooth on $\mathcal{Y} \times [0,1] \times D(p^\dagger)$, and each $q^{\mathrm{in}}_{y,s}$ is a symplectic embedding with

$$q^{\mathrm{in}}_{y,0}(z) = b_y + \mathbf{A}_y z, \qquad q^{\mathrm{in}}_{y,1} = q_{\mathrm{in},y}|_{D(p^\dagger)}, \qquad q^{\mathrm{in}}_{y,s}(0) = b_y, \qquad Dq^{\mathrm{in}}_{y,s}(0) = \mathbf{A}_y.$$

(d) (Ranges.) $|q^{\mathrm{in}}_{y,s}(z) - b_y| \le B^*$ and $|q^{\mathrm{in}}_{y,1}(z)| \le R_q$ for $z \in D(p^\dagger)$.

Proof. (a) By Lemma 4.10(d), (D5) and monotonicity of $\mathcal{A}_{y,s}$, $u^2 + v^2 = E_{y,s}(\pi|z|^2) \le \bar E$. Since $X_{y,s}(u)$ is a convex combination of $a_0 u$ and $X_y(u)$, (D5) gives $|X_{y,s}(u)| \le \bar X$; likewise $|Y_{y,s}(v)| \le \bar Y$.

(b) By Lemma 4.10(b) and (D4), $\mathcal{A}_{y,s}(E_c) > p^\dagger \ge \pi|z|^2$, so $u^2, v^2 < E_c$ and (D2) applies. By (D3), (D4), $|X_{y,s}(u) - X_y(u)| = (1-s)|a_0 u - X_y(u)| \le \delta_s\delta_x \le \frac12\Delta P$; similarly $|Y_{y,s}(v) - Y_y(v)| \le \frac12\Delta\theta$.

(c) For $s \le 1 - \delta_s$, $\lambda(s) \le \lambda_0$, and (a) with (D6) puts the radial coordinate in $[\frac12 p^*, \frac32 p^*]$ with $|t\lambda Y_{y,s}| \le \frac14$. Otherwise it lies between p^* and $p^* + X_{y,s}(u) \in [\frac12\Delta P, h - \frac12\Delta P]$, and $|t\lambda Y_{y,s}| \le$

$\frac{1}{2} - \frac{1}{2}\Delta\theta$ by (b). Either way $(y, \lambda(s), f_{y,s}(z)) \in \mathcal{W}$, with radial coordinate in $[P_{\min}, P_{\max}]$. Smoothness follows from Lemma 4.10(d) and smoothness of λ and of C on $\mathcal{W}$; $\det Dq^{\mathrm{in}}_{y,s} = \det DC_y(\lambda f_{y,s}) \det Df_{y,s} = 1$ and injectivity follow from Lemma 4.9 and injectivity of $f_{y,s}$. At $s = 0$, $\lambda = 0$ and $f_{y,0} = \mathbf{L}_y$ give $q^{\mathrm{in}}_{y,0}(z) = b_y + DC_y(0)\mathbf{L}_y z$. At $s = 1$, $C_{y,1} = C_y$, $f_{y,1} = f_y$ and $p^\dagger < I_{\mathrm{rect}}(y)$ (Lemma 4.12(b)) give $q^{\mathrm{in}}_{y,1} = q_{\mathrm{in},y}$ on $D(p^\dagger)$. The values at 0 follow from $f_{y,s}(0) = 0$, $C_{y,\lambda}(0) = b_y$, $Df_{y,s}(0) = \mathbf{L}_y$ and $DC_y(0)\mathbf{L}_y = \mathbf{A}_y$.

(d) Let $w = f_{y,s}(z) = (x, \theta)$; then $|x| \le \bar{X}$ and $|\theta| \le \bar{Y}$ by (a). At radial coordinate P the columns of DC_y are orthogonal, of lengths $1/(2\sqrt{\pi P})$ and $2\sqrt{\pi P}$, so $\|DC_y\| \le M_C$ on $[P_{\min}, P_{\max}]$, which by (c) contains the segment of Definition 4.8(b). So $|C_{y,\lambda}(w) - b_y| \le M_C\sqrt{\bar{X}^2 + \bar{Y}^2}$.

For $\lambda > 0$, put $P = p^* + \lambda x$, $\varphi = 2\pi\lambda\theta$ and $r_b = r_b(y)$. Then $C_y(\lambda w) - b_y = r_b(1 - e^{i\varphi}) + (r_b - \sqrt{P/\pi})e^{i\varphi}$, with $|r_b - \sqrt{P/\pi}| \le \lambda|x|/\sqrt{\pi p^*}$ and $|1 - e^{i\varphi}| \le |\varphi|$. Dividing by λ gives $|C_{y,\lambda}(w) - b_y| \le \bar{X}/\sqrt{\pi p^*_{\mathrm{lo}}} + 2\pi r_b^+ \bar{Y}$. The same bound holds for $|DC_y(0)w|$ when $\lambda = 0$, since $2\pi r_b = 2\sqrt{\pi p^*}$.

Hence $|q^{\mathrm{in}}_{y,s}(z) - b_y| \le B^*$ by (D7). Finally, $\pi|q^{\mathrm{in}}_{y,1}(z)|^2 = p^* + X_y(u) \le \pi R_q^2$ by (b) and (D2). □

4.6. **The five-segment ordinary path.** Since B^* may exceed $R_h - r_b$, we shrink by c_{sh}, run the inner family, and expand back. The flow and area correction below apply to any ordinary path.

Definition 4.14 (Ordinary path)**.** Let $\mathcal{Y} \subset \mathbb{R}^d$ be open and let $p^\dagger > 0$, $0 < P_{\mathrm{supp}} < h$ and $0 < c_{\mathrm{sh}} \le 1$. An *ordinary path* with these constants is a smooth map $\mathfrak{e}\colon \mathcal{Y} \times \mathbb{R} \times D(p^\dagger) \to \mathbb{C}$, $(y, t, z) \mapsto \mathfrak{e}_{y,t}(z)$, together with a smooth function $\mu\colon \mathbb{R} \to (0, \infty)$, such that for all $y \in \mathcal{Y}$ and $t \in \mathbb{R}$:

(O1) $\mathfrak{e}_{y,t} = \mathrm{id}$ for $t \le 0$ and $\mathfrak{e}_{y,t} = \mathfrak{e}_{y,1}$ for $t \ge 1$;
(O2) $\mathfrak{e}_{y,t}$ is injective, $\det D\mathfrak{e}_{y,t}(z) = \mu_t$ for all z, $\mu_t \ge c_{\mathrm{sh}}^2$ and $\mu_1 = 1$;
(O3) $\pi|\mathfrak{e}_{y,t}(z)|^2 \le P_{\mathrm{supp}}$ for all $z \in D(p^\dagger)$.

By (O1) and $\mu_1 = 1$, $\mu_t = 1$ and $\partial_t \mathfrak{e}_{y,t} = 0$ for $t \notin (0, 1)$.

Construction 4.15 (The five-segment path)**.** Let (U, V) be an admissible potential pair over $\mathcal{Y}$ with isotopy data, and c_{sh} the constant of the data. For $y \in \mathcal{Y}$ and $\sigma \in [0, 1]$ let $\mathbf{M}_{y,\sigma} = \mathrm{Rot}_{\pi\sigma}\,\mathrm{diag}\big(|(\mathbf{A}_y)_{11}|^\sigma, |(\mathbf{A}_y)_{22}|^\sigma\big)$, and for $z \in D(p^\dagger)$ define the five segments

$$\mathfrak{e}^0_{y,\sigma}(z) = c_{\mathrm{sh}}^\sigma z, \qquad \mathfrak{e}^1_{y,\sigma}(z) = \sigma b_y + c_{\mathrm{sh}} z, \qquad \mathfrak{e}^2_{y,\sigma}(z) = b_y + c_{\mathrm{sh}}\,\mathbf{M}_{y,\sigma} z,$$
$$\mathfrak{e}^3_{y,\sigma}(z) = b_y + c_{\mathrm{sh}}\big(q^{\mathrm{in}}_{y,\sigma}(z) - b_y\big), \quad \mathfrak{e}^4_{y,\sigma}(z) = b_y + c_{\mathrm{sh}}^{1-\sigma}\big(q^{\mathrm{in}}_{y,1}(z) - b_y\big).$$

Put $\mathfrak{e}_{y,t} = \mathfrak{e}^j_{y,S(5t-j)}$ for $t \in [\frac{j}{5}, \frac{j+1}{5}]$, $j = 0, \dots, 4$, $\mathfrak{e}_{y,t} = \mathrm{id}$ for $t \le 0$ and $\mathfrak{e}_{y,t} = \mathfrak{e}_{y,1}$ for $t \ge 1$, and $\mu_t = c_{\mathrm{sh}}^{2S(5t)-2S(5t-4)}$.

By (4), $\mathbf{A}_y = \mathrm{Rot}_\pi\,\mathrm{diag}(\alpha_1, \alpha_2)$ with $\alpha_i = |(\mathbf{A}_y)_{ii}|$ and $\alpha_1\alpha_2 = 1$. Hence $\mathbf{M}_{y,0} = \mathrm{Id}$, $\mathbf{M}_{y,1} = \mathbf{A}_y$, $\det \mathbf{M}_{y,\sigma} = 1$ and $\|\mathbf{M}_{y,\sigma}\| \le \max(1, \alpha_1, \alpha_2)$.

Lemma 4.16 (The five-segment path is an ordinary path)**.** *The map $\mathfrak{e}$ of Construction 4.15, with μ, is an ordinary path with constants $p^\dagger$, P_{supp}, h, c_{sh}, and $\mathfrak{e}_{y,1} = q^{\mathrm{in}}_{y,1} = q_{\mathrm{in},y}$ on $D(p^\dagger)$.*

Proof. Junctions and smoothness. Consecutive segments agree at the junctions ($\mathfrak{e}^2_{y,1} = \mathfrak{e}^3_{y,0}$ by $\lambda(0) = 0$), $\mathfrak{e}^0_{y,0} = \mathrm{id}$ and $\mathfrak{e}^4_{y,1} = q^{\mathrm{in}}_{y,1}$. Each segment is smooth (Proposition 4.13(c) for $j = 3, 4$), so Lemma 2.5 gives smoothness of the concatenation and (O1).

(O2). As $\det \mathbf{M}_{y,\sigma} = \det Dq^{\mathrm{in}}_{y,\sigma} = 1$, the segment Jacobians are $c_{\mathrm{sh}}^{2\sigma}$, c_{sh}^2, c_{sh}^2, c_{sh}^2, $c_{\mathrm{sh}}^{2(1-\sigma)}$, so $\mu_t \in [c_{\mathrm{sh}}^2, 1]$ and $\mu_1 = 1$. Segments $0, 1, 2$ are affine bijections; segments $3, 4$ are $q^{\mathrm{in}}_{y,\sigma}$ followed by affine bijections.

(O3). Let $|z| < R^\dagger$. Then:

- $|\mathfrak{e}^0| < R^\dagger$;
- $|\mathfrak{e}^1|, |\mathfrak{e}^2| \le r_b^+ + c_{\mathrm{sh}} B_{\mathrm{lin}}$, by (D7) and the bound on $\|\mathbf{M}_{y,\sigma}\|$;
- $|\mathfrak{e}^3| \le r_b^+ + c_{\mathrm{sh}} B^*$, by Proposition 4.13(d);
- $\mathfrak{e}^4_{y,\sigma}(z) = (1-\beta)b_y + \beta q^{\mathrm{in}}_{y,1}(z)$ with $\beta = c_{\mathrm{sh}}^{1-\sigma} \in [c_{\mathrm{sh}}, 1]$, so $|\mathfrak{e}^4| \le \max(r_b^+, R_q)$ by convexity and Proposition 4.13(d).

All are at most R_{supp}. Finally, $P_{\mathrm{supp}} < h$ and $0 < c_{\mathrm{sh}} \le \frac{1}{4}$ by Lemma 4.12(b) and (D7). □

Segments $2, 3$ stay within $(R_h - r_b^+)/8$ of b_y, segment 0 in $\{|x| < R^\dagger\}$, and segment 1 in $\{|x| \le r_b^+ + c_{\mathrm{sh}} R^\dagger\}$ by convexity of the disc. As μ_t is independent of z and $\mu_0 = \mu_1 = 1$, the area correction of §4.8 acts on the core as $z \mapsto \mu_t^{-1/2} z$, the identity at $t = 0, 1$.

4.7. **The cut-off ambient flow.** Fix $0 < p_{\mathrm{on}} < p_{\mathrm{zero}} < p^\dagger$, $r_{\mathrm{on}} = \sqrt{p_{\mathrm{on}}/\pi}$ and $r_{\mathrm{zero}} = \sqrt{p_{\mathrm{zero}}/\pi}$. The cutoff of Definition 2.6 is

$$\chi_{r_{\mathrm{on}}, r_{\mathrm{zero}}}(z) = 1 - S\big((\pi|z|^2 - p_{\mathrm{on}})/(p_{\mathrm{zero}} - p_{\mathrm{on}})\big).$$

By Lemma 2.7(a) it is C^∞, equal to 1 for $\pi|z|^2 \le p_{\mathrm{on}}$ and to 0 for $\pi|z|^2 \ge p_{\mathrm{zero}}$.

Lemma 4.17 (The ambient flow)**.** *Let $\mathfrak{e}$ be an ordinary path and $0 < p_{\mathrm{on}} < p_{\mathrm{zero}} < p^\dagger$. Let $\mathcal{O} = \{(y, t, x) : x \in \mathfrak{e}_{y,t}(D(p^\dagger))\}$ and define*

$$W_{y,t}(x) = \chi_{r_{\mathrm{on}}, r_{\mathrm{zero}}}\big(\mathfrak{e}_{y,t}^{-1}(x)\big)\,\big(\partial_t \mathfrak{e}_{y,t}\big)\big(\mathfrak{e}_{y,t}^{-1}(x)\big) \quad \text{on } \mathcal{O}, \qquad W_{y,t}(x) = 0 \quad \text{otherwise.}$$

Let $\phi_{y,t}$ be the flow of the time-dependent field $W_{y,t}$ with $\phi_{y,0} = \mathrm{id}$.

(a) W is C^∞ on $\mathcal{Y} \times \mathbb{R} \times \mathbb{C}$ and vanishes for $t \notin (0,1)$ and for $\pi|x|^2 > P_{\mathrm{supp}}$. Hence $\phi_{y,t}$ is defined for all t, each $\phi_{y,t}$ is a diffeomorphism of $\mathbb{C}$, and $(y, t, x) \mapsto \phi_{y,t}(x)$ is C^∞.

(b) $\phi_{y,t}(z) = \mathfrak{e}_{y,t}(z)$ for $\pi|z|^2 \le p_{\mathrm{on}}$, and $\det D\phi_{y,t}(z) = \mu_t$ there.

(c) $\phi_{y,t} = \mathrm{id}$ on $\{\pi|x|^2 > P_{\mathrm{supp}}\}$; $\phi_{y,t}$ maps $\bar{D}(P)$ onto itself for every $P \ge P_{\mathrm{supp}}$ and $D(P)$ onto itself for every $P > P_{\mathrm{supp}}$, in particular $D(h)$ onto $D(h)$; and $\det D\phi_{y,t} > 0$ everywhere. For $t \le 0$, $\phi_{y,t} = \mathrm{id}$; for $t \ge 1$, $\phi_{y,t} = \phi_{y,1}$.

Proof. (a) The map $\hat{\mathfrak{e}}(y, t, z) = (y, t, \mathfrak{e}_{y,t}(z))$ is smooth and injective with Jacobian $\mu_t > 0$, hence a diffeomorphism onto the open set $\mathcal{O}$, so W is smooth on $\mathcal{O}$. There W vanishes outside $K = \hat{\mathfrak{e}}(\mathcal{Y} \times [0,1] \times \bar{D}(p_{\mathrm{zero}})) \subset \mathcal{O}$, which is closed by compactness of $\bar{D}(p_{\mathrm{zero}})$. As $\mathcal{O}$ and K^c cover $\mathcal{Y} \times \mathbb{R} \times \mathbb{C}$, W is smooth. By (O3), W vanishes for $\pi|x|^2 > P_{\mathrm{supp}}$, so its flow is global, by diffeomorphisms, and smooth in (y, t, x).

(b) For $\pi|z|^2 \le p_{\mathrm{on}}$, $\frac{d}{dt}\mathfrak{e}_{y,t}(z) = W_{y,t}(\mathfrak{e}_{y,t}(z))$ and $\mathfrak{e}_{y,0}(z) = z$, so $\phi_{y,t}(z) = \mathfrak{e}_{y,t}(z)$ by uniqueness; the determinant identity follows on $D(p_{\mathrm{on}})$ and, by continuity, on its boundary.

(c) Points with $\pi|x|^2 > P_{\mathrm{supp}}$ are stationary and $\phi_{y,t}$ is a bijection of $\mathbb{C}$, giving the claims on $\bar{D}(P)$ and $D(P)$ ($h > P_{\mathrm{supp}}$). $\det D\phi_{y,t}$ is continuous, nonvanishing and 1 at $t = 0$, hence positive. As $W_{y,t} = 0$ for $t \notin (0,1)$, the flow is constant on $(-\infty, 0]$ and on $[1, \infty)$. □

4.8. **Area normalisation of the ambient flow.** The normalisations of §4.3 are applied to $G_{y,t}(w) = \phi_{y,t}(w/\sqrt{\mu_t})$.

Lemma 4.18 (Normalisation of the whole path)**.** *Let $\mathfrak{e}$ be an ordinary path, $0 < p_{\mathrm{on}} < p_{\mathrm{zero}} < p^{\dagger}$, and ϕ the ambient flow of Lemma 4.17. For $(y,t) \in \mathcal{Y} \times \mathbb{R}$ put*

$$G_{y,t}(w) = \phi_{y,t}\big(w/\sqrt{\mu_t}\big), \qquad \hat{J}_{y,t}(w) = \det DG_{y,t}(w) = \frac{\det D\phi_{y,t}(w/\sqrt{\mu_t})}{\mu_t},$$

let $\hat{T}_{y,t}$, $\hat{D}_{y,t}$ be the rotation and area normalisations of $\hat{J}_{y,t}$ (parameters (y,t)), and define

$$\mathcal{N}_{y,t} = \mu_t^{-1/2}\, \hat{T}_{y,t}^{-1} \circ \hat{D}_{y,t}, \qquad \hat{q}_{y,t} = \phi_{y,t} \circ \mathcal{N}_{y,t} = G_{y,t} \circ \hat{T}_{y,t}^{-1} \circ \hat{D}_{y,t}.$$

(a) $\hat{J}$ is C^{∞} and positive on $\mathcal{Y} \times \mathbb{R} \times \mathbb{C}$; $\hat{J}_{y,t}(w) = 1$ for $\pi|w|^2 \le \mu_t p_{\mathrm{on}}$, in particular for $|w| \le \rho_c = c_{\mathrm{sh}}\sqrt{p_{\mathrm{on}}/\pi}$; $\hat{J}_{y,t}(w) = 1/\mu_t$ for $\pi|w|^2 > \mu_t P_{\mathrm{supp}}$; and $\mathcal{A}_{\hat{J}_{y,t}}(\mu_t P/\pi) = \operatorname{area} \phi_{y,t}(\bar{D}(P))$, which equals $\mu_t P$ for $P \le p_{\mathrm{on}}$ and P for $P \ge P_{\mathrm{supp}}$. In particular $\mathcal{A}_{\hat{J}_{y,t}}(\infty) = \infty$.

(b) Each $\hat{q}_{y,t}$ is a symplectic diffeomorphism of $\mathbb{C}$; it maps $\bar{D}(P)$ onto itself for every $P \ge P_{\mathrm{supp}}$ and $D(P)$ onto itself for every $P > P_{\mathrm{supp}}$, in particular $D(h)$ and $\bar{D}(h)$.

(c) (Core.) $\hat{q}_{y,t}(z) = \mathfrak{e}_{y,t}(z/\sqrt{\mu_t})$ for $\pi|z|^2 \le \mu_t p_{\mathrm{on}}$, in particular on the common core $\pi|z|^2 \le c_{\mathrm{sh}}^2 p_{\mathrm{on}}$.

(d) (Collar.) $\hat{q}_{y,t}(z) = z$ for $\pi|z|^2 > P_{\mathrm{supp}}$.

(e) (Smoothness.) $(y,t,z) \mapsto \hat{q}_{y,t}(z)$ and $(y,t,x) \mapsto \hat{q}_{y,t}^{-1}(x)$ are C^{∞} on $\mathcal{Y} \times \mathbb{R} \times \mathbb{C}$, including $z = 0$; $\hat{q}_{y,t} = \mathrm{id}$ for $t \le 0$ and $\hat{q}_{y,t} = \hat{q}_{y,1}$ for $t \ge 1$.

(f) (Endpoint.) $\hat{q}_{y,1} = \mathfrak{e}_{y,1}$ on $\bar{D}(p_{\mathrm{on}})$.

Proof. (a) The formula for $\hat{J}$ is the chain rule; smoothness and positivity follow from Lemma 4.17 and $\mu_t \ge c_{\mathrm{sh}}^2 > 0$. If $\pi|w|^2 \le \mu_t p_{\mathrm{on}}$, then $x = w/\sqrt{\mu_t}$ has $\pi|x|^2 \le p_{\mathrm{on}}$, where $\det D\phi_{y,t} = \mu_t$ (Lemma 4.17(b)); so $\hat{J}_{y,t}(w) = 1$, and $\mu_t \ge c_{\mathrm{sh}}^2$ gives the uniform radius ρ_c. If $\pi|w|^2 > \mu_t P_{\mathrm{supp}}$, then $\phi_{y,t} = \mathrm{id}$ near $w/\sqrt{\mu_t}$, so $\hat{J}_{y,t}(w) = 1/\mu_t$. Change of variables gives $\mathcal{A}_{\hat{J}_{y,t}}(\mu_t P/\pi) = \operatorname{area} \phi_{y,t}(\bar{D}(P))$, which is $\mu_t P$ for $P \le p_{\mathrm{on}}$ (there $\phi_{y,t} = \mathfrak{e}_{y,t}$ has Jacobian μ_t) and P for $P \ge P_{\mathrm{supp}}$ (Lemma 4.17(c)).

Setting (N) applies with $\mathcal{P} = \mathcal{Y} \times \mathbb{R}$, $\mathcal{U} = \mathcal{P} \times \mathbb{R}^2$ and $J = \hat{J}$, so by Lemmas 4.4 and 4.5: $\hat{T}^{\pm 1}$ and $\hat{D}$ are smooth in all variables; $\hat{T}_{y,t}^{\pm 1} = \mathrm{id}$ on $|w| \le \rho_c$; and $\hat{q}_{y,t}$ is a symplectic diffeomorphism of $\mathbb{C}$ onto $G_{y,t}(\mathbb{C}) = \mathbb{C}$ (Lemma 4.5(iv) with $F = G_{y,t}$).

(d) Let $\pi|z|^2 > P_{\mathrm{supp}}$. By (a), $\hat{D}_{y,t}(z) = \sqrt{\mu_t}\, z$, and on the circle through $w = \sqrt{\mu_t} z$ the density is the constant $1/\mu_t$, so $\mathcal{M}_{\hat{J}_{y,t}}(w) = 0$ and $\hat{T}_{y,t}^{-1}(w) = w$. Hence $\hat{q}_{y,t}(z) = \phi_{y,t}(z) = z$.

(b) follows from (d), as $\hat{q}_{y,t}$ is a bijection of $\mathbb{C}$ fixing every point outside $\bar{D}(P_{\mathrm{supp}})$.

(c) By (a), $\hat{J}_{y,t} = 1$ on $|w| \le \sqrt{\mu_t p_{\mathrm{on}}/\pi}$, so Lemma 4.5(v) gives $\hat{q}_{y,t}(z) = \phi_{y,t}(z/\sqrt{\mu_t})$ there, which is $\mathfrak{e}_{y,t}(z/\sqrt{\mu_t})$ by Lemma 4.17(b).

(e) For $t \le 0$, $\phi_{y,t} = \mathrm{id}$ and $\mu_t = 1$, so $\hat{J} = 1$ and $\hat{q}_{y,t} = \mathrm{id}$. For $t \ge 1$, $\phi_{y,t} = \phi_{y,1}$ and $\mu_t = 1$, so $\hat{q}_{y,t} = \hat{q}_{y,1}$. The map $(y,t,z) \mapsto (y,t,\hat{q}_{y,t}(z))$ is a smooth bijection with Jacobian 1, so its inverse is smooth by the inverse function theorem.

(f) is (c) at $t = 1$, where $\mu_1 = 1$. □

4.9. **The Hamiltonian.**

Construction 4.19 (The Hamiltonian)**.** For an ordinary path with normalised isotopy $\hat{q}$, $y \in \mathcal{Y}$, $t \in \mathbb{R}$ and $z, w \in \mathbb{C}$, put

$$\mathfrak{a}_{y,t}(z)[w] = \omega_{\mathbb{C}}\big(\partial_t \hat{q}_{y,t}(z),\, D\hat{q}_{y,t}(z)w\big), \qquad \mathfrak{h}_{y,t}(z) = \int_0^1 \mathfrak{a}_{y,t}(rz)[z]\, dr,$$

$$z_{\mathrm{bd}} = \big(\sqrt{p_{\mathrm{bd}}/\pi}, 0\big), \quad p_{\mathrm{bd}} = \tfrac{1}{2}(h + P_{\mathrm{supp}}), \qquad Q_{y,t}(x) = -\big[\mathfrak{h}_{y,t} - \mathfrak{h}_{y,t}(z_{\mathrm{bd}})\big]\big(\hat{q}_{y,t}^{-1}(x)\big).$$

Proposition 4.20 (Q generates $\hat{q}$)**.** *Q is C^∞ on $\mathcal{Y}\times\mathbb{R}\times\mathbb{C}$; $Q_{y,t}$ and all its derivatives vanish on $\{\pi|x|^2 > P_{\mathrm{supp}}\}$ for every (y,t), and $Q_{y,t}=0$ for $t\notin(0,1)$. With $\xi_{y,t}=(\partial_t\hat{q}_{y,t})\circ\hat{q}_{y,t}^{-1}$,*

$$\iota_{\xi_{y,t}}\omega_{\mathbb{C}} = -dQ_{y,t}, \qquad \partial_t\hat{q}_{y,t} = \xi_{y,t}\circ\hat{q}_{y,t},$$

so $\hat{q}_{y,\cdot}$ is the isotopy generated by $Q_{y,\cdot}$ in the convention $\iota_X\omega_{\mathbb{C}}=-dQ$.

Proof. By definition $\mathfrak{a}_{y,t}=\hat{q}_{y,t}^*(\iota_{\xi_{y,t}}\omega_{\mathbb{C}})$. Differentiating $\hat{q}_{y,t}^*\omega_{\mathbb{C}}=\omega_{\mathbb{C}}$ in t gives $d\iota_{\xi_{y,t}}\omega_{\mathbb{C}} = \mathcal{L}_{\xi_{y,t}}\omega_{\mathbb{C}}=0$, so $\mathfrak{a}_{y,t}$ is closed, and as the plane is star-shaped its radial primitive, smooth in (y,t,z), satisfies $d\mathfrak{h}_{y,t}=\mathfrak{a}_{y,t}$. Hence $-dQ_{y,t}=(\hat{q}_{y,t}^{-1})^*d\mathfrak{h}_{y,t}=\iota_{\xi_{y,t}}\omega_{\mathbb{C}}$.

On the connected set $\{\pi|z|^2>P_{\mathrm{supp}}\}$, $\hat{q}_{y,t}=\mathrm{id}$ for all t (Lemma 4.18(d)), so $\mathfrak{a}_{y,t}=0$ and $\mathfrak{h}_{y,t}$ is constant there, equal to $\mathfrak{h}_{y,t}(z_{\mathrm{bd}})$ as $P_{\mathrm{supp}}<p_{\mathrm{bd}}$. So $Q_{y,t}$ vanishes with all derivatives on $\hat{q}_{y,t}(\{\pi|z|^2>P_{\mathrm{supp}}\})=\{\pi|x|^2>P_{\mathrm{supp}}\}$. For $t\notin(0,1)$, $\partial_t\hat{q}_{y,t}=0$, so $Q_{y,t}=0$. Smoothness of Q follows from Lemma 4.18(e). □

4.10. **The whole-disc isotopy.**

Theorem 4.21 (Whole-disc Hamiltonian isotopy with prescribed endpoint on the core)**.** *Let (U,V) be an admissible potential pair over $\mathcal{Y}$ with disc area h, and let isotopy data be given (Definition 4.11). Let $\hat{q}$ and Q be obtained from the five-segment path of Construction 4.15 by Lemma 4.18, with the levels $p_{\mathrm{on}}<p_{\mathrm{zero}}$ of the data, and Construction 4.19. Then $P_{\mathrm{supp}}<h$, and for every $y\in\mathcal{Y}$:*

- *(a) each $\hat{q}_{y,t}$ is a symplectic diffeomorphism of $\mathbb{C}$ which is the identity on $\{\pi|z|^2>P_{\mathrm{supp}}\}$ and maps $\bar{D}(P)$ onto itself for every $P\ge P_{\mathrm{supp}}$, in particular $\bar{D}(h)$ and $D(h)$;*
- *(b) $\hat{q}_{y,t}=\mathrm{id}$ for $t\le 0$ and $\hat{q}_{y,t}=\hat{q}_{y,1}$ for $t\ge 1$;*
- *(c) (same endpoint on the core) $\hat{q}_{y,1}(z)=q_{\mathrm{in},y}(z)$ for $\pi|z|^2\le p_{\mathrm{on}}$;*
- *(d) $Q_{y,t}$ vanishes with all derivatives on $\{\pi|x|^2>P_{\mathrm{supp}}\}$ and for $t\notin(0,1)$, and generates $\hat{q}_{y,t}$: $\partial_t\hat{q}_{y,t}=\xi_{y,t}\circ\hat{q}_{y,t}$ with $\iota_{\xi_{y,t}}\omega_{\mathbb{C}}=-dQ_{y,t}$;*
- *(e) (core formula) $\hat{q}_{y,t}(z)=\mathfrak{e}_{y,t}(z/\sqrt{\mu_t})$ for $\pi|z|^2\le\mu_t p_{\mathrm{on}}$.*

Proof. By Lemma 4.16 the five-segment path is ordinary, with constants $p^\dagger$, P_{supp}, h, c_{sh}; $p_{\mathrm{on}}<p_{\mathrm{zero}}<p^\dagger$ by (D1) and $P_{\mathrm{supp}}<h$ by Lemma 4.12(b). Items (a), (b), (e) are Lemma 4.18(b)–(e), and (d) is Proposition 4.20. For (c), Lemma 4.18(f) gives $\hat{q}_{y,1}=\mathfrak{e}_{y,1}$ on $\bar{D}(p_{\mathrm{on}})$, and $\mathfrak{e}_{y,1}=q_{y,1}^{\mathrm{in}}=q_{\mathrm{in},y}$ on $D(p^\dagger)$ by Lemma 4.16. □

Corollary 4.22 (Comparison on the whole closed disc)**.** *In the situation of Theorem 4.21, let $y\in\mathcal{Y}$, let $\Upsilon\colon[0,h]\to\mathbb{R}$ be any function, $g\colon[0,h]\to\mathbb{R}$ nondecreasing and $\varepsilon\ge 0$, and suppose that $\Upsilon(\pi|q_{\mathrm{in},y}(z)|^2)\le g(\pi|z|^2)+\varepsilon$ whenever $\pi|z|^2\le p_{\mathrm{on}}$, and that $\sup_{[0,h]}\Upsilon\le g(p_{\mathrm{on}})+\varepsilon$. Then*

$$\Upsilon\big(\pi|\hat{q}_{y,1}(z)|^2\big)\le g(\pi|z|^2)+\varepsilon \qquad \text{for all } z\in\bar{D}(h).$$

Proof. For $\pi|z|^2\le p_{\mathrm{on}}$ use Theorem 4.21(c). For $p_{\mathrm{on}}<\pi|z|^2\le h$, $\hat{q}_{y,1}(z)\in\bar{D}(h)$ by Theorem 4.21(a), so the left side is at most $g(p_{\mathrm{on}})+\varepsilon\le g(\pi|z|^2)+\varepsilon$. □

Theorem 4.23 (Parametrised version)**.** *In the situation of Theorem 4.21:*

- *(a) the maps $(y,t,z)\mapsto\hat{q}_{y,t}(z)$, $(y,t,x)\mapsto\hat{q}_{y,t}^{-1}(x)$ and $(y,t,x)\mapsto Q_{y,t}(x)$ are C^∞ on $\mathcal{Y}\times\mathbb{R}\times\mathbb{C}$, and $(y,s,z)\mapsto q_{y,s}^{\mathrm{in}}(z)$ is smooth on $\mathcal{Y}\times[0,1]\times D(p^\dagger)$; the numbers p_{on}, P_{supp}, c_{sh}, and hence the collar $\{\pi|z|^2>P_{\mathrm{supp}}\}$, the core $\bar{D}(p_{\mathrm{on}})$ and the common core $\bar{D}(c_{\mathrm{sh}}^2p_{\mathrm{on}})$, do not depend on y;*
- *(b) if $\mathcal{Y}_0\subset\mathcal{Y}$ is compact, then the derivatives of order ≥ 1 of $\hat{q}$ and $\hat{q}^{-1}$, and all derivatives (including order 0) of $\hat{q}-\mathrm{id}$, $\hat{q}^{-1}-\mathrm{id}$ and Q, are bounded on $\mathcal{Y}_0\times\mathbb{R}\times\mathbb{C}$.*

Proof. (a) is Lemma 4.18(e), Proposition 4.20 and Proposition 4.13(c); the data are numbers independent of y.

(b) The maps are smooth and constant in t outside $[0,1]$, and $\hat{q}-\mathrm{id}$, $\hat{q}^{-1}-\mathrm{id}$ and Q vanish outside $\bar{D}(P_{\mathrm{supp}})$. So outside the compact set $\mathcal{Y}_0\times[0,1]\times\bar{D}(P_{\mathrm{supp}})$ each listed derivative equals its value at $t=0$ or $t=1$, or 0, or the identity matrix (first derivatives of $\hat{q}^{\pm1}$), and is bounded. □

Remark 4.24 (Explicitness). Every object of this section is a defined term in the sense of Definition M2.5: finite expressions in the potentials and their derivatives; integrals over compact domains ($\bar{J}$, $\mathcal{M}$, $\mathcal{G}$, the charts $C_{y,\lambda}$, the radial primitive); inverses of explicit bijections with nonvanishing derivative (X_y, Y_y, $\mathcal{A}_{y,s}$, Ψ_σ, T_σ, $(y,t,z)\mapsto(y,t,\mathfrak{e}_{y,t}(z))$, $\hat{q}_{y,t}$); and the flow of an explicit compactly supported vector field. The base point z_{bd}, the schedule $\lambda(s)$, the clocks $S(5t-j)$ of Lemma 2.5 and the intervals $[\frac{j}{5},\frac{j+1}{5}]$ are fixed; the isotopy data (§5) are finite expressions with square roots and compact extrema.

5. The explicit inner map for equal balls

We construct the fibre isotopy (q_t) of Proposition M4.4, whose endpoint q satisfies $\mathcal{R}_{\mathrm{tent}}(\pi|q(z)|^2)\le\pi|z|^2+4e_q$ on $\bar{D}(h)$ (Theorem 5.34). It is §4 applied to one potential pair $(\tilde{R},\tilde{V})$: a hyperbolic smoothing of $\mathcal{R}_{\mathrm{tent}}$, and a quadratic angular potential with a steep step at $\theta=\pm\frac12$. Both are exactly quadratic near their minima, so the inner map equals $\mathbf{L}=\mathrm{diag}(c_x,c_y)$ on an explicit disc (Proposition 5.18). The error $4e_q$ comes from the smoothing and angular mean on the core (Proposition 5.24) and from an annulus of area $4e_q$ outside it, where only $q(\bar{D}(h))=\bar{D}(h)$ is used.

5.1. **Setting and conventions.** Fix an integer $k\ge10$ and $t_0\in(\frac78,1)$; in the equal-capacity branch, $t_0=\frac12(1+\rho\sqrt{k})$ (Remark 10.25). Let $a=t_0/\sqrt{k}$, $h=t_0/3$, $A=h/a$ and $B=h/(h-a)$, as in §M4.2. Then $0<a<h$, $A,B>0$, $A+B=AB$, $h/A=a$ and $h/B=h-a$ (Lemma 3.3). The tent is $\mathcal{R}_{\mathrm{tent}}(x)=A(a-x)$ for $x\le a$ and $B(x-a)$ for $x\ge a$; this is the convex function Λ of §M4.2, with $\mathcal{R}_{\mathrm{tent}}(0)=\mathcal{R}_{\mathrm{tent}}(h)=h$ and $\mathcal{R}_{\mathrm{tent}}(a)=0$. By Lemma M4.2, for $\beta\ge0$,

$$\{x\in\mathbb{R}:\mathcal{R}_{\mathrm{tent}}(x)\le\beta\}=\Big[a-\frac{\beta}{A},\ a+\frac{\beta}{B}\Big], \tag{5}$$

an interval of length β, contained in $[0,h]$ if $\beta\le h$; in particular $\{\mathcal{R}_{\mathrm{tent}}<h\}=(0,h)$. Throughout, $e=e_q$ (Definition 3.4), and only $0<e\le h/100$ is used.

The fibre is $\mathbb{C}$ with area form ω_2 and moment $p(\zeta)=\pi|\zeta|^2$; $D(P_0)=\{\pi|\zeta|^2<P_0\}$, and $\bar{D}(P_0)$ is its closure. A planar map is symplectic iff its Jacobian is $\equiv1$. As in §4, z is the source plane, $w=(u,v)$ the Morse plane and (x,θ) rectangle coordinates; here $d=0$, $I_U=I_V=\mathbb{R}$, $U(p)=\tilde{R}(p-p^*)$, $V=\tilde{V}$ (Construction 5.9), and $x=p-p^*$. Fibre Hamiltonians use $\iota_\Xi\omega_2=-dQ$. S is the flat step of Definition 2.3; by Lemma 2.4 it is C^∞ and nondecreasing, 0 on $(-\infty,0]$, 1 on $[1,\infty)$, flat at 0 and 1, and $S(x)+S(1-x)=1$, so $S(\frac12)=\frac12$. Rot_ϕ is rotation by ϕ.

5.2. **Constants and the hyperbola.**

Definition 5.1 (Constants)**.**

$$m_-=\frac{B-A}{2},\quad m_+=\frac{A+B}{2},\quad r=\sqrt{AB},$$
$$\eta_{\mathrm{w}}=\frac{e}{8m_+},\quad x^*=-\frac{m_-\eta_{\mathrm{w}}}{r},\quad p^*=a+x^*,\quad \delta=\frac{e}{8}.$$

Definition 5.2 (Curvatures and linear constants)**.**

$$\alpha_0 = \frac{r^3}{\eta_{\mathrm{w}} m_+^2}, \qquad a_0 = \sqrt{2/\alpha_0}, \qquad b_0 = \sqrt{2/e}, \qquad j_0 = a_0 b_0,$$
$$c_x = \sqrt{a_0/b_0}, \qquad c_y = 1/c_x, \qquad \mathbf{L} = \operatorname{diag}(c_x, c_y).$$

Lemma 5.3 (Relations between the constants)**.** *$r^2 = 2m_+ = m_+^2 - m_-^2$, hence $0 < r \le m_+$ and $|m_-| < m_+$; moreover $\alpha_0 = 8r^3/(em_+) = 16r/e$, $\eta_{\mathrm{w}} m_+ = \delta$, $a_0 = \sqrt{e/(8r)}$, $j_0 = 1/(2\sqrt{r})$ and $c_x^4 = e^2/(16r)$.*

Proof. By Lemma 3.3, $r^2 = AB = A + B = 2m_+$, and $m_+^2 - m_-^2 = AB = r^2$. As $A, B > 0$, $|m_-| < m_+$ and $r \le m_+$. Then $\alpha_0 = r^3 \cdot 8m_+/(em_+^2) = 8r^3/(em_+) = 8r \cdot r^2/(em_+) = 16r/e$, and $\eta_{\mathrm{w}} m_+ = e/8$. Hence $a_0^2 = 2/\alpha_0 = e/(8r)$, $j_0^2 = a_0^2 b_0^2 = (e/(8r))(2/e) = 1/(4r)$ and $c_x^4 = a_0^2/b_0^2 = (e/(8r))(e/2) = e^2/(16r)$. □

Definition 5.4 (The hyperbola and its inverse)**.** For real x and u put

$$R(x) = m_-(x^* + x) + m_+\sqrt{(x^* + x)^2 + \eta_{\mathrm{w}}^2} - \eta_{\mathrm{w}} r, \qquad X(u) = \frac{-m_- u^2 + m_+ u\sqrt{u^2 + 2\eta_{\mathrm{w}} r}}{r^2}.$$

Lemma 5.5 (The hyperbola)**.** *R is C^∞ and strictly convex, $R(0) = R'(0) = 0$, $R''(0) = \alpha_0 > 0$, $R \ge 0$, and $R(x) \to +\infty$ as $x \to \pm\infty$. In particular 0 is the unique minimum point of R, and it is nondegenerate.*

Proof. Write $\xi = x^* + x$.

- $R(0) = 0$ follows from $(x^*)^2 + \eta_{\mathrm{w}}^2 = (\eta_{\mathrm{w}} m_+/r)^2$ and $m_+^2 - m_-^2 = r^2$.
- $R'(x) = m_- + m_+\xi/\sqrt{\xi^2 + \eta_{\mathrm{w}}^2}$, so $R'(0) = 0$.
- $R''(x) = m_+\eta_{\mathrm{w}}^2(\xi^2 + \eta_{\mathrm{w}}^2)^{-3/2} > 0$, so $R''(0) = \alpha_0$.
- $R \ge 0$ by Cauchy–Schwarz: $\eta_{\mathrm{w}} r - m_-\xi \le \sqrt{m_-^2 + r^2}\sqrt{\xi^2 + \eta_{\mathrm{w}}^2} = m_+\sqrt{\xi^2 + \eta_{\mathrm{w}}^2}$.
- $R(x) \ge (m_+ - |m_-|)|\xi| - \eta_{\mathrm{w}} r$, with $m_+ - |m_-| > 0$, so $R \to +\infty$.

□

Lemma 5.6 (X inverts R)**.** *X is a C^∞ diffeomorphism of $\mathbb{R}$ with $X' > 0$, $\operatorname{sign} X(u) = \operatorname{sign} u$ and $R(X(u)) = u^2$. For $u \ne 0$, $X(u)$ is the only solution of $R(x) = u^2$ with the sign of u. Moreover $X'(0) = \sqrt{2/\alpha_0} = a_0$.*

Proof. By Lemma 5.5, for $u \ne 0$ the equation $R(x) = u^2$ has exactly one positive and one negative root. With $\xi = x^* + x$, any root satisfies $m_+\sqrt{\xi^2 + \eta_{\mathrm{w}}^2} = u^2 + \eta_{\mathrm{w}} r - m_-\xi$. Squaring, and using $m_+^2 - m_-^2 = r^2$ and $(u^2 + \eta_{\mathrm{w}} r)^2 - r^2\eta_{\mathrm{w}}^2 = u^2(u^2 + 2\eta_{\mathrm{w}} r)$, the roots are $x_\pm = [-m_- u^2 \pm m_+|u|\sqrt{u^2 + 2\eta_{\mathrm{w}} r}]/r^2$, and $x_+ > 0 > x_-$. So $X(u)$ is the root with the sign of u.

X is C^∞ because $2\eta_{\mathrm{w}} r > 0$. Differentiating $R(X(u)) = u^2$ gives $R'(X(u))X'(u) = 2u$, so $X' > 0$ for $u \ne 0$; also $X'(0)^2 = 2/\alpha_0$. Finally $X(u) \to \pm\infty$ as $u \to \pm\infty$, since $X(u) \sim (m_+ \mp m_-)\, u|u|/r^2$. □

Lemma 5.7 (Comparison of R with the tent)**.** *$\mathcal{R}_{\mathrm{tent}}(p) = m_-(p - a) + m_+|p - a|$ for all real p, and*

$$\left|R(p - p^*) - \mathcal{R}_{\mathrm{tent}}(p)\right| \le \delta = \frac{e}{8} \qquad (p \in \mathbb{R}). \tag{6}$$

Proof. The formula for $\mathcal{R}_{\mathrm{tent}}$ is checked separately on $p \le a$ and $p \ge a$. With $\xi = p - a = x^* + (p - p^*)$,

$$R(p - p^*) - \mathcal{R}_{\mathrm{tent}}(p) = m_+\big(\sqrt{\xi^2 + \eta_{\mathrm{w}}^2} - |\xi|\big) - \eta_{\mathrm{w}} r \in \big[-\eta_{\mathrm{w}} r,\ \eta_{\mathrm{w}}(m_+ - r)\big].$$

Since $0 < r \le m_+$, its absolute value is at most $\eta_{\mathrm{w}} m_+ = \delta$. □

Lemma 5.8 (Position of the minimum)**.** $0 < p^* < h$.

Proof. By (6) and $R(0) = 0$, $\mathcal{R}_{\text{tent}}(p^*) \le e/8 < h$; now apply (5). □

5.3. **Potentials exactly quadratic near the origin.** $\tilde R$ equals $\frac12\alpha_0 x^2$ on $|x| \le \varepsilon_R$. $\tilde V$ is $e\theta^2/2$ plus a step of height $2h$ at $\theta = \pm\frac12$. This step confines the relevant level sets to $|\theta| < \frac12$ (Lemma 5.14(c)) and adds at most $e/5$ to the mean (Lemma 5.22).

Construction 5.9 (The potential pair)**.** *Radial potential.* Put

$$\psi_R(t) = 1 - S\Big(\frac{t^2-1}{3}\Big) \quad (t \in \mathbb{R}), \qquad C_\psi = \sup_{t\in\mathbb{R}} |\psi_R'(t)|, \qquad K_3 = \frac{m_+}{\eta_{\mathrm w}^2},$$

$$\varepsilon_R = \frac{e\, r^3}{256\, m_+^4\,(1 + C_\psi)}, \qquad \kappa_R = \frac{e}{196608} = \frac{e}{3\cdot 2^{16}}, \qquad \delta' = \delta + \kappa_R = \frac{e}{8} + \kappa_R,$$

$g_R(x) = \frac12\alpha_0 x^2 - R(x)$ and

$$\tilde R(x) = R(x) + \psi_R(x/\varepsilon_R)\, g_R(x) = \tfrac12\alpha_0 x^2 - \big(1 - \psi_R(x/\varepsilon_R)\big) g_R(x) \qquad (x \in \mathbb{R}).$$

Let $\Sigma_R(x) = \operatorname{sign}(x)\sqrt{\tilde R(x)}$, let $\tilde X = \Sigma_R^{-1}$ (a bijection of $\mathbb{R}$ by Lemma 5.13), and put $u_\varepsilon = \sqrt{\alpha_0/2}\,\varepsilon_R$.

Angular potential. Put $w_V = e/(20h)$, $c_V = 1 - 2w_V$, and for real θ

$$\operatorname{st}(\theta) = S\Big(\frac{4\theta^2 - c_V^2}{1 - c_V^2}\Big), \qquad \tilde V(\theta) = \frac{e\,\theta^2}{2} + 2h\operatorname{st}(\theta), \qquad \theta_c = \frac12\sqrt{c_V^2 + \frac{1-c_V^2}{2}}\,.$$

Let $\sigma_V(\theta) = \operatorname{st}(\theta)/\theta^2$ for $\theta \ne 0$ and $\sigma_V(0) = 0$, put $\Sigma_V(\theta) = \theta\sqrt{e/2 + 2h\,\sigma_V(\theta)}$, let $\tilde Y = \Sigma_V^{-1}$ (a bijection of $\mathbb{R}$ by Lemma 5.14), and put $v_\varepsilon = \sqrt{e/2}\,(\frac12 - w_V)$ and $\rho_T = \min(u_\varepsilon, v_\varepsilon)$.

ψ_R is C^∞, even, nonincreasing in $|t|$ and $[0,1]$-valued, with $\psi_R = 1$ for $|t| \le 1$ and $\psi_R = 0$ for $|t| \ge 2$, so $C_\psi = \max_{[-2,2]} |\psi_R'| < \infty$ (only $|\psi_R'| \le C_\psi$ is used); $\tilde R = \frac12\alpha_0 x^2$ for $|x| \le \varepsilon_R$ and $\tilde R = R$ for $|x| \ge 2\varepsilon_R$. As $e \le h/100$, $w_V \le 1/2000$ and $c_V \in (0,1)$. The function st is even and nondecreasing in $|\theta|$, with $\operatorname{st} = 0$ on $|\theta| \le \frac12 - w_V$ and $\operatorname{st} = 1$ on $|\theta| \ge \frac12$.

Lemma 5.10 (Third derivative and Taylor bounds)**.** *(i) With $\xi = x^* + x$,*

$$R'''(x) = -3m_+\eta_{\mathrm w}^2\,\xi\,(\xi^2 + \eta_{\mathrm w}^2)^{-5/2}, \qquad |R'''(x)| \le K_3 \quad (x \in \mathbb{R}).$$

(ii) $g_R(0) = g_R'(0) = g_R''(0) = 0$ and $|g_R'''| \le K_3$; hence $|g_R'(x)| \le K_3 x^2/2$ and $|g_R(x)| \le K_3|x|^3/6$.

(iii) $K_3\,\varepsilon_R = \alpha_0/\big(32(1 + C_\psi)\big)$.

Proof. (i) Differentiate $R''(x) = m_+\eta_{\mathrm w}^2(\xi^2 + \eta_{\mathrm w}^2)^{-3/2}$. Substituting $\xi = \eta_{\mathrm w}\tau$ gives

$$|R'''(x)| = (3m_+/\eta_{\mathrm w}^2)\varphi(|\tau|), \qquad \varphi(\tau) = \tau(1+\tau^2)^{-5/2} \quad (\tau \ge 0).$$

Since

$$\varphi'(\tau) = (1+\tau^2)^{-7/2}(1 - 4\tau^2),$$

φ is maximal at $\tau = \frac12$, where

$$\varphi(\tfrac12) = \tfrac12(\tfrac45)^{5/2} = \frac{16}{25\sqrt5}.$$

Hence $3\varphi \le \frac{48}{25\sqrt5} < 1$, because $48^2 = 2304 < 3125 = 25^2\cdot 5$, and $|R'''| \le K_3$.

(ii) By Lemma 5.5, $R(0) = R'(0) = 0$ and $R''(0) = \alpha_0$, so $g_R(0) = g_R'(0) = g_R''(0) = 0$, and $g_R''' = -R'''$. Integrating three times from 0: $|g_R''(x)| \le K_3|x|$, $|g_R'(x)| \le K_3 x^2/2$, $|g_R(x)| \le K_3|x|^3/6$.

(iii) $\eta_{\mathrm{w}} = e/(8m_+)$ gives $K_3 = 64m_+^3/e^2$, and $\alpha_0 = 8r^3/(em_+)$ by Lemma 5.3. So

$$K_3\varepsilon_R = \frac{64m_+^3}{e^2}\cdot\frac{e\,r^3}{256\,m_+^4(1+C_\psi)} = \frac{r^3}{4em_+(1+C_\psi)} = \frac{8r^3}{em_+}\cdot\frac{1}{32(1+C_\psi)} = \frac{\alpha_0}{32(1+C_\psi)}. \quad \square$$

Lemma 5.11 (Closeness)**.** *For all real x and p,*

$$|\tilde R(x) - R(x)| \le \kappa_R, \qquad \big|\tilde R(p-p^*) - \mathcal{R}_{\mathrm{tent}}(p)\big| \le \delta' = \frac{e}{8} + \kappa_R. \tag{7}$$

Proof. $\tilde R - R = \psi_R(x/\varepsilon_R)g_R$ vanishes for $|x| \ge 2\varepsilon_R$. For $|x| \le 2\varepsilon_R$, Lemma 5.10(ii),(iii) give

$$|\tilde R - R| \le K_3(2\varepsilon_R)^3/6 = \alpha_0\varepsilon_R^2/(24(1+C_\psi)).$$

Substituting α_0 and ε_R, the right-hand side equals $\frac{e}{3\cdot 2^{16}}(r/m_+)^9(1+C_\psi)^{-3}$, which is $\le \kappa_R$ since $r \le m_+$. The second inequality then follows from (6). $\square$

Lemma 5.12 (Monotonicity)**.** *$\tilde R$ is C^∞, and $\operatorname{sign}(x)\,\tilde R'(x) > 0$ for every $x \ne 0$; more precisely $\operatorname{sign}(x)\tilde R'(x) \ge \frac{91}{96}\alpha_0|x|$ for $0 < |x| \le 2\varepsilon_R$. Consequently $\tilde R$ is strictly decreasing on $(-\infty, 0]$ and strictly increasing on $[0,\infty)$, $\tilde R(0) = 0$, $\tilde R(x) > 0$ for $x \ne 0$, and $\tilde R(x) \to +\infty$ as $x \to \pm\infty$.*

Proof. Differentiating,

$$\tilde R'(x) = \alpha_0 x - (1 - \psi_R(x/\varepsilon_R))g_R'(x) + \varepsilon_R^{-1}\psi_R'(x/\varepsilon_R)g_R(x).$$

For $0 < |x| \le 2\varepsilon_R$, Lemma 5.10 bounds the last two terms by $\alpha_0|x|/32$ and $\alpha_0|x|/48$, giving $\frac{91}{96}\alpha_0|x|$. For $|x| > 2\varepsilon_R$, $\tilde R = R$ near x, and $\operatorname{sign}(x)R'(x) > 0$ by strict convexity and $R'(0) = 0$. The remaining claims follow, using $\tilde R = R \to \infty$. $\square$

Lemma 5.13 ($\tilde X$ inverts $\tilde R$ and is linear near 0)**.** *(i) Σ_R is a C^∞ diffeomorphism of $\mathbb{R}$ with $\Sigma_R' > 0$, and $\Sigma_R(x) = \sqrt{\alpha_0/2}\,x$ for $|x| \le \varepsilon_R$.*

(ii) $\tilde X$ is a C^∞ diffeomorphism of $\mathbb{R}$ with $\tilde X' > 0$, $\operatorname{sign}\tilde X(u) = \operatorname{sign} u$ and $\tilde R(\tilde X(u)) = u^2$.

(iii) $\tilde X(u) = a_0 u$ for $|u| \le u_\varepsilon$; in particular $\tilde X'(u) = a_0$ there.

(iv) If $|X(u)| \ge 2\varepsilon_R$ or $|\tilde X(u)| \ge 2\varepsilon_R$, then $\tilde X(u) = X(u)$. Hence

$$|\tilde X(u)| \le \max\big(|X(u)|, 2\varepsilon_R\big) \le |X(u)| + 2\varepsilon_R \qquad (u \in \mathbb{R}).$$

Proof. (i) On $|x| \le \varepsilon_R$, $\Sigma_R = \sqrt{\alpha_0/2}\,x$. On $\pm x > 0$, $\Sigma_R = \pm\sqrt{\tilde R}$ has $\Sigma_R' = \pm\tilde R'/(2\sqrt{\tilde R}) > 0$ by Lemma 5.12. Since $\Sigma_R^2 = \tilde R \to \infty$ and Σ_R has the sign of x, it is a diffeomorphism of $\mathbb{R}$.

(ii) $\tilde X' = 1/\Sigma_R'(\tilde X) > 0$ and $\tilde R(\tilde X(u)) = \Sigma_R(\tilde X(u))^2 = u^2$.

(iii) For $|u| \le u_\varepsilon$, $|a_0 u| \le \varepsilon_R$, so $\Sigma_R(a_0 u) = u$.

(iv) If $|X(u)| \ge 2\varepsilon_R$, then $\tilde R(X(u)) = R(X(u)) = u^2$, so $\Sigma_R(X(u)) = u$. If $|\tilde X(u)| \ge 2\varepsilon_R$, then $R(\tilde X(u)) = u^2$, so $\tilde X(u) = X(u)$ by uniqueness in Lemma 5.6. The bound follows by cases on $|\tilde X(u)|$. $\square$

Lemma 5.14 (The angular wall and its signed inverse)**.** *(a) $\tilde V$ is C^∞, even and nonnegative, $\tilde V(\theta) = e\theta^2/2$ for $|\theta| \le \frac12 - w_V$, $\tilde V(\theta) \ge e\theta^2/2$ for all θ, and $\tilde V'(\theta) \ge e\theta > 0$ for $\theta > 0$; in particular $\tilde V$ is strictly increasing in $|\theta|$.*

(b) $\tilde V(\pm\frac12) = e/8 + 2h > h$.

(c) $\frac12 - w_V < \theta_c < \frac12$, and if $\tilde V(\theta) \le h - 2e$, then $\operatorname{st}(\theta) < \frac12$ and $|\theta| < \theta_c$.

(d) Σ_V is an odd C^∞ diffeomorphism of $\mathbb{R}$ with $\Sigma_V' > 0$, $\Sigma_V^2 = \tilde V$, $|\Sigma_V(\theta)| \ge \sqrt{e/2}\,|\theta|$, and $\Sigma_V(\theta) = \sqrt{e/2}\,\theta$ for $|\theta| \le \frac12 - w_V$.

(e) $\tilde Y$ is an odd C^∞ diffeomorphism of $\mathbb{R}$, $\tilde Y' > 0$ is even, $\tilde V(\tilde Y(v)) = v^2$, $|\tilde Y(v)| \le b_0|v|$, and $\tilde Y(v) = b_0 v$ for $|v| \le v_\varepsilon$.

Proof. (a) st is C^∞ and even, vanishes on $|\theta| \le \frac{1}{2} - w_V$, and has $\mathrm{st}' \ge 0$ for $\theta > 0$, so $\tilde V' = e\theta + 2h\,\mathrm{st}' \ge e\theta$ for $\theta > 0$.

(b) $\mathrm{st}(\pm\frac{1}{2}) = S(1) = 1$.

(c) $\theta_c^2 = \frac{c_V^2+1}{8}$ lies strictly between $(\frac{1}{2} - w_V)^2 = c_V^2/4$ and $\frac{1}{4}$, since $0 < c_V < 1$. If $\tilde V(\theta) \le h - 2e$, then $2h\,\mathrm{st}(\theta) < h$, so $\mathrm{st}(\theta) < \frac{1}{2}$. As S is nondecreasing with $S(\frac{1}{2}) = \frac{1}{2}$, this gives $4\theta^2 < c_V^2 + (1 - c_V^2)/2 = 4\theta_c^2$.

(d) σ_V is C^∞, vanishing on $|\theta| < \frac{1}{2} - w_V$ and a quotient of smooth functions on $\theta \ne 0$; so Σ_V is odd and C^∞, and the stated identities and bounds are immediate. For $\theta > 0$, $\Sigma_V' = \tilde V'/(2\sqrt{\tilde V}) > 0$ by (a); by oddness $\Sigma_V' > 0$ for $\theta < 0$, and $\Sigma_V'(0) = \sqrt{e/2}$. Being proper, Σ_V is a diffeomorphism.

(e) Invert (d). $\tilde Y' = 1/\Sigma_V'(\tilde Y)$ is even, and $|v| = |\Sigma_V(\tilde Y(v))| \ge \sqrt{e/2}\,|\tilde Y(v)|$. For $|v| \le v_\varepsilon$, $|b_0 v| \le \frac{1}{2} - w_V$, so $\Sigma_V(b_0 v) = v$. □

Lemma 5.15 (The equal-ball potential pair)**.** *Let $d = 0$, $I_U = I_V = \mathbb{R}$, $U(p) = \tilde R(p - p^*)$ and $V = \tilde V$. Then (U, V) is an admissible potential pair with disc area h (Definition 4.1), with minimum point p^* and minimum value $m = 0$, and it is exactly quadratic near the minimum with $\varepsilon_U = \varepsilon_R$ and $\varepsilon_V = \frac{1}{2} - w_V$. Its objects in Lemma 4.2 are $X_y = \tilde X$, $Y_y = \tilde Y$, $a_0(y) = a_0$, $b_0(y) = b_0$, $j_0(y) = j_0$, $\rho_T(y) = \rho_T$, and $\mathbf{L}_y = \mathbf{L}$ in Definition 4.6.*

Proof. (P1): $p^* \in (0, h)$ by Lemma 5.8. By Lemma 5.12, $U'(p)(p - p^*) = \tilde R'(x)\,x > 0$ for $x \ne 0$, and $U''(p^*) = \alpha_0 > 0$. By Lemma 5.14(a), $\tilde V(0) = 0$, $\tilde V''(0) = e > 0$, and $\tilde V'(\theta)\theta > 0$ for $\theta \ne 0$.

(P2): $\tilde R \to +\infty$ (Lemma 5.12) and $\tilde V \ge e\theta^2/2$. Moreover $m = \tilde R(0) = 0$, and the quadratic regions are $|p - p^*| \le \varepsilon_R$ and $|\theta| \le \frac{1}{2} - w_V$.

The signed square roots of $U(p^* + \cdot)$ and $\tilde V$ are Σ_R and Σ_V (Lemma 5.14(d)). Hence $X_y = \tilde X$ and $Y_y = \tilde Y$, and so $a_0(y) = a_0$, $b_0(y) = b_0$, $j_0(y) = j_0$ and $\mathbf{L}_y = \mathbf{L}$. Finally, $\varepsilon_R/a_0 = u_\varepsilon$ and $(\frac{1}{2} - w_V)/b_0 = v_\varepsilon$ give $\rho_T(y) = \rho_T$. □

5.4. **The inner map.**

Lemma 5.16 (The Morse chart and its density)**.** *Put $F_{\mathrm{mor}}(u, v) = (\tilde X(u), \tilde Y(v))$, $J(u, v) = \tilde X'(u)\tilde Y'(v)$ and $\mathcal{A}_J(E) = \int_{|w|^2 \le E} J(w)\,dw$ for $E \ge 0$. Then F_{mor} is a C^∞ diffeomorphism of $\mathbb{R}^2$ with inverse (Σ_R, Σ_V), $J > 0$ is C^∞, $J(w) = j_0$ for $|w| \le \rho_T$, $F_{\mathrm{mor}}^*(dx \wedge d\theta) = J\,du \wedge dv$, $\tilde R(\tilde X(u)) + \tilde V(\tilde Y(v)) = u^2 + v^2$, and $\mathcal{A}_J(E) = \mathrm{area}\{(x, \theta) : \tilde R(x) + \tilde V(\theta) \le E\} \to \infty$ as $E \to \infty$.*

Proof. By Lemma 5.15, F_{mor} and J are the Morse chart F_y and density J_y of Lemma 4.2(iv), with Morse rectangle $\mathbb{R}^2$; parts (iv), (v) of that lemma give all claims except the limit, which holds as $\tilde R, \tilde V$ are proper and nonnegative. □

Let T_J, D_J, E_J be the rotation normalisation, the area normalisation and the inverse of $\mathcal{A}_J$ for J (Definition 4.3), and let $\bar J$ be the circle mean of J.

Definition 5.17 (The inner map)**.** $f = F_{\mathrm{mor}} \circ T_J^{-1} \circ D_J$. By Lemma 5.15 this is the map f_y of Definition 4.6 for the pair (U, V).

Proposition 5.18 (The inner map; exactly linear at the origin)**.**

(a) The map f is a smooth diffeomorphism of the plane, and

$$f^*(dx \wedge d\theta) = \omega_2.$$

If $f(z) = (x, \theta)$, then $\tilde R(x) + \tilde V(\theta) = E_J(\pi|z|^2)$.

(b) On $\{|w| \le \rho_T\}$, $\bar J = j_0$ and $T_J = T_J^{-1} = \mathrm{id}$; $\mathcal{A}_J(E) = \pi j_0 E$ for $E \le \rho_T^2$, $E_J(I) = I/(\pi j_0)$ for $I \le \pi j_0 \rho_T^2$, and $D_J(z) = z/\sqrt{j_0}$ for $|z| \le \sqrt{j_0}\,\rho_T$.

(c) For $|z| \le \sqrt{j_0}\,\rho_T$,

$$f(z) = \operatorname{diag}\big(a_0/\sqrt{j_0},\, b_0/\sqrt{j_0}\big)z = \operatorname{diag}(c_x, c_y)\, z = \mathbf{L}z,$$

a linear map with $\det \mathbf{L} = 1$.

Proof. Theorem 4.7 applies by Lemmas 5.15 and 5.16. (a) is Theorem 4.7(a),(b) with $m = 0$.

(b) $J = j_0$ on $\{|w| \le \rho_T\}$, so $\bar{J} = j_0$ there. Lemma 4.4(iv) gives $T_J = T_J^{-1} = \mathrm{id}$, Lemma 4.5(v) gives $\mathcal{A}_J$ and D_J, and inverting gives E_J.

(c) is Theorem 4.7(d). Explicitly, by (b) and Lemmas 5.13(iii) and 5.14(e), $f(z) = (a_0 u, b_0 v)$ with $(u, v) = z/\sqrt{j_0}$; moreover $a_0/\sqrt{j_0} = c_x$ and $b_0/\sqrt{j_0} = 1/c_x = c_y$. □

5.5. **The physical chart and the comparison.**

Definition 5.19 (Physical chart)**.** Let $\mathcal{U} = \{(x,\theta) \in \mathbb{R}^2 : p^* + x > 0,\ |\theta| < \tfrac{1}{2}\}$ and

$$C_{\mathrm{ph}}(x,\theta) = -\sqrt{\frac{p^* + x}{\pi}}\, e^{2\pi i\theta} \qquad ((x,\theta) \in \mathcal{U}). \tag{8}$$

Put $r_b = \sqrt{p^*/\pi}$ and $b = C_{\mathrm{ph}}(0) = (-r_b, 0)$. For the pair of Lemma 5.15 these are the chart C_y, its domain $\mathcal{C}_y$, and $r_b(y)$, b_y of (3).

Lemma 5.20 (The physical chart)**.** *$\mathcal{U}$ is convex and contains* 0*. C_{ph} is a C^∞ injective map on $\mathcal{U}$ with* $\det DC_{\mathrm{ph}} = 1$*, i.e.* $C_{\mathrm{ph}}^*\omega_2 = dx \wedge d\theta$*, and* $p(C_{\mathrm{ph}}(x,\theta)) = p^* + x$*. In particular C_{ph} maps $\mathcal{U} \cap \{x < h - p^*\}$ into $D(h)$. At a point with $p^* + x = P$ the columns of DC_{ph} are orthogonal, of lengths $1/(2\sqrt{\pi P})$ and $2\sqrt{\pi P}$, and $DC_{\mathrm{ph}}(0) = \operatorname{diag}\big(-\frac{1}{2\pi r_b},\, -2\pi r_b\big)$.*

Proof. $0 \in \mathcal{U}$ by Lemma 5.8, and $\mathcal{U}$ is an intersection of half-planes. C_{ph} has polar radius $\rho_C = \sqrt{(p^* + x)/\pi}$ and polar angle $2\pi\theta + \pi \in (0, 2\pi)$, which gives injectivity and the moment. The area form is $\rho_C\, d\rho_C \wedge d(2\pi\theta) = \frac{1}{2\pi} d(p^* + x) \wedge 2\pi\, d\theta = dx \wedge d\theta$. With $P = p^* + x$, $\partial_x C_{\mathrm{ph}} = -\frac{1}{2\sqrt{\pi P}} e^{2\pi i\theta}$ and $\partial_\theta C_{\mathrm{ph}} = -2\pi i\sqrt{P/\pi}\, e^{2\pi i\theta}$, which are orthogonal with the stated lengths; at 0, $e^{2\pi i\theta} = 1$ and $\sqrt{\pi p^*} = \pi r_b$. □

Definition 5.21 (Comparison constants)**.** $\mu_V = \int_{-1/2}^{1/2} \tilde{V}(\theta)\, d\theta$, $d_0 = \delta' + \mu_V$, $p_+ = h - 15e/4$, and $E_c = h - 2e$.

Lemma 5.22 (Size of the comparison constants)**.** $\mu_V \le e/24 + e/5 = 29e/120 < 7e/24$*. Moreover* $d_0 < 5e/12$*,* $2\delta' + \mu_V < 13e/24$*,* $p_+ > 0$*, and* $E_c - d_0 > p_+$.

Proof. $\int_{-1/2}^{1/2} e\theta^2/2\, d\theta = e/24$. The function st takes values in $[0,1]$ and vanishes on $|\theta| \le \frac{1}{2} - w_V$, so $\int_{-1/2}^{1/2} 2h\,\mathrm{st} \le 2h \cdot 2w_V = e/5$. Hence $\mu_V \le 5e/120 + 24e/120 = 29e/120 < 7e/24$. Since $\kappa_R = e/196608 < e/40$,

$$d_0 \le \frac{15e}{120} + \frac{29e}{120} + \kappa_R < \frac{50e}{120} = \frac{5e}{12}, \qquad 2\delta' + \mu_V \le \frac{59e}{120} + 2\kappa_R < \frac{65e}{120} = \frac{13e}{24}.$$

As $e \le h/100$, $15e/4 < h$, so $p_+ > 0$; and $E_c - d_0 > h - 2e - 5e/12 = h - 29e/12 > h - 45e/12 = p_+$. □

Lemma 5.23 (Area lower bound)**.** $\mathcal{A}_J(E) \ge E - d_0$ *for all* $E \ge 0$.

Proof. Fix $|\theta| \le \frac{1}{2}$. By (7) and (5), the x-section of $\{\tilde{R} + \tilde{V} \le E\}$ contains an interval of length $(E - \delta' - \tilde{V}(\theta))_+$. Integrating over θ gives $\mathcal{A}_J(E) \ge E - d_0$. □

Proposition 5.24 (Comparison and chart containment)**.** *Let $z \in \mathbb{C}$, $I = \pi|z|^2$, and suppose that $f(z) \in \mathcal{U}$. Then $E_J(I) \le I + d_0$ and*

$$\mathcal{R}_{\mathrm{tent}}\big(p(C_{\mathrm{ph}}f(z))\big) \le I + 2\delta' + \mu_V < I + \frac{13e}{24}. \tag{9}$$

Moreover, if $I \le p_+$ then $E_J(I) \le p_+ + d_0 < h - \delta'$ and $f(z) \in \{(x,\theta) : -p^* < x < h - p^*,\ |\theta| < \frac{1}{2}\}$; thus $C_{\mathrm{ph}}f$ is defined on $\bar{D}(p_+)$, takes values in $\{0 < p < h\}$, and satisfies (9) *there.*

Proof. Let $f(z) = (x,\theta)$ and $E = E_J(I)$. By Lemma 5.23, $I = \mathcal{A}_J(E) \ge E - d_0$. By Proposition 5.18(a), $\tilde{R}(x) \le \tilde{R}(x) + \tilde{V}(\theta) = E$. By Lemma 5.20 and (7), $\mathcal{R}_{\mathrm{tent}}(p(C_{\mathrm{ph}}f(z))) = \mathcal{R}_{\mathrm{tent}}(p^* + x) \le \tilde{R}(x) + \delta' \le I + d_0 + \delta' = I + 2\delta' + \mu_V$, and $2\delta' + \mu_V < 13e/24$ by Lemma 5.22.

Let $I \le p_+$. Then $E \le p_+ + d_0 < h - 15e/4 + 5e/12 = h - 10e/3 < h - \delta'$. At $x = -p^*$ and $x = h - p^*$ (moments 0 and h) we have $\mathcal{R}_{\mathrm{tent}} = h$, hence $\tilde{R} \ge h - \delta' > E$ by (7). Since $\tilde{R}$ decreases strictly on $(-\infty, 0]$ and increases strictly on $[0,\infty)$ (Lemma 5.12) and $0 \in (-p^*, h - p^*)$ (Lemma 5.8), the sublevel set $\{\tilde{R} \le E\}$ is an interval contained in $(-p^*, h - p^*)$. At the angular boundary $\tilde{V}(\pm\frac{1}{2}) = e/8 + 2h > E$, and $\tilde{V}$ is even and increasing in $|\theta|$ (Lemma 5.14), so $\{\tilde{V} \le E\} \subset (-\frac{1}{2}, \frac{1}{2})$. Since $\tilde{R}(x) \le E$ and $\tilde{V}(\theta) \le E$, $f(z)$ lies in the stated rectangle, and the first part applies. □

5.6. **The data of the whole-disc isotopy.**

Definition 5.25 (The density family)**.** For $s \in \mathbb{R}$ let

$$X_s(u) = (1-s)a_0 u + s\tilde{X}(u), \qquad Y_s(v) = (1-s)b_0 v + s\tilde{Y}(v), \qquad J_s(u,v) = X_s'(u)\,Y_s'(v),$$

$F_s = (X_s, Y_s)$, and let $\mathcal{A}_s$, $E_s = \mathcal{A}_s^{-1}$, T_{J_s}, D_{J_s} and $f_s = F_s \circ T_{J_s}^{-1} \circ D_{J_s}$ be the objects of Definition 4.8(a) for the pair of Lemma 5.15. Put $K_0 = \pi j_0$ and $\kappa_J = K_0/(1 + K_0)$.

Lemma 5.26 (Positivity and area growth)**.** *For all $s \in [0,1]$:*

(a) $J_s = (1-s)^2 j_0 + s(1-s)\big(a_0\tilde{Y}'(v) + b_0\tilde{X}'(u)\big) + s^2 J > 0$;
(b) $J_s(w) = j_0$ for $|w| \le \rho_T$ (this holds for every $s \in \mathbb{R}$);
(c) for all $E \ge 0$, $\mathcal{A}_s(E) \ge [K_0(1-s)^2 + s^2]E - s^2 d_0 \ge \kappa_J E - d_0$.

In particular $\mathcal{A}_s$ is an increasing bijection of $[0,\infty)$, and for every $I_0 \ge 0$, $E_s(I) \le (I_0 + d_0)/\kappa_J$ whenever $0 \le I \le I_0$.

Proof. (a) is Lemma 4.10(a). (b) For $|w| \le \rho_T$, $\tilde{X}'(u) = a_0$ and $\tilde{Y}'(v) = b_0$ (Lemma 5.16), hence $X_s'(u) = a_0$, $Y_s'(v) = b_0$. (c) By Lemma 4.10(b), $\mathcal{A}_s(E) \ge K_0(1-s)^2 E + s^2 \mathcal{A}_J(E)$, and $\mathcal{A}_J(E) \ge E - d_0$ by Lemma 5.23. The minimum of $K_0(1-s)^2 + s^2$ over $s \in \mathbb{R}$ is $K_0/(1+K_0)$, and $s^2 d_0 \le d_0$. $\mathcal{A}_s$ is continuous and strictly increasing with $\mathcal{A}_s(0) = 0$ because $J_s > 0$, and $\mathcal{A}_s(E) \to \infty$ by (c). If $\mathcal{A}_s(E) = I \le I_0$, then $\kappa_J E - d_0 \le I_0$. □

Definition 5.27 (Constants of the common source disc)**.** Put $c_P = 2e - \delta'$ and, for $E \ge 0$,

$$\Delta P = \frac{c_P}{\max(A,B)}, \qquad \Delta\theta = \tfrac{1}{2} - \theta_c,$$

$$M_X(E) = \frac{|m_-|E + m_+\sqrt{E(E + 2\eta_{\mathrm{w}} r)}}{r^2}, \quad \tilde{M}_X(E) = M_X(E) + 2\varepsilon_R,$$

$$D_x = a_0\sqrt{E_c} + \tilde{M}_X(E_c), \qquad D_y = b_0\sqrt{E_c} + \theta_c, \qquad R_q = \sqrt{\big(h - c_P/B\big)/\pi},$$

$$\delta_s = \min\Big(\frac{1}{4},\ \frac{1}{2}\Big[1 - \sqrt{\tfrac{p_+}{E_c - d_0}}\Big],\ \frac{\Delta P}{2D_x},\ \frac{\Delta\theta}{2D_y}\Big), \qquad p^\dagger = \tfrac{1}{2}\big[p_+ + (1-\delta_s)^2(E_c - d_0)\big],$$

$$\bar{E} = \frac{p^\dagger + d_0}{\kappa_J}, \qquad \bar{X} = \max\big(a_0\sqrt{\bar{E}},\ \tilde{M}_X(\bar{E})\big), \qquad \bar{Y} = b_0\sqrt{\bar{E}}.$$

Lemma 5.28 (Order of the constants)**.** *$0 < c_P < h$, all four entries of the minimum defining δ_s are positive, so $0 < \delta_s \le \frac{1}{4}$, and*

$$p_+ < p^\dagger < (1-\delta_s)^2(E_c - d_0) < h.$$

Proof. We have $c_P = 15e/8 - \kappa_R \in (0,h)$, and $\Delta\theta > 0$ since $\theta_c < \frac{1}{2}$ (Lemma 5.14(c)). Lemma 5.22 gives $E_c - d_0 > p_+ > 0$. Moreover $1-\delta_s \ge \frac{1}{2}(1+\sqrt{p_+/(E_c-d_0)}) > \sqrt{p_+/(E_c-d_0)}$, so $p_+ < (1-\delta_s)^2(E_c-d_0)$, and $p^\dagger$ is their midpoint. Finally $(1-\delta_s)^2(E_c-d_0) < E_c < h$. □

Lemma 5.29 (Bounds for the instance)**.** *(a) For all real u, $|X(u)| \le M_X(u^2)$ and $|\tilde X(u)| \le \tilde M_X(u^2)$; $\tilde M_X$ is increasing. For all u, v with $u^2 \le \bar E$, $v^2 \le \bar E$:*

$$\max\bigl(a_0|u|, |\tilde X(u)|\bigr) \le \bar X, \qquad \max\bigl(b_0|v|, |\tilde Y(v)|\bigr) \le \bar Y.$$

(b) For all u, v with $u^2 < E_c$ and $v^2 < E_c$:

$$\Delta P \le \frac{c_P}{A} \le p^* + \tilde X(u) \le h - \frac{c_P}{B} = \pi R_q^2 \le h - \Delta P, \qquad |\tilde Y(v)| < \theta_c = \tfrac{1}{2} - \Delta\theta.$$

(c) For all u, v with $u^2 < E_c$ and $v^2 < E_c$: $|a_0 u - \tilde X(u)| \le D_x$ and $|b_0 v - \tilde Y(v)| \le D_y$.

(d) $(1-\delta_s)^2 \mathcal{A}_J(E_c) > p^\dagger$, and $\mathcal{A}_s(\bar E) \ge p^\dagger$ for every $s \in [0,1]$.

Proof. (a) From the formula for X, $|X(u)| \le (|m_-|u^2 + m_+|u|\sqrt{u^2+2\eta_{\mathrm w} r})/r^2 = M_X(u^2)$, and $|\tilde X(u)| \le |X(u)| + 2\varepsilon_R \le \tilde M_X(u^2)$ by Lemma 5.13(iv); M_X is increasing. If $u^2 \le \bar E$, then $a_0|u| \le a_0\sqrt{\bar E} \le \bar X$ and $|\tilde X(u)| \le \tilde M_X(\bar E) \le \bar X$. If $v^2 \le \bar E$, then $|\tilde Y(v)| \le b_0|v| \le b_0\sqrt{\bar E} = \bar Y$ by Lemma 5.14(e).

(b) $\tilde R(\tilde X(u)) = u^2 < E_c$, so (7) gives $\mathcal{R}_{\mathrm{tent}}(p^* + \tilde X(u)) < E_c + \delta' = h - c_P$. As $0 < c_P < h$ (Lemma 5.28), (5) with $\beta = h - c_P$ and $h/A = a$, $h/B = h - a$ (Lemma 3.3) gives

$$p^* + \tilde X(u) \in \bigl[a - (h-c_P)/A,\ a + (h-c_P)/B\bigr] = \bigl[c_P/A,\ h - c_P/B\bigr].$$

Since $\Delta P = c_P/\max(A,B)$ is at most c_P/A and c_P/B, the outer inequalities follow, and $h - c_P/B = \pi R_q^2$ by definition. Next $\tilde V(\tilde Y(v)) = v^2 < E_c = h - 2e$, so $|\tilde Y(v)| < \theta_c$ by Lemma 5.14(c).

(c) $|a_0 u - \tilde X(u)| \le a_0|u| + |\tilde X(u)| \le a_0\sqrt{E_c} + \tilde M_X(E_c) = D_x$ by (a), and $|b_0 v - \tilde Y(v)| \le b_0\sqrt{E_c} + \theta_c = D_y$ by (b).

(d) By Lemmas 5.23 and 5.28, $(1-\delta_s)^2\mathcal{A}_J(E_c) \ge (1-\delta_s)^2(E_c - d_0) > p^\dagger$. By Lemma 5.26(c), $\mathcal{A}_s(\bar E) \ge \kappa_J \bar E - d_0 = p^\dagger$. □

Definition 5.30 (Cutoff levels)**.** Put $p_{\mathrm{on}} = h - \frac{31e}{8}$ and $p_{\mathrm{zero}} = h - \frac{61e}{16}$. The source cutoff of §4.7 is then $\chi(I) = 1 - S\bigl((I - p_{\mathrm{on}})/(p_{\mathrm{zero}} - p_{\mathrm{on}})\bigr)$.

Lemma 5.31 (Cutoff levels)**.** *$0 < h - 4e < p_{\mathrm{on}} < p_{\mathrm{zero}} < p_+ < p^\dagger$; $\chi = 1$ on $I \le p_{\mathrm{on}}$ and $\chi = 0$ on $I \ge p_{\mathrm{zero}}$.*

Proof. $h - 4e > 0$ since $e \le h/100$, and $4 > \frac{31}{8} > \frac{61}{16} > \frac{15}{4}$; Lemma 5.28 gives $p_+ < p^\dagger$. The values of χ follow from $S = 0$ on $(-\infty, 0]$ and $S = 1$ on $[1, \infty)$. □

Definition 5.32 (Constants of the path)**.** Put $P_{\min} = \frac{1}{2}\min(p^*, \Delta P)$ and $P_{\max} = \max(\frac{3}{2}p^*, h)$, and let M_C, $R^\dagger$, R_h, r_b^+, R_{supp}, P_{supp} and p_{bd} be the derived constants of Definition 4.11, formed with $p^*_{\mathrm{lo}} = p^*_{\mathrm{hi}} = p^*$ (so $r_b^+ = r_b$), the numbers ΔP, $p^\dagger$, R_q of Definition 5.27, and the numbers B_3, B^*, $c = c_{\mathrm{new}}$ below. Let $\mathbf{A}_0 = DC_{\mathrm{ph}}(0)\,\mathbf{L} = \mathrm{diag}\bigl(-\frac{c_x}{2\pi r_b}, -2\pi r_b c_y\bigr)$ and put

$$\lambda_0 = \min\Bigl(\frac{1}{2}, \frac{p^*}{2\bar X}, \frac{1}{4\bar Y}\Bigr), \qquad B_3 = R^\dagger \max\bigl(1, |(\mathbf{A}_0)_{11}|, |(\mathbf{A}_0)_{22}|\bigr),$$

$$B^* = \min\Bigl(M_C\sqrt{\bar X^2 + \bar Y^2},\ \frac{\bar X}{\sqrt{\pi p^*}} + 2\pi r_b \bar Y\Bigr), \qquad c_{\mathrm{new}} = \min\Bigl(\frac{1}{4},\ \frac{R_h - r_b}{8\max(R^\dagger, B_3, B^*)}\Bigr).$$

Lemma 5.33 (Isotopy data for the equal-ball pair)**.** *For the admissible potential pair $U(p) = \tilde{R}(p-p^*)$, $V = \tilde{V}$ of Lemma 5.15, the numbers*

$$p_{\mathrm{on}},\ p_{\mathrm{zero}},\ p^{\dagger},\ p^*_{\mathrm{lo}} = p^*_{\mathrm{hi}} = p^*,\ E_c,\ \Delta P,\ \Delta\theta,\ R_q,\ D_x,\ D_y,\ \delta_s,\ \bar{E},\ \bar{X},\ \bar{Y},\ \lambda_0,\ B_3,\ B^*,\ c = c_{\mathrm{new}}$$

of Definitions 5.21, 5.27, 5.30 and 5.32 are isotopy data in the sense of Definition 4.11. In the notation of that definition, $X_y = \tilde{X}$, $Y_y = \tilde{Y}$, $\mathcal{A}_y = \mathcal{A}_J$, $\mathcal{A}_{y,s} = \mathcal{A}_s$ and $\mathbf{A}_y = \mathbf{A}_0$.

Proof. By Lemma 5.15, $X_y = \tilde{X}$, $Y_y = \tilde{Y}$, $a_0(y) = a_0$, $b_0(y) = b_0$ and $\mathbf{L}_y = \mathbf{L}$. Hence $J_{y,s} = J_s$, so $\mathcal{A}_{y,s} = \mathcal{A}_s$ and $\mathcal{A}_{y,1} = \mathcal{A}_J$. By (4) and Definition 5.19, $\mathbf{A}_y = DC_{\mathrm{ph}}(0)\mathbf{L} = \mathbf{A}_0$.

Positivity. We have $p^{\dagger} > p_{\mathrm{zero}} > p_{\mathrm{on}} > h - 4e > 0$ (Lemma 5.31), $p^* > 0$ (Lemma 5.8) and $E_c > 0$. Lemma 5.28 gives $\Delta P, \Delta\theta, D_x, D_y, \delta_s > 0$. Next, $R_q > 0$ since $c_P < h$ and $B > 1$. Also $\bar{E}, \bar{X}, \bar{Y} > 0$, hence $\lambda_0 > 0$, $B_3 \geq R^{\dagger} > 0$ and $B^* > 0$. Finally $c_{\mathrm{new}} > 0$ since $R_h > r_b$, i.e. $p^* < h$.

The conditions. (D1) follows from Lemmas 5.31 and 5.8. (D2) and (D3) are Lemma 5.29(b) and (c). (D4) follows from the definition of δ_s and Lemma 5.29(d). (D5) is Lemma 5.29(d),(a). (D6) follows from the definition of λ_0. For (D7), take $r_b^+ = r_b$, $B_{\mathrm{lin}} = B_3$ and $\mathbf{A}_y = \mathbf{A}_0$. Then the three conditions hold with equality by the definitions of B_3, B^* and c_{new}, and $P_{\min}, P_{\max}$ satisfy the range conditions of Definition 4.11 with equality. □

5.7. **The isotopy and the comparison on the closed disc.** Let $(\hat{q}_t)_{t\in\mathbb{R}}$ and $(Q_t)_{t\in\mathbb{R}}$ be the isotopy and Hamiltonian of Theorem 4.21 for the pair of Lemma 5.15 with the data of Lemma 5.33 (Lemma 4.18, Construction 4.19), and P_{supp} the constant of Definition 5.32.

Theorem 5.34 (The explicit inner map and its isotopy)**.** *Let $k \geq 10$ and $t_0 \in (\frac{7}{8}, 1)$, with h, $\mathcal{R}_{\mathrm{tent}}$ and e_q as defined in §5.1. Let $q_t = \hat{q}_t$ ($t \in \mathbb{R}$) and $q = q_1$. The family (q_t), its Hamiltonian (Q_t) and the number P_{supp} have the following properties.*

- *(a) Each q_t is a symplectic diffeomorphism of $\mathbb{C}$, and $(t,z) \mapsto q_t(z)$, $(t,\zeta) \mapsto q_t^{-1}(\zeta)$ are C^∞ on $\mathbb{R}\times\mathbb{C}$; $q_t = \mathrm{id}$ for $t \leq 0$ and $q_t = q_1$ for $t \geq 1$.*
- *(b) $P_{\mathrm{supp}} < h$; for every t, $q_t = \mathrm{id}$ on $\{\pi|z|^2 > P_{\mathrm{supp}}\}$, and q_t maps $\bar{D}(P')$ onto itself for every $P' \geq P_{\mathrm{supp}}$, in particular $\bar{D}(h)$ and $D(h)$ onto themselves.*
- *(c) Q is C^∞ on $\mathbb{R}\times\mathbb{C}$, Q_t vanishes with all its derivatives on $\{\pi|\zeta|^2 > P_{\mathrm{supp}}\}$ for every t, $Q_t = 0$ for $t \notin (0,1)$, and q_t is the isotopy generated by Q_t: $\partial_t q_t = \Xi_t \circ q_t$ with $\iota_{\Xi_t}\omega_2 = -dQ_t$.*
- *(d) q equals $C_{\mathrm{ph}} \circ f$ on $\bar{D}(p_{\mathrm{on}})$, $p_{\mathrm{on}} = h - \frac{31}{8}e_q$; in particular on $\bar{D}(h - 4e_q)$.*
- *(e) For every $z \in \bar{D}(h)$, with $I = \pi|z|^2$,*

$$\mathcal{R}_{\mathrm{tent}}\big(p(q(z))\big) \leq I + 4e_q. \tag{10}$$

Proof. Theorem 4.21 applies with $d = 0$ (Lemmas 5.15, 5.33).

(a) Theorem 4.21(a),(b); joint smoothness is Lemma 4.18(e).

(b) $P_{\mathrm{supp}} < h$ is Lemma 4.12(b); the rest is Theorem 4.21(a).

(c) Smoothness of Q is Proposition 4.20; the rest is Theorem 4.21(d), with $\Xi_t = \xi_{y,t}$.

(d) By Theorem 4.21(c), $q = q_{\mathrm{in},y}$ on $\{\pi|z|^2 \leq p_{\mathrm{on}}\}$, and by Definitions 4.6, 5.17, 5.19, $q_{\mathrm{in},y} = C_{\mathrm{ph}} \circ f$ on $D(I_{\mathrm{rect}}(y)) \supset \bar{D}(p_{\mathrm{on}})$, as $p_{\mathrm{on}} < p^{\dagger} < I_{\mathrm{rect}}(y)$ (Lemmas 5.31, 4.12(b)). Also $h - 4e_q < p_{\mathrm{on}}$ (Lemma 5.31).

(e) Apply Corollary 4.22 with $\Upsilon = \mathcal{R}_{\mathrm{tent}}|_{[0,h]}$, $g(I) = I + 4e_q$, $\varepsilon = 0$. If $\pi|z|^2 \leq p_{\mathrm{on}}$, then $\pi|z|^2 < p_+$ (Lemma 5.31), so by (d) and Proposition 5.24, $\Upsilon(\pi|q_{\mathrm{in},y}(z)|^2) < \pi|z|^2 + 13e_q/24 \leq g(\pi|z|^2)$. By convexity, $\sup_{[0,h]}\Upsilon = \max(\mathcal{R}_{\mathrm{tent}}(0), \mathcal{R}_{\mathrm{tent}}(h)) = h < p_{\mathrm{on}} + 4e_q = g(p_{\mathrm{on}})$. The corollary gives (10). □

On the core the error is below $13e_q/24$; $4e_q$ is needed only on the annulus $\{p_{\mathrm{on}} < \pi|z|^2 \le h\}$. With c_s, g as in §M4.2, it costs $4c_se_q \le g/32$ in Proposition M4.4 (Lemma 3.5): a relative loss of at most $1/32$, uniformly in k and t_0.

6. Dimension four: the cubic model

Here $n = 2$, the target is $B^\circ(1)$, and the model is the degree-9 normal disc bundle of a smooth cubic in $\mathbb{CP}^2$ (Biran's decomposition [Bir01]), written on the cover $\mathbb{R}^2 \times \mathbb{C}$ of a hexagonal torus; the coefficient $1 - 3p$ forces $p < 1/3$, giving $3c_i < C$.

6.1. **The model.** Let $\zeta = e^{2\pi i/3}$ and $\Gamma = (\mathbb{R}/\mathbb{Z})^2 \cong \mathbb{C}/(\mathbb{Z} + \zeta\mathbb{Z})$ via $(u, v) \mapsto u + \zeta v$. By §7.2 there is a smooth $Z : \mathbb{R}^2 \to \mathbb{C}^3$ with $|Z| = 1$, $Z_0^3 + Z_1^3 + Z_2^3 = 0$, $Z(u + 1, v) = -Z(u, v)$ and $Z(u, v + 1) = \phi(u, v)Z(u, v)$, where $|\phi| = 1$. Representatives related by $(Z, U) \mapsto (e^{i\chi}Z, e^{3i\chi}U)$ are gauge equivalent; forms are evaluated on the cover and on horizontal unit representatives.

Definition 6.1 (The cubic cover model)**.** Use the deck maps

$$T_u(u, v, w) = (u + 1, v, -w), \qquad T_v(u, v, w) = (u, v + 1, \phi(u, v)^3 w)$$

on $\widetilde{N} = \mathbb{R}^2 \times \mathbb{C}$; N denotes these cover data with their deck identification. Put

$$p = \pi|w|^2, \quad \eta = \frac{1}{2\pi} \operatorname{Im} \sum_j \overline{Z_j(u, v)}\, dZ_j(u, v), \quad \sigma = d\eta, \quad \alpha = \frac{d \arg w}{2\pi} - 3\eta,$$

$$\Omega_0 = \sigma + d(p\alpha) = \omega_V + (1 - 3p)\sigma - 3dp \wedge \eta, \quad \Delta = [0, h], \quad 0 < h < 1/3, \quad H_{\mathrm{b}}(p) = 3(1 - 3p).$$

The expression $p\, d \arg w/(2\pi) = \frac{1}{2} \operatorname{Im}(\bar{w}\, dw)$ makes the form smooth at $w = 0$. The marked set $\mathcal{Z}$ consists of the three torus points at which $Z_0 = 0$; its inverse image in the cover is $\widetilde{\mathcal{Z}} = \mathcal{Z} + \mathbb{Z}^2$, using the three representatives given below. The cover-to-representative map is $\mathsf{N}(u, v, w) = (Z(u, v), w)$.

The deck maps preserve p and Ω_0 and send N to gauge-equivalent pairs. Since $\sigma = \sigma_\Gamma\, du \wedge dv$ with σ_Γ smooth, periodic, strictly positive and $\int_0^1 \int_0^1 \sigma_\Gamma\, du\, dv = 3$ (Lemma 7.10), Ω_0 is nondegenerate for $p < 1/3$.

Lemma 6.2 (Integrated profile)**.** *For $0 < h < 1/3$,*

$$\int_0^h H_{\mathrm{b}}(p)\, dp = 3h - \frac{9}{2}h^2, \qquad \frac{1}{2} - \int_0^h H_{\mathrm{b}}(p)\, dp = \frac{1}{2}(1 - 3h)^2.$$

Thus this scalar budget tends to $1/2$ as $h \uparrow 1/3$.

Proof. Integrate $3 - 9p$ and complete the square. □

6.2. **Handle data.** *Torus coordinates.* Let $\Lambda_0 = \mathbb{Z} + \zeta\mathbb{Z}$, $\wp = \wp_{\Lambda_0}$, $g_3 = g_3(\Lambda_0) \neq 0$, $c_F = g_3^{-1/3}$, $d_F = (3g_3)^{-1/2}$. The Weierstrass parametrisation $\Phi_\Gamma(z) = [2c_F\wp(z) : -(1 + d_F\wp'(z)) : -(1 - d_F\wp'(z))]$, $z = u + \zeta v$, with $\Phi_\Gamma(0) = [0 : 1 : -1]$, is injective modulo the lattice into gauge classes of Fermat vectors; a pole-free theta lift $Z(u, v)$ makes σ smooth on $(\mathbb{R}/\mathbb{Z})^2$ and $Z^*\eta_H$ smooth on $\mathbb{R}^2$ with explicit lattice periods (§7). Put $\nu = du \wedge dv$ ($\int \nu = 1$) and $\mathcal{Z} = \{b_1, b_2, b_3\}$ with $b_1 = (0, 0) = \Phi_\Gamma(0)$, $b_2 = (\frac{1}{3}, \frac{2}{3})$, $b_3 = (\frac{2}{3}, \frac{1}{3})$, the zeros of $\wp$ (non-lattice fixed points of $z \mapsto \zeta z$).

Proposition 6.3 (Handle data on the cubic)**.** *In the torus coordinates (u, v) the cubic model $(\Gamma, N, [0, h], \Omega_0, \mathcal{Z})$ carries the following data:*

(a) the annuli $\mathcal{A} = \{\frac{1}{8} \le u \le \frac{5}{24}\} \subset \mathcal{A}^+ = \{\frac{1}{12} \le u \le \frac{1}{4}\}$ with handle coordinates $(s, v) = (12u - \frac{3}{2},\, v)$, so that $\mathcal{A} = [0, 1] \times \mathbb{R}/\mathbb{Z}$ and $\mathcal{A}^+ = [-\frac{1}{2}, \frac{3}{2}] \times \mathbb{R}/\mathbb{Z}$;

(b) *the dual band* $\mathcal{B} = \{\frac{1}{9} < v < \frac{2}{9}\}$*, containing the closed clock band* $[\frac{1}{8}, \frac{5}{24}]$*; its active intersection with* $\mathcal{A}^+$ *is* $[\frac{1}{12}, \frac{1}{4}] \times [\frac{1}{8}, \frac{5}{24}]$*, with* $v_- = \frac{1}{8}$*,* $v_+ = \frac{5}{24}$*;*

(c) *the closed one-form* $\beta = b(v)\,dv$*,* $b = \varphi / \int_0^1 \varphi$*, where* $\varphi(v) = a_{\mathrm{flat}}(v - \frac{1}{8})\, a_{\mathrm{flat}}(\frac{5}{24} - v)$ *on* $[0,1]$*, extended* 1*-periodically, with* $a_{\mathrm{flat}}(x) = e^{-1/x}$ *for* $x > 0$ *and* $a_{\mathrm{flat}}(x) = 0$ *for* $x \le 0$*;*

(d) *the area form* $\nu = du \wedge dv = \frac{1}{12}\, ds \wedge dv$*, so* $c_\nu = \frac{1}{12}$*;*

(e) *the primitive operator* P_Γ*: for a smooth function* $\mathfrak{b}$ *on* $(\mathbb{R}/\mathbb{Z})^2$ *with* $\int \mathfrak{b}\,\nu = 0$*,*

$$\begin{aligned} P_\Gamma(\mathfrak{b}\,\nu) &= A_0\,dv - B_0\,du, \\ A_0(u,v) &= \int_0^u \big(\mathfrak{b}(s',v) - \mathfrak{b}_0(v)\big) ds', \\ \mathfrak{b}_0(v) &= \int_0^1 \mathfrak{b}(s',v)\,ds', \qquad B_0(v) = \int_0^v \mathfrak{b}_0(t)\,dt; \end{aligned}$$

(f) *the unitary frame* $(Z_0/|Z_0|)^3$ *of* N *over* $\mathcal{A}^+$*, with fibre coordinate* $w_{\mathcal{A}}$ *defined by* $U = (Z_0/|Z_0|)^3 w_{\mathcal{A}}$*, angle* $\theta_{\mathcal{A}} = \arg w_{\mathcal{A}}/2\pi$*, and the toric one-form*

$$\Gamma_{\mathcal{A}}(p) = (1 - 3p)\,\lambda_\Gamma, \qquad \lambda_\Gamma = \eta_H - \frac{1}{2\pi}\, d\arg Z_0,$$

so that $\Omega_0 = \omega_V + d\Gamma_{\mathcal{A}}$ *on* $N|_{\mathcal{A}^+}$*, with* $\omega_V = dp \wedge d\theta_{\mathcal{A}}$*;*

(g) *the coordinate discs* U_j *of radius* $\frac{1}{100}$ *about* b_j *in the centred coordinates* $(u - u(b_j), v - v(b_j))$*.*

Proof. (a), (b) The maps $u \mapsto 12u - \frac{3}{2}$ and $v \mapsto v$ identify the closed bands as stated, and $ds \wedge dv = 12\,du \wedge dv$ is positive. Modulo 1, the u-coordinates $0, \frac{1}{3}, \frac{2}{3}$ of the b_j have distance at least $\frac{1}{12}$ from $[\frac{1}{12}, \frac{1}{4}]$, and their v-coordinates $0, \frac{2}{3}, \frac{1}{3}$ have positive distance from $[\frac{1}{9}, \frac{2}{9}]$; so $\mathcal{A}^+$ and $\mathcal{B}$ avoid $\mathcal{Z}$. The base is closed, so there is no boundary collar. (c) φ is smooth on $\mathbb{R}$ because a_{flat} is flat at 0; it vanishes outside $[\frac{1}{8}, \frac{5}{24}] \subset (0,1)$ and is positive exactly on $(\frac{1}{8}, \frac{5}{24})$. Hence $b \ge 0$ is smooth and 1-periodic, $b > 0$ exactly on (v_-, v_+), $\int_0^1 b = 1$, $\operatorname{supp}\beta \subset \mathcal{B}$, β is closed because it depends on v only, and $\int_{\{s\}\times\mathbb{R}/\mathbb{Z}} \beta = 1$. (d) is the positivity of ν (§7). (e) A_0 and B_0 are 1-periodic because $\int_0^1 (\mathfrak{b} - \mathfrak{b}_0)\,ds' = 0$ and $\int_0^1 \mathfrak{b}_0 = \int \mathfrak{b}\,\nu = 0$, and $d(A_0\,dv - B_0\,du) = (\mathfrak{b} - \mathfrak{b}_0)\,du \wedge dv + \mathfrak{b}_0\,du \wedge dv = \mathfrak{b}\,\nu$; the operator is linear and preserves smooth dependence on parameters, with all jets, by differentiation under the integral sign. No support condition is needed on a closed base. (f) $\mathcal{Z} = \{b_1, b_2, b_3\}$ is disjoint from $\mathcal{A}^+$ by (a), so $Z_0 \neq 0$ on $\mathcal{A}^+$ and the frame is defined there. The identity $\Omega_0 = \omega_V + d\Gamma_{\mathcal{A}}$ is Lemma 7.13: λ_Γ is basic with $d\lambda_\Gamma = \sigma$, and $\arg U = 3\arg Z_0 + \arg w_{\mathcal{A}}$ gives $\alpha = d\theta_{\mathcal{A}} - 3\lambda_\Gamma$. The form $\Gamma_{\mathcal{A}}$ is a smooth family of base one-forms depending on p, hence toric. (g) The b_j have mutual distance at least $\frac{1}{3}$ in the flat metric of the torus, and by (a), (b) distance at least $\frac{1}{12}$ from $\mathcal{A}^+ \cup \mathcal{B}$; both bounds exceed $\frac{2}{100}$. So the closed discs $\overline{U}_j$ are disjoint from each other and from $\mathcal{A}^+ \cup \mathcal{B}$. □

As the base is closed, $\chi_\partial = 0$, no atlas is needed, P_Γ of (e) is global, and the correction region is the whole torus, where $\kappa(p) = (1 - 3p)\sigma \ge (1 - 3h)\sigma$ for $p \le h$. The background is $e_h\nu$ (Definition 7.17); twist, paths, Moser and layer maps are in §8.1.

6.3. **Transfer into the ball.** The transfer is a deformation to the normal cone, following the scheme of [Ope26]. Let $\mathcal{F}(Z) = Z_0^3 + Z_1^3 + Z_2^3$ and $X_T = \{\mathcal{F}(Z) = TW\}$, W of weight three: X_1 is a graph over $\mathbb{CP}^2$ and X_0 is the total space of $\mathcal{O}(3)|_E$ over the cubic E. One potential Φ_E restricts to every X_T; the transport (Ω_E-orthogonal lift of ∂_T on a bounded region) carries X_0 to X_1 preserving restricted forms, and the Moser return removes the exact term of $\omega_{\mathrm{FS}} + d_n\, dd^c \log(1+\mathfrak{u})$; both fields are explicit with denominators bounded below for $0 \le T, s \le 1$.

Every step commutes with $\mathsf{r}\colon Z_0 \mapsto \zeta Z_0$, whose fixed set is $\{Z_0 = 0\} \cup \{[1:0:0]\}$, so by injectivity modulo gauge the model over $E \setminus \{Z_0 = 0\}$ avoids the line (Proposition 10.19).

Construction 6.4 (The cubic transfer)**.** Fix $h < d_n < 1/3$. Use unit pairs (Z, W) with the gauge relation $(Z, W) \sim (e^{i\chi}Z, e^{3i\chi}W)$, the ambient form $\Omega_E = dd^c\Phi_E$, $\Phi_E = \log|Z|^2 + d_n \log(1 + |W|^2/|Z|^6)$ (Definition 10.1), and the constraint $X_T = \{(Z, W) : |Z| = 1,\ \mathcal{F}(Z) = TW\}$. The transfer is the composite of the following four maps.

(1) The *weight-three injection* $\iota[Z, U] = [Z, \sqrt{\pi}\, U/\sqrt{d_n - p}]$ for unit pairs with $p < d_n$ (Construction 10.4).
(2) The *transport* τ_1, the time-one map, from $T = 0$ to $T = 1$, of the field (20) on unit representatives in the gauge $Z^\dagger \dot{Z} = 0$ (Construction 10.7).
(3) The projection $\mathrm{pr}(Z, W) = Z$ from the graph constraint at $T = 1$.
(4) The *Moser return* ψ_1^{-1}, where ψ_s is the unit-representative flow of the field Y_s of (21) for the graph forms $\omega_s^{\mathrm{gr}} = \omega_{\mathrm{FS}} + s\, d_n\, dd^c \log(1 + \mathfrak{u})$, $\mathfrak{u} = |\mathcal{F}(Z)|^2/\|Z\|^6$, $0 \le s \le 1$ (Construction 10.11).

Finally set $\mathsf{r}Z = (\zeta Z_0, Z_1, Z_2)$ and $\mathsf{r}(Z, W) = (\mathsf{r}Z, W)$. This preserves the central constraint and the gauge relation.

With the ball chart b_{ball} (Lemma 10.20), the cover transfer $\mathcal{T} = b_{\mathrm{ball}} \circ \psi_1^{-1} \circ \mathrm{pr} \circ \tau_1 \circ \iota \circ \mathsf{N}$ is defined on $\{p < h\}$ off the marked fibres, pulls ω_0 back to Ω_0, identifies exactly deck-equivalent points (Theorem 10.21 in §10.6), and avoids $\{Z_0 = 0\}$ (Proposition 10.19).

6.4. **The packing.** The packing uses valid cubic data (Definition 8.1): auxiliary capacities $a_i > \rho_i$, a height $h < \frac{1}{3}$ and a comparison datum for the profile $P_{\mathrm{b}} = H_{\mathrm{b}} - \sum_i (a_i - p)_+$ on $[0, h]$, whose integral $\frac{1}{2}(1 - \sum_i a_i^2 - (1 - 3h)^2)$ (Lemma 6.2) must be positive. The four-dimensional embeddings are Theorem 10.23 in §10.7; the equal-capacity parameters are in Remark 10.25.

6.5. **The four-dimensional argument in outline.** For valid cubic data (Definition 8.1) with $d_n = \frac{1}{2}(h + \frac{1}{3})$ the i-th ball is embedded by

$$f_i = b_{\mathrm{ball}} \circ \psi_1^{-1} \circ \mathrm{pr} \circ \tau_1 \circ \iota \circ \mathsf{N} \circ \Phi^{-1} \circ \chi_{\mathcal{A}} \circ \mathcal{D}^{-1} \circ E_i \qquad \text{on } B^\circ(\hat{\rho}_i). \tag{11}$$

Here E_i places the ball in the i-th handle layer; $\mathcal{D}^{-1}$ undoes the twist (the comparison isotopy $q_{f(u)}$ fibrewise over the annulus $\mathcal{A}$); $\chi_{\mathcal{A}}$ is the frame chart; Φ^{-1} the inverse handle Moser map; N passes to unit Fermat representatives; $\iota, \tau_1, \mathrm{pr}, \psi_1^{-1}$ are the weight-three injection, transport from the cone X_0 to the graph X_1, projection to $\mathbb{CP}^2$ and inverse Moser return (Construction 6.4); $b_{\mathrm{ball}}\colon \mathbb{CP}^2 \setminus \{Z_0 = 0\} \to B^\circ(1)$ is the affine chart. With $\bar{\rho}_i = \frac{1}{2}(\rho_i + a_i)$, $\delta_i = a_i - \bar{\rho}_i$, $\hat{\rho}_i = a_i - \frac{3}{4}\delta_i$: $\rho_i < \rho_i^+ \le \bar{\rho}_i < \hat{\rho}_i < a_i < h < d_n < \frac{1}{3}$. Margins are in Table M1.

(i) Vector fields. Each flow is generated by a globally smooth field, compactly supported or of bounded speed, equal to the geometric field on a set its trajectories never leave.

- q_t: smooth Hamiltonian vanishing for $t \notin (0, 1)$ and outside $D(P_{\mathrm{supp}})$, $P_{\mathrm{supp}} < h$ (Theorem 5.34 in the equal case).
- Handle Moser field Y_t, $\iota_{Y_t}\Omega_t^{\mathrm{M}} = -\mathfrak{v}_t$: by $\Omega_\lambda^2 = 2\,\omega_V \wedge \kappa_\lambda$, nondegeneracy holds on $U = \{p < h + 2\delta_m\}$, $\delta_m = \min((d_n - h)/4, g/100)$, where $H^* > \frac{15}{16}g - 18\delta_m > 0$ (Lemmas 7.28 and 8.3(d), Propositions 8.9 and 8.13). It is cut by $\mathfrak{c}(p)$, $= 1$ for $p \le h + \frac{3}{2}\delta_m$, $= 0$ for $p \ge h + \frac{7}{4}\delta_m$; above P_{supp} it preserves p ($\iota_R\Omega_t^{\mathrm{M}} = -dp$, $\mathfrak{v}_t(R) = 0$), its deck-invariant base components are bounded by C_b (maximum over a fundamental domain), and on $\{p \le h + \delta_m\}$ the flow is the uncut one (Proposition 8.16).
- Transport field (equation 20): denominators $A\,N_{\mathrm{den}}$, $B\,N_{\mathrm{den}}$, $A = 1 - 3p_W > 1 - 3d_n > 0$, $B > 0$, $N_{\mathrm{den}} = G/A + T^2/B$. On the set $\mathcal{C}_*$ of points of X_T, $0 \le T \le 1$, $|W|^2 \le w_* = h/(d_n - h)$, one has $N_{\mathrm{den}} \ge c_* = \min(\frac{3}{2}, T_0^2/d_n)$, $T_0 = \min\big(1, (\sqrt{6}\max(1, \sqrt{w_*}))^{-1}\big)$:

$G \geq 3$ at $T = 0$ by the Fermat equation, $G \geq \frac{3}{2}$ for $T \leq T_0$ via $|\mathcal{F}(Z)| = T|W|$, and T^2/B suffices for $T \geq T_0$ (Lemma 10.9). Cut off where $N_{\mathrm{den}} \leq c_*/2$ and outside a compact set; since $|Z|$, $\mathcal{F}(Z) - TW$ are constant, $|W|^2$ is nonincreasing and $\iota(\{p \leq h\}) \subset \{|W|^2 \leq w_*\}$, trajectories from $X_0 \cap \{p \leq h\}$ stay in $\mathcal{C}_*$ (Lemma 10.5, Proposition 10.10).

- Return field: denominator $(1 + \mathfrak{u})(A_s + B_s G)$, $A_s + B_s G \geq 1 - \frac{3}{2} d_n > \frac{1}{2}$ on the unit sphere for $s \in [0, 1]$ as $\mathfrak{u} \leq 1$ (Lemma 10.12); its cutoff at half this bound is inactive (Proposition 10.15).

The other factors are closed formulas: ι needs $p < d_n$; the layer root exists as x_i increases strictly in u onto $(0, h_i(p))$; b_{ball} needs $Z_0 \neq 0$, see (ii).

(ii) Injectivity and the deleted set. Only $\{p < h\}$ is used, so the skeleton never enters; the set to avoid is the line $L = \{Z_0 = 0\}$, meeting the cubic in points b_j with marked fibres $\widetilde{\mathcal{F}}$ over them.

(a) The E_i are injective with images over disjoint intervals $l_i < u < r_i$, and $\chi_{\mathcal{A}} \mathcal{D}^{-1}$, Φ^{-1} are common diffeomorphisms, so $f_i^N = \Phi^{-1} \chi_{\mathcal{A}} \mathcal{D}^{-1} E_i$ are injective with disjoint images (Proposition 8.17(b)).

(b) Their points lie over $(\frac{1}{8}, \frac{5}{24})^2$, disjoint from its integral translates, and Φ^{-1} commutes with deck maps, so no two image points are deck equivalent (Proposition 8.17(c)).

(c) The layers lie over $\mathcal{A}$, missing the b_j; the variation primitives vanish on $\widetilde{\mathcal{F}}$ (Lemma 8.14), so Φ fixes $\widetilde{\mathcal{F}}$ and $Z_0 \neq 0$ on the image of f_i^N.

(d) $\mathcal{T}_{\mathrm{u}} = \psi_1^{-1} \circ \mathrm{pr} \circ \tau_1 \circ \iota$ is injective modulo gauge and commutes with $\mathsf{r}\colon Z_0 \mapsto \zeta Z_0$, so inputs with $Z_0 \neq 0$ map off L, and $b_{\mathrm{ball}} \circ \mathcal{T}_{\mathrm{u}} \circ \mathsf{N}$ identifies exactly deck-equivalent points (§M6.2; Lemmas 10.18, 10.17, 10.16, Proposition 10.19, Theorem 10.21).

Hence the f_i are injective with disjoint images; avoidance of L comes from symmetry, not an estimate.

(iii) Closed balls. No area-preserving map takes a closed disc of area a into $\{0 < x < a\} \times (0, 1)$, and Ψ_δ has excess $\frac{\delta}{2}$ (Proposition 8.20); so the domain is $B^\circ(\hat{\rho}_i) \supset B(\bar{\rho}_i) \supset B(\rho_i^+)$, where $0 < x < a_i - p - \frac{\delta_i}{4} < h_i(p)$. Images lie in $\{p < a_i\} \subset \{p < h\}$, inside the Moser region $\{p \leq h + \delta_m\}$. The compact $\mathcal{K} = \bigcup_i f_i(B(\rho_i^+))$ lies in $B(1 - \mu_{\mathrm{ball}})$, $\mu_{\mathrm{ball}} = \frac{1}{2} \min_{\mathcal{K}}(1 - \pi|z|^2) > 0$ (Construction 10.22). In Section M7, $\rho_i^+, \rho_i^{\mathrm{dom}}$ give the radii $r_i < r_{1,i} < r_{2,i} < r_{3,i} < r_{4,i} < r_i' < \bar{r}_i$ of Definition M7.4: the Hamiltonian field equals the isotopy velocity on $\mathbb{B}^\circ(r_{1,i}) \supset \mathbb{B}(r_i)$, the cutoff acts between $r_{1,i}$ and $r_{2,i}$, disjointness holds on $\mathbb{B}(r_{3,i})$, all stages are smooth on $\mathbb{B}^\circ(r_{4,i})$ (Proposition M7.6, Lemma M7.12). So each closed $B(s_i, c_i)$ goes onto $f_i(B(c_i)) \subset f_i(B(c_i'))$, compact in $B^\circ(C)$; the room to $\partial B(C)$ is used by the finite-output algorithm of [Son26].

7. The Fermat torus cover

The base Γ is the hexagonal torus $(\mathbb{R}/\mathbb{Z})^2$, $z = u + \zeta v$, $\zeta = e^{2\pi i/3}$. This section gives the unit Fermat lift $Z(u, v)$, the deck maps, the model form, the handle data for §6, and the first Moser primitive α_1. Write $\mathcal{F}(Z) = Z_0^3 + Z_1^3 + Z_2^3$, $S_\Gamma = \{|Z| = 1,\ \mathcal{F}(Z) = 0\} \subset \mathbb{C}^3$, and $\eta_H = \mathrm{Im}(Z^\dagger dZ)/(2\pi)$ on the unit sphere.

7.1. The representative and cover data.

Lemma 7.1 (Unit Fermat representatives)**.** *The theta lift defined below is smooth, has unit norm and satisfies $Z_0(u, v)^3 + Z_1(u, v)^3 + Z_2(u, v)^3 = 0$ for every $(u, v) \in \mathbb{R}^2$.*

Proof. Off the lattice $\hat{Z}$ is a nonzero multiple of the Weierstrass vector of Lemma 7.5, and at the lattice a nonzero multiple of $(0, 1, -1)$. Both are Fermat and normalization preserves this. Z is smooth since $\hat{Z} \neq 0$ (Lemma 7.7). □

We use the cover model of Definition 6.1: $\widetilde{N} = \mathbb{R}^2 \times \mathbb{C}$ with deck maps T_u, T_v, $\mathsf{N}(u,v,w) = (Z(u,v), w)$, and gauge $(Z,U) \sim (e^{i\chi}Z, e^{3i\chi}U)$. Put $p = \pi|w|^2$, $\eta = \operatorname{Im}\sum_j \bar{Z}_j dZ_j/(2\pi)$ and $\sigma = d\eta$.

Lemma 7.2 (Angular and model forms)**.** *On $w \neq 0$, the form $\alpha = d\arg w/(2\pi) - 3\eta$ is deck invariant, $d\alpha = -3\sigma$, and α has integral one on each fibre circle. The product $p\alpha$ extends smoothly across $w = 0$.*

Proof. Under $Z \mapsto e^{i\chi}Z$, the changes $d\chi/(2\pi)$ of η and $3d\chi/(2\pi)$ of $d\arg w/(2\pi)$ cancel in α. The rest follows from $d\eta = \sigma$ and $p\,d\arg w/(2\pi) = \frac{1}{2}\operatorname{Im}(\bar{w}\,dw)$. □

Lemma 7.3 (The coordinate model form)**.** *On the cover,*

$$\Omega_0 = \sigma + d(p\alpha) = \omega_V + (1-3p)\sigma - 3dp \wedge \eta. \tag{12}$$

It is closed, deck invariant and smooth at the fibre axis, with square $2(1-3p)\omega_V \wedge \sigma$, so nondegenerate for $p < 1/3$.

Proof. Differentiate the smooth primitive $\frac{1}{2}\operatorname{Im}(\bar{w}\,dw) + (1-3p)\eta$. In the square, every other mixed term contains a repeated vertical differential or three base differentials. Positivity of σ is Lemma 7.10. □

7.2. **An explicit parametrisation and atlas.** We uniformise Γ by $\mathbb{C}/\Lambda_0$, $\Lambda_0 = \mathbb{Z}+\zeta\mathbb{Z}$, and lift the curve to the unit sphere without poles; with the flat cutoffs of §7.3, all base data are explicit smooth functions of (u,v). Since $\zeta_6 := e^{i\pi/3} = 1+\zeta$, also $\Lambda_0 = \mathbb{Z} + \zeta_6\mathbb{Z}$, and $\zeta_6\Lambda_0 = \zeta\Lambda_0 = \Lambda_0$.

Definition 7.4 (Torus parametrisation)**.** Let $\wp = \wp_{\Lambda_0}$ be the Weierstrass function of Λ_0, $w_\wp = \wp'(z)$, and g_2, g_3 the invariants of Λ_0, so that $w_\wp^2 = 4\wp^3 - g_2\wp - g_3$. Put

$$c_F = g_3^{-1/3}, \qquad d_F = (3g_3)^{-1/2} \qquad \text{(principal powers)},$$

so that $c_F^3 g_3 = 1$ and $3d_F^2 g_3 = 1$ ($g_3 \neq 0$ by Lemma 7.5). Define Φ_Γ on the quotient by the lattice, with brackets denoting nonzero complex vectors modulo a scalar by

$$\Phi_\Gamma(z) = \big[\, 2c_F\,\wp(z) : -(1 + d_F\,w_\wp) : -(1 - d_F\,w_\wp) \,\big] \quad (z \notin \Lambda_0), \qquad \Phi_\Gamma(0) = [0:1:-1].$$

The torus is oriented by $\nu = du \wedge dv$, so $\int \nu = 1$.

The form ν gives the complex orientation, as $d\operatorname{Re} z \wedge d\operatorname{Im} z = \frac{\sqrt{3}}{2}\,du \wedge dv$. The constants c_F, d_F normalise the map, not the lattice.

Lemma 7.5 (Injectivity of the Fermat representative map)**.** *The invariants of Λ_0 are $g_2 = 0$ and $g_3 \neq 0$, so $w_\wp^2 = 4\wp^3 - g_3$. The map Φ_Γ takes values in gauge classes of nonzero Fermat vectors, and it is injective on $\mathbb{C}/\Lambda_0$.*

Proof. Invariants. From $G_k(\lambda\Lambda) = \lambda^{-k}G_k(\Lambda)$ and $\zeta_6\Lambda_0 = \Lambda_0$ we get $G_4 = \zeta_6^{-4}G_4$, and $\zeta_6^4 \neq 1$ gives $g_2 = 0$. Suppose $g_3 = 0$, so $w_\wp^2 = 4\wp^3$. Then $\wp'(\frac{1}{2}) = 0$, as $\wp'$ is odd and periodic, which forces $\wp(\frac{1}{2}) = 0$; a zero of order $n \geq 1$ there would give $2(n-1) = 3n$, impossible.

Image. Off Λ_0, $(2c_F\wp)^3 - (1 + d_F w_\wp)^3 - (1 - d_F w_\wp)^3 = (8\wp^3 - 2g_3 - 2w_\wp^2)/g_3 = 0$; at the lattice point, $0^3 + 1^3 + (-1)^3 = 0$.

Injectivity. The last two coordinates sum to -2 off Λ_0 and to 0 at $[0:1:-1]$, so only the lattice point maps to $[0:1:-1]$. For $z, z' \notin \Lambda_0$, equal classes give $\wp(z) = \wp(z')$ and $\wp'(z) = \wp'(z')$. Since $\wp - \wp(z)$ has a single double pole, $z' \equiv \pm z$. If $z' \equiv -z$, then $\wp'(z) = 0$; as $\wp'$ has a single triple pole and vanishes at the three half periods, those are all its zeros, so z is a half period and $-z \equiv z$. □

Definition 7.6 (Theta lift)**.** Let $x = \pi z = \pi(u + \zeta v)$ and

$$\vartheta(x) = 2\sum_{n\geq 0}(-1)^n e^{i\pi\zeta(n+1/2)^2}\sin\big((2n+1)x\big),$$

the Jacobi function $\vartheta_1(x\,|\,\zeta)$; primes denote x-derivatives. Put $r_\vartheta = \vartheta'''(0)/(3\vartheta'(0))$,

$$P_\vartheta = \vartheta'^2\vartheta - \vartheta''\vartheta^2 + r_\vartheta\vartheta^3, \qquad Q_\vartheta = -\vartheta'''\vartheta^2 + 3\vartheta'\vartheta''\vartheta - 2\vartheta'^3,$$

$$\hat{Z} = \big(2c_F\pi^2 P_\vartheta,\ -\vartheta^3 - d_F\pi^3 Q_\vartheta,\ -\vartheta^3 + d_F\pi^3 Q_\vartheta\big), \qquad Z = \frac{\hat{Z}}{\|\hat{Z}\|},$$

all evaluated at $x = \pi z$.

Equivalently, $\vartheta(x) = -i\,e^{i\pi\zeta/4+ix}\,\Theta(x/\pi + \frac{1}{2} + \frac{\zeta}{2}, \zeta)$ with $\Theta(y,\zeta) = \sum_{n\in\mathbb{Z}} e^{2\pi i n y + i\pi n^2\zeta}$.

Lemma 7.7 (The lift has no zero)**.** *The zeros of ϑ are exactly the points of $\pi\Lambda_0$, and they are simple. Off Λ_0,*

$$\wp(z) = -\pi^2(\log\vartheta)''(\pi z) + \pi^2 r_\vartheta,$$

and $\hat{Z} = \vartheta^3\cdot(2c_F\wp, -1 - d_F w_\wp, -1 + d_F w_\wp)$ wherever $\vartheta \neq 0$. $\hat{Z}$ is an entire function of z without zeros, so Z is a smooth map $\mathbb{R}^2 \to \mathbb{C}^3$ with $\|Z\| = 1$. Under $(u,v)\mapsto(u+1,v)$ and $(u,v)\mapsto(u,v+1)$, $\hat{Z}$ is multiplied by -1 and by $\mathfrak{m}(\pi z)^3$, where $\mathfrak{m}(x) = -e^{-i\pi\zeta}e^{-2ix}$.

Proof. Quasi-periodicity. ϑ is entire and odd, and the identities of Θ give $\vartheta(x+\pi) = -\vartheta(x)$ and $\vartheta(x+\pi\zeta) = \mathfrak{m}(x)\vartheta(x)$. Hence $\ell = \vartheta'/\vartheta$ satisfies $\ell(x+\pi) = \ell(x)$ and $\ell(x+\pi\zeta) = \ell(x) - 2i$.

Zeros. Take a parallelogram $P = a + [0,1]\pi + [0,1]\pi\zeta$ with no zeros on ∂P. In the argument principle the edges parallel to $\pi\zeta$ cancel, and the other two contribute $\int_a^{a+\pi}(\ell(x) - \ell(x+\pi\zeta))\,dx = 2\pi i$, so P contains exactly one zero, and it is simple. Since $\vartheta(0) = 0$ and P contains exactly one point of $\pi\Lambda_0$, the zero set is $\pi\Lambda_0$, and $\vartheta' \neq 0$ there.

Formula for $\wp$. $F(z) = \pi^2(\vartheta'^2 - \vartheta\vartheta'')/\vartheta^2$, evaluated at πz, is Λ_0-periodic because ℓ changes only by constants, and $F(z) = z^{-2} - \pi^2 r_\vartheta + O(z^2)$ at 0. By Liouville, and since $\wp$ has zero constant Laurent term, $\wp - F = \pi^2 r_\vartheta$, so $\vartheta^3\wp = \pi^2 P_\vartheta$. Differentiating gives $w_\wp = \pi^3 Q_\vartheta/\vartheta^3$, hence the formula for $\hat{Z}$.

No zero. Where $\vartheta \neq 0$, the last two components sum to $-2\vartheta^3 \neq 0$; where $\vartheta = 0$, $Q_\vartheta = -2\vartheta'^3$, so the second component is $2d_F\pi^3\vartheta'^3 \neq 0$. $\hat{Z}$ is entire, being polynomial in $\vartheta, \ldots, \vartheta'''$.

Automorphy. Under $x \mapsto x+\pi$, all of $\vartheta, \ldots, \vartheta'''$ change sign and $P_\vartheta, Q_\vartheta, \vartheta^3$ are cubic in them, so $\hat{Z} \mapsto -\hat{Z}$. Under $x \mapsto x + \pi\zeta$, periodicity of $\wp, w_\wp$ and the previous step show that $P_\vartheta, Q_\vartheta, \vartheta^3$ gain the factor $\mathfrak{m}^3$ off $\pi\Lambda_0$, hence everywhere by continuity. □

At lattice points $P_\vartheta = 0$ and $Q_\vartheta = -2\vartheta'^3$, so $[\hat{Z}] = [0:1:-1]$; off the lattice $[\hat{Z}] = \Phi_\Gamma(z)$. Hence Z is a smooth unit Fermat lift of that gauge class.

Construction 7.8 (The atlas of N)**.**

(a) *Base.* The chart $(u,v)\mapsto\Phi_\Gamma(u+\zeta v)$ is the injective Fermat representative map of the base torus $\Gamma = (\mathbb{R}/\mathbb{Z})^2$; all base objects are written in (u,v), and $\nu = du\wedge dv$.

(b) *The torus cover of N.* The map $\mathbb{R}^2\times\mathbb{C}\to N$, $(u,v,w)\mapsto[Z(u,v),w]$, with $w = \xi + i\eta$, is the cover-to-representative map. Let $\phi(u,v) = \mathfrak{m}(\pi z)^3/|\mathfrak{m}(\pi z)|^3$, so that $Z(u+1,v) = -Z(u,v)$ and $Z(u,v+1) = \phi(u,v)Z(u,v)$ (Lemma 7.7). The deck maps

$$T_u(u,v,w) = (u+1, v, -w), \qquad T_v(u,v,w) = (u, v+1, \phi(u,v)^3 w)$$

generate a free $\mathbb{Z}^2$-action, and $(u,v,w)\mapsto[Z(u,v),w]$ is exactly the deck relation defining N.

(c) *Algebraic frames.* For $j = 0, 1, 2$, $Z\bar{Z}_j/|Z_j|$ is a local unitary frame over $\Gamma \cap \{Z_j \neq 0\}$ in the sense of Definition 6.1. Its fibre coordinate w_j is given by $U = (Z_j/|Z_j|)^3 w_j$, so its unit section is $[Z] \mapsto [Z, (Z_j/|Z_j|)^3]$, and its connection form is $-3\eta_H + \frac{3}{2\pi} d \arg Z_j$. These three sets cover Γ, and the transition functions $w_j = (Z_k|Z_j|/(|Z_k|Z_j))^3 w_k$ have weight three.

(d) *The model form in the torus frame.* In the frame (b), $p = \pi|w|^2$, $\alpha = d\theta - 3Z^*\eta_H$, and

$$\Omega_0 = d\Big((1-3p)\, Z^*\eta_H + \tfrac{1}{2}(\xi\, d\eta - \eta\, d\xi)\Big) = \omega_V + (1-3p)\sigma + dp \wedge (-3Z^*\eta_H).$$

Verification. (b) Since $(e^{i\pi}Z, e^{3i\pi}U) \sim (Z, U)$, $[Z(u+1, v), -w] = [Z(u, v), w]$, and likewise $[\phi Z, \phi^3 w] = [Z, w]$. As $\mathfrak{m}(x + \pi) = \mathfrak{m}(x)$, ϕ is u-periodic, so $T_u T_v = T_v T_u$. By projective injectivity, gauge-equivalent unit pairs have base coordinates differing by a lattice vector; after the corresponding deck map, unit gauge equivalence fixes the fibre coordinate.

(c) The frame is gauge invariant, and $[Z, (Z_j/|Z_j|)^3 w_j] = [e^{-i \arg Z_j} Z, w_j]$. From $\arg U = 3 \arg Z_j + \arg w_j$ we get $\alpha = d \arg w_j/(2\pi) + \frac{3}{2\pi} d \arg Z_j - 3\eta_H$, which is the connection form. The sets cover Γ since $Z \neq 0$, and comparing sections on overlaps gives the transitions.

(d) Apply Lemmas 7.2 and 7.3 with $a_{\mathrm{loc}} = -3Z^*\eta_H$ and $d(Z^*\eta_H) = \sigma$; the primitive is $Z^*\eta_H + p\alpha$. □

Lemma 7.9 (The deck phase)**.** *The two fibre factors are -1 and ϕ^3; they commute with the lattice translations. On $v = 0$, $\phi(u, 0) = \phi_0 e^{-6\pi i u}$ for a constant $|\phi_0| = 1$.*

Proof. $\mathfrak{m}$ is u-periodic, so the deck maps commute. On $v = 0$, $\mathfrak{m}(\pi u) = -e^{-i\pi\zeta} e^{-2\pi i u}$; cubing its unit phase gives the claim. □

Lemma 7.10 (The area of the base)**.** *Write $Z^*\eta_H = \mathsf{h}_u\, du + \mathsf{h}_v\, dv$ and $\sigma = d(Z^*\eta_H) = \sigma_\Gamma(u, v)\, du \wedge dv$ on $\mathbb{R}^2$. Then σ_Γ is smooth, 1-periodic in u and in v, strictly positive, and $\int_0^1 \int_0^1 \sigma_\Gamma\, du\, dv = 3$.*

Proof. Positivity. σ_Γ is the pull-back of ω_{FS} by $z \mapsto [\hat{Z}(z)]$: $\sigma_\Gamma = \operatorname{Im}\zeta \cdot (\|\hat{Z}\|^2 \|\hat{Z}'\|^2 - |\hat{Z}^\dagger \hat{Z}'|^2)/(\pi\|\hat{Z}\|^4)$, where $\hat{Z}' = d\hat{Z}/dz$; this is positive unless $\hat{Z}' \parallel \hat{Z}$. Off the lattice, $\hat{Z} = \vartheta^3 R$ with $R = (2c_F \wp, -1 - d_F w_\wp, -1 + d_F w_\wp)$ and $R' = (2c_F w_\wp, -d_F \wp'', d_F \wp'')$. If $R' = aR$, summing the last two components gives $a = 0$, so $w_\wp = \wp'' = 0$, impossible since $\wp'' = 6\wp^2$ and $w_\wp = 0$ forces $4\wp^3 = g_3 \neq 0$. At lattice points, the first component of $\hat{Z}$ is 0 while that of $\hat{Z}'$ is $2c_F \pi^3 \vartheta'^3 \neq 0$.

Periodicity. Under deck translations Z changes by unit phases ϕ (Lemma 7.7), so η_H changes by the closed form $(2\pi i)^{-1} \bar{\phi}\, d\phi$.

Integral. By Stokes, $\iint \sigma_\Gamma = \iint (\partial_u \mathsf{h}_v - \partial_v \mathsf{h}_u)$. The constant phase -1 gives $\mathsf{h}_v(1, v) = \mathsf{h}_v(0, v)$, so the first term contributes 0; by Lemma 7.9, $\mathsf{h}_u(u, 1) = \mathsf{h}_u(u, 0) - 3$. Hence $\iint \sigma_\Gamma = 3$. □

Lemma 7.11 (The three points on L)**.** *The set $\mathcal{Z} = \{(u, v) \in \Gamma : Z_0(u, v) = 0\}$ of base points on the line $L = \{Z_0 = 0\}$ consists of*

$$b_1 = (0, 0), \qquad b_2 = (\tfrac{1}{3}, \tfrac{2}{3}), \qquad b_3 = (\tfrac{2}{3}, \tfrac{1}{3}).$$

Proof. $Z_0 = 0$ iff $P_\vartheta = 0$, i.e. iff $\vartheta = 0$ (the point b_1) or $\wp = 0$. As $\zeta\Lambda_0 = \Lambda_0$, $\wp(\zeta z) = \zeta\wp(z)$, so $\wp$ vanishes at non-lattice fixed points of $z \mapsto \zeta z$. By $\zeta^2 = -1 - \zeta$, these include $z_2 = \frac{1}{3} + \frac{2}{3}\zeta$ and $z_3 = \frac{2}{3} + \frac{1}{3}\zeta \equiv -z_2$. Since $\wp$ has exactly two zeros counted with multiplicity and $z_2 \not\equiv z_3$, these are all its zeros. □

7.3. **Handle data.** We use the handle data of Proposition 6.3: $\mathcal{A} = \{\frac{1}{8} \le u \le \frac{5}{24}\} \subset \mathcal{A}^+ = \{\frac{1}{12} \le u \le \frac{1}{4}\}$; the dual band $\mathcal{B} = \{\frac{1}{9} < v < \frac{2}{9}\} \supset \{\frac{1}{8} \le v \le \frac{5}{24}\}$, carrying the closed one-form

$\beta = b(v)\,dv$ and meeting $\mathcal{A}^+$ in one square; and small discs U_j about the points b_j, which lie far from both bands.

Definition 7.12 (Frame over $\mathcal{A}^+$)**.** On $\Gamma \cap \{Z_0 \neq 0\}$ let $w_{\mathcal{A}}$ be the fibre coordinate of the local unitary frame $Z\bar{Z}_0/|Z_0|$ of Construction 7.8(c), i.e. $U = (Z_0/|Z_0|)^3 w_{\mathcal{A}}$ (the frame $(Z_0/|Z_0|)^3$, after its unit section). Let $\theta_{\mathcal{A}} = \arg w_{\mathcal{A}}/2\pi$, and put

$$\lambda_\Gamma = \eta_H - \frac{1}{2\pi}\operatorname{Im}\frac{dZ_0}{Z_0}, \qquad a_{\mathcal{A}} = -3\lambda_\Gamma, \qquad \Gamma_{\mathcal{A}}(p) = (1-3p)\,\lambda_\Gamma,$$

so that $a_{\mathcal{A}}$ is the connection form a_{loc} of this frame.

Lemma 7.13 (The frame over the annulus)**.** *λ_Γ is a smooth one-form on $\Gamma \cap \{Z_0 \neq 0\}$ (no branch of* arg *is needed), $d\lambda_\Gamma = \sigma$, and on $N|_{\Gamma\cap\{Z_0\neq 0\}}$*

$$\alpha = d\theta_{\mathcal{A}} + a_{\mathcal{A}}, \qquad \Omega_0 = \omega_V + d\Gamma_{\mathcal{A}}, \quad \omega_V = dp \wedge d\theta_{\mathcal{A}}.$$

Proof. Since $\operatorname{Im}(dZ_0/Z_0) = d\arg Z_0$ is smooth and closed on $\{Z_0 \neq 0\}$, λ_Γ is invariant on S_Γ and vanishes on iZ, hence is basic with $d\lambda_\Gamma = d\eta_H = \sigma$. From $\arg U = 3\arg Z_0 + \arg w_{\mathcal{A}}$ we get $\alpha = d\theta_{\mathcal{A}} - 3\lambda_\Gamma$, and a direct computation gives $\Omega_0 = \omega_V + d\big((1-3p)\lambda_\Gamma\big)$. □

Definition 7.14 (Torus primitive operator)**.** This is the operator of Proposition 6.3(e). For a smooth function $\mathfrak{b}$ on $(\mathbb{R}/\mathbb{Z})^2$ with $\int \mathfrak{b}\,\nu = 0$ put

$$\mathfrak{b}_0(v) = \int_0^1 \mathfrak{b}(s',v)\,ds', \quad A_0(u,v) = \int_0^u \big(\mathfrak{b}(s',v) - \mathfrak{b}_0(v)\big)\,ds', \quad B_0(v) = \int_0^v \mathfrak{b}_0(t)\,dt,$$

and $P_\Gamma(\mathfrak{b}\,\nu) = A_0\,dv - B_0\,du$.

Lemma 7.15. *A_0 and B_0 are 1-periodic in both variables, so $P_\Gamma(\mathfrak{b}\nu)$ is a smooth one-form on the torus, and $d_\Sigma P_\Gamma(\mathfrak{b}\nu) = \mathfrak{b}\nu$. P_Γ is linear, and if $\mathfrak{b}$ depends smoothly on parameters then so does $P_\Gamma(\mathfrak{b}\nu)$, with all jets; in particular it preserves affine dependence on p.*

Proof. $A_0(u+1,v) - A_0(u,v) = \int_u^{u+1}(\mathfrak{b} - \mathfrak{b}_0)\,ds' = 0$ and $B_0(v+1) - B_0(v) = \int \mathfrak{b}\,\nu = 0$; v-periodicity of A_0 is inherited from $\mathfrak{b}, \mathfrak{b}_0$. Directly, $d(A_0\,dv - B_0\,du) = \partial_u A_0\,du\wedge dv - \mathfrak{b}_0\,dv\wedge du = \mathfrak{b}\,\nu$. Parameter dependence follows by differentiation under the integral sign. □

Proposition 7.16 (Cutoffs on the cubic handle)**.** *Let $\mathcal{A} \subset \mathcal{A}^+$, $\mathcal{B}$ and β be the handle data of Proposition 6.3, and take the discs of its item (g) closed: $U_j = \{x_j^2 + y_j^2 \le 10^{-4}\}$, of radius $\frac{1}{100}$ about b_j in the centred coordinates $(x_j, y_j) = (u - u(b_j), v - v(b_j))$, reduced to $[-\frac12, \frac12)^2$. With the flat step S and the cutoffs of Definition 2.6, the cutoffs*

$$\zeta_{\mathcal{A}}(u) = \chi^{\mathrm{I}}_{9/96,\,23/96,\,2/96}(u), \qquad \zeta_j = \chi_{1/200,\,1/100}(x_j, y_j)$$

(extended by zero) satisfy: $\zeta_{\mathcal{A}} = 1$ on $\{\frac{11}{96} \le u \le \frac{21}{96}\}$, a neighbourhood of $\mathcal{A}$, and $\operatorname{supp}\zeta_{\mathcal{A}} = \{\frac{9}{96} \le u \le \frac{23}{96}\} \subset \operatorname{int}\mathcal{A}^+$; $\zeta_j = 1$ on the disc of radius $\frac{1}{200}$ about b_j, and $\operatorname{supp}\zeta_j = U_j$. The discs U_j are pairwise disjoint and disjoint from $\mathcal{A}^+ \cup \mathcal{B}$, and $\operatorname{supp}\beta \cap U_j = \varnothing$. All these data are independent of the input.

Proof. By Lemma 2.7(b), $\chi^{\mathrm{I}}_{9/96,23/96,2/96} = 1$ on $[\frac{11}{96}, \frac{21}{96}] \supset \mathcal{A} = [\frac{12}{96}, \frac{20}{96}]$ and is positive exactly on $(\frac{9}{96}, \frac{23}{96})$, so its support lies in $\operatorname{int}\mathcal{A}^+ = (\frac{8}{96}, \frac{24}{96})$. By Lemma 2.7(a), $\chi_{1/200,1/100}$ is 1 on the closed disc of radius $\frac{1}{200}$, vanishes outside the open disc of radius $\frac{1}{100}$ and is positive on it (as $S < 1$ on $(-\infty, 1)$), so its support is U_j. The b_j have mutual flat distance $\ge \frac13$ and distance $\ge \frac{1}{12}$ from $\mathcal{A}^+ \cup \mathcal{B}$ (proof of Proposition 6.3(a),(b)), both exceeding $\frac{2}{100}$; and $\operatorname{supp}\beta \subset \mathcal{B}$. □

7.4. **A global primitive vanishing on three whole fibres.** We write the first path of the handle construction (§M5) for the cubic in closed form, for all valid cubic data, correcting an explicit primitive α^0 of $\kappa_1-\kappa$ by d_Σ-closed forms supported over $\mathcal{A}^+$ and near the b_j (disjoint by Proposition 7.16).

Definition 7.17 (Admissible auxiliary data and base area forms)**.** Let $0<h<1/3$ and $0<g\le 3(1-3h)$. Set

$$e_h=g/16,\qquad H^*(p)=3(1-3p)-e_h,\qquad f(s)=S(2s-1/2),\qquad \Gamma_{\mathrm{sm}}=e_h(u-1/6)\,dv.$$

Thus $0<e_h\le H_{\mathrm{b}}(h)/2$, f is nondecreasing, 0 for $s\le 1/4$, 1 for $s\ge 3/4$, and $d\Gamma_{\mathrm{sm}}=e_h\nu$. Put $\chi=f'(s)\,ds\wedge\beta$ (on $\mathcal{A}$, extended by zero) and

$$\kappa(p)=(1-3p)\,\sigma,\qquad \kappa_1(p)=e_h\nu+H^*(p)\chi.$$

Lemma 7.18 (Base area forms)**.** *$\chi\ge 0$, $\int\chi=1$, and $\int\kappa(p)=\int\kappa_1(p)=H_{\mathrm{b}}(p)$. On $[0,h]$, $H^*\ge\frac12 H_{\mathrm{b}}(h)>0$. For every $p<\frac13$ with $H^*(p)\ge 0$, both $\kappa(p)$ and $\kappa_1(p)$ are positive area forms, and $\kappa_1(p)\ge e_h\nu$.*

Proof. $\chi=f'(s)b(v)\,ds\wedge dv\ge 0$ and $\int\chi=\int_0^1 f'\,ds\int_0^1 b\,dv=1$. By Lemma 7.10, $\int\kappa=3(1-3p)$ and $\sigma>0$; $\int\kappa_1=e_h+H^*=H_{\mathrm{b}}$. On $[0,h]$, $H_{\mathrm{b}}\ge H_{\mathrm{b}}(h)$ and $e_h\le\frac12 H_{\mathrm{b}}(h)$. Finally $\kappa_1-e_h\nu=H^*\chi\ge 0$. □

Definition 7.19 (The annular difference and its period)**.** Let $\alpha^0(p)=P_\Gamma\big(\kappa_1(p)-\kappa(p)\big)$, a base one-form with coefficients depending on the base point and affinely on p, with $d_\Sigma\alpha^0=\kappa_1-\kappa$ (Lemmas 7.15 and 7.18). On $\mathcal{A}^+$ put

$$\Delta_0(p)=\Gamma_{\mathcal{A}}(p)+\alpha^0(p)-\Gamma_{\mathrm{sm}}-f(s)H^*(p)\beta,$$

and let $c(p)=\oint_{\{u=\mathrm{const}\}}\Delta_0(p)$ be its period over a v-circle in $\mathcal{A}^+$.

Lemma 7.20. *For each p, $\Delta_0(p)$ is d_Σ-closed on $\mathcal{A}^+$; hence $c(p)$ does not depend on the circle, and c is affine in p.*

Proof. By Lemma 7.13, $d_\Sigma\Gamma_{\mathcal{A}}=\kappa$; with $d_\Sigma\alpha^0=\kappa_1-\kappa$, $d_\Sigma\Gamma_{\mathrm{sm}}=e_h\nu$ and $d_\Sigma(fH^*\beta)=H^*\chi$ this gives $d_\Sigma\Delta_0=0$. Two v-circles bound an annulus in $\mathcal{A}^+$, so Stokes gives independence; every term is affine in p since P_Γ is linear and e_h,Γ_{sm} do not depend on p. □

In general $c(p)\neq 0$ (holonomy of the frame $(Z_0/|Z_0|)^3$ around the core circle and periods of the primitives); since $\oint\beta=1$, $\Delta_0-c(p)\beta$ has period zero and is d_Σ-exact on $\mathcal{A}^+$.

Definition 7.21 (Annular primitive)**.** Write $\Delta_0(p)-c(p)\beta=D_u\,du+D_v\,dv$ on $\mathcal{A}^+$ and put

$$F_{\mathcal{A}}(u,v,p)=\int_{1/6}^{u}D_u(t,0,p)\,dt+\int_0^{v}D_v(u,t,p)\,dt\qquad(\tfrac{1}{12}\le u\le\tfrac14,\ v\in\mathbb{R}),$$

the integral of $\Delta_0-c\beta$ along the path from $(\frac16,0)$ horizontally and then vertically.

Lemma 7.22. *$F_{\mathcal{A}}$ is a smooth function on $\mathcal{A}^+$, 1-periodic in v, affine in p, and $d_\Sigma F_{\mathcal{A}}=\Delta_0-c(p)\beta$.*

Proof. $\partial_v F_{\mathcal{A}}=D_v$, and closedness ($\partial_u D_v=\partial_v D_u$) gives $\partial_u F_{\mathcal{A}}=D_u(u,0)+\int_0^v\partial_t D_u\,dt=D_u$. Periodicity: $\int_0^1 D_v(u,t)\,dt=c(p)-c(p)=0$ by Proposition 6.3(c). Smoothness and affinity are inherited from D_u,D_v. □

Construction 7.23 (The global primitive α_1)**.** Define the base one-form, with $\zeta_{\mathcal{A}}F_{\mathcal{A}}$ extended by zero outside $\mathcal{A}^+$,

$$\alpha_{\mathrm{pre}}(p) = \alpha^0(p) - c(p)\beta - d_\Sigma\big(\zeta_{\mathcal{A}}F_{\mathcal{A}}(\cdot,p)\big),$$

write its value at b_j as $A_j(p)\,dx_j + B_j(p)\,dy_j$ (A_j, B_j affine in p), and put

$$\alpha_1(p) = \alpha_{\mathrm{pre}}(p) - \sum_{j=1}^{3} d_\Sigma\Big[\zeta_j\big(A_j(p)\,x_j + B_j(p)\,y_j\big)\Big]. \tag{13}$$

We use α_1 as the toric one-form $\pi_N^*\alpha_1(p(\cdot))$ on N. For the data of §M5 this is the primitive α_1 of the first path, with the base point $(\frac{1}{6}, 0)$ for $F_{\mathcal{A}}$ and the cutoffs $\zeta_{\mathcal{A}}$, ζ_j of Proposition 7.16.

Lemma 7.24 (Properties of α_1)**.** *(i) α_1 is a smooth one-form on N, horizontal (no dp and no fibre-angle component), affine in p, and $d_\Sigma\alpha_1 = \kappa_1 - \kappa$.*

(ii) Over $\mathcal{A}$, in the frame $(Z_0/|Z_0|)^3$,

$$\Gamma_{\mathcal{A}} + \alpha_1 = \Gamma_{\mathrm{sm}} + f(s)H^*(p)\beta, \qquad \textit{hence} \qquad \Omega_0 + d\alpha_1 = \omega_V + e_h\nu + d\big(f(s)H^*(p)\beta\big), \tag{14}$$

with the full exterior derivative, which contains the mixed term $f(s)H^{*\prime}(p)\,dp\wedge\beta = -9f(s)\,dp\wedge\beta$.

Proof. (i) Each term is a base one-form smooth in the base point and affine in p, so its pull-back is a smooth horizontal one-form; $c\beta$, $d_\Sigma(\zeta_{\mathcal{A}}F_{\mathcal{A}})$ and the corrections at the b_j are d_Σ-closed, so $d_\Sigma\alpha_1 = d_\Sigma\alpha^0 = \kappa_1 - \kappa$.

(ii) On $\mathcal{A}$, $\zeta_{\mathcal{A}} = 1$ nearby and all $\zeta_j = 0$ (Proposition 7.16), so $\alpha_1 = \alpha^0 - \Delta_0$, the first identity. By Lemma 7.13, $\Omega_0 + d\alpha_1 = \omega_V + d\Gamma_{\mathrm{sm}} + d(fH^*\beta)$, with $d\Gamma_{\mathrm{sm}} = e_h\nu$ as Γ_{sm} is p-independent. □

Remark 7.25 (The period is subtracted in α_1). The term $-c(p)\beta$ is part of α_1, not only of $F_{\mathcal{A}}$: otherwise the left side of (14) gains $c(p)\beta$, whose ambient differential $c'(p)\,dp\wedge\beta$ is in general nonzero. As β is closed and $\operatorname{supp}\beta\cap U_j = \varnothing$, the subtraction changes only the mixed term, not the base curvature or the values at the b_j, and the mixed term does not enter (15).

Lemma 7.26 (Vanishing on three whole fibres)**.** *At every point of each of the three fibres $N_{b_1}, N_{b_2}, N_{b_3}$, α_1 is the zero covector of T^*N.*

Proof. Near b_j, $\zeta_j = 1$ and $\zeta_i = 0$ for $i \neq j$ (Proposition 7.16), so $\alpha_1(p)_{b_j} = \alpha_{\mathrm{pre}}(p)_{b_j} - A_j(p)\,dx_j - B_j(p)\,dy_j = 0$ for every p. At a point of N_{b_j} with moment p_0, the toric form α_1 is the pull-back of $\alpha_1(p_0)_{b_j}$ by $d\pi_N$, hence zero. □

Remark 7.27 (Ambient covector versus pull-back). The pull-back of α_1 to a fibre vanishes automatically, as α_1 is horizontal; that would only make the Moser field Y ($\iota_Y\Omega_\lambda = -\alpha_1$) Ω_λ-orthogonal to the fibre, hence in general transverse to it. Lemma 7.26 gives $Y = 0$ at every point of N_{b_j}, so the Moser flow fixes these fibres pointwise and preserves their complement, which the $\mathbb{Z}/3$-equivariant transfer maps into $\{Z_0 \neq 0\}$ (Lemma 7.26 and Proposition 10.19).

7.5. **Positivity including the mixed terms.** The corrections change α_1 by base forms with p-dependent coefficients, whose differentials contain mixed terms $dp\wedge a$ (a a base one-form). These do not affect nondegeneracy: locally the space is a two-dimensional base times a fibre, and $dp\wedge a$ wedges to zero with every base two-form, with $\omega_V = dp\wedge d\theta$, and with itself.

Lemma 7.28 (Positivity)**.** *For $\lambda\in[0,1]$ let $\Omega_\lambda = \Omega_0 + \lambda\,d\alpha_1$ and $\kappa_\lambda = (1-\lambda)\kappa + \lambda\kappa_1$. In every local unitary frame,*

$$\Omega_\lambda = \omega_V + \kappa_\lambda + dp\wedge a_\lambda, \qquad a_\lambda = a_{\mathrm{loc}} + \lambda\,\partial_p\alpha_1,$$

with a_λ a base one-form, and

$$\Omega_\lambda^2 = 2\,\omega_V\wedge\kappa_\lambda, \tag{15}$$

smoothly across the zero section. Hence Ω_λ is a circle-invariant symplectic form on the open set $\{p < \frac{1}{3},\ H^*(p) > 0\} = \{p < \frac{1}{3} - \frac{e_h}{9}\}$, which contains $N_{p\le h}$; moreover $\iota_{X_p}\Omega_\lambda = -dp$ for the generator X_p of the circle action, and $\partial_\lambda\Omega_\lambda = d\alpha_1$.

Proof. By Lemma 7.24(i), $d\alpha_1 = \kappa_1 - \kappa + dp \wedge \partial_p\alpha_1$; with (12) this gives the local form, and the preceding paragraph (Lemma 2.8(a) with $m = 1$) gives $\Omega_\lambda^2 = 2\,\omega_V \wedge \kappa_\lambda$. Write $\kappa_\lambda = k_\lambda\nu$. Where $p < \frac{1}{3}$ and $H^*(p) \ge 0$, $k_\lambda > 0$ by Lemma 7.18, and $\omega_V \wedge \nu$ is a positive volume form; $H^*(p) > 0$ iff $p < \frac{1}{3} - \frac{e_h}{9}$, and $h < \frac{1}{3} - \frac{e_h}{9}$ since $e_h \le \frac{3}{2}(1 - 3h) < 9(\frac{1}{3} - h)$. In Cartesian fibre coordinates $\omega_V = d\xi \wedge d\eta$ and $dp = 2\pi(\xi\, d\xi + \eta\, d\eta)$ are smooth, so the decomposition and (15) hold across $w = 0$. Finally Ω_0, α_1 are circle-invariant, $\iota_{X_p}\Omega_0 = -dp$ (Lemma 2.9(c)), and $\iota_{X_p}d\alpha_1 = \mathcal{L}_{X_p}\alpha_1 - d(\alpha_1(X_p)) = 0$ as α_1 is invariant and horizontal. □

The remaining paths, primitives and Moser map are in §8.1.

Remark 7.29 (Scalar data of valid cubic data). The equal branch uses $g = 3(1-t_0)$ and $h = t_0/3$, so $g = H_{\mathrm{b}}(h)$; the scalar comparison datum of [Son26] uses $g = 3d/4$. In both cases the data are valid in the sense of Definition 8.1: $g \le H_{\mathrm{b}}(h)$, and (16) holds with constant $\frac{31}{32}g$. With $e_h = g/16$ and the cap smoothing scale $\mu_h = g/(16k)$, the smoothing loss is at most $g/32$ and the residual gap is

$$\frac{31g}{32} - \frac{g}{16} - \frac{g}{32} = \frac{7g}{8} > 0.$$

This is Lemma 8.3(a),(c); the constructions above apply to all valid cubic data.

8. The handle construction

The construction is an explicit version of the surface packing lemma of [Ope26, §3.2]. Here we treat the periodic four-dimensional instance outlined in §M5; the higher-dimensional inputs are treated in [Son26].

8.1. **The four-dimensional handle.** The base is the torus Γ of §6, with global coordinates (u, v) on the cover $\widetilde{N} = \mathbb{R}^2 \times \mathbb{C}$ of Definition 6.1, fibre coordinate w, $p = \pi|w|^2$; deck invariance replaces compactness. A *deck map* is an element $T_u^m T_v^n$ of the group generated by T_u, T_v; it shifts (u, v) by (m, n) and preserves p (Construction 7.8(b)). Write $\Omega_t = \Omega_0 + t\, d\alpha_1$ ($t \in \mathbb{R}$), $\nu = du \wedge dv$, $\beta = b(v)\, dv$, with α_1 of Construction 7.23, β of Proposition 6.3(c) and S the flat step of Definition 2.3.

Definition 8.1 (Cubic data)**.** *Valid cubic data* consist of an integer $k \ge 1$, numbers ρ_i, a_i ($1 \le i \le k$), h, d_n, g, a smooth function K on $\mathbb{R}$, and a smooth isotopy $(q_t)_{t\in\mathbb{R}}$ of $(\mathbb{C}, \omega_{\mathbb{C}})$ with smooth Hamiltonian (Q_t), $\partial_t q_t = \Xi_t \circ q_t$ and $\iota_{\Xi_t}\omega_{\mathbb{C}} = -dQ_t$, such that:

(a) $0 < \rho_i < a_i < h < d_n < \frac{1}{3}$ and $0 < g \le H_{\mathrm{b}}(h) = 3(1 - 3h)$;
(b) $q_0 = \mathrm{id}$, $Q_t = 0$ for $t \notin (0, 1)$, and for a number $P_{\mathrm{supp}} < h$ one has $q_t = \mathrm{id}$ and $Q_t = 0$ on $\{\pi|z|^2 > P_{\mathrm{supp}}\}$ for every t;
(c) with $P_{\mathrm{b}}(p) = H_{\mathrm{b}}(p) - \sum_i (a_i - p)_+$ and $F = K - P_{\mathrm{b}}$,

$$K(\pi|z|^2) - F(\pi|q_1(z)|^2) \ge \tfrac{31}{32}\, g \qquad (\pi|z|^2 \le h). \tag{16}$$

Both the linear datum and the scalar datum of [Son26] take for (q_t) the whole-disc isotopy $\hat{q}$ of an ordinary path of disc area h (Definition 4.14, Lemma 4.18, Construction 4.19), which satisfies (b) by Theorem 4.21; hence $q_t^{\pm 1}$ maps $\bar{D}(c)$ onto itself for $c \ge P_{\mathrm{supp}}$ and $D(c)$ onto itself for $c > P_{\mathrm{supp}}$. Valid cubic data come from the linear datum for equal capacities (Proposition M4.4, Remark 10.25) and, for all other admissible data, from the scalar comparison datum of [Son26].

Definition 8.2 (Derived scalars)**.** For valid cubic data put $q = q_1$ and

$$e_h = \frac{g}{16}, \qquad H^* = H_{\mathrm{b}} - e_h, \qquad J = K - H^*, \qquad \mu_h = \frac{g}{16k}, \qquad \delta_m = \min\Big(\frac{d_n - h}{4}, \frac{g}{100}\Big),$$

$$h_i(p) = \tfrac{1}{2}\Big(a_i - p + \sqrt{(a_i - p)^2 + \mu_h^2}\Big), \qquad R^*(w) = K\big(p(q^{-1}w)\big) - J\big(p(w)\big) - \sum_{i=1}^{k} h_i\big(p(w)\big).$$

The caps h_i smooth the tents $(a_i - p)_+$ from above; for the budget see Remark 7.29.

Lemma 8.3 (Upper caps and the residual budget)**.** *For valid cubic data:*

- *(a) each h_i is smooth on $\mathbb{R}$, $(a_i - p)_+ < h_i(p) \le (a_i - p)_+ + \mu_h/2$ for all p, and $\sum_i h_i \le \sum_i (a_i - p)_+ + g/32$;*
- *(b) $H^*(p) \ge 15g/16$ for $p \le h$;*
- *(c) $R^*(w) \ge 7g/8$ whenever $\pi|w|^2 \le h$;*
- *(d) if $p < h + 2\delta_m$, then $p < d_n$ and $H^*(p) \ge 15g/16 - 9(p-h)_+ > 15g/16 - 18\delta_m > 0$; if $\pi|w|^2 < h + 2\delta_m$, then $R^*(w) > 0$.*

Proof. (a) With $x = a_i - p$, $h_i - x_+ = \frac{1}{2}(\sqrt{x^2 + \mu_h^2} - |x|) \in (0, \mu_h/2]$, and $k\mu_h/2 = g/32$. (b) $H^*(p) \ge H_{\mathrm{b}}(h) - g/16$. (c) $z = q^{-1}(w) \in \bar{D}(h)$, and by (a) $R^*(w) \ge K(\pi|z|^2) - F(\pi|q(z)|^2) - \frac{3g}{32} \ge \frac{7g}{8}$ by (16). (d) $p < h + (d_n - h)/2 < d_n$; $H^*(p) = H^*(h) - 9(p-h)$ for $p > h$, and $18\delta_m \le 18g/100 < 15g/16$. If $h < \pi|w|^2 < h + 2\delta_m$, then $q^{-1}w = w$ and $h_i \le \mu_h/2$, so $R^*(w) > 29g/32 - 18g/100 > 0$; the case $\pi|w|^2 \le h$ is (c). □

Definition 8.4 (Frame chart, clock and layers)**.** On the cover let $\mathcal{A} = \{\frac{1}{8} < u < \frac{5}{24}\} \subset \mathcal{A}^+ = \{\frac{1}{12} < u < \frac{1}{4}\}$ (single strips, not taken modulo one); in this subsection $\mathcal{A}$ and $\mathcal{A}^+$ denote these interiors of the closed annuli of Proposition 6.3(a). On $\mathcal{A}^+$ we have $Z_0 \neq 0$ (Proposition 6.3(a),(f)); put $\mathsf{F} = (Z_0/|Z_0|)^3$ and define the *frame chart*

$$\chi_{\mathcal{A}}(u, v, w_{\mathcal{A}}) = \big(u, v, \mathsf{F}(u, v)\, w_{\mathcal{A}}\big) \qquad (u \in \mathcal{A}^+).$$

The *clock* is $f(u) = S\big((u - u_0)/(u_1 - u_0)\big)$ with $u_0 = \frac{7}{48}$, $u_1 = \frac{3}{16}$. The *layers* are

$$N_\ell = k + 1, \quad \ell_s = \frac{u_1 - u_0}{N_\ell}, \quad l_j = u_0 + (j-1)\ell_s + \frac{\ell_s}{4}, \quad r_j = l_j + \frac{\ell_s}{2}, \quad \varsigma_j(u) = S\Big(\frac{u - l_j}{r_j - l_j}\Big)$$

for $1 \le j \le N_\ell$; the last index $j = N_\ell$ is denoted by $*$.

$\chi_{\mathcal{A}}$ is a diffeomorphism of $\mathcal{A}^+ \times \mathbb{C}$ preserving base points and p, and $\chi_{\mathcal{A}}(u, v+1, w_{\mathcal{A}}) = T_v \chi_{\mathcal{A}}(u, v, w_{\mathcal{A}})$ since $Z_0(u, v+1) = \phi Z_0(u, v)$, $|\phi| = 1$. In the coordinate $s = 12u - \frac{3}{2}$ of Proposition 6.3(a), f is the clock $S(2s - \frac{1}{2})$ of Definition 7.17; $f = 0$ for $u \le u_0$, $f = 1$ for $u \ge u_1$. Also $u_0 < l_1 < r_1 < l_2 < \cdots < r_* = u_1 - \ell_s/4 < u_1$, $\varsigma_j = 0$ for $u \le l_j$, $\varsigma_j = 1$ for $u \ge r_j$, $\varsigma_j' \ge 0$. Over $\mathcal{A}^+$ we use frame coordinates $(u, v, w_{\mathcal{A}})$, q_t acting on $w_{\mathcal{A}}$ (Figure 1).

Lemma 8.5 (The first path in the frame)**.** *Over $\mathcal{A}^+$, $\chi_{\mathcal{A}}^* \Omega_t = \omega_V + \kappa_t + dp \wedge (a_{\mathcal{A}} + t\, \partial_p \alpha_1)$ with the positive forms κ_t of Lemma 7.28. Over $\mathcal{A}$,*

$$\chi_{\mathcal{A}}^* \Omega_1 = \omega_V + e_h \nu + d\big(f(u) H^*(p) \beta\big).$$

Proof. Lemma 7.28 in the frame of Lemma 7.13, and Lemma 7.24(ii) with $d\Gamma_{\mathrm{sm}} = e_h \nu$. □

The twist. For $\lambda \in \mathbb{R}$ define on $\mathcal{A} \times \mathbb{C}$

$$\begin{aligned} \mathcal{D}_\lambda(u, v, w_{\mathcal{A}}) &= \big(u, v, q_{\lambda f(u)}(w_{\mathcal{A}})\big), \qquad \mathcal{D}_\lambda^{-1}(u, v, w_{\mathcal{A}}) = \big(u, v, q_{\lambda f(u)}^{-1}(w_{\mathcal{A}})\big), \\ Q_u^\lambda &= \lambda f'(u)\, Q_{\lambda f(u)}, \qquad G_\lambda(u, w_{\mathcal{A}}) = f(u)\, H^*\big(p(q_\lambda^{-1} w_{\mathcal{A}})\big), \end{aligned} \tag{17}$$

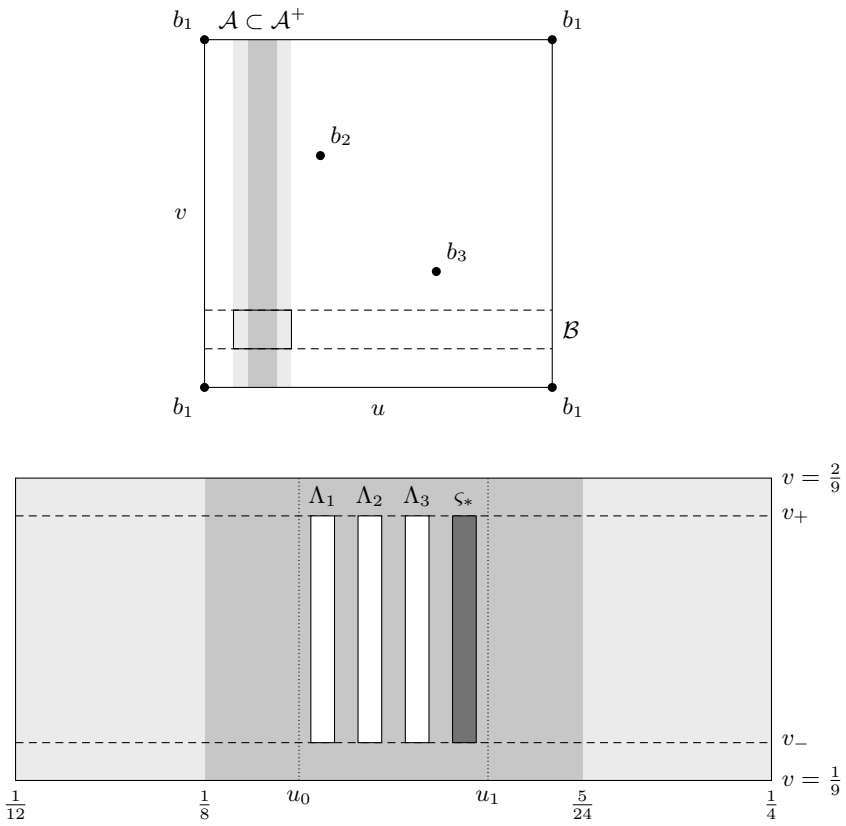


FIGURE 1. The handle data on the base torus $\Gamma = (\mathbb{R}/\mathbb{Z})^2$ in the coordinates (u, v) (Proposition 6.3, Definition 8.4). Top: the annuli $\mathcal{A} \subset \mathcal{A}^+$ (dark and light grey), the dual band $\mathcal{B}$ (between $v = \frac{1}{9}$ and $v = \frac{2}{9}$, dashed) and the marked points b_1, b_2, b_3, where $Z_0 = 0$. Bottom: $\mathcal{A}^+ \cap \mathcal{B}$ enlarged, dashed at $v = v_\pm$; the clock f rises on $[u_0, u_1]$; the layers Λ_i (white) and the residual layer where ς_* rises (black) lie over $(l_j, r_j) \times (v_-, v_+)$, where $b > 0$. Here $k = 3$.

and $\mathcal{D} = \mathcal{D}_1$; these are smooth in $(\lambda, u, v, w_{\mathcal{A}})$, mutually inverse, fix base points and preserve $\mathcal{A} \times D(c)$ for $c > P_{\mathrm{supp}}$.

Lemma 8.6 (Suspension)**.** *On $\mathcal{A} \times \mathbb{C}$, $\mathcal{D}_\lambda^* \omega_V = \omega_V + d\big[(Q_u^\lambda \circ q_{\lambda f(u)})\, du\big]$ for every λ.*

Proof. Lemma M5.2(i) with $m = 1$ and u for s: $D\mathcal{D}_\lambda$ fixes ∂_v, maps vertical ξ to $Dq\,\xi$ and ∂_u to $\partial_u + X_{Q_u^\lambda}$, and $q_{\lambda f(u)}$ is symplectic. □

Periodisation. For a one-form α on $\widetilde{N}$ supported over $\{u_0 \le u \le u_1\}$ put $\mathrm{per}(\alpha) = \sum_{m \in \mathbb{Z}} (T_u^m)^* \alpha$ (locally finite). It is T_u-invariant, equals α over $\{u_1 - 1 < u < 1 + u_0\}$, vanishes near points with $u \notin [u_0, u_1] + \mathbb{Z}$, and for horizontal α its value at a point with $\lfloor u \rfloor = m$ is the covector of α at T_u^{-m} of that point.

Construction 8.7 (The second-path primitive)**.** On $\mathcal{A} \times \mathbb{C}$ let

$$\upsilon_\lambda^{\mathcal{A}} = \big(Q_u^\lambda \circ q_{\lambda f(u)}\big)\, du + f(u)\Big[H^*\big(p(q_\lambda^{-1} q_{\lambda f(u)} w_{\mathcal{A}})\big) - H^*\big(p(w_{\mathcal{A}})\big)\Big]\beta.$$

Let υ_λ be the one-form on $\widetilde{N}$ equal to $(\chi_{\mathcal{A}}^{-1})^*\upsilon_\lambda^{\mathcal{A}}$ over $\mathcal{A}$ and to zero elsewhere, let $\upsilon_\lambda^{\mathrm{per}} = \mathrm{per}(\upsilon_\lambda)$, and put

$$\Omega_\lambda^{(1)} = \Omega_1 + d\upsilon_\lambda^{\mathrm{per}}, \qquad \dot{\upsilon}_\lambda = \partial_\lambda \upsilon_\lambda^{\mathrm{per}}.$$

Lemma 8.8 (Properties of the second-path primitive). *(a) $\upsilon_\lambda^{\mathcal{A}}$ vanishes for $u \le u_0$, for $u \ge u_1$, and for $p > P_{\mathrm{supp}}$; $\upsilon_0^{\mathcal{A}} = 0$.*

(b) υ_λ and $\upsilon_\lambda^{\mathrm{per}}$ are horizontal one-forms on $\widetilde{N}$, smooth in (λ, x); υ_λ is T_v-invariant, $\upsilon_\lambda^{\mathrm{per}}$ is invariant under all deck maps, and $\upsilon_0^{\mathrm{per}} = 0$. Both $\upsilon_\lambda^{\mathrm{per}}$ and $\dot{\upsilon}_\lambda$ vanish near every point with $p > P_{\mathrm{supp}}$ or with $u \notin [u_0, u_1] + \mathbb{Z}$.

(c) Over $\mathcal{A}$, $\chi_{\mathcal{A}}^* \Omega_\lambda^{(1)} = \mathcal{D}_\lambda^* \big(\omega_V + e_h \nu + d(G_\lambda \beta)\big)$.

Proof. (a) For $u \le u_0$, $f = f' = 0$; for $u \ge u_1$, $f' = 0$ and $q_\lambda^{-1} q_{\lambda f(u)} = \mathrm{id}$; for $p > P_{\mathrm{supp}}$, $Q_t = 0$ near $w_{\mathcal{A}}$ and $q_t^{\pm 1}$ fix it; at $\lambda = 0$ both terms vanish. (b) By (a) the extension by zero is smooth, and horizontal since $\chi_{\mathcal{A}}$ preserves base points. Its coefficients are 1-periodic in v, so $T_v^* \upsilon_\lambda = \upsilon_\lambda$ by the T_v-equivariance of $\chi_{\mathcal{A}}$; per gives T_u-invariance, preserving T_v-invariance as T_u, T_v commute. Supports follow from (a) and the properties of per, also for ∂_λ. (c) Over $\mathcal{A}$, $\upsilon_\lambda^{\mathrm{per}} = \upsilon_\lambda$, $\mathcal{D}_\lambda$ preserves ν, β and $G_\lambda \circ \mathcal{D}_\lambda = f(u) H^*(p(q_\lambda^{-1} q_{\lambda f(u)} w_{\mathcal{A}}))$; add Lemma 8.6 to Lemma 8.5. □

Proposition 8.9 (The second path). *For valid cubic data and $\lambda \in [0,1]$, the form $\Omega_\lambda^{(1)}$ is closed, invariant under all deck maps, equal to Ω_1 for $\lambda = 0$, and nondegenerate at every point of $\{p < h + 2\delta_m\}$.*

Proof. Closedness, invariance and $\Omega_0^{(1)} = \Omega_1$ follow from Lemma 8.8(b) and Lemma 7.28. By invariance we check nondegeneracy over $\{u_1 - 1 < u < 1 + u_0\}$. Near $u \notin [u_0, u_1]$ the form is Ω_1, symplectic on $\{p < \frac{1}{3} - e_h/9\} \supset \{p < h + 2\delta_m\}$ (Lemmas 7.28, 8.3(d)). Over $\mathcal{A}$, Lemma 8.8(c) and Lemma M5.1(b) ($n = 2$, $\sigma_{\mathcal{A}} = \nu$, $e = e_h$) give $(\omega_V + e_h \nu + d(G_\lambda \beta))^2 = 2(e_h + b(v)\, \partial_u G_\lambda)\, \omega_V \wedge \nu$ with $\partial_u G_\lambda(u, w') = f'(u) H^*(p(q_\lambda^{-1} w'))$. If $p(w_{\mathcal{A}}) < h + 2\delta_m$, then $w' = q_{\lambda f(u)}(w_{\mathcal{A}})$ and $q_\lambda^{-1} w'$ lie in $D(h + 2\delta_m)$, where $H^* > 0$ (Lemma 8.3(d)); with $f', b \ge 0$, $e_h > 0$ the coefficient is positive. □

The final coefficient. Put $G_{\mathrm{f}}(u, w) = J(p(w)) + \sum_{i=1}^k \varsigma_i(u)\, h_i(p(w)) + \varsigma_*(u)\, R^*(w)$ and $A_i = J + \sum_{j<i} h_j$.

Lemma 8.10 (The final coefficient). *For valid cubic data, G_{f} is smooth on $\mathbb{R} \times \mathbb{C}$ and*

(a) $\partial_u G_{\mathrm{f}} = \sum_i \varsigma_i' h_i \circ p + \varsigma_*' R^* \ge 0$ at every (u, w) with $R^*(w) \ge 0$;

(b) $G_{\mathrm{f}} = J \circ p$ for $u \le l_1$, and $G_{\mathrm{f}} = K \circ p \circ q^{-1}$ for $u \ge r_*$;

(c) $G_{\mathrm{f}}(u, w) = A_i(p(w)) + \varsigma_i(u) h_i(p(w))$ for $l_i < u < r_i$, $1 \le i \le k$.

Proof. (a) Differentiate; $h_i > 0$ (Lemma 8.3(a)), $\varsigma_j' \ge 0$. (b) For $u \le l_1$ all clocks vanish; for $u \ge r_*$ all equal one and $J + \sum_i h_i + R^* = K \circ p \circ q^{-1}$. (c) On (l_i, r_i), $\varsigma_j = 1$ for $j < i$ and $\varsigma_j = 0$ for $j > i$, including $j = *$. □

Construction 8.11 (The third-path primitive). On $\mathcal{A} \times \mathbb{C}$ let $\xi^{\mathcal{A}} = \big((G_{\mathrm{f}} - G_1) \circ \mathcal{D}\big)\beta$. Let ξ be the one-form on $\widetilde{N}$ equal to $J(p)\beta + (\chi_{\mathcal{A}}^{-1})^* \big(\xi^{\mathcal{A}} - J(p)\beta\big)$ over $\mathcal{A}$ and to $J(p)\beta$ elsewhere, and put

$$\xi^{\mathrm{per}} = J(p)\beta + \mathrm{per}\big(\xi - J(p)\beta\big), \qquad \Omega_\tau^{(2)} = \Omega_1^{(1)} + \tau\, d\xi^{\mathrm{per}},$$
$$\Omega_{\mathrm{fin}} = \Omega_1^{(2)}, \qquad \hat{G}_\tau = (1 - \tau) G_1 + \tau G_{\mathrm{f}}.$$

Lemma 8.12 (Properties of the third-path primitive). *(a) $\xi^{\mathcal{A}} = J(p)\beta$ for $u \le u_0$ and for $u \ge u_1$.*

(b) ξ^{per} *is a smooth one-form on* $\widetilde{N}$*, invariant under all deck maps, of the form* $\Xi\,\beta$ *with* Ξ *smooth; on* $\{p > P_{\mathrm{supp}}\}$*,* Ξ *is a function of* $(u \bmod 1, p)$*.*

(c) Over $\mathcal{A}$*,* $\chi_{\mathcal{A}}^{*}\Omega_{\tau}^{(2)} = \mathcal{D}^{*}\big(\omega_V + e_h\nu + d(\hat{G}_\tau\beta)\big)$*; near points with* $u \notin [u_0, u_1] + \mathbb{Z}$*,* $\Omega_{\tau}^{(2)} = \Omega_1 + \tau J'(p)\, b(v)\, dp \wedge dv$*.*

Proof. (a) For $u \le u_0$: $\mathcal{D} = \mathrm{id}$, $G_1 = 0$, $G_{\mathrm{f}} = J \circ p$. For $u \ge u_1 > r_*$: $G_1 \circ \mathcal{D} = H^*(p)$, $G_{\mathrm{f}} \circ \mathcal{D} = K(p)$, with difference $J(p)$.

(b) By (a), $\xi - J(p)\beta$ is supported over $[u_0, u_1]$, so ξ and ξ^{per} are smooth multiples of β ($\chi_{\mathcal{A}}$ preserves base points). Invariance follows as in Lemma 8.8(b), $J(p)\beta$ being invariant as deck maps preserve p and $v \bmod 1$. On $\{p > P_{\mathrm{supp}}\}$, $\mathcal{D} = \mathrm{id}$ and G_1, G_{f} depend only on (u, p).

(c) Over $\mathcal{A}$, $\chi_{\mathcal{A}}^{*}\xi^{\mathrm{per}} = \mathcal{D}^{*}((G_{\mathrm{f}} - G_1)\beta)$; add Lemma 8.8(c) at $\lambda = 1$. Near $u \notin [u_0, u_1] + \mathbb{Z}$, $v_1^{\mathrm{per}} = 0$, $\xi^{\mathrm{per}} = J(p)\beta$ and $d(J(p)\beta) = J'(p) b(v)\, dp \wedge dv$. □

Proposition 8.13 (The third path and the final form)**.** *For valid cubic data and* $\tau \in [0, 1]$*,* $\Omega_{\tau}^{(2)}$ *is closed, invariant under all deck maps and nondegenerate at every point of* $\{p < h + 2\delta_m\}$*. Over* $\mathcal{A}$*,*

$$\chi_{\mathcal{A}}^{*}\Omega_{\mathrm{fin}} = \mathcal{D}^{*}\Omega_{\mathrm{pre}}, \qquad (\chi_{\mathcal{A}} \circ \mathcal{D}^{-1})^{*}\Omega_{\mathrm{fin}} = \Omega_{\mathrm{pre}}, \qquad \Omega_{\mathrm{pre}} = \omega_V + e_h\nu + d(G_{\mathrm{f}}\beta),$$

and $\chi_{\mathcal{A}} \circ \mathcal{D}^{-1}$ *maps* $\mathcal{A} \times D(h)$ *diffeomorphically onto* $\mathcal{A} \times D(h)$*.*

Proof. Closedness and invariance follow from Proposition 8.9 and Lemma 8.12(b), so it suffices to work over $\{u_1 - 1 < u < 1 + u_0\}$.

Over $\mathcal{A}$, Lemma 8.12(c) and Lemma M5.1(b) give the square at $\mathcal{D}(x)$ as $2(e_h + b\,\partial_u\hat{G}_\tau)\,\omega_V \wedge \nu$, where $\partial_u\hat{G}_\tau = (1-\tau)\partial_u G_1 + \tau\partial_u G_{\mathrm{f}}$. If $p(x) < h + 2\delta_m$, the image fibre point w' has $p(w') < h + 2\delta_m$, so $\partial_u G_1(u, w') \ge 0$ as in Proposition 8.9, and $\partial_u G_{\mathrm{f}}(u, w') \ge 0$ by Lemmas 8.10(a) and 8.3(d).

Near $u \notin [u_0, u_1]$, Lemma 8.12(c) adds $dp \wedge$ (base one-form) to Ω_1, which changes only the mixed part in the local form of Lemma 7.28; the square remains $2\omega_V \wedge \kappa_1 > 0$ (Lemma 2.8).

The identities are Lemma 8.12(c) at $\tau = 1$. The last claim holds because $\mathcal{D}^{\pm 1}$ fix base points, $q_t^{\pm 1}$ preserve $D(h)$, and $\chi_{\mathcal{A}}$ preserves p. □

Lemma 8.14 (Vanishing on the marked fibres)**.** *Let* $\widetilde{\mathcal{F}}$ *be the union of all lifts of the fibres* $N_{b_1}, N_{b_2}, N_{b_3}$*, that is, the set of points* (u, v, w) *with* $(u, v) \in b_j + \mathbb{Z}^2$ *for some* j*. At every point of* $\widetilde{\mathcal{F}}$ *the covectors* α_1*,* $\dot{v}_\lambda$ $(\lambda \in \mathbb{R})$ *and* ξ^{per} *of* $T^*\widetilde{N}$ *are zero.*

Proof. For α_1, lift Lemma 7.26 to the cover, where α_1 is periodic. The u-coordinates $0, \frac{1}{3}, \frac{2}{3}$ of the b_j avoid $[u_0, u_1] = [\frac{7}{48}, \frac{9}{48}]$ mod one, so $\dot{v}_\lambda$ vanishes near $\widetilde{\mathcal{F}}$ by Lemma 8.8(b). Finally, $\xi^{\mathrm{per}} = \Xi\, b(v)\, dv$ by Lemma 8.12(b). The v-coordinates $0, \frac{2}{3}, \frac{1}{3}$ lie outside $[\frac{1}{8}, \frac{5}{24}] + \mathbb{Z} \supset \operatorname{supp} b$ (Proposition 6.3(c)), so $\beta = 0$ on $\widetilde{\mathcal{F}}$. □

Construction 8.15 (The handle Moser map)**.** The three stages are

$$(\Omega_s^{(0)}, \mathfrak{v}_s^{(0)}) = (\Omega_s, \alpha_1), \qquad (\Omega_s^{(1)}, \mathfrak{v}_s^{(1)}) = (\Omega_s^{(1)}, \dot{v}_s), \qquad (\Omega_s^{(2)}, \mathfrak{v}_s^{(2)}) = (\Omega_s^{(2)}, \xi^{\mathrm{per}}),$$

with matching endpoints $\Omega_1^{(0)} = \Omega_0^{(1)}$ and $\Omega_1^{(1)} = \Omega_0^{(2)}$. With the clocks $\sigma_i(t) = S(9t - 3i - 1)$, $i = 0, 1, 2$, which move only on $[\frac{3i+1}{9}, \frac{3i+2}{9}]$, put

$$\Omega_t^{\mathrm{M}} = \Omega_{\sigma_0(t)}^{(0)} + \sum_{i=1,2} \big(\Omega_{\sigma_i(t)}^{(i)} - \Omega_0^{(i)}\big), \qquad \mathfrak{v}_t = \sum_{i=0}^{2} \sigma_i'(t)\, \mathfrak{v}_{\sigma_i(t)}^{(i)}.$$

On $U=\{p<h+2\delta_m\}$ let Y_t be the field with $\iota_{Y_t}\Omega_t^{\mathrm{M}}=-\mathfrak{v}_t$; in the coordinates $(\operatorname{Re}w,\operatorname{Im}w,u,v)$ it is the adjugate solution of a 4×4 skew system. With the moment cutoff

$$\mathfrak{c}(p)=S\Big(\frac{h+\frac{7}{4}\delta_m-p}{\delta_m/4}\Big),$$

which equals one for $p\le h+\frac{3}{2}\delta_m$ and zero for $p\ge h+\frac{7}{4}\delta_m$, let $\widehat{Y}_t=\mathfrak{c}(p)Y_t$, extended by zero. The handle Moser map Φ is the time-one map of $\widehat{Y}$ from time 0, and Φ^{-1} is its inverse.

Proposition 8.16 (The handle Moser map)**.** *For valid cubic data, Φ and Φ^{-1} are mutually inverse C^∞ diffeomorphisms of $\widetilde{N}$, and:*

(a) $\Phi^*\Omega_{\mathrm{fin}}=\Omega_0$ and $(\Phi^{-1})^*\Omega_0=\Omega_{\mathrm{fin}}$ at every point of $\{p\le h+\delta_m\}$;
(b) for every $c>P_{\mathrm{supp}}$, Φ and Φ^{-1} map each of $\{p\le c\}$, $\{p<c\}$, $\{p=c\}$ onto itself;
(c) Φ and Φ^{-1} fix every point of $\widetilde{\mathcal{F}}$;
(d) Φ and Φ^{-1} commute with every deck map.

Moreover $\Phi=\Phi_2\circ\Phi_1\circ\Phi_0$, where Φ_i is the time-one map of the cut stage field $\mathfrak{c}(p)Y_s^{(i)}$, $\iota_{Y_s^{(i)}}\Omega_s^{(i)}=-\mathfrak{v}_s^{(i)}$.

Proof. Hypotheses. On $[\frac{3i+1}{9},\frac{3i+2}{9}]$ only σ_i moves, so $\Omega_t^{\mathrm{M}}=\Omega^{(i)}_{\sigma_i(t)}$ there; $\Omega_t^{\mathrm{M}}=\Omega_0$ for $t\le\frac19$ and $\Omega_t^{\mathrm{M}}=\Omega_{\mathrm{fin}}$ for $t\ge\frac89$. Hence Ω_t^{M} is closed and nondegenerate on U, by Lemma 7.28, Lemma 8.3(d) and Propositions 8.9, 8.13.

The pair is smooth on $\mathbb{R}\times\widetilde{N}$, with $\partial_t\Omega_t^{\mathrm{M}}=d\mathfrak{v}_t$ and $\mathfrak{v}_t=0$ for $t\notin(0,1)$. By Lemma 2.10(a), Y is smooth on $\mathbb{R}\times U$, hence $\widehat{Y}$ is smooth on $\mathbb{R}\times\widetilde{N}$. All stage data and $\mathfrak{c}(p)$ are deck invariant.

Toric band. Let $R=2\pi(-y\partial_x+x\partial_y)$ with $w=x+iy$, so that $\iota_R\Omega_0=-dp$ (Lemma 7.28). On $U'=\{P_{\mathrm{supp}}<p<h+2\delta_m\}$ we have $\iota_R\Omega_t^{\mathrm{M}}=-dp$ and $\mathfrak{v}_t(R)=0$: α_1 is horizontal with $\iota_R d\alpha_1=0$, v_s^{per} and $\dot v_s$ vanish near U' (Lemma 8.8(b)), and $\xi^{\mathrm{per}}=\Xi(u \bmod 1,p)\,b(v)\,dv$ on U'. Lemma 2.10(d) then gives $dp(\widehat{Y}_t)=0$ on $\{p>P_{\mathrm{supp}}\}$. Moreover $\widehat{Y}=0$ for $p\ge h+\frac74\delta_m$.

Completeness. By Lemma 2.10(g), $du(\widehat{Y}_t)$ and $dv(\widehat{Y}_t)$ are deck invariant, hence $|du(\widehat{Y}_t)|+|dv(\widehat{Y}_t)|\le C_b$, the maximum over $[0,1]\times[0,1]^2\times\bar D(h+\frac74\delta_m)$; also $\widehat{Y}_t=0$ for $t\notin[0,1]$. By the toric band, an integral curve through x stays in $\{p\le\max(p(x),h)\}$ with base displacement at most C_b, so the flow is global and smooth.

Conclusions. (b) follows from $dp(\widehat{Y}_t)=0$ on $\{p>P_{\mathrm{supp}}\}$. (a) Since $P_{\mathrm{supp}}<h$, curves from $\{p\le h+\delta_m\}$ stay there, and $\{p<h+\frac32\delta_m\}$ is invariant. There $\mathfrak{c}=1$, so Lemma 2.10(b) gives $(\Phi^*\Omega_1^{\mathrm{M}})_x=(\Omega_0^{\mathrm{M}})_x$; the inverse identity follows as Φ^{-1} preserves $\{p\le h+\delta_m\}$. (c) By Lemma 8.14, $\widehat{Y}_t=0$ on $\widetilde{\mathcal{F}}$; apply Lemma 2.10(c). (d) is Lemma 2.10(g).

Decomposition. Each stage pair is deck invariant and satisfies $\iota_R\Omega_s^{(i)}=-dp$ and $\mathfrak{v}_s^{(i)}(R)=0$ on U', so the same argument gives a global flow $\Phi_s^{(i)}$ of $\mathfrak{c}(p)Y_s^{(i)}$; set $\Phi_i=\Phi_1^{(i)}$. On $[\frac{3i+1}{9},\frac{3i+2}{9}]$ we have $\widehat{Y}_t=\sigma_i'(t)\mathfrak{c}(p)Y^{(i)}_{\sigma_i(t)}$, so $t\mapsto\Phi^{(i)}_{\sigma_i(t)}(x)$ is an integral curve of $\widehat{Y}$; elsewhere $\widehat{Y}_t=0$. Uniqueness and Lemma 2.10(f) give $\Phi=\Phi_2\circ\Phi_1\circ\Phi_0$. □

Equivalently, let Φ_A ($A\ge0$) be the time-one map of $\widehat{Y}$ times a product cutoff in (u,v) equal to one on $\{|u|,|v|\le A+C_b+2\}$. Then $\Phi(x)=\Phi_A(x)$ for $A=|u(x)|+|v(x)|$, by the base speed bound.

Layer maps. Work on the layer $\Lambda_i=\{l_i<u<r_i,\ \frac18<v<\frac{5}{24},\ p<h\}$ (frame coordinates). Let $T(v)=\int_{1/8}^v b$, a diffeomorphism $(\frac18,\frac{5}{24})\to(0,1)$ with $dT=\beta$ and inverse $v(T)$. Set $\ell(T)=e_h/b(v(T))$ and

$$x_i(u,T,p)=(u-l_i)\,\ell(T)+\varsigma_i(u)\,h_i(p).$$

Since $\partial_u x_i = \ell + \varsigma_i' h_i > 0$, $x_i = 0$ at $u = l_i$ and $x_i > h_i(p)$ at $u = r_i$, there is a unique root $u = \sigma_i(x,T,p) \in (l_i, r_i)$ for $0 < x < h_i(p)$, $0 < T < 1$.

For $1 \le i \le k$ put $\bar\rho_i = (\rho_i + a_i)/2$, $\delta_i = a_i - \bar\rho_i$ and $\hat\rho_i = \bar\rho_i + \delta_i/4$. For $(z_{\mathrm b}, z_{\mathrm f}) \in B^\circ(\hat\rho_i) \subset \mathbb{C}^2$, write $(x,T) = \Psi_{\delta_i}(z_{\mathrm b})$ (Proposition 8.20) and $p = \pi|z_{\mathrm f}|^2$, and define

$$E_i(z_{\mathrm b}, z_{\mathrm f}) = \Big(\sigma_i(x,T,p),\ v(T),\ e^{-2\pi i T A_i'(p)} z_{\mathrm f}\Big), \qquad f_i^N = \Phi^{-1} \circ \chi_{\mathcal A} \circ \mathcal D^{-1} \circ E_i.$$

Proposition 8.17 (The four-dimensional embeddings into the cover)**.** *Let the cubic data be valid and $1 \le i, j \le k$. Then $\rho_i < \bar\rho_i < \hat\rho_i < a_i$, and:*

(a) f_i^N is defined and smooth on $B^\circ(\hat\rho_i)$, with values in $\{p < h\} \setminus \widetilde{\mathcal F}$, and $(f_i^N)^*\Omega_0 = \omega_0$;
(b) f_i^N is injective, $f_i^N(B^\circ(\hat\rho_i)) \cap f_j^N(B^\circ(\hat\rho_j)) = \varnothing$ for $i \ne j$, and $f_i^N(B(\bar\rho_i))$ is compact;
(c) if $f_i^N(X) = T\, f_j^N(X')$ for a deck map T, then $T = \mathrm{id}$, $i = j$ and $X = X'$.

Proof. Let $\varpi = \pi|z_{\mathrm b}|^2$, so $\varpi + p < \hat\rho_i$. By Proposition 8.20 and Lemma 8.3(a),

$$0 < x < \varpi + \tfrac{\delta_i}{2} < a_i - p - \tfrac{\delta_i}{4} < h_i(p), \qquad p < a_i < h,$$

so E_i maps $B^\circ(\hat\rho_i)$ into Λ_i.

Since $\ell\, b = e_h$, $e_h \nu = \ell\, du \wedge dT$, and by Lemma 8.10(c), $d(G_{\mathrm f}\beta) = (\varsigma_i' h_i\, du + dA_i + \varsigma_i\, dh_i) \wedge dT$ on Λ_i; hence $\Omega_{\mathrm{pre}} = \omega_V + dx_i \wedge dT + dA_i(p) \wedge dT$ there.

Now E_i is the composite of $\Psi_{\delta_i} \times \mathrm{id}$, the shear of Lemma 8.19 with $A = A_i$, and the inverse of $(u, v, w_{\mathcal A}) \mapsto (x_i, T, w_{\mathcal A})$ (the periodic counterpart of the layer maps of [Son26]). By Proposition 8.20 and Lemma 8.19, E_i is a diffeomorphism onto an open subset of Λ_i with $E_i^*\Omega_{\mathrm{pre}} = \omega_0$.

(a) By Proposition 8.13, $\chi_{\mathcal A}\mathcal D^{-1}$ maps Λ_i into $\mathcal A \times D(h)$ and pulls Ω_{fin} back to Ω_{pre}; by Proposition 8.16(a),(b), Φ^{-1} preserves $\{p < h\}$ and pulls Ω_0 back to Ω_{fin} there. Composing gives $(f_i^N)^*\Omega_0 = \omega_0$. The point $\chi_{\mathcal A}\mathcal D^{-1}E_i(X)$ lies over $\mathcal A$, which misses $\widetilde{\mathcal F}$, and $\Phi^{\pm 1}$ fix $\widetilde{\mathcal F}$ pointwise (Proposition 8.16(c)), so f_i^N misses $\widetilde{\mathcal F}$.

(b) E_i is injective, and the images of different E_i lie over the disjoint intervals (l_i, r_i). The maps $\chi_{\mathcal A}\mathcal D^{-1}$ and Φ^{-1} are injective and common to all i. The closed ball is compact.

(c) By Proposition 8.16(d), $y = \chi_{\mathcal A}\mathcal D^{-1}E_i(X)$ and $y' = \chi_{\mathcal A}\mathcal D^{-1}E_j(X')$ satisfy $y = Ty'$. Both lie over $(\frac18, \frac{5}{24})^2$, so $T = \mathrm{id}$, and (b) gives $i = j$ and $X = X'$. □

Remark 8.18 (The four-dimensional embeddings). By Proposition 8.17(a),(c) the maps f_i^N satisfy the hypotheses of Theorem 10.21, and the four-dimensional embeddings are $f_i^{\mathrm{cap}} = b_{\mathrm{ball}} \circ \mathcal T_{\mathrm u} \circ \mathsf N \circ f_i^N$ on $B^\circ(\hat\rho_i)$. The equal branch uses these maps for the data of Remark 10.25: $\rho_i = \rho$, $a_i = a$, $K(p) = (3\sqrt{k} - 9)p$, and (q_t) the isotopy of Theorem 5.34, with the recorded capacities of Remark 10.24; the valid cubic data of [Son26] use the same maps. The strip constants are the four-dimensional values of Definition 9.8; only the disc-to-strip map of Proposition 9.7 is taken from §9.

8.2. **The shear and the strip map.** The layer maps of §8.1, and those of the higher-dimensional handle [Son26], use the following two maps.

Lemma 8.19 (Multi-angle shear)**.** *Let A be a smooth function on an open set $W \subset \mathbb{R}^m$ and $\mathrm{Sh}_A(x,T,w) = (x, T, \mathrm{Sh}_{A,T}(w))$, where*

$$\mathrm{Sh}_{A,T}(w)_j = \exp\big(-2\pi i\, T\, \partial_j A(p(w))\big)\, w_j \qquad (1 \le j \le m),$$

on $\{(x,T,w) : p(w) \in W\} \subset \mathbb{R}^2 \times \mathbb{C}^m$. Then Sh_A is a diffeomorphism of this set onto itself, preserves p, x and T, and

$$\mathrm{Sh}_A^*\big(\omega_V + dx \wedge dT + dA(p) \wedge dT\big) = \omega_V + dx \wedge dT.$$

Proof. $\mathrm{Sh}_{A,T}$ multiplies each w_j by a unimodular factor smooth in $(T, p(w))$, so it preserves p; the same formula with $+2\pi i$ is a smooth inverse, also on the coordinate axes. Where all $w_j \neq 0$, write $w_j = \sqrt{p_j/\pi}\, e^{2\pi i \theta_j}$; then $\mathrm{Sh}_{A,T}$ is $\theta_j \mapsto \theta_j - T\partial_j A(p)$ and

$$\mathrm{Sh}_A^* \omega_V = \omega_V - dA \wedge dT - T \sum_{j,k} \partial_{jk} A \, dp_j \wedge dp_k = \omega_V - dA \wedge dT.$$

The Hessian term vanishes by symmetry, and $dx \wedge dT$ and $dA(p) \wedge dT$ are unchanged. The identity extends to the axes by continuity. □

Proposition 8.20 (Disc-to-strip map)**.** *For every $\delta > 0$ there is an explicit smooth diffeomorphism $\Psi_\delta \colon \mathbb{C} \to (0,\infty) \times (0,1)$, $z \mapsto (x,T)$, with $\Psi_\delta^*(dx \wedge dT) = \omega_{\mathbb{C}}$ and*

$$0 < x < \pi|z|^2 + \frac{\delta}{2} \qquad \text{for all } z \in \mathbb{C}.$$

The map is $\Psi_\delta = F \circ \mathsf{T}^{-1} \circ \mathsf{D}$, with $\varepsilon = \delta/\pi$: $F(u,v) = (X(u), Y(v))$ inverts the square roots of the Morse functions $x + \varepsilon^2/(4x) - \varepsilon$ on $(0,\infty)$ and $\varepsilon\,[(2\sqrt{T(1-T)})^{-1} - 1]$ on $(0,1)$, and T, D are the explicit rotation and radial area normalisations of the density $X'(u)Y'(v)$. The estimate holds because the level set of energy $\mathcal{E}$ encloses area at least $\mathcal{E} - \varepsilon(\frac{\pi}{2} - 1)$ and lies in $\{x < \mathcal{E} + \varepsilon\}$ (Proposition 9.7). Some excess is unavoidable, since no area-preserving map takes the closed disc of area a into $\{0 < x < a\} \times (0,1)$; the excess $\delta/2$ serves all concentric discs at once.

9. The disc-to-strip map and the layer maps

This section constructs the planar disc-to-strip map and proves Proposition 8.20. The higher-dimensional layer maps, one ball per layer, are in [Son26].

Conventions. On $\mathbb{C} \ni z = x + iy$, $\omega_{\mathbb{C}} = dx \wedge dy = dI \wedge d\theta$ with $I = \pi|z|^2$, $\theta = \arg z/2\pi$. The strip $(0,\infty) \times (0,1)$ has coordinates (x,T) and area form $dx \wedge dT$. The plane $\mathbb{R}^2 \ni \xi = (\xi_1, \xi_2) \cong \mathbb{C}$ ($\xi \mapsto \xi_1 + i\xi_2$) carries $d\xi = d\xi_1 \wedge d\xi_2$; Rot_ϕ is rotation by ϕ. In §9.3 (and [Son26]) $\mathbb{C}^n = \mathbb{C} \times \mathbb{C}^m \ni (z_{\mathrm{b}}, z_{\mathrm{f}})$, $\varpi = \pi|z_{\mathrm{b}}|^2$, $p = p(z_{\mathrm{f}})$, so $\omega_0 = \omega_{\mathbb{C}}(z_{\mathrm{b}}) + \omega_V(z_{\mathrm{f}})$, $B(c) = \{\varpi + |p| \le c\}$, $B^\circ(c) = \{\varpi + |p| < c\}$.

9.1. **The strip Morse functions.** Ψ_δ sends the circle of area I onto a level set of $R_{\mathrm{st}}(x) + V_{\mathrm{st}}(T)$ inside $\{x < I + \frac{\delta}{2}\}$ (Figure 2); it is the product F_{st} of signed square-root inverses $X_{\mathrm{st}}, Y_{\mathrm{st}}$ of $R_{\mathrm{st}}, V_{\mathrm{st}}$, followed by two circle-preserving density normalisations.

Construction 9.1 (Strip Morse functions)**.** Let $\delta > 0$ and $b_{\mathrm{st}} = \delta/\pi$. For $x > 0$, $0 < T < 1$ and $\xi = (\xi_1, \xi_2) \in \mathbb{R}^2$ put

$$R_{\mathrm{st}}(x) = x + \frac{b_{\mathrm{st}}^2}{4x} - b_{\mathrm{st}}, \qquad V_{\mathrm{st}}(T) = b_{\mathrm{st}}\Big[\frac{1}{2\sqrt{T(1-T)}} - 1\Big],$$

$$X_{\mathrm{st}}(\xi_1) = \tfrac{1}{2}\Big[\xi_1^2 + b_{\mathrm{st}} + \xi_1\sqrt{\xi_1^2 + 2b_{\mathrm{st}}}\Big], \qquad Y_{\mathrm{st}}(\xi_2) = \frac{1}{2} + \frac{\xi_2\sqrt{\xi_2^2 + 2b_{\mathrm{st}}}}{2(\xi_2^2 + b_{\mathrm{st}})},$$

and $F_{\mathrm{st}}(\xi) = \big(X_{\mathrm{st}}(\xi_1), Y_{\mathrm{st}}(\xi_2)\big)$, $\mathcal{J}_{\mathrm{st}}(\xi) = X_{\mathrm{st}}'(\xi_1)\, Y_{\mathrm{st}}'(\xi_2)$.

Lemma 9.2 (Smooth Morse inverses)**.** *(i) $R_{\mathrm{st}} \ge 0$ on $(0,\infty)$ with unique zero $x = b_{\mathrm{st}}/2$, and $V_{\mathrm{st}} \ge 0$ on $(0,1)$ with unique zero $T = \frac{1}{2}$.*

(ii) $R_{\mathrm{st}}(X_{\mathrm{st}}(\xi_1)) = \xi_1^2$ and $V_{\mathrm{st}}(Y_{\mathrm{st}}(\xi_2)) = \xi_2^2$ for all $\xi_1, \xi_2 \in \mathbb{R}$.

(iii) X_{st} and Y_{st} are smooth on $\mathbb{R}$, with

$$X_{\mathrm{st}}'(\xi_1) = \frac{\big(\xi_1 + \sqrt{\xi_1^2 + 2b_{\mathrm{st}}}\big)^2}{2\sqrt{\xi_1^2 + 2b_{\mathrm{st}}}} > 0, \qquad Y_{\mathrm{st}}'(\xi_2) = \frac{b_{\mathrm{st}}^2}{\sqrt{\xi_2^2 + 2b_{\mathrm{st}}}\,(\xi_2^2 + b_{\mathrm{st}})^2} > 0.$$

(iv) X_{st} *is an increasing bijection* $\mathbb{R}\to(0,\infty)$ *and* Y_{st} *an increasing bijection* $\mathbb{R}\to(0,1)$. F_{st} *is a smooth diffeomorphism of* $\mathbb{R}^2$ *onto* $(0,\infty)\times(0,1)$ *with* $F_{\mathrm{st}}^*(dx\wedge dT)=\mathcal{J}_{\mathrm{st}}\,d\xi$ *and* $R_{\mathrm{st}}\circ X_{\mathrm{st}}+V_{\mathrm{st}}\circ Y_{\mathrm{st}}=|\xi|^2$.

Proof. (i) $R_{\mathrm{st}}(x)=(\sqrt{x}-b_{\mathrm{st}}/(2\sqrt{x}))^2$, which vanishes only at $x=b_{\mathrm{st}}/2$. Since $T(1-T)\le\frac14$ with equality only at $T=\frac12$, $V_{\mathrm{st}}\ge0$ with unique zero $T=\frac12$.

(ii) For $x>0$, $R_{\mathrm{st}}(x)=\xi_1^2$ is equivalent to $x^2-(\xi_1^2+b_{\mathrm{st}})x+b_{\mathrm{st}}^2/4=0$, with discriminant $(\xi_1^2+b_{\mathrm{st}})^2-b_{\mathrm{st}}^2=\xi_1^2(\xi_1^2+2b_{\mathrm{st}})$ and roots $x_\pm=\frac12[\xi_1^2+b_{\mathrm{st}}\pm|\xi_1|\sqrt{\xi_1^2+2b_{\mathrm{st}}}]$, both positive because $x_+x_-=b_{\mathrm{st}}^2/4$. $X_{\mathrm{st}}(\xi_1)$ is x_+ for $\xi_1\ge0$ and x_- for $\xi_1\le0$. For the second identity write $Y_{\mathrm{st}}=\frac12+\tau$ with $\tau=\xi_2\sqrt{\xi_2^2+2b_{\mathrm{st}}}/(2(\xi_2^2+b_{\mathrm{st}}))$. Then $\tau^2=\frac14-\frac{b_{\mathrm{st}}^2}{4(\xi_2^2+b_{\mathrm{st}})^2}<\frac14$, so $0<Y_{\mathrm{st}}<1$, $T(1-T)=\frac14-\tau^2=b_{\mathrm{st}}^2/(4(\xi_2^2+b_{\mathrm{st}})^2)$ for $T=Y_{\mathrm{st}}(\xi_2)$, hence $1/(2\sqrt{T(1-T)})=(\xi_2^2+b_{\mathrm{st}})/b_{\mathrm{st}}$ and $V_{\mathrm{st}}(T)=\xi_2^2$.

(iii) Smoothness holds because $\xi_1^2+2b_{\mathrm{st}}>0$ and $\xi_2^2+b_{\mathrm{st}}>0$. With $r=\sqrt{\xi_1^2+2b_{\mathrm{st}}}$, $2X'_{\mathrm{st}}=2\xi_1+r+\xi_1^2/r=(\xi_1+r)^2/r$, and $\xi_1+r>0$ since $r>|\xi_1|$. With $\varphi(\xi_2)=\xi_2\sqrt{\xi_2^2+2b_{\mathrm{st}}}/(\xi_2^2+b_{\mathrm{st}})$,

$$\varphi'(\xi_2)=\frac{(2\xi_2^2+2b_{\mathrm{st}})(\xi_2^2+b_{\mathrm{st}})-2\xi_2^2(\xi_2^2+2b_{\mathrm{st}})}{\sqrt{\xi_2^2+2b_{\mathrm{st}}}\,(\xi_2^2+b_{\mathrm{st}})^2}=\frac{2b_{\mathrm{st}}^2}{\sqrt{\xi_2^2+2b_{\mathrm{st}}}\,(\xi_2^2+b_{\mathrm{st}})^2},$$

and $Y'_{\mathrm{st}}=\varphi'/2$.

(iv) Both maps are continuous and strictly increasing by (iii). As $\xi_1\to+\infty$, $X_{\mathrm{st}}\to\infty$; for $\xi_1<0$, $X_{\mathrm{st}}(\xi_1)=x_-=b_{\mathrm{st}}^2/(4x_+)$ with $x_+\to\infty$ as $\xi_1\to-\infty$, so $X_{\mathrm{st}}\to0^+$. Since φ is odd and $\varphi^2=1-b_{\mathrm{st}}^2/(\xi_2^2+b_{\mathrm{st}})^2\to1$, Y_{st} increases from 0^+ to 1^-. Thus F_{st} is a smooth bijection with Jacobian $\mathcal{J}_{\mathrm{st}}>0$, hence a smooth diffeomorphism by the inverse function theorem; the last identity is (ii). □

9.2. **The strip map.** The two normalisations are those of Definition 4.3 for one density, in Cartesian coordinates; we repeat the proofs since $\mathcal{J}_{\mathrm{st}}$ is not constant near the origin, as setting (N) of §4.3 assumes.

Definition 9.3 (Rotation and area normalisation)**.** Let $\mathcal{J}$ be a positive smooth function on $\mathbb{R}^2$. Put

$$\begin{aligned}\bar{\mathcal{J}}(|\xi|^2)&=\frac{1}{2\pi}\int_0^{2\pi}\mathcal{J}(\mathrm{Rot}_\phi\xi)\,d\phi,\\ \mathcal{M}(\xi)&=\frac{1}{2\pi}\int_0^{2\pi}(\phi-\pi)\,\mathcal{J}(\mathrm{Rot}_\phi\xi)\,d\phi,\\ \mathsf{T}_{\mathcal{J}}(\xi)&=\mathrm{Rot}_{\mathcal{M}(\xi)/\bar{\mathcal{J}}(|\xi|^2)}\,\xi.\end{aligned}$$

and $\mathfrak{a}_{\mathcal{J}}(\mathcal{E})=\int_{|\xi|^2\le\mathcal{E}}\mathcal{J}\,d\xi$ for $\mathcal{E}\ge0$. In polar coordinates $\mathfrak{a}_{\mathcal{J}}(\mathcal{E})=\int_0^{\sqrt{\mathcal{E}}}\int_0^{2\pi}\mathcal{J}\,d\psi\,r\,dr=\int_0^{\sqrt{\mathcal{E}}}2\pi r\bar{\mathcal{J}}(r^2)\,dr=\pi\int_0^{\mathcal{E}}\bar{\mathcal{J}}(t)\,dt$ with $\bar{\mathcal{J}}>0$, so $\mathfrak{a}_{\mathcal{J}}$ is continuous and strictly increasing with $\mathfrak{a}_{\mathcal{J}}(0)=0$. If $\mathfrak{a}_{\mathcal{J}}(\mathcal{E})\to\infty$ as $\mathcal{E}\to\infty$, it is a bijection of $[0,\infty)$; then let $\mathcal{E}_{\mathcal{J}}=\mathfrak{a}_{\mathcal{J}}^{-1}$ and, for $z\in\mathbb{C}$ with $I=\pi|z|^2$,

$$\mathsf{D}_{\mathcal{J}}(z)=\sqrt{\pi\,\mathcal{E}_{\mathcal{J}}(I)/I}\;z\quad(z\neq0),\qquad\mathsf{D}_{\mathcal{J}}(0)=0.$$

Lemma 9.4 (Normalisation)**.** *Let* $\mathcal{J}$ *be a positive smooth function on* $\mathbb{R}^2$.

(i) $\mathsf{T}_{\mathcal{J}}$ *is a smooth diffeomorphism of* $\mathbb{R}^2$ *which maps every circle* $\{|\xi|=r\}$ *onto itself, and* $\mathsf{T}_{\mathcal{J}}^*\big(\bar{\mathcal{J}}(|\xi|^2)\,d\xi\big)=\mathcal{J}\,d\xi$.

(ii) If $\mathfrak{a}_{\mathcal{J}}(\mathcal{E})\to\infty$, *then* $\mathsf{D}_{\mathcal{J}}$ *is a smooth diffeomorphism* $\mathbb{C}\to\mathbb{R}^2$ *with* $|\mathsf{D}_{\mathcal{J}}(z)|^2=\mathcal{E}_{\mathcal{J}}(\pi|z|^2)$ *and* $\mathsf{D}_{\mathcal{J}}^*\big(\bar{\mathcal{J}}(|\xi|^2)\,d\xi\big)=\omega_{\mathbb{C}}$.

(iii) If $\mathfrak{a}_{\mathcal{J}}(\mathcal{E}) \to \infty$ and $F\colon \mathbb{R}^2 \to U$ is a smooth diffeomorphism onto an open set $U \subset \mathbb{R}^2$ with $F^*(dx \wedge dT) = \mathcal{J}\, d\xi$, then $F \circ \mathsf{T}_{\mathcal{J}}^{-1} \circ \mathsf{D}_{\mathcal{J}}\colon \mathbb{C} \to U$ is a smooth diffeomorphism which pulls $dx \wedge dT$ back to $\omega_{\mathbb{C}}$ and maps the circle $\{\pi|z|^2 = I\}$ onto $F(\{|\xi|^2 = \mathcal{E}_{\mathcal{J}}(I)\})$.

Proof. Differentiating under the integral, $\mathcal{M}$ and $\bar{\mathcal{J}}(|\xi|^2) > 0$ are smooth; bijections with non-vanishing Jacobian are diffeomorphisms.

(i) Fix $r > 0$, $\mathcal{J}_r(\psi) = \mathcal{J}(re^{i\psi})$. Integrating by parts in $\mathcal{M}(re^{i\psi}) = \frac{1}{2\pi}\int_0^{2\pi}(\phi - \pi)\mathcal{J}_r(\psi + \phi)\, d\phi$ gives $\partial_\psi \mathcal{M} = \mathcal{J}_r(\psi) - \bar{\mathcal{J}}(r^2)$. So $\mathsf{T}_{\mathcal{J}}(re^{i\psi}) = re^{i\Theta(\psi)}$ with $\Theta' = \mathcal{J}/\bar{\mathcal{J}} > 0$, $\Theta(\psi + 2\pi) = \Theta(\psi) + 2\pi$; hence $\mathsf{T}_{\mathcal{J}}$ is a smooth bijection preserving each circle, and off 0, $\mathsf{T}_{\mathcal{J}}^*(\bar{\mathcal{J}}\, r\, dr \wedge d\psi) = \bar{\mathcal{J}}\, r\, dr \wedge (\partial_r \Theta\, dr + \Theta'\, d\psi) = \mathcal{J}\, r\, dr \wedge d\psi$. Thus $\det D\mathsf{T}_{\mathcal{J}} = \mathcal{J}/\bar{\mathcal{J}} > 0$, also at 0 by continuity.

(ii) $G(\xi) = \int_0^1 \bar{\mathcal{J}}(t|\xi|^2)\, dt = \frac{1}{2\pi}\int_0^1\int_0^{2\pi} 2r\, \mathcal{J}(\mathrm{Rot}_\phi(r\xi))\, d\phi\, dr$ is smooth, positive, and $\pi|\xi|^2 G(\xi) = \mathfrak{a}_{\mathcal{J}}(|\xi|^2)$. Hence $\mathsf{D}'(\xi) = \sqrt{G(\xi)}\,\xi$ preserves the argument, satisfies $\pi|\mathsf{D}'(\xi)|^2 = \mathfrak{a}_{\mathcal{J}}(|\xi|^2)$, is a bijection, and inverts $\mathsf{D}_{\mathcal{J}}$. With $\mathcal{E} = |\xi|^2$, off 0, $(\mathsf{D}')^*\omega_{\mathbb{C}} = (\mathsf{D}')^*(dI \wedge d\psi/2\pi) = \frac{1}{2}\bar{\mathcal{J}}(\mathcal{E})\, d\mathcal{E} \wedge d\psi = \bar{\mathcal{J}}\, d\xi$, so $\det D\mathsf{D}' = \bar{\mathcal{J}} > 0$ (also at 0). Thus $\mathsf{D}_{\mathcal{J}} = (\mathsf{D}')^{-1}$ is a diffeomorphism with $\mathsf{D}_{\mathcal{J}}^*(\bar{\mathcal{J}}\, d\xi) = \omega_{\mathbb{C}}$, and $|\mathsf{D}_{\mathcal{J}}(z)|^2 = \mathcal{E}_{\mathcal{J}}(I)$.

(iii) By (i),(ii) the composite is a diffeomorphism onto U with pullback $\mathsf{D}_{\mathcal{J}}^*(\mathsf{T}_{\mathcal{J}}^{-1})^*(\mathcal{J}\, d\xi) = \mathsf{D}_{\mathcal{J}}^*(\bar{\mathcal{J}}\, d\xi) = \omega_{\mathbb{C}}$; $\mathsf{D}_{\mathcal{J}}$ maps $\{\pi|z|^2 = I\}$ onto $\{|\xi|^2 = \mathcal{E}_{\mathcal{J}}(I)\}$, preserved by $\mathsf{T}_{\mathcal{J}}^{-1}$. □

For $\mathcal{J} = \mathcal{J}_{\mathrm{st}}$ write $\bar{\mathcal{J}}_{\mathrm{st}}$ and $\mathfrak{a}_{\mathrm{st}} = \mathfrak{a}_{\mathcal{J}_{\mathrm{st}}}$.

Lemma 9.5 (Sublevel sets and area)**.** *Let $b_{\mathrm{st}} = \delta/\pi$.*

(i) For $c \geq 0$, $\{x > 0 : R_{\mathrm{st}}(x) \leq c\} = [x_-(c), x_+(c)]$ with $x_\pm(c) = \frac{1}{2}\big[c + b_{\mathrm{st}} \pm \sqrt{c(c + 2b_{\mathrm{st}})}\big]$. Its length is $\sqrt{c(c + 2b_{\mathrm{st}})} \geq c$, and $0 < x_-(c) \leq x_+(c) < c + b_{\mathrm{st}}$. For $c < 0$ the set is empty.

(ii) $\int_0^1 V_{\mathrm{st}}(T)\, dT = b_{\mathrm{st}}(\frac{\pi}{2} - 1)$.

(iii) $\mathfrak{a}_{\mathrm{st}}(\mathcal{E})$ is the $dx\, dT$-area of $\{(x, T) : R_{\mathrm{st}}(x) + V_{\mathrm{st}}(T) \leq \mathcal{E}\}$, and $\mathfrak{a}_{\mathrm{st}}(\mathcal{E}) \geq \mathcal{E} - b_{\mathrm{st}}(\frac{\pi}{2} - 1)$.

(iv) $\mathfrak{a}_{\mathrm{st}}$ is a continuous strictly increasing bijection of $[0, \infty)$, with $\mathfrak{a}'_{\mathrm{st}}(\mathcal{E}) = \pi\bar{\mathcal{J}}_{\mathrm{st}}(\mathcal{E}) > 0$.

Proof. (i) As in Lemma 9.2(ii) with $\xi_1^2 = c$; $x_+ - x_- = \sqrt{c(c + 2b_{\mathrm{st}})} \geq c$, and $(c + b_{\mathrm{st}})^2 - c(c + 2b_{\mathrm{st}}) = b_{\mathrm{st}}^2 > 0$ gives $x_- > 0$, $x_+ < c + b_{\mathrm{st}}$. For $c < 0$ use $R_{\mathrm{st}} \geq 0$. (ii) $T = \sin^2\phi$ gives $\int_0^1 dT/\sqrt{T(1 - T)} = \pi$. (iii) The first claim is change of variables via F_{st} (Lemma 9.2(iv)). By Fubini, (i), (ii): $\mathfrak{a}_{\mathrm{st}}(\mathcal{E}) = \int_0^1 |\{x > 0 : R_{\mathrm{st}}(x) \leq \mathcal{E} - V_{\mathrm{st}}(T)\}|\, dT \geq \int_0^1 (\mathcal{E} - V_{\mathrm{st}})_+\, dT \geq \mathcal{E} - b_{\mathrm{st}}(\frac{\pi}{2} - 1)$. (iv) From $\mathfrak{a}_{\mathrm{st}} = \pi\int_0^{\mathcal{E}} \bar{\mathcal{J}}_{\mathrm{st}}$ (Definition 9.3) and (iii) for $\mathfrak{a}_{\mathrm{st}} \to \infty$. □

Construction 9.6 (The strip map)**.** For $\delta > 0$ let $\mathsf{T}_{\mathrm{st}} = \mathsf{T}_{\mathcal{J}_{\mathrm{st}}}$, $\mathcal{E}_{\mathrm{st}} = \mathfrak{a}_{\mathrm{st}}^{-1}$ (defined on $[0, \infty)$ by Lemma 9.5(iv)) and $\mathsf{D}_{\mathrm{st}} = \mathsf{D}_{\mathcal{J}_{\mathrm{st}}}$, and define

$$\Psi_\delta = F_{\mathrm{st}} \circ \mathsf{T}_{\mathrm{st}}^{-1} \circ \mathsf{D}_{\mathrm{st}}\colon \mathbb{C} \to (0, \infty) \times (0, 1), \qquad \Psi_\delta(z) = (x, T).$$

Proposition 9.7 (Disc-to-strip map)**.** *Let $\delta > 0$ and $b_{\mathrm{st}} = \delta/\pi$. Then:*

(i) Ψ_δ is a smooth diffeomorphism of $\mathbb{C}$ onto $(0, \infty) \times (0, 1)$ with $\Psi_\delta^*(dx \wedge dT) = \omega_{\mathbb{C}}$;

(ii) $R_{\mathrm{st}}(x) + V_{\mathrm{st}}(T) = \mathcal{E}_{\mathrm{st}}(\pi|z|^2)$ for $(x, T) = \Psi_\delta(z)$;

(iii) for every $z \in \mathbb{C}$, with $(x, T) = \Psi_\delta(z)$,

$$0 < x < \pi|z|^2 + \frac{\delta}{2}, \qquad 0 < T < 1; \tag{18}$$

(iv) for $I_{\max} \geq 0$, $\mathcal{E}_{\max} = I_{\max} + b_{\mathrm{st}}(\frac{\pi}{2} - 1)$ and $\pi|z|^2 \leq I_{\max}$,

$$x \geq \tfrac{1}{2}\Big[\mathcal{E}_{\max} + b_{\mathrm{st}} - \sqrt{\mathcal{E}_{\max}(\mathcal{E}_{\max} + 2b_{\mathrm{st}})}\Big] > 0, \qquad \big|T - \tfrac{1}{2}\big| \leq \tfrac{1}{2}\sqrt{1 - (1 + \mathcal{E}_{\max}/b_{\mathrm{st}})^{-2}} < \tfrac{1}{2}.$$

In particular Ψ_δ has the properties stated in Proposition 8.20.

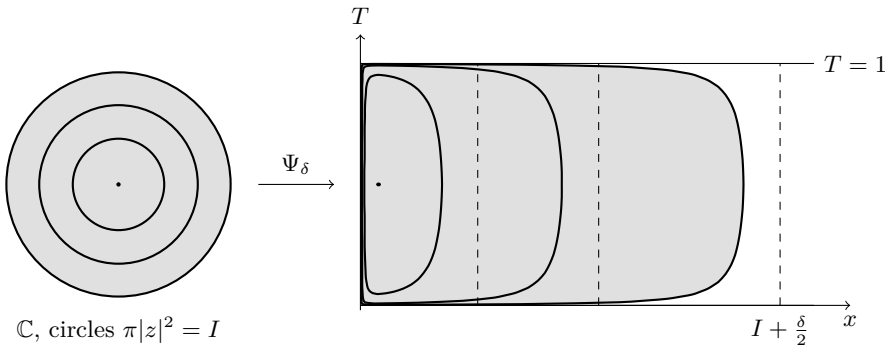


FIGURE 2. The disc-to-strip map Ψ_δ of Proposition 9.7: computed level sets for $b_{\mathrm{st}} = \delta/\pi = 0.12$ and $I = 0.2, 0.6, 1.2$. The circle of area I goes to the level set $\{R_{\mathrm{st}} + V_{\mathrm{st}} = \mathcal{E}_{\mathrm{st}}(I)\}$, of area I, inside $\{x < I + \frac{\delta}{2}\}$ (dashed); the dot is the common minimum $(b_{\mathrm{st}}/2, \frac{1}{2})$. Axes not to the same scale.

Proof. (i), (ii) Lemma 9.4(iii) applied to $\mathcal{J}_{\mathrm{st}}$ and F_{st} (Lemma 9.2(iv)), with $\mathfrak{a}_{\mathrm{st}} \to \infty$ by Lemma 9.5(iii); the level identity follows from $R_{\mathrm{st}} \circ X_{\mathrm{st}} + V_{\mathrm{st}} \circ Y_{\mathrm{st}} = |\xi|^2$. (iii) Let $I = \pi|z|^2$, $\mathcal{E} = \mathcal{E}_{\mathrm{st}}(I)$; by Lemma 9.5(iii), $I \geq \mathcal{E} - b_{\mathrm{st}}(\frac{\pi}{2} - 1)$. By (ii) and $V_{\mathrm{st}} \geq 0$, $R_{\mathrm{st}}(x) \leq \mathcal{E}$, so Lemma 9.5(i) gives $x < \mathcal{E} + b_{\mathrm{st}} \leq I + b_{\mathrm{st}}\frac{\pi}{2} = I + \frac{\delta}{2}$; the remaining bounds hold since Ψ_δ maps into the strip. (iv) Since $\mathcal{E}_{\mathrm{st}}$ increases and $I_{\max} \geq \mathcal{E}_{\mathrm{st}}(I_{\max}) - b_{\mathrm{st}}(\frac{\pi}{2} - 1)$ (Lemma 9.5(iii)), $\mathcal{E} = \mathcal{E}_{\mathrm{st}}(\pi|z|^2) \leq \mathcal{E}_{\max}$. By (ii), $R_{\mathrm{st}}(x) \leq \mathcal{E}_{\max}$, so $x \geq x_-(\mathcal{E}_{\max}) > 0$ (Lemma 9.5(i)); and $V_{\mathrm{st}}(T) \leq \mathcal{E}_{\max}$, i.e. $2\sqrt{T(1-T)} \geq (1 + \mathcal{E}_{\max}/b_{\mathrm{st}})^{-1}$, equivalently $(T - \frac{1}{2})^2 = \frac{1}{4} - T(1-T) \leq \frac{1}{4}[1 - (1 + \mathcal{E}_{\max}/b_{\mathrm{st}})^{-2}]$. □

9.3. **Enlarged capacities.** The higher-dimensional layer maps of [Son26] and the four-dimensional ones of §8.1 use the following capacities between ρ_i and the auxiliary capacities a_i.

Definition 9.8 (Enlarged capacities)**.** For $1 \leq i \leq k$ put

$$\delta_i = \frac{a_i - \rho_i}{2}, \qquad \bar{\rho}_i = a_i - \frac{3}{4}\delta_i, \qquad \hat{\rho}_i = a_i - \frac{\delta_i}{2}.$$

($\bar{\rho}_i$: closed enlarged source capacity; $\hat{\rho}_i$: open domain capacity.) Then $\rho_i < \frac{\rho_i + a_i}{2} = a_i - \delta_i < \bar{\rho}_i < \hat{\rho}_i < a_i$. Put $\varepsilon_i = (a_i - \hat{\rho}_i)/4 = \delta_i/8$; the i-th ball is placed by Ψ_{ε_i}. In the four-dimensional branch (§8.1) the same letters denote different values: there $\bar{\rho}_i = (\rho_i + a_i)/2 = a_i - \delta_i$, $\hat{\rho}_i = a_i - \frac{3}{4}\delta_i$, and the i-th ball is placed by Ψ_{δ_i} (Proposition 8.17). The two branches never occur together.

10. Return to the four-ball

We use the torus cover of §7 and unit representatives in $\mathbb{C}^3 \times \mathbb{C}$. Write $\mathcal{F}(Z) = Z_0^3 + Z_1^3 + Z_2^3$, fix $0 < h < d_n < 1/3$, and put $\mathcal{N}_h = \{(Z, U) : |Z| = 1,\ \mathcal{F}(Z) = 0,\ \pi|U|^2 \leq h\}$. The cover map $\mathsf{N}(u, v, w) = (Z(u, v), w)$ takes $\{p \leq h\}$ into $\mathcal{N}_h$. The three marked base points are the zeros of Z_0 on the torus; on the cover, the zero set is their $\mathbb{Z}^2$-periodic inverse image (Lemma 7.11). Pairs are gauge equivalent if $(Z', U') = (cZ, c^3U)$ with $|c| = 1$.

10.1. **Conventions and the ambient form.** For $Z \neq 0$ put $\eta_H = \mathrm{Im}(Z^* dZ)/(2\pi|Z|^2)$ and $\omega_{\mathrm{FS}} = d\eta_H = dd^c \log|Z|^2$. For unit Z put $H_Z = \{\xi : Z^*\xi = 0\}$. On H_Z one has $\omega_{\mathrm{FS}}(\xi_1, \xi_2) = \pi^{-1}\,\mathrm{Im}(\xi_1^*\xi_2)$, because the terms of $d\eta_H$ containing $d|Z|^2 = 2\,\mathrm{Re}(Z^* \cdot)$ vanish there. All forms below are gauge invariant and annihilate the gauge generator, so they descend to the quotient; we evaluate them on horizontal representatives.

Definition 10.1 (Ambient potential on representatives)**.** On $(\mathbb{C}^3 \setminus 0) \times \mathbb{C}$ set

$$q(Z,W) = \frac{|W|^2}{|Z|^6}, \quad \Phi_E^{(\delta)} = \log|Z|^2 + \delta \log(1+q), \quad \Omega_E^{(\delta)} = dd^c \Phi_E^{(\delta)}, \quad \Omega_E = \Omega_E^{(d_n)}.$$

On unit representatives put $p_W = d_n|W|^2/(1+|W|^2)$ and $X_T = \{(Z,W) : |Z| = 1,\ \mathcal{F}(Z) = TW\}$. Use E_3 as shorthand for these representative data with gauge $(Z,W) \sim (cZ, c^3W)$. For $W \neq 0$, write $\alpha = d\arg W/(2\pi) - 3\eta_H$.

A unit phase $e^{i\chi}$ shifts η_H by $d\chi/(2\pi)$ and the fibre angular form by $3d\chi/(2\pi)$. Hence α is gauge invariant and vanishes on the gauge generator.

Lemma 10.2 (d^c of the norm)**.** *On $\{W \neq 0\}$, $d^c \log\|W\|^2 = \alpha$. On E_3,*

$$d^c \log\left(1 + \|W\|^2\right) = \frac{\|W\|^2}{1+\|W\|^2}\,\alpha,$$

where the right side extends smoothly across $\{W = 0\}$.

Proof. Choose a holomorphic lift $\hat{Z}$ and a fibre coordinate υ with $[Z,W] = [\hat{Z}, \upsilon]$. Then $\|W\|^2 = |\upsilon|^2\|\hat{Z}\|^{-6}$ and $\arg W = \arg \upsilon$. Moreover $\eta_H = d^c \log\|\hat{Z}\|^2$, since $\mathrm{Re}(\hat{Z}^\dagger d\hat{Z}) = \frac{1}{2} d\|\hat{Z}\|^2$. Hence $d^c \log\|W\|^2 = d\arg\upsilon/2\pi - 3\, d^c \log\|\hat{Z}\|^2 = \alpha$.

The second identity follows from $d^c\phi(\mu) = \phi'(\mu)\mu\, d^c \log\mu$ with $\mu = \|W\|^2$. Smoothness follows from $\|W\|^2\alpha = \|\hat{Z}\|^{-6}[(\mathrm{Re}\,\upsilon\, d\,\mathrm{Im}\,\upsilon - \mathrm{Im}\,\upsilon\, d\,\mathrm{Re}\,\upsilon)/2\pi - 3|\upsilon|^2 d^c \log\|\hat{Z}\|^2]$. □

Lemma 10.3 (The horizontal Hermitian blocks)**.** *At $|Z| = 1$, for $(\xi, \varsigma) \in H_Z \times \mathbb{C}$,*

$$\Omega_E^{(\delta)}((\xi_1,\varsigma_1),(\xi_2,\varsigma_2)) = \frac{1}{\pi}\,\mathrm{Im}(A_\delta \xi_1^* \xi_2 + B_\delta \bar{\varsigma}_1 \varsigma_2), \quad A_\delta = 1 - \frac{3\delta|W|^2}{1+|W|^2}, \quad B_\delta = \frac{\delta}{(1+|W|^2)^2}. \tag{19}$$

For $\delta = d_n$, $A > 1 - 3d_n > 0$ and $B > 0$. The form is closed, and its restriction to every complex horizontal constraint kernel $\{(\xi,\varsigma) : d\mathcal{F}_Z \xi = T\varsigma\}$ is nondegenerate.

Proof. Since q is $(\lambda Z, \lambda^3 W)$-invariant for $\lambda \in \mathbb{C}^*$, it suffices to work on $H_Z \times \mathbb{C}$ at $|Z| = 1$. There $\partial q = \bar{W}\, dW$, and $\partial\bar{\partial}\log(1+q) = \frac{dW \wedge d\bar{W}}{(1+|W|^2)^2} - \frac{3|W|^2}{1+|W|^2}\sum_j dZ_j \wedge d\bar{Z}_j$, with no mixed terms. Applying $dd^c = \frac{i}{2\pi}\partial\bar{\partial}$ and adding ω_{FS} gives (19). For $\delta = d_n$ the Hermitian form $A\xi_1^*\xi_2 + B\bar{\varsigma}_1\varsigma_2$ is positive definite, so its imaginary part is nondegenerate on every complex subspace. Closedness is $d\, dd^c = 0$. □

10.2. **The central normal injection.**

Construction 10.4 (Weight-three injection)**.** For a unit pair (Z,U) with $p < d_n$ put

$$\iota[Z,U] = [Z,W], \qquad W = \frac{\sqrt{\pi}\,U}{\sqrt{d_n - p}}, \qquad p = \pi|U|^2.$$

Lemma 10.5 (The fibre injection)**.** *At each unit Z, the radial maps*

$$U \mapsto \frac{\sqrt{\pi}\,U}{\sqrt{d_n - \pi|U|^2}}, \qquad W \mapsto W\sqrt{\frac{d_n}{\pi(1+|W|^2)}}$$

are smooth inverses from $\{\pi|U|^2 < d_n\}$ to $\mathbb{C}$ and back. They commute with the weight-three gauge action and preserve Z. Moreover $|W|^2 = p/(d_n - p)$, $p_W = p$, and on $\mathcal{N}_h$ one has $|W|^2 \le h/(d_n - h)$.

Proof. Substitute each radial expression into the other and use positivity of its denominator. The norm and moment identities follow by squaring the positive radial factor. The last bound follows from the increasing function $p \mapsto p/(d_n - p)$. □

Proposition 10.6 (The central pullback on the cover)**.** *On $\{p < d_n\}$ in the torus cover, $(\iota \circ \mathsf{N})^*\Omega_E = \Omega_0$.*

Proof. By Lemma 10.2, $d^c\Phi_E = \eta_H + p_W\alpha$. Along ι we have $p_W = p$ and $\arg W = \arg U$ (Lemma 10.5). Along N we have $Z^*\eta_H = \eta$ and $U = w$. Hence $(\iota \circ \mathsf{N})^* d^c\Phi_E = (1-3p)\eta + \frac{1}{2}\operatorname{Im}(\bar{w}\,dw)$, which is smooth across $w = 0$ and has exterior derivative Ω_0 by Lemma 7.3. □

10.3. **Transport to $T = 1$.** The transport field is the lift of ∂_T that preserves the constraint and is Ω_E-orthogonal to $TX_T = \ker d(\mathcal{F}(Z) - TW)$.

Construction 10.7 (Transport field)**.** For $(Z, W, T) \in S^5 \times \mathbb{C} \times \mathbb{R}$ put $f = \mathcal{F}(Z)$, $\bar{Z}^2 = (\bar{Z}_0^2, \bar{Z}_1^2, \bar{Z}_2^2)$,

$$n = 3\bar{Z}^2 - 3\bar{f}Z, \qquad G = \|n\|^2, \qquad N_{\mathrm{den}} = \frac{G}{A} + \frac{T^2}{B},$$

with A, B as in Lemma 10.3. On the open set $\mathcal{O} = \{N_{\mathrm{den}} > 0\}$ define

$$\dot{Z} = \frac{W\,n}{A\,N_{\mathrm{den}}}, \qquad \dot{W} = -\frac{T\,W}{B\,N_{\mathrm{den}}}, \qquad \dot{T} = 1. \tag{20}$$

Since $A, B > 0$, $N_{\mathrm{den}} \geq 0$ with equality only where $T = 0$ and $n = 0$; so $\mathcal{O}$ is the complement of $\{T = 0,\ n = 0\}$, on which the field is smooth.

Lemma 10.8 (Properties of the transport field)**.** *(i) $Z^\dagger n = 0$, $n^\dagger\xi = d\mathcal{F}_Z(\xi)$ for $\xi \in H_Z$, and $G = 9\sum_j |Z_j|^4 - 9|f|^2$.*

(ii) The field is equivariant under $(Z, W) \mapsto (e^{i\chi}Z, e^{3i\chi}W)$ and satisfies $Z^\dagger\dot{Z} = 0$; hence it is tangent to $S^5 \times \mathbb{C} \times \mathbb{R}$. Its horizontal component is $V = (\dot{Z}, \dot{W}) \in H_Z \times \mathbb{C}$.

(iii) Along the field, $\frac{d}{dT}\mathcal{F}(Z) = \frac{WG}{AN_{\mathrm{den}}} = \frac{d}{dT}(TW)$, so $\mathcal{F}(Z) - TW$ is constant; $\|Z\|$ is constant; and $\frac{d}{dT}|W|^2 = -\frac{2T|W|^2}{BN_{\mathrm{den}}} \leq 0$ for $T \geq 0$.

(iv) At a point of $X_T \times \{T\}$ in $\mathcal{O}$, V is Ω_E-orthogonal to $\ker(d\mathcal{F}_Z, -T) \subset H_Z \times \mathbb{C}$.

Proof. (i) Since $Z^\dagger(3\bar{Z}^2) = 3\bar{f}$, n is the orthogonal projection of $3\bar{Z}^2$ to H_Z. This gives $Z^\dagger n = 0$ and $G = \|3\bar{Z}^2\|^2 - |3\bar{f}|^2$. For $\xi \in H_Z$, $d\mathcal{F}_Z(\xi) = (3\bar{Z}^2)^\dagger\xi = n^\dagger\xi$.

(ii) Under the gauge action $f \mapsto e^{3i\chi}f$ and $n \mapsto e^{-2i\chi}n$, while $G, A, B, N_{\mathrm{den}}$ are invariant. Tangency follows from (i). Finally, $H_Z \times \mathbb{C}$ is a complement to the kernel $\langle (iZ, 3iW) \rangle$ of $T(S^5 \times \mathbb{C}) \to TE_3$.

(iii) By (i), $\frac{d}{dT}\mathcal{F}(Z) = n^\dagger\dot{Z}$ and $\frac{d}{dT}(TW) = W + T\dot{W}$. The remaining claims follow from $\frac{d}{dT}\|Z\|^2 = 2\operatorname{Re}(Z^\dagger\dot{Z})$ and $\frac{d}{dT}|W|^2 = 2\operatorname{Re}(\bar{W}\dot{W})$.

(iv) By (i), $\ker(d\mathcal{F}_Z, -T) \cap (H_Z \times \mathbb{C})$ is the kernel of $\ell(\xi, \varsigma) = n^\dagger\xi - T\varsigma$. Let $\mathfrak{h} = A\,\xi_1^\dagger\xi_2 + B\,\bar{\varsigma}_1\varsigma_2$ and $\nu_\ell = (n/A, -T/B)$. Then $\mathfrak{h}(\nu_\ell, \kappa) = \ell(\kappa)$. Since $V = (W/N_{\mathrm{den}})\nu_\ell$ and $\mathfrak{h}$ is conjugate linear in its first slot, $\mathfrak{h}(V, \kappa) = 0$ on $\ker\ell$, so $\Omega_E(V, \kappa) = \frac{1}{\pi}\operatorname{Im}\mathfrak{h}(V, \kappa) = 0$. Also $\ell(V) = W$. Hence the field is the unique Ω_E-orthogonal lift of ∂_T preserving the constraint. □

Lemma 10.9 (Denominator)**.** *Let $\mathcal{C} = \{(Z, W, T) : \|Z\| = 1,\ \mathcal{F}(Z) = TW,\ 0 \leq T \leq 1\}$. On $\mathcal{C}$, $N_{\mathrm{den}} > 0$: for $T = 0$ one has $G \geq 3$ and $N_{\mathrm{den}} \geq 3$, and for $T > 0$, $N_{\mathrm{den}} \geq T^2/B > 0$. If moreover $|W| \leq W_{\max}$ and $T_0 = \min\big(1, 1/(\sqrt{6}\max(1, W_{\max}))\big)$, then $N_{\mathrm{den}} \geq \min\big(\frac{3}{2}, T_0^2/d_n\big)$.*

Proof. By Cauchy–Schwarz $\sum_j |Z_j|^4 \geq \frac{1}{3}$; since $0 < A \leq 1$, $N_{\mathrm{den}} \geq G$; and $|f| = T|W|$ on $\mathcal{C}$. If $T = 0$, $G = 9\sum_j |Z_j|^4 \geq 3$. If $T \leq T_0$, $9|f|^2 \leq 9T_0^2W_{\max}^2 \leq \frac{3}{2}$, so $G \geq \frac{3}{2}$. If $T \geq T_0$, $N_{\mathrm{den}} \geq T^2/B \geq T_0^2/d_n$, since $B \leq d_n$. □

Proposition 10.10 (Protected transport)**.** *Fix $w_* = h/(d_n - h)$ and let τ_T be the flow, from $T = 0$, of the cut field defined in the proof. It is a smooth injective map, commuting with the gauge action. For every unit pair $(Z, W) \in X_0$ with $|W|^2 \leq w_*$ the trajectory solves* (20) *on $[0, 1]$,*

stays in $X_T \cap \{|W|^2 \le w_*\}$, and τ_1 maps $X_0 \cap \{|W|^2 \le w_*\}$ into X_1 with $\tau_1^*(\Omega_E|_{X_1}) = \Omega_E|_{X_0}$ there.

Proof. Cut field. Let $T_0 = T_0(\sqrt{w_*})$ and $c_* = \min(\frac{3}{2}, T_0^2/d_n)$. For $a < b$ write $S_{a,b}(x) = S((x-a)/(b-a))$ and $P_{a,b,c,d}(x) = S_{a,b}(x)(1 - S_{c,d}(x))$, and put

$$\chi = S_{c_*/2,3c_*/4}(N_{\mathrm{den}})\, P_{1/2,3/4,5/4,3/2}(|Z|^2)\, S_{0,1}(w_* + 2 - |W|^2)\, P_{-1,-1/2,3/2,2}(T).$$

Off the unit sphere, n is replaced by $3\bar{Z}^2 - 3\bar{f}Z/|Z|^2$ in computing N_{den}. The field $\chi \cdot (\dot{Z}, \dot{W}, 1)$ is smooth, since its first factor vanishes where $N_{\mathrm{den}} \le c_*/2$. It is supported in $\{\frac{1}{2} \le |Z|^2 \le \frac{3}{2},\ |W|^2 \le w_* + 2,\ |T| \le 2\}$, so its flow is complete, with diffeomorphic time-T maps. The flow is gauge equivariant because χ is gauge invariant and the field is equivariant (Lemma 10.8(ii)).

Trajectories. On $\mathcal{C}_* = \{|Z| = 1,\ \mathcal{F}(Z) = TW,\ |W|^2 \le w_*,\ 0 \le T \le 1\}$ we have $\chi = 1$, by Lemma 10.9 with $W_{\max} = \sqrt{w_*}$. By Lemma 10.8(iii), uncut trajectories from $X_0 \cap \{|W|^2 \le w_*\}$ stay in $\mathcal{C}_*$, so by uniqueness they coincide with cut trajectories on $[0,1]$.

Pullback. Let $F(\sigma, T) = \tau_T(x(\sigma))$ for a smooth family $x(\sigma)$ of initial points. Since $\mathcal{F}(Z) - TW \equiv 0$ along F, each tangent vector (κ, t) satisfies $\ell(\kappa) = tW$, where $\ell = (d\mathcal{F}_Z, -T)$. Replacing κ by its horizontal representative κ_H modulo $G_{\mathrm{gauge}} = (iZ, 3iW)$ changes neither the constraint, since $\ell(G_{\mathrm{gauge}}) = 3i(\mathcal{F}(Z) - TW) = 0$, nor the pairing with Ω_E. Now $\partial_T F = (V, 1)$ with $\ell(V) = W$, so $\kappa_H - tV \in \ker \ell$. Lemma 10.8(iv) then gives $\Omega_E(V, \kappa) = 0$, i.e. $\iota_{\partial_T} F^*\Omega_E = 0$. Since $F^*\Omega_E$ is closed, Cartan's formula gives $\partial_T(F_T^*\Omega_E) = 0$, hence $F_1^*\Omega_E = F_0^*\Omega_E$. □

10.4. **Return to the Fubini–Study form.** On X_1 we have $W = \mathcal{F}(Z)$, so pr has inverse $\mathrm{gr}[Z] = [Z, \mathcal{F}(Z)]$. Let $u = |\mathcal{F}(Z)|^2/\|Z\|^6$, which is smooth on unit representatives. For $s \in \mathbb{R}$ put $\omega_s^{\mathrm{gr}} = \omega_{\mathrm{FS}} + s\, d_n\, dd^c \log(1+u) = \mathrm{gr}^*\Omega_E^{(sd_n)}$, where the equality uses $\mathrm{gr}^*\|W\|^2 = u$. Then $\omega_0^{\mathrm{gr}} = \omega_{\mathrm{FS}}$ and $\omega_1^{\mathrm{gr}} = \mathrm{gr}^*\Omega_E = (\mathrm{pr}^{-1})^*(\Omega_E|_{X_1})$. The return is the Moser flow of this path, which requires $\iota_{Y_s}\omega_s^{\mathrm{gr}} = -d_n\, d^c \log(1+u)$; this is solved in closed form because n is an eigenvector of M_s.

Construction 10.11 (Return field)**.** In unit representatives, with f, n, G as in Construction 10.7, put

$$A_s = 1 - 3s\, d_n \frac{u}{1+u}, \qquad B_s = \frac{s\, d_n}{(1+u)^2}, \qquad M_s = A_s I + B_s\, nn^\dagger \ \text{ on } H_Z,$$

$$Y_s = -\frac{d_n}{2}\, \frac{f\, n}{(1+u)(A_s + B_s G)} \in H_Z. \tag{21}$$

Lemma 10.12 (Graph forms)**.** *For $s \in [0,1]$ and $\xi_1, \xi_2 \in H_Z$, $\omega_s^{\mathrm{gr}}(\xi_1, \xi_2) = \frac{1}{\pi}\mathrm{Im}(\xi_1^\dagger M_s \xi_2)$. Moreover $u \le 1$, $A_s \ge 1 - \frac{3}{2}d_n > \frac{1}{2}$ and $B_s \ge 0$; hence $M_s \ge \frac{1}{2}I$ and every ω_s^{gr} is positive on H_Z.*

Proof. By Lemma 10.8(i), $d\,\mathrm{gr}(\xi)$ is represented by $(\xi, n^\dagger\xi)$ in the gauge of Lemma 10.3. Since $|W|^2 = u$ at $W = \mathcal{F}(Z)$, (19) gives the Hermitian form $A_s\xi_1^\dagger\xi_2 + B_s\overline{n^\dagger\xi_1}\, n^\dagger\xi_2 = \xi_1^\dagger M_s \xi_2$. For $\|Z\| = 1$, $|f| \le \max_j |Z_j| \sum_j |Z_j|^2 \le 1$. Hence $u \le 1$, so $u/(1+u) \le \frac{1}{2}$ and $A_s \ge 1 - \frac{3}{2}d_n > \frac{1}{2}$, because $d_n < \frac{1}{3}$. □

Lemma 10.13 (d^c along the graph)**.** *For $\xi \in H_Z$ (unit representative), $d^c \log(1+u)(\xi) = \dfrac{\mathrm{Im}\,(\bar{f}\, n^\dagger\xi)}{2\pi(1+u)}$.*

Proof. Where $f \ne 0$, as in Lemma 10.2, $d^c \log(1+u) = \frac{u}{1+u}\big(\frac{d\arg\mathcal{F}}{2\pi} - 3\eta_H\big)$. On H_Z we have $\eta_H(\xi) = 0$ and $d\arg\mathcal{F}(\xi) = \mathrm{Im}(n^\dagger\xi/f)$, and $u/f = \bar{f}$. Both sides are smooth, so the identity extends to $\{f = 0\}$. □

Lemma 10.14 (The return field)**.** *Y_s is smooth in (s, Z) for s in a neighbourhood of $[0,1]$, gauge equivariant and tangent to S^5, hence descends to a time-dependent smooth vector field on unit representatives, and*

$$\iota_{Y_s}\omega_s^{\mathrm{gr}} = -d_n\, d^c \log(1+u) \qquad (s \in [0,1]).$$

Proof. Under the gauge action $f \mapsto e^{3i\chi}f$ and $n \mapsto e^{-2i\chi}n$, so $Y_s \mapsto e^{i\chi}Y_s$; also $Z^\dagger Y_s = 0$. The denominator is positive for $-\frac{1}{9d_n} < s < \frac{2}{3d_n}$, an interval containing $[0,1]$: for $s \geq 0$, $A_s + B_s G \geq A_s \geq 1 - \frac{3}{2}s\, d_n$; for $s < 0$, $A_s \geq 1$ and $B_s G \geq -9|s|d_n$, because $0 \leq G \leq 9$. Since $M_s n = (A_s + B_s G)n$, we get $M_s Y_s = -\frac{d_n}{2}\frac{fn}{1+u}$. Lemmas 10.12 and 10.13 then give, for $\xi \in H_Z$, $\omega_s^{\mathrm{gr}}(Y_s, \xi) = \frac{1}{\pi}\operatorname{Im}((M_s Y_s)^\dagger \xi) = -d_n\, d^c \log(1+u)(\xi)$. □

Proposition 10.15 (Unit-representative return)**.** *Let ψ_s be the flow from $s = 0$ of the cut field of the proof. It is defined for all s, preserves $|Z| = 1$, commutes with the gauge action, and its time-s maps are diffeomorphisms. On the unit sphere it is the flow of Y_s for $s \in [0,1]$, and $\psi_s^*\omega_s^{\mathrm{gr}} = \omega_{\mathrm{FS}}$. Consequently $(\psi_1^{-1})^*\omega_{\mathrm{FS}} = \omega_1^{\mathrm{gr}}$, and $(\psi_1^{-1} \circ \mathrm{pr})^*\omega_{\mathrm{FS}} = \Omega_E|_{X_1}$.*

Proof. Put $m = 1 - \frac{3}{2}d_n > 0$ and $D_s = A_s + B_s G$, with n extended off the sphere as in Proposition 10.10. The cut field is

$$S_{m/2,3m/4}(D_s)\, P_{1/2,3/4,5/4,3/2}(|Z|^2)\, P_{-1,-1/2,3/2,2}(s)\, Y_s.$$

It is smooth, since the first factor vanishes where $D_s \leq m/2$, and compactly supported, so its flow is complete with diffeomorphic time maps. The flow is gauge equivariant by Lemma 10.14, and it preserves the sphere because $2\operatorname{Re}(Z^\dagger Y_s) = 0$. For $|Z| = 1$ and $s \in [0,1]$, Lemma 10.12 gives $D_s \geq m$, so the cut field equals Y_s there. Since $\partial_s \omega_s^{\mathrm{gr}} = d(d_n\, d^c \log(1+u))$, Cartan's formula and Lemma 10.14 give $\frac{d}{ds}\psi_s^*\omega_s^{\mathrm{gr}} = 0$. Finally, $\mathrm{gr} \circ \mathrm{pr} = \mathrm{id}$ on X_1 gives $\mathrm{pr}^*\omega_1^{\mathrm{gr}} = \Omega_E|_{X_1}$. □

10.5. **Equivariant injectivity and the ball chart.** From now on, $\mathcal{T}_{\mathrm{u}}$ uses the protected cutoff $w_* = h/(d_n - h)$ and is considered on $\mathcal{N}_h$. Let $\mathsf{r}Z = (\zeta Z_0, Z_1, Z_2)$ and $\mathsf{r}(Z, W) = (\mathsf{r}Z, W)$; r preserves the norm, the Fermat equation and the source moment, and commutes with the unit gauge action.

Lemma 10.16 (Fixed representatives modulo gauge)**.** *For a nonzero vector, $\mathsf{r}Z$ is a scalar multiple of Z exactly when $Z_0 = 0$ or $Z_1 = Z_2 = 0$. For a central unit pair (Z, U), the equation $\mathsf{r}(Z, U) = (cZ, c^3U)$ is possible exactly when $Z_0 = 0$.*

Proof. A nonzero last coordinate forces $c = 1$ and then $Z_0 = 0$. Otherwise both last coordinates vanish, which $\mathcal{F}(Z) = 0$ excludes at a central unit vector. If $Z_0 = 0$, the pair is fixed with $c = 1$. □

Lemma 10.17 (Equivariance)**.** *r preserves $\mathcal{F}$, ω_{FS}, $\|W\|$, Ω_E, u, each X_T and $\{(Z,U) : |Z| = 1, \mathcal{F}(Z) = 0, p < d_n\}$, and it maps the fields* (20) *and* (21) *to themselves. Consequently ι, τ_1,* pr*, ψ_1^{-1} and $\mathcal{T}_{\mathrm{u}}$ commute with r.*

Proof. r is unitary and $\mathcal{F} \circ \mathsf{r} = \mathcal{F}$, so it preserves all the listed quantities and also p, A, B, A_s, B_s. Since $\bar{\zeta}^2 = \zeta$, we have $n(\mathsf{r}Z) = \mathsf{r}\, n(Z)$. Hence G and N_{den} are invariant, both fields are r-equivariant, and by uniqueness so are their flows. ι commutes with r because p is invariant, and pr does because the lift of r to E_3 covers r. □

Lemma 10.18 (Injectivity modulo gauge)**.** *For $x, x' \in \mathcal{N}_h$ and $|c| = 1$, the equality $\mathcal{T}_{\mathrm{u}}(x) = c\mathcal{T}_{\mathrm{u}}(x')$ implies $x = (cZ', c^3U')$.*

Proof. Each factor of $\mathcal{T}_{\mathrm{u}} = \psi_1^{-1} \circ \mathrm{pr} \circ \tau_1 \circ \iota$ is injective on representatives: ι by the explicit inverse of Lemma 10.5, τ_1 and ψ_1^{-1} as time maps of complete flows, and pr on X_1, where $W = \mathcal{F}(Z)$. Each factor also commutes with $(Z, W) \mapsto (cZ, c^3W)$. Hence $c\,\mathcal{T}_{\mathrm{u}}(x') = \mathcal{T}_{\mathrm{u}}(cZ', c^3U')$, and injectivity concludes. □

Proposition 10.19 (Avoidance and compact margin)**.** *If $x = (Z, U) \in \mathcal{N}_h$ and $Z_0 \neq 0$, then $(\mathcal{T}_{\mathrm{u}}(x))_0 \neq 0$ and its last two coordinates do not vanish together. On every nonempty compact family of these inputs, $m_{\mathcal{K}} = \min_{x\in\mathcal{K}} |(\mathcal{T}_{\mathrm{u}}(x))_0|^2 > 0$. For $\mathcal{K} = \varnothing$ set $m_{\mathcal{K}} = 1$.*

Proof. Let $Y = \mathcal{T}_{\mathrm{u}}(x)$. If $Y_0 = 0$, then $\mathsf{r}Y = Y$, and Lemmas 10.17 and 10.18 (with $c = 1$) give $\mathsf{r}x = x$. If $Y_1 = Y_2 = 0$, then $\mathsf{r}Y = \zeta Y$, and the same lemmas give $\mathsf{r}x = (\zeta Z, \zeta^3 U)$. In both cases Lemma 10.16 gives $Z_0 = 0$, a contradiction. The compact margin holds because $x \mapsto |(\mathcal{T}_{\mathrm{u}}(x))_0|^2$ is continuous and positive on $\mathcal{K}$. □

Lemma 10.20 (The ball chart on representatives)**.** *On $S^5_* = \{Z \in \mathbb{C}^3 : |Z| = 1,\ Z_0 \neq 0\}$ put $b_{\mathrm{ball}}(Z) = \frac{\bar{Z}_0}{|Z_0|}\frac{(Z_1,Z_2)}{\sqrt{\pi}}$. This map is gauge invariant, $\pi|b_{\mathrm{ball}}(Z)|^2 = 1 - |Z_0|^2 < 1$, and in the affine coordinates $z = (Z_1/Z_0, Z_2/Z_0)$ it is the chart $\mathrm{ch}(z) = z/\sqrt{\pi(1+|z|^2)}$ of Lemma 2.2(b). Moreover $b^*_{\mathrm{ball}}\omega_0 = \frac{1}{\pi}\sum_{j=0}^{2} dx_j \wedge dy_j\big|_{S^5_*} = \omega_{\mathrm{FS}}|_{S^5_*}$, where $Z_j = x_j + iy_j$. Hence $(b_{\mathrm{ball}} \circ \varphi)^*\omega_0 = \varphi^*\omega_{\mathrm{FS}}$ for every smooth map φ of a manifold into S^5_*.*

Proof. The unit phases cancel, and substituting $Z = e^{i\chi}(1, z)/\sqrt{1+|z|^2}$ gives the affine formula and the norm identity. Write $e = \bar{Z}_0/|Z_0| = e^{i\phi}$, $\phi = -\arg Z_0$, and $y_j = eZ_j/\sqrt{\pi}$ $(j = 1, 2)$. Then $\bar{y}_j dy_j = \frac{1}{\pi}\big(|Z_j|^2\,\bar{e}\,de + \bar{Z}_j dZ_j\big)$ with $\bar{e}\,de = i\,d\phi$, so the Liouville form $\lambda_0 = \frac{1}{2}\sum_{j=1}^{2}\mathrm{Im}(\bar{y}_j dy_j)$ pulls back to $\frac{1}{2\pi}\big((1-|Z_0|^2)d\phi + \sum_{j=1}^{2}\mathrm{Im}(\bar{Z}_j dZ_j)\big)$ on S^5_*. Since $\mathrm{Im}(\bar{Z}_0 dZ_0) = |Z_0|^2 d\arg Z_0 = -|Z_0|^2 d\phi$, this equals $\frac{1}{2\pi}\big(d\phi + \mathrm{Im}(Z^\dagger dZ)\big)$, whose differential is $\frac{1}{2\pi}\sum_{j=0}^{2} d(x_j dy_j - y_j dx_j) = \frac{1}{\pi}\sum_j dx_j \wedge dy_j$. As $d\lambda_0 = \omega_0$, this proves the first identity; the second holds because $\frac{1}{2\pi}\mathrm{Im}(Z^\dagger dZ) = \eta_H$ on the sphere and $d\eta_H = \omega_{\mathrm{FS}}$. □

For $y \in B^\circ(1)$, $(\sqrt{1-\pi|y|^2}, \sqrt{\pi}\,y_1, \sqrt{\pi}\,y_2)$ is the unique gauge representative with positive first coordinate of every unit vector with nonzero first coordinate and ball-chart image y.

10.6. **The transfer theorem.**

Theorem 10.21 (Transfer on the cover)**.** *Let $w_* = h/(d_n - h)$ and $\mathcal{T}_{\mathrm{u}} = \psi_1^{-1} \circ \mathrm{pr} \circ \tau_1 \circ \iota$ on $\mathcal{N}_h$. The cover map*

$$\mathcal{T} = b_{\mathrm{ball}} \circ \mathcal{T}_{\mathrm{u}} \circ \mathsf{N} \quad \text{on} \quad \{p < h,\ (u, v) \notin \mathcal{Z}\}$$

is smooth, invariant under the deck maps, takes values in $B^\circ(1)$ and satisfies $\mathcal{T}^*\omega_0 = \Omega_0$. Two points have the same image if and only if they are deck equivalent. For every compact $\mathcal{K}$ in the domain, $\mathcal{T}(\mathcal{K}) \subset B(1 - m_{\mathcal{K}})$ with $m_{\mathcal{K}} > 0$ as in Proposition 10.19. In particular the restriction of $\mathcal{T}$ to an open set on which it is injective is a symplectic embedding, and $\mathcal{T}$ carries pairwise disjoint subsets of its domain, no two points of whose union are deck equivalent, to pairwise disjoint sets.

Proof. On $\{p < h\}$, N takes values in $\mathcal{N}_h$ with $p < d_n$, and $\tau_1 \circ \iota$ lands in $X_1 \cap \{|W|^2 \leq w_*\}$ (Lemma 10.5, Proposition 10.10). Propositions 10.6, 10.10 and 10.15 give $(\mathcal{T}_{\mathrm{u}} \circ \mathsf{N})^*\omega_{\mathrm{FS}} = \Omega_0$. For $(u, v) \notin \mathcal{Z}$, $\mathcal{T}_{\mathrm{u}}(\mathsf{N}(x)) \in S^5_*$ (Proposition 10.19), so Lemma 10.20 gives $\mathcal{T}^*\omega_0 = \Omega_0$ and $\pi|\mathcal{T}|^2 = 1 - |(\mathcal{T}_{\mathrm{u}}\mathsf{N})_0|^2 < 1$, whence the margin $m_{\mathcal{K}}$ on $\mathcal{K}$. Deck maps change $\mathsf{N}(x)$ to a gauge-equivalent pair (Construction 7.8(b)), $\mathcal{T}_{\mathrm{u}}$ is gauge equivariant and b_{ball} gauge invariant, so $\mathcal{T}$ is deck invariant. Conversely, if $\mathcal{T}(x) = \mathcal{T}(x')$, the ball chart determines a unit vector with nonzero first coordinate up to gauge, so Lemma 10.18 gives $\mathsf{N}(x) \sim \mathsf{N}(x')$; hence $[Z(u, v)] = [Z(u', v')]$ in $\mathbb{CP}^2$, and Lemma 7.5 gives $(u, v) - (u', v') \in \mathbb{Z}^2$. For the corresponding deck map T, the unit vectors $Z(u, v)$ agree, so the gauge factor is one and $x = Tx'$. Injective restrictions are injective symplectic immersions in equal dimension, hence open symplectic embeddings; the last assertion follows from the characterisation of equal images. □

All objects are explicit; ψ_1^{-1} is obtained by integrating (21) backwards from $s = 1$, and the transport by (20).

10.7. **The packing in dimension four.** The transfer depends on the data only through h.

Construction 10.22 (Data of the four-dimensional packing)**.** Let $(k, \rho_i, a_i, h, d_n, g, K, q)$ be valid cubic data (Definition 8.1) with $d_n = \frac{1}{2}(h + \frac{1}{3})$. Let $f_i = f_i^{\mathrm{cap}}$ be the maps of Remark 8.18, built by the handle construction of §8.1, put $\bar\rho_i = (\rho_i + a_i)/2$ and $\hat\rho_i = \bar\rho_i + (a_i - \bar\rho_i)/4$, and let the open domain be $B^\circ(\hat\rho_i)$. Let $\rho_i^+ < \rho_i^{\mathrm{dom}}$ be the recorded capacities of Remark 10.24. Put $\mathcal{K} = \bigcup_i f_i(B(\rho_i^+))$ and $m = \frac{1}{2}\min_{z\in\mathcal{K}}(1 - \pi|z|^2)$.

The equal case. Let $k \geq 10$, $\rho_1 = \cdots = \rho_k = \rho$ and $\frac{3}{4} < \rho\sqrt{k} < 1$. Put $t_0 = \frac{1}{2}(1 + \rho\sqrt{k})$, $a_i = a = t_0/\sqrt{k}$, $h = t_0/3$, $g = 3(1-t_0)$, and let (K, q) be the comparison datum of Proposition M4.4, with gap $\frac{31}{32}g$. These are valid cubic data (Remark 10.25). For $k = 9$ the same formulas would give $a = t_0/3 = h$, violating $a < h$; valid cubic data for $k = 9$ and for unequal capacities are constructed in the companion paper [Son26] from a scalar comparison datum.

Theorem 10.23 (Packing in dimension four)**.** *Let $(k, \rho_i, a_i, h, d_n, g, K, q)$ be valid cubic data with $d_n = \frac{1}{2}(h + \frac{1}{3})$. Then there are explicit numbers $\rho_i < \rho_i^+ < \rho_i^{\mathrm{dom}}$ and smooth maps $f_i\colon B^\circ(\rho_i^{\mathrm{dom}}) \to \mathbb{C}^2$ with symplectic derivative, injective on $B(\rho_i^+)$, such that the sets $f_i(B(\rho_i^+))$ are pairwise disjoint and contained in $B(1 - \mu_{\mathrm{ball}})$ for some $\mu_{\mathrm{ball}} \in (0, 1)$. More precisely, for the data of Construction 10.22:*

- *(i) each f_i is a smooth symplectic embedding of $B^\circ(\hat\rho_i)$ into $(B^\circ(1), \omega_0)$, and $\rho_i < \rho_i^+ < \rho_i^{\mathrm{dom}} \leq \hat\rho_i$;*
- *(ii) the sets $f_i(B^\circ(\hat\rho_i))$, $1 \leq i \leq k$, are pairwise disjoint;*
- *(iii) $m > 0$ and $f_i(B(\rho_i^+)) \subset B(1 - m)$ for every i;*
- *(iv) the comparison isotopy q, the Moser map Φ, the handle map $\mathcal{D}$ and the transfer $\mathcal{T}$ are the same for all i; only the layer map E_i depends on i.*

Proof. The data are valid cubic data by hypothesis (in the equal case by Proposition M4.4 and Remark 10.25), and $h < d_n < \frac{1}{3}$. Here $f_i = \mathcal{T} \circ f_i^N$ with f_i^N of Proposition 8.17 (Remark 8.18).

(i) By Proposition 8.17(a), f_i^N is smooth on $B^\circ(\hat\rho_i)$ into $\{p < h\} \setminus \widetilde{\mathcal{F}}$, the domain of $\mathcal{T}$, with $(f_i^N)^*\Omega_0 = \omega_0$; Theorem 10.21 gives f_i smooth into $B^\circ(1)$ with $f_i^*\omega_0 = \omega_0$. If $f_i(X) = f_i(X')$, Theorem 10.21 gives $f_i^N(X) = Tf_i^N(X')$ for a deck map T, and Proposition 8.17(c) gives $X = X'$; so f_i is an open embedding. The capacity inequalities are in Remark 10.24. (ii) If $f_i(X) = f_j(X')$, the same argument gives $i = j$, $X = X'$. (iii) $\mathcal{K}$ is compact in $B^\circ(1)$, so $1 - \pi|z|^2 \geq 2m > m > 0$ on $\mathcal{K}$. (iv) q, $\mathcal{D}$, Φ, $\chi_{\mathcal{A}}$ and $\mathcal{T}$ do not depend on i.

The first statement follows with $\mu_{\mathrm{ball}} = m < 1$, restricting f_i to $B^\circ(\rho_i^{\mathrm{dom}})$. □

Remark 10.24 (Recorded capacities). Since the proof embeds the open balls $B^\circ(\hat\rho_i)$ with pairwise disjoint images, Theorem 10.23 holds for every choice of $\rho_i < \rho_i^+ < \rho_i^{\mathrm{dom}} \leq \hat\rho_i$. The choice enters only the radii of the common isotopy (Definition M7.4) and hence the Hamiltonian. In the equal case the construction records $(\rho_i^+, \rho_i^{\mathrm{dom}}) = (\rho\varrho_3^2, \rho\varrho_4^2)$, where $\varrho_4 = \sqrt{\bar\rho/\rho}$ and $\varrho_3 = (1 + \varrho_4)/2$; then $1 < \varrho_3 < \varrho_4$ gives $\rho < \rho\varrho_3^2 < \rho\varrho_4^2 = \bar\rho < \hat\rho$. For target capacity C and given capacity $c = C\rho$ the physical map $u \mapsto \sqrt{C}\, f_i(u/\sqrt{C})$ has the capacities $C\rho_i^+$ and $C\rho_i^{\mathrm{dom}}$, that is, $c\varrho_3^2$ and $c\varrho_4^2$. For the valid cubic data of the companion paper [Son26] the construction records $(\rho_i^+, \rho_i^{\mathrm{dom}}) = (\bar\rho_i, \hat\rho_i)$.

Remark 10.25 (The equal-capacity branch). In the equal case of Construction 10.22 let $\eta_\star = 1 - \rho\sqrt{k} \in (0, \frac{1}{4})$ be the volume defect. Then

$$t_0 = 1 - \frac{\eta_\star}{2}, \quad a = \frac{t_0}{\sqrt{k}}, \quad h = \frac{t_0}{3}, \quad d_n = \frac{1}{3} - \frac{\eta_\star}{12}, \qquad a - \rho = \frac{\eta_\star}{2\sqrt{k}}, \quad d_n - h = \frac{\eta_\star}{12}, \quad 1 - 3d_n = \frac{\eta_\star}{4},$$

the comparison function is $K(p) = (3\sqrt{k} - 9)p$, and the comparison gap of Proposition M4.4 is $\frac{31}{32}g$ with $g = 3(1 - t_0) = \frac{3\eta_\star}{2}$. With the isotopy of Theorem 5.34 and the handle construction of

§8.1, the maps of this branch are

$$f_i = b_{\mathrm{ball}} \circ \psi_1^{-1} \circ \mathrm{pr} \circ \tau_1 \circ \iota \circ \mathsf{N} \circ \Phi^{-1} \circ \chi_{\mathcal{A}} \circ \mathcal{D}^{-1} \circ E_i \qquad (1 \le i \le k).$$

For the input of Corollary C, $\rho = \sqrt{(1-\varepsilon/2)/k}$ and $\eta_\star = 1 - \sqrt{1-\varepsilon/2} \in (\varepsilon/4, \varepsilon/2)$; the equal case applies exactly when $\rho\sqrt{k} > 3/4$, i.e. $\varepsilon < \frac{7}{8}$. For unit balls (capacity π) the target capacity is $C = \pi/\rho$, and the physical rescaling of Definition M7.1 makes the target the open Euclidean ball of radius $R = \rho^{-1/2}$, so $k/R^4 = k\rho^2 = (1-\eta_\star)^2$. For the data of Theorem A, $R^4 = k/(1-\varepsilon/4)$, so $\rho\sqrt{k} = \sqrt{1-\varepsilon/4} > \frac{3}{4}$ and $\eta_\star = 1 - \sqrt{1-\varepsilon/4} \in (\varepsilon/8, \varepsilon/4)$; the equal case applies for every $\varepsilon \in (0,1)$ when $k \ge 10$. The strip constants are those of Remark 8.18 with $\delta = (a-\rho)/2$.

The defect $1 - \rho\sqrt{k} = \eta_\star$ splits as $(1-t_0) + \sqrt{k}\,(a-\rho) = \frac{\eta_\star}{2} + \frac{\eta_\star}{2}$. The fibre part $1 - t_0 = 1 - 3h$ pays for $h < \frac{1}{3}$, for the comparison gap g and for the margins $d_n - h$ and $1 - 3d_n$ of the transfer; the strip part $\sqrt{k}\,(a-\rho)$ pays for the enlargement from ρ to $\bar\rho$ and for the excess of the strip map. The transfer, the ball chart and the dilation are symplectic embeddings and lose no volume beyond these margins.

11. The normalised embeddings of the linear branch

This section fixes the normalised embeddings used in §M7.1.

Construction 11.1 (The normalised embeddings)**.** Let $k \ge 10$ and $\rho > 0$ with $\frac{3}{4} < \rho\sqrt{k} < 1$, and put $\rho_1 = \cdots = \rho_k = \rho$ (the *linear branch*). The *normalised embeddings* $f_i^\natural$ are the maps of Theorem 10.23 for the valid cubic data of the equal case of Construction 10.22 (Remark 10.25); the enlarged capacity and the domain capacity are $(\rho_i^+, \rho_i^{\mathrm{dom}}) = (\rho\varrho_3^2, \rho\varrho_4^2)$, as in Remark 10.24.

The data satisfy $3\rho < 1$ and $k\rho^2 < 1$, so $(2, k, 1, \rho)$ is admissible.

12. The common isotopy and the pushforward of the source primitive

For k disjoint balls in $\mathbb{R}^{2n}$ we construct one smooth family $F_{i,t}$ of symplectic embeddings from the translations $s_i + u$ to the f_i, with disjoint images at all times (Proposition 12.14). The balls are translated to far centres along a route on which they never meet, deformed linearly to $A_i = Df_i(0)$, then to $u \mapsto \tau^{-1} f_i(\tau u)$, and finally dilated by a common parameter running from τ to 1. Disjointness comes from separation in the first three stages and from the common dilation in the tail. All ingredients are explicit functions of the input: the maxima defining L, the minimum defining δ_c, the positive square root and least admissible angle in the linear path, and the ranks in the route. The pushforward of a cut-off source primitive along F is smooth with compact support (Lemma 12.16), with closed forms on each stage (Lemma 12.15).

Conventions (§2.1). We write $\omega_0(\xi, \xi') = \langle \xi, J_0 \xi' \rangle$ with $J_0 = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$ in (x, y), and $\mathrm{Sp}(2n) = \{A : A^{\mathsf{T}} J_0 A = J_0\}$. The matrix $J_{\mathbb{C}} = -J_0$ is multiplication by i on $\mathbb{C}^n$. Matrices are written in $n \times n$ blocks with respect to (x, y), and linear maps are identified with their matrices. Vector norms are Euclidean and $\|\cdot\|$ is the operator norm. $\mathbb{B}(y, r)$ and $\mathbb{B}^\circ(y, r)$ denote the closed and open Euclidean balls, and $\mathbb{B}(r) = \mathbb{B}(0, r)$. S is the flat step of Definition 2.3. A matrix-valued function of $\sigma \in \mathbb{R}$ has *smooth entries* if every entry is C^∞ in σ. Symplectic matrices are invertible ($A^{-1} = J_0^{-1} A^{\mathsf{T}} J_0$), so a map with symplectic derivative is a local diffeomorphism.

12.1. The input.

Definition 12.1 (Isotopy input)**.** Let $n \ge 1$ and $k \ge 1$. An *isotopy input* consists of centres $s_1, \dots, s_k \in \mathbb{R}^{2n}$, radii $0 < r_{1,i} < r_{2,i} < r_{3,i} < r_{4,i} < \bar r_i$ for $1 \le i \le k$, a number $R_0 > 0$, and maps $f_i \colon \mathbb{B}^\circ(\bar r_i) \to \mathbb{R}^{2n}$ such that

(i) f_i is C^∞ and $Df_i(u) \in \mathrm{Sp}(2n)$ for every $u \in \mathbb{B}^\circ(\bar r_i)$;

(ii) f_i is injective on $\mathbb{B}(r_{4,i})$;
(iii) the sets $f_i(\mathbb{B}(r_{4,i}))$, $1 \le i \le k$, are pairwise disjoint and contained in $\mathbb{B}^\circ(R_0)$;
(iv) $|s_i - s_j| > r_{3,i} + r_{3,j}$ for $i \neq j$.

We write $y_i = f_i(0)$ and $A_i = Df_i(0) \in \mathrm{Sp}(2n)$.

By Lemma M7.5, an embedding datum with the radii of Definition M7.4 and $R_0 = \sqrt{C/\pi}$ is an isotopy input; we fix one. Stages are defined on $\mathbb{B}^\circ(r_{4,i})$, disjointness is kept on $\mathbb{B}(r_{3,i})$, and the Hamiltonian cutoff interpolates between $r_{1,i}$ and $r_{2,i}$.

12.2. **An explicit path in** $\mathrm{Sp}(2n)$**.** We factor A as a unitary matrix times a block upper triangular symplectic matrix and deform each factor; the unitary factor is deformed by a Cayley transform after a rotation by one of $n+1$ fixed angles, so no matrix logarithm is needed.

Definition 12.2 (The linear path)**.** For $n \times n$ matrices T, a, e and $\theta, \sigma \in \mathbb{R}$ let

$$\mathrm{Up}(T) = \begin{pmatrix} I & T \\ 0 & I \end{pmatrix}, \qquad \mathrm{Dg}(a,e) = \begin{pmatrix} a & 0 \\ 0 & e \end{pmatrix}, \qquad \mathrm{rot}(\theta) = \cos\theta\, I + \sin\theta\, J_{\mathbb{C}},$$

and, for a $2n \times 2n$ matrix X with $X^{\mathsf{T}} = -X$, $\mathrm{Cay}(X,\sigma) = (I+\sigma X)(I-\sigma X)^{-1}$. Let $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{Sp}(2n)$. Put

$$G_A = a^{\mathsf{T}}a + c^{\mathsf{T}}c, \qquad P_A = \mathrm{the}\{P : \ P \text{ symmetric positive semidefinite, } P^2 = G_A\},$$

$$U_A = \begin{pmatrix} aP_A^{-1} & -cP_A^{-1} \\ cP_A^{-1} & aP_A^{-1} \end{pmatrix}, \qquad T_A = P_A^{-1}\left(U_A^{\mathsf{T}}A\right)_{12},$$

where $(\cdot)_{12}$ is the upper right block. Let $\theta_m = \pi m/(n+1)$ for $m = 0, \dots, n$, let m_A be the least $m \in \{0, \dots, n\}$ with $\det\left(\mathrm{rot}(\theta_m)U_A + I\right) \neq 0$, $\theta_A = \theta_{m_A}$, $V_A = \mathrm{rot}(\theta_A)U_A$, and $X_A = (V_A - I)(V_A + I)^{-1}$. Finally, for $\sigma \in \mathbb{R}$,

$$P_{A,\sigma} = I + \sin^2\!\left(\tfrac{\pi\sigma}{2}\right)(P_A - I), \qquad M_{A,\sigma} = \mathrm{rot}(-\sigma\theta_A)\ \mathrm{Cay}(X_A,\sigma)\ \mathrm{Dg}\left(P_{A,\sigma}, P_{A,\sigma}^{-1}\right)\ \mathrm{Up}(\sigma T_A),$$

and $M_{i,\sigma} = M_{A_i,\sigma}$. By Lemma 12.3(a),(b), P_A and $V_A + I$ are invertible and m_A exists.

The index m_A is obtained by $n+1$ case distinctions on propositions about the input, an operation of level L1 (Definition M2.5).

A $2n \times 2n$ matrix Q is of *unitary type* if $Q^{\mathsf{T}}Q = I$ and $QJ_{\mathbb{C}} = J_{\mathbb{C}}Q$. A matrix commutes with $J_{\mathbb{C}}$ exactly when it has the form $\left(\begin{smallmatrix} \mathfrak{w}_1 & -\mathfrak{w}_2 \\ \mathfrak{w}_2 & \mathfrak{w}_1 \end{smallmatrix}\right)$, the realification of $\mathfrak{w} = \mathfrak{w}_1 + i\mathfrak{w}_2$. Realification is an injective ring homomorphism taking $\mathfrak{w}^*$ to the transpose, so matrices of unitary type are the realifications of unitary matrices; $\mathrm{rot}(\theta)$ realifies $e^{i\theta}I$. Conjugating by $\left(\begin{smallmatrix} I & 0 \\ iI & I \end{smallmatrix}\right)$ over $\mathbb{C}$ gives the block triangular matrix $\left(\begin{smallmatrix} \mathfrak{w}_1 + i\mathfrak{w}_2 & -\mathfrak{w}_2 \\ 0 & \mathfrak{w}_1 - i\mathfrak{w}_2 \end{smallmatrix}\right)$, so

$$\det \begin{pmatrix} \mathfrak{w}_1 & -\mathfrak{w}_2 \\ \mathfrak{w}_2 & \mathfrak{w}_1 \end{pmatrix} = \left|\det(\mathfrak{w}_1 + i\mathfrak{w}_2)\right|^2. \tag{22}$$

Lemma 12.3 (The linear path)**.** *Let* $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{Sp}(2n)$.

(a) G_A *is positive definite,* P_A *is well defined, symmetric and positive definite,* U_A *is of unitary type,* T_A *is symmetric, and* $A = U_A\,\mathrm{Dg}(P_A, P_A^{-1})\,\mathrm{Up}(T_A)$.
(b) m_A *exists.* X_A *is skew-symmetric and commutes with* $J_{\mathbb{C}}$*; for every* $\sigma \in \mathbb{R}$ *the matrix* $I - \sigma X_A$ *is invertible and* $\mathrm{Cay}(X_A,\sigma)$ *is of unitary type; and* $\mathrm{Cay}(X_A,1) = V_A$.
(c) $M_{A,\sigma} \in \mathrm{Sp}(2n)$ *for every* $\sigma \in \mathbb{R}$, $M_{A,0} = I$, $M_{A,1} = A$, *and* $M_{A,\sigma}$ *has smooth entries; hence* $(\sigma, u) \mapsto M_{A,\sigma}u$ *is* C^∞ *on* $\mathbb{R} \times \mathbb{R}^{2n}$.

Proof. Expanding $A^{\mathsf T}J_0A$ in blocks shows that $A\in\mathrm{Sp}(2n)$ if and only if

$$a^{\mathsf T}c=c^{\mathsf T}a,\qquad b^{\mathsf T}d=d^{\mathsf T}b,\qquad a^{\mathsf T}d-c^{\mathsf T}b=I. \tag{23}$$

Hence $\mathrm{Up}(T)$ is symplectic for symmetric T, and $\mathrm{Dg}(P,P^{-1})$ for symmetric invertible P. A matrix Q of unitary type is symplectic, since $Q^{\mathsf T}J_0Q=-Q^{\mathsf T}QJ_{\mathbb C}=J_0$, and products of such matrices are of unitary type.

(a) With $\iota=\left(\begin{smallmatrix}I\\0\end{smallmatrix}\right)$ we have $\xi^{\mathsf T}G_A\xi=|A\iota\xi|^2>0$ for $\xi\neq0$, so G_A is positive definite. Its positive semidefinite symmetric square root exists by orthogonal diagonalisation, is unique because any such root commutes with G_A and so acts on each eigenspace as the positive root of the eigenvalue, and is positive definite.

Put $\hat a=aP_A^{-1}$ and $\hat c=cP_A^{-1}$. Then $\hat a^{\mathsf T}\hat a+\hat c^{\mathsf T}\hat c=I$, and $\hat a^{\mathsf T}\hat c$ is symmetric by (23). Hence $U_A^{\mathsf T}U_A=I$, and since U_A commutes with $J_{\mathbb C}$, it is of unitary type. By the same identities, $U_A^{\mathsf T}A=\left(\begin{smallmatrix}P_A & B'\\ 0 & D'\end{smallmatrix}\right)$, which is symplectic; (23) applied to it gives $D'=P_A^{-1}$ and shows that $T_A=P_A^{-1}B'$ is symmetric. Thus $\mathrm{Dg}(P_A,P_A^{-1})\,\mathrm{Up}(T_A)=U_A^{\mathsf T}A$, the factorisation.

(b) Let $\mathfrak u$ be the unitary matrix realified by U_A. By (22), $\det(\mathrm{rot}(\theta)U_A+I)=|\det(e^{i\theta}\mathfrak u+I)|^2$, which vanishes exactly when $-e^{-i\theta}$ is an eigenvalue of $\mathfrak u$. The $n+1$ numbers $-e^{-i\theta_m}$ are pairwise distinct because $0\le\theta_m<\pi$, while $\mathfrak u$ has at most n eigenvalues; so m_A exists.

Let $V=V_A$, which is of unitary type with $V+I$ invertible. The matrices V, $V^{\mathsf T}=V^{-1}$ and $(V+I)^{-1}$ commute with each other and with $J_{\mathbb C}$, so X_A commutes with $J_{\mathbb C}$. Moreover $X_A^{\mathsf T}=(V^{-1}+I)^{-1}(V^{-1}-I)=-X_A$. For skew X we have $\langle\xi,(I-\sigma X)\xi\rangle=|\xi|^2$, so $I-\sigma X$ is invertible. Since $I\pm\sigma X_A$ commute, $\mathrm{Cay}(X_A,\sigma)$ is orthogonal and commutes with $J_{\mathbb C}$, hence is of unitary type. Finally, $I+X_A=2V(V+I)^{-1}$ and $I-X_A=2(V+I)^{-1}$ give $\mathrm{Cay}(X_A,1)=V$.

(c) $\mathrm{rot}(\theta)$ is of unitary type, and $\mathrm{rot}(\alpha)\,\mathrm{rot}(\beta)=\mathrm{rot}(\alpha+\beta)$ because $J_{\mathbb C}^2=-I$. With $\beta_\sigma=\sin^2(\pi\sigma/2)\in[0,1]$, the matrix $P_{A,\sigma}=(1-\beta_\sigma)I+\beta_\sigma P_A$ is symmetric with eigenvalues between 1 and those of P_A, so it is positive definite. By the block criterion, (a) and (b), all four factors of $M_{A,\sigma}$ are symplectic; at $\sigma=0$ each is I; at $\sigma=1$, $M_{A,1}=\mathrm{rot}(-\theta_A)V_A\,\mathrm{Dg}(P_A,P_A^{-1})\,\mathrm{Up}(T_A)=A$ by (a). The factors have smooth entries, using the adjugate formula for the inverses of $I-\sigma X_A$ and $P_{A,\sigma}$, whose determinants never vanish; products preserve smooth entries, and $(\sigma,u)\mapsto M_{A,\sigma}u$ is bilinear in the entries and u. □

12.3. **Parameters and the stage maps.**

Definition 12.4 (Parameters)**.** Put

$$L=1+\sum_{i=1}^{k}\Big(\max_{u\in\mathbb B(r_{3,i})}\|Df_i(u)\|+\max_{(\sigma,u)\in[0,1]\times\mathbb B(r_{3,i})}|M_{i,\sigma}u|\Big),\qquad r_{\mathrm{tot}}=1+\sum_{i=1}^{k}r_{3,i},$$

$\delta_c=\frac12\min_{i\neq j}|y_i-y_j|$ if $k\ge2$ and $\delta_c=1$ if $k=1$, and

$$\tau=\min\Big(\frac12,\ \frac{\delta_c}{2Lr_{\mathrm{tot}}}\Big),\qquad Y_i=\frac{y_i}{\tau}.$$

The maxima exist since Df_i and $(\sigma,u)\mapsto M_{i,\sigma}u$ (Lemma 12.3(c)) are continuous on the compact sets $\mathbb B(r_{3,i})\subset\mathbb B^\circ(\bar r_i)$ and $[0,1]\times\mathbb B(r_{3,i})$.

Lemma 12.5 (Parameters)**.** *(a) $L\ge1$ and $r_{\mathrm{tot}}\ge1$.*
(b) $\|Df_i(u)\|\le L$ and $|M_{i,\sigma}u|\le L$ for $\sigma\in[0,1]$ and $u\in\mathbb B(r_{3,i})$.
(c) $r_{3,i}+r_{3,j}<r_{\mathrm{tot}}$ for $i\neq j$.
(d) The y_i are pairwise distinct, $\delta_c>0$, and $\delta_c\le\frac12|y_i-y_j|$ for $i\neq j$.
(e) $0<\tau\le\frac12$, and $|Y_i-Y_j|\ge4Lr_{\mathrm{tot}}>r_{3,i}+r_{3,j}$ for $i\neq j$.

Proof. All summands of L and r_{tot} are nonnegative, which gives (a), (b) and $r_{\mathrm{tot}} \geq 1+r_{3,i}+r_{3,j}$, hence (c). (d): $0 \in \mathbb{B}(r_{4,i})$ and the sets $f_i(\mathbb{B}(r_{4,i}))$ are pairwise disjoint, so $y_i \neq y_j$. (e): $\tau \leq \delta_c/(2Lr_{\mathrm{tot}})$ gives $|Y_i - Y_j| \geq 2\delta_c/\tau \geq 4Lr_{\mathrm{tot}} > r_{\mathrm{tot}} > r_{3,i} + r_{3,j}$ by (a), (c). □

Definition 12.6 (The stage maps)**.** For $1 \leq i \leq k$ define:

(i) the *linear stage* $G^{\mathrm{lin}}_{i,\sigma}(u) = Y_i + M_{i,\sigma}u$ for $\sigma \in \mathbb{R}$, $u \in \mathbb{R}^{2n}$;

(ii) the *shape family* $\kappa_{i,\rho}(u) = \int_0^1 Df_i(\phi\rho u)\, u\, d\phi$ for $(\rho, u) \in \Omega_i = (-1,1) \times \mathbb{B}^\circ(r_{4,i})$, and the *shape stage* $G^{\mathrm{sh}}_{i,\sigma}(u) = Y_i + \kappa_{i,\tau\sigma}(u)$ for $\sigma \in (-2,2)$, $u \in \mathbb{B}^\circ(r_{4,i})$;

(iii) the *tail* $T_{i,v}(u) = v^{-1}f_i(vu)$ for $v > 0$ and $u \in \mathbb{R}^{2n}$ with $vu \in \mathbb{B}^\circ(\bar{r}_i)$.

The shape stage and $\kappa_{i,\rho}$ are defined since $|\tau\sigma| < 1$ and $|\phi\rho u| < r_{4,i} < \bar{r}_i$.

Lemma 12.7 (The linear stage)**.** *$(\sigma, u) \mapsto G^{\mathrm{lin}}_{i,\sigma}(u)$ is C^∞, $G^{\mathrm{lin}}_{i,0}(u) = Y_i + u$ and $G^{\mathrm{lin}}_{i,1}(u) = Y_i + A_iu$. For $\sigma \in [0,1]$, $G^{\mathrm{lin}}_{i,\sigma}$ is an affine symplectic isomorphism of $\mathbb{R}^{2n}$ which maps $\mathbb{B}(r_{3,i})$ into $\mathbb{B}(Y_i, L) \subset \mathbb{B}(Y_i, Lr_{\mathrm{tot}})$, and the balls $\mathbb{B}(Y_i, Lr_{\mathrm{tot}})$, $1 \leq i \leq k$, are pairwise disjoint.*

Proof. Smoothness and the endpoints are Lemma 12.3(c); $M_{i,\sigma}$ is symplectic, hence invertible; the inclusion follows from $|M_{i,\sigma}u| \leq L \leq Lr_{\mathrm{tot}}$ (Lemma 12.5(a),(b)); the balls are disjoint since their centres are at least $4Lr_{\mathrm{tot}}$ apart (Lemma 12.5(e)). □

12.4. **The shape stage and the tail.**

Lemma 12.8 (The shape stage)**.** *(a) $(\rho, u) \mapsto \kappa_{i,\rho}(u)$ is C^∞ on Ω_i.*

(b) $\kappa_{i,0}(u) = A_iu$, and $\rho\,\kappa_{i,\rho}(u) = f_i(\rho u) - y_i$; in particular $\kappa_{i,\rho}(u) = \rho^{-1}(f_i(\rho u) - y_i)$ for $\rho \neq 0$.

(c) $D_u\kappa_{i,\rho}(u) = Df_i(\rho u)$; in particular it is symplectic.

(d) $|\kappa_{i,\rho}(u)| \leq L|u|$ for $|\rho| < 1$ and $u \in \mathbb{B}(r_{3,i})$.

(e) $\partial_\rho\kappa_{i,\rho}(u) = \int_0^1 \phi\, D^2f_i(\phi\rho u)[u,u]\, d\phi$.

(f) For $\sigma \in [0,1]$, $G^{\mathrm{sh}}_{i,\sigma}$ is injective on $\mathbb{B}(r_{3,i})$ and maps it into $\mathbb{B}(Y_i, Lr_{\mathrm{tot}})$; $G^{\mathrm{sh}}_{i,0} = G^{\mathrm{lin}}_{i,1}$ and $G^{\mathrm{sh}}_{i,1} = T_{i,\tau}$ on $\mathbb{B}^\circ(r_{4,i})$.

Proof. (a), (e) The integrand $Df_i(\phi\rho u)u$ is C^∞ on the open set $\{|\phi\rho u| < \bar{r}_i\} \supset [0,1] \times \Omega_i$, with (ρ, u)-derivatives bounded on $[0,1] \times N$ for compact $N \subset \Omega_i$; so we may differentiate under the integral, and ∂_ρ of the integrand is $D^2f_i(\phi\rho u)[\phi u, u]$.

(b) At $\rho = 0$ the integrand is A_iu; in general $\frac{d}{d\phi}f_i(\phi\rho u) = \rho\, Df_i(\phi\rho u)u$, and the fundamental theorem of calculus gives $f_i(\rho u) - f_i(0) = \rho\kappa_{i,\rho}(u)$.

(c) $\kappa_{i,0} = A_i$ is linear; for $\rho \neq 0$ differentiate (b), and use Definition 12.1(i).

(d) $|\phi\rho u| \leq r_{3,i}$, where $\|Df_i\| \leq L$ (Lemma 12.5(b)).

(f) $G^{\mathrm{sh}}_{i,0} = Y_i + A_iu = G^{\mathrm{lin}}_{i,1}$ is injective. For $\sigma \in (0,1]$ and $\rho = \tau\sigma \in (0, \frac{1}{2}]$, (b) gives $G^{\mathrm{sh}}_{i,\sigma}(u) = Y_i + \rho^{-1}(f_i(\rho u) - y_i)$, injective on $\mathbb{B}(r_{3,i})$ as $\rho\mathbb{B}(r_{3,i}) \subset \mathbb{B}(r_{4,i})$; by (d), $|G^{\mathrm{sh}}_{i,\sigma}(u) - Y_i| \leq Lr_{3,i} \leq Lr_{\mathrm{tot}}$. At $\sigma = 1$, (b) and $Y_i = \tau^{-1}y_i$ give $G^{\mathrm{sh}}_{i,1} = T_{i,\tau}$. □

Lemma 12.9 (The tail stage)**.** *The map $(v, u) \mapsto T_{i,v}(u)$ is C^∞ on the open set $\{(v,u) : v > 0,\ |vu| < \bar{r}_i\}$, which contains $(0,1] \times \mathbb{B}^\circ(r_{4,i})$. For $v \in (0,1]$:*

(a) $D_uT_{i,v}(u) = Df_i(vu)$ is symplectic, and $\partial_vT_{i,v}(u) = v^{-2}\big(v\,Df_i(vu)u - f_i(vu)\big)$;

(b) $T_{i,v}$ is injective on $\mathbb{B}(r_{3,i})$, and the sets $T_{i,v}(\mathbb{B}(r_{3,i}))$, $1 \leq i \leq k$, are pairwise disjoint;

(c) $|T_{i,v}(u)| < R_0/v$ for $u \in \mathbb{B}(r_{3,i})$;

(d) $T_{i,1} = f_i$, and $T_{i,\tau}(u) = G^{\mathrm{sh}}_{i,1}(u) = Y_i + \kappa_{i,\tau}(u)$ for $u \in \mathbb{B}^\circ(r_{4,i})$.

Proof. Smoothness is clear since $|vu| < r_{4,i} < \bar{r}_i$ for $0 < v \leq 1$, $|u| < r_{4,i}$; (a) is the chain and product rules. (b) Injectivity follows from $v\mathbb{B}(r_{3,i}) \subset \mathbb{B}(r_{4,i})$ and Definition 12.1(ii); the sets

$T_{i,v}(\mathbb{B}(r_{3,i}))$ are images of the pairwise disjoint sets $f_i(v\mathbb{B}(r_{3,i})) \subset f_i(\mathbb{B}(r_{4,i}))$ (Definition 12.1(iii)) under the single injective map $y \mapsto v^{-1}y$. (c) $vu \in \mathbb{B}(r_{4,i})$, so $|f_i(vu)| < R_0$ by Definition 12.1(iii). (d) is Lemma 12.8(b),(f). □

12.5. **The translation route.** Let e_1, e_{n+1} be the unit vectors in the x_1- and y_1-directions. For $\gamma_1, \dots, \gamma_k \in \mathbb{R}$ let $\mathrm{rk}_\gamma(i) = \#\{j : \gamma_j < \gamma_i \text{ or } (\gamma_j = \gamma_i \text{ and } j < i)\} \in \{0, \dots, k-1\}$, the rank of i in the lexicographic order of (γ_j, j), defined by finitely many case distinctions (Definition M2.5).

Definition 12.10 (The route)**.** Let $\nu_i = \mathrm{rk}_\gamma(i)$ for $\gamma_j = \langle s_j, e_1\rangle$, and $\nu'_i = \mathrm{rk}_{\gamma'}(i)$ for $\gamma'_j = \langle Y_j, e_{n+1}\rangle$. Put

$$\Lambda_{\mathrm{rt}} = 1 + \sum_{j=1}^{k} 2\big(r_{3,j} + |s_j| + |Y_j|\big),$$

and, for $\boldsymbol{\sigma} = (\sigma_0, \dots, \sigma_4) \in \mathbb{R}^5$,

$$z_i(\boldsymbol{\sigma}) = s_i + \sigma_2(Y_i - s_i) + (\sigma_0 - \sigma_3)\Lambda_{\mathrm{rt}}\nu_i\, e_1 + (\sigma_1 - \sigma_4)\Lambda_{\mathrm{rt}}\nu'_i\, e_{n+1}.$$

A vector $\boldsymbol{\sigma}$ is *on the route* if there is $m \in \{0, \dots, 4\}$ with $\sigma_l = 1$ for $l < m$, $\sigma_l = 0$ for $l > m$, and $\sigma_m \in [0, 1]$ (*phase* m).

Lemma 12.11 (The route)**.** *(a) For $\gamma \in \mathbb{R}^k$ and $i \neq j$: $\mathrm{rk}_\gamma(i) \neq \mathrm{rk}_\gamma(j)$, so $|\mathrm{rk}_\gamma(i) - \mathrm{rk}_\gamma(j)| \geq 1$, and $(\gamma_i - \gamma_j)\big(\mathrm{rk}_\gamma(i) - \mathrm{rk}_\gamma(j)\big) \geq 0$.*

(b) $\boldsymbol{\sigma} \mapsto z_i(\boldsymbol{\sigma})$ is affine, $z_i(0, \dots, 0) = s_i$ and $z_i(1, \dots, 1) = Y_i$.

(c) If $\boldsymbol{\sigma}$ is on the route, then $|z_i(\boldsymbol{\sigma}) - z_j(\boldsymbol{\sigma})| > r_{3,i} + r_{3,j}$ for all $i \neq j$.

(d) If $\boldsymbol{\sigma} \in [0, 1]^5$, then $|z_i(\boldsymbol{\sigma})| \leq \max(|s_i|, |Y_i|) + 2(k-1)\Lambda_{\mathrm{rt}}$.

Proof. (a) $j \prec i$ iff $\gamma_j < \gamma_i$, or $\gamma_j = \gamma_i$ and $j < i$, is a strict total order with $\mathrm{rk}_\gamma(i) = \#\{j \prec i\}$. If $j \prec i$, every $l \prec j$ has $l \prec i$, and $j \prec i$, $j \not\prec j$; so $\mathrm{rk}_\gamma(j) < \mathrm{rk}_\gamma(i)$ and $\gamma_j \leq \gamma_i$.

(b) is immediate; at $(1, \dots, 1)$ the offsets cancel.

(c) For a unit vector e and $x\langle \xi, e\rangle \geq 0$,

$$|\xi + xe|^2 = |\xi|^2 + 2x\langle \xi, e\rangle + x^2 \geq |\xi|^2; \tag{24}$$

also $|\xi| \geq |\langle \xi, e\rangle|$. Fix $i \neq j$, put $d = r_{3,i} + r_{3,j}$, $\Delta s = s_i - s_j$, $\Delta Y = Y_i - Y_j$, $\Delta\nu = \nu_i - \nu_j$, $\Delta\nu' = \nu'_i - \nu'_j$, $\Sigma = |s_i| + |s_j| + |Y_i| + |Y_j|$. By Definition 12.1(iv) and Lemma 12.5(e), $|\Delta s|, |\Delta Y| > d$; by (a), $|\Delta\nu|, |\Delta\nu'| \geq 1$, $\langle \Delta s, e_1\rangle\Delta\nu \geq 0$, $\langle \Delta Y, e_{n+1}\rangle\Delta\nu' \geq 0$; and $\Lambda_{\mathrm{rt}} \geq 1 + 2d + 2\Sigma$, so $\Lambda_{\mathrm{rt}} - \Sigma > d$. Let $w = z_i(\boldsymbol{\sigma}) - z_j(\boldsymbol{\sigma})$ with $\boldsymbol{\sigma}$ in phase m. *Phase 0:* $w = \Delta s + \sigma_0\Lambda_{\mathrm{rt}}\Delta\nu\, e_1$, $\sigma_0 \geq 0$, so $|w| \geq |\Delta s| > d$ by (24). *Phase 1:* $\langle w, e_1\rangle = \langle \Delta s, e_1\rangle + \Lambda_{\mathrm{rt}}\Delta\nu$ (as $e_{n+1} \perp e_1$), two terms not of opposite signs, so $|w| \geq \Lambda_{\mathrm{rt}} > d$. *Phase 2:* $\langle w, e_1\rangle = (1 - \sigma_2)\langle \Delta s, e_1\rangle + \sigma_2\langle \Delta Y, e_1\rangle + \Lambda_{\mathrm{rt}}\Delta\nu$, the convex combination having absolute value $\leq \Sigma$, so $|w| \geq \Lambda_{\mathrm{rt}} - \Sigma > d$. *Phase 3:* $\langle w, e_{n+1}\rangle = \langle \Delta Y, e_{n+1}\rangle + \Lambda_{\mathrm{rt}}\Delta\nu'$, so $|w| \geq \Lambda_{\mathrm{rt}} > d$ as in phase 1. *Phase 4:* $w = \Delta Y + (1 - \sigma_4)\Lambda_{\mathrm{rt}}\Delta\nu'\, e_{n+1}$, $1 - \sigma_4 \geq 0$, so $|w| \geq |\Delta Y| > d$ by (24).

(d) $s_i + \sigma_2(Y_i - s_i)$ is a convex combination of s_i, Y_i; $|\sigma_0 - \sigma_3|, |\sigma_1 - \sigma_4| \leq 1$ and $\nu_i, \nu'_i \leq k-1$. □

12.6. **Concatenation and the common isotopy.** The stages (five route phases, linear, shape, tail) run in eight slots of length $\frac{1}{8}$. Slot m is driven by the flat clock $\sigma_m(t) = S(8t - m)$, $m = 0, \dots, 7$, which is 0 for $t \leq m/8$ and 1 for $t \geq (m+1)/8$. Write $\boldsymbol{\sigma}(t) = (\sigma_0(t), \dots, \sigma_4(t))$ and $v(t) = \tau + (1 - \tau)\sigma_7(t)$.

Construction 12.12 (The common isotopy)**.** Each bracket in (25) below vanishes while its clock is 0, and once its clock is 1 its first term cancels the subtracted term of the next bracket, so on slot m only the m-th stage moves. For $t \in \mathbb{R}$ and $u \in \mathbb{B}^\circ(r_{4,i})$ let

$$F_{i,t}(u) = z_i(\boldsymbol{\sigma}(t)) + u + \big(M_{i,\sigma_5(t)}u - u\big) + \big(\kappa_{i,\tau\sigma_6(t)}(u) - A_i u\big) + \big(T_{i,v(t)}(u) - T_{i,\tau}(u)\big). \tag{25}$$

The stage maps are $G_{i,m}(\sigma,u) = z_i(1,\dots,1,\sigma,0,\dots,0) + u$ (σ in position m) for $m \le 4$, $G_{i,5}(\sigma,u) = G^{\mathrm{lin}}_{i,\sigma}(u)$, $G_{i,6}(\sigma,u) = G^{\mathrm{sh}}_{i,\sigma}(u)$ and $G_{i,7}(\sigma,u) = T_{i,\tau+(1-\tau)\sigma}(u)$.

Every term of (25) is defined, since $\tau\sigma_6(t) \in [0,\frac12]$ and $v(t) \in [\tau,1]$.

Lemma 12.13 (Flat concatenation)**.** *(a) Each σ_m is C^∞ with values in $[0,1]$. If $t \in [m/8,(m+1)/8]$, then $\sigma_l(t) = 1$ and $\dot\sigma_l(t) = 0$ for $l < m$, $\sigma_l(t) = 0$ and $\dot\sigma_l(t) = 0$ for $l > m$, and $\boldsymbol\sigma(t)$ is on the route if $m \le 4$. All clocks and their derivatives vanish for $t \le 0$; all clocks equal 1 and their derivatives vanish for $t \ge 1$.*

(b) For $t \in [m/8,(m+1)/8]$ and $u \in \mathbb{B}^\circ(r_{4,i})$, $F_{i,t}(u) = G_{i,m}(\sigma_m(t),u)$. Explicitly: $F_{i,t}(u) = z_i(\boldsymbol\sigma(t)) + u$ if $m \le 4$; $Y_i + M_{i,\sigma_5(t)}u$ if $m = 5$; $Y_i + \kappa_{i,\tau\sigma_6(t)}(u)$ if $m = 6$; $T_{i,v(t)}(u)$ if $m = 7$. Moreover $G_{i,m}(1,\cdot) = G_{i,m+1}(0,\cdot)$ for $m < 7$.

(c) $(t,u) \mapsto F_{i,t}(u)$ is C^∞ on $\mathbb{R} \times \mathbb{B}^\circ(r_{4,i})$.

(d) $F_{i,t} = F_{i,0}$ for $t \le 0$, $F_{i,t} = F_{i,1}$ for $t \ge 1$, $F_{i,0}(u) = s_i + u$ and $F_{i,1} = f_i$ on $\mathbb{B}^\circ(r_{4,i})$.

(e) For $t \in [m/8,(m+1)/8]$,

$$\partial_t F_{i,t}(u) = \dot\sigma_m(t)\,\partial_\sigma G_{i,m}(\sigma_m(t),u),$$
$$D_u F_{i,t}(u) = D_u G_{i,m}(\sigma_m(t),u).$$

In particular $\partial_t F_{i,t} = 0$ for $t \notin (0,1)$.

Proof. (a) By Lemma 2.4, S is C^∞ and $[0,1]$-valued, with $S = 0$ on $(-\infty,0]$, $S = 1$ on $[1,\infty)$, and $S' = 0$ on both half-lines. On slot m, $8t - l \ge 1$ for $l < m$ and $8t - l \le 0$ for $l > m$. Also $8t - l \le 0$ for $t \le 0$, and $8t - l \ge 1$ for $t \ge 1$, $l \le 7$.

(b) Insert these clock values into (25). Use $M_{i,0} = I$ and $M_{i,1} = A_i$ (Lemma 12.3(c)), $\kappa_{i,0} = A_i$ (Lemma 12.8(b)) and $z_i(1,\dots,1) = Y_i$ (Lemma 12.11(b)). For $m \le 4$ the last three brackets vanish; for $m = 5$ the last two; for $m = 6$, $u + (A_iu - u) + (\kappa_{i,\tau\sigma_6}(u) - A_iu) = \kappa_{i,\tau\sigma_6}(u)$; for $m = 7$, $Y_i + \kappa_{i,\tau}(u) + T_{i,v(t)}(u) - T_{i,\tau}(u) = T_{i,v(t)}(u)$ by Lemma 12.9(d). The matching conditions are trivial for $m < 4$; for $m = 4$, $z_i(1,\dots,1) = Y_i = G^{\mathrm{lin}}_{i,0}(u) - u$; they are Lemma 12.8(f) for $m = 5$ and Lemma 12.9(d) for $m = 6$.

(c) Each term is a smooth map composed with $(t,u) \mapsto$ (clock values, u): z_i is affine; $(\sigma,u) \mapsto M_{i,\sigma}u$ is smooth (Lemma 12.3(c)); κ is smooth on $\Omega_i \ni (\tau\sigma_6(t),u)$ (Lemma 12.8(a)); and T is smooth near $(v(t),u)$ and (τ,u) (Lemma 12.9).

(d) The clocks are constant for $t \le 0$ and for $t \ge 1$. By (b), $F_{i,0}(u) = z_i(0) + u = s_i + u$, and $F_{i,1} = T_{i,1} = f_i$ (Lemma 12.9(d)).

(e) Chain rule and (a): only σ_m has nonzero derivative on the closed slot, and D_u is computed from (b) on the open ball. □

Proposition 12.14 (The common symplectic isotopy)**.** *For every isotopy input, the maps $F_{i,t}$ of Construction 12.12 satisfy:*

(a) $(t,u) \mapsto F_{i,t}(u)$ is C^∞ on $\mathbb{R} \times \mathbb{B}^\circ(r_{4,i})$; $F_{i,0}(u) = s_i + u$, $F_{i,1} = f_i$; $F_{i,t} = F_{i,0}$ for $t \le 0$ and $F_{i,t} = F_{i,1}$ for $t \ge 1$;

(b) $D_uF_{i,t}(u) \in \mathrm{Sp}(2n)$ for all $t \in \mathbb{R}$ and $u \in \mathbb{B}^\circ(r_{4,i})$;

(c) each $F_{i,t}$ is injective on $\mathbb{B}(r_{3,i})$, and for every t the sets $F_{i,t}(\mathbb{B}(r_{3,i}))$, $1 \le i \le k$, are pairwise disjoint;

(d) $F_{i,t}(\mathbb{B}(r_{3,i})) \subset \mathbb{B}(R_F)$ for all i and t, where $R_F = 2k\Lambda_{\mathrm{rt}} + Lr_{\mathrm{tot}} + R_0/\tau$.

Proof. (a) is Lemma 12.13(c),(d). Outside $[0,1]$, $F_{i,t}$ equals $F_{i,0}$ or $F_{i,1}$, so by Lemma 12.13(b),(e) it suffices to check (b)–(d) for $G_{i,m}(\sigma,\cdot)$ with $\sigma = \sigma_m(t) \in [0,1]$, and with $u \in \mathbb{B}(r_{3,i})$ in (c),(d).

Route slots, $m \le 4$. The map is the translation by $z_i = z_i(\boldsymbol\sigma(t))$, with derivative I; its images $\mathbb{B}(z_i,r_{3,i})$ are disjoint since $|z_i - z_j| > r_{3,i} + r_{3,j}$ (Lemma 12.11(c)). Lemma 12.11(d) and $\max(|s_i|,|Y_i|) + r_{3,i} \le \Lambda_{\mathrm{rt}}$ give $|F_{i,t}(u)| \le \Lambda_{\mathrm{rt}} + 2(k-1)\Lambda_{\mathrm{rt}} \le 2k\Lambda_{\mathrm{rt}}$.

Linear slot, $m = 5$. By Lemma 12.7, $M_{i,\sigma}$ is symplectic, the map is injective and its images lie in the disjoint balls $\mathbb{B}(Y_i, Lr_{\mathrm{tot}})$, so $|F_{i,t}(u)| \le \Lambda_{\mathrm{rt}} + Lr_{\mathrm{tot}}$.

Shape slot, $m = 6$. By Lemma 12.8(c),(f), the derivative $Df_i(\tau\sigma u)$ is symplectic and the map is injective on $\mathbb{B}(r_{3,i})$. The images and the bound are as for $m = 5$.

Tail slot, $m = 7$. By Lemma 12.9 with $v = v(t) \in [\tau, 1]$, the derivative is symplectic, the maps are injective with disjoint images, and $|F_{i,t}(u)| < R_0/v \le R_0/\tau$.

Each bound is at most R_F, since $k \ge 1$ and all terms are positive. □

12.7. **The source primitive on each stage and the pushforward.** For the isotopy of Construction 12.12, let $\mathfrak{a}_{i,t}(u)[\xi] = \omega_0(\partial_t F_{i,t}(u), D_u F_{i,t}(u)\xi)$ and $\mathfrak{h}_{i,t}(u) = \int_0^1 \mathfrak{a}_{i,t}(\vartheta u)[u]\, d\vartheta$ be the source one-form and primitive of Definition M7.7.

Lemma 12.15 (The primitive on each stage)**.** *Let $t \in [m/8, (m+1)/8]$, $\sigma = \sigma_m(t)$ and $u \in \mathbb{B}^\circ(r_{4,i})$. Then $\mathfrak{h}_{i,t}(u) = \dot\sigma_m(t)\, \hat h_{i,m,\sigma}(u)$, where:*

(a) for $m \le 4$, $\hat h_{i,m,\sigma}(u) = \omega_0(w_{i,m}, u)$ with the constant vectors $w_{i,0} = -w_{i,3} = \Lambda_{\mathrm{rt}}\nu_i e_1$, $w_{i,1} = -w_{i,4} = \Lambda_{\mathrm{rt}}\nu_i' e_{n+1}$ and $w_{i,2} = Y_i - s_i$;

(b) for $m = 5$, $\hat h_{i,5,\sigma}(u) = \frac12\,\omega_0\big(\dot M_{i,\sigma}u, M_{i,\sigma}u\big)$, $\dot M_{i,\sigma} = \frac{d}{d\sigma}M_{i,\sigma}$;

(c) for $m = 6$, $\hat h_{i,6,\sigma} = \tau\, h^{\mathrm{sh}}_{i,\tau\sigma}$, where for $|\rho| < 1$

$$h^{\mathrm{sh}}_{i,\rho}(u) = \int_0^1\!\!\int_0^1 \vartheta^2\phi\;\omega_0\big(D^2 f_i(\phi\rho\vartheta u)[u,u],\; Df_i(\rho\vartheta u)u\big)\, d\phi\, d\vartheta;$$

(d) for $m = 7$, $\hat h_{i,7,\sigma} = (1-\tau)\, h^{\mathrm{tl}}_{i,v}$ with $v = \tau + (1-\tau)\sigma$, where for $0 < v \le 1$

$$h^{\mathrm{tl}}_{i,v}(u) = \frac{1}{v^2}\int_0^1 \omega_0\big(Df_i(v\vartheta u)u,\; f_i(v\vartheta u)\big)\, d\vartheta.$$

Proof. By Lemma 12.13(e), $\mathfrak{h}_{i,t} = \dot\sigma_m(t)\hat h$, where $\hat h(u) = \int_0^1 \omega_0\big(\partial_\sigma G_{i,m}(\sigma,\vartheta u), D_u G_{i,m}(\sigma,\vartheta u)u\big)\, d\vartheta$.

(a) $\partial_\sigma G_{i,m} = w_{i,m}$ (Definition 12.10) and $D_u G = I$.

(b) $\partial_\sigma G(\vartheta u) = \vartheta \dot M_{i,\sigma}u$ and $D_u G = M_{i,\sigma}$, and $\int_0^1 \vartheta\, d\vartheta = \frac12$.

(c) Put $\rho = \tau\sigma$. Lemma 12.8(e) gives $\partial_\sigma G(x) = \tau\int_0^1 \phi D^2 f_i(\phi\rho x)[x,x]\, d\phi$, which yields the factor ϑ^2 at $x = \vartheta u$. Also $D_u G(\vartheta u)u = Df_i(\rho\vartheta u)u$ (Lemma 12.8(c)).

(d) $\partial_\sigma G = (1-\tau)\partial_v T_{i,v}$. Lemma 12.9(a) with $\xi = Df_i(v\vartheta u)u$ gives $\partial_v T_{i,v}(\vartheta u) = v^{-2}(v\vartheta\xi - f_i(v\vartheta u))$ and $D_u T_{i,v}(\vartheta u)u = \xi$. Then use $\omega_0(\xi,\xi) = 0$. □

In (c) the difference quotient of Lemma 12.8(b) has disappeared, so $h^{\mathrm{sh}}_{i,\rho}$ is smooth through $\rho = 0$.

Lemma 12.16 (The pushforward of a cut-off source function)**.** *Let $0 < r_2 < r_3 < r_4$, and let $F\colon \mathbb{R}\times\mathbb{B}^\circ(r_4) \to \mathbb{R}^{2n}$, $(t,u) \mapsto F_t(u)$, be C^∞ with $D_u F_t(u) \in \mathrm{Sp}(2n)$, F_t injective on $\mathbb{B}(r_3)$ for every t, $F_t = F_0$ for $t \le 0$ and $F_t = F_1$ for $t \ge 1$. Let $g\colon \mathbb{R}\times\mathbb{B}^\circ(r_4) \to \mathbb{R}$ be C^∞ with $g(t,u) = 0$ whenever $|u| \ge r_2$, and define*

$$\mathrm{Push}(g)(t,y) = \begin{cases} g(t,u), & \text{if } u \text{ is the unique point of } \mathbb{B}(r_3) \text{ with } F_t(u) = y,\\ 0, & \text{if there is no such unique point.}\end{cases}$$

Let $\Gamma(t,u) = (t, F_t(u))$, $W = \mathbb{R}\times\mathbb{B}^\circ(r_3)$ and $Z = \Gamma(\mathbb{R}\times\mathbb{B}(r_2))$.

(a) $O = \Gamma(W)$ is open in $\mathbb{R}\times\mathbb{R}^{2n}$, and Γ restricts to a C^∞ diffeomorphism $W \to O$.

(b) Z is closed, $Z \subset O$, $\mathrm{Push}(g) = g \circ (\Gamma|_W)^{-1}$ on O, and $\mathrm{Push}(g) = 0$ outside Z.

(c) $\mathrm{Push}(g)$ is C^∞ on $\mathbb{R}\times\mathbb{R}^{2n}$.

(d) For $t \in \mathbb{R}$, $|u| < r_3$ and $\xi \in \mathbb{R}^{2n}$: $d_y\,\mathrm{Push}(g)(t, F_t(u))\big[D_u F_t(u)\xi\big] = d_u g(t,u)[\xi]$.

(e) $\mathrm{Push}(g)(t,\cdot)$ *vanishes outside* $F_t(\mathbb{B}(r_2))$*, a subset of the compact set*

$$K = \bigcup_{t\in[0,1]} F_t(\mathbb{B}(r_2)).$$

If $g(t,\cdot) = 0$ *for* $t \notin (0,1)$*, then* $\mathrm{Push}(g)(t,\cdot) = 0$ *for* $t \notin (0,1)$*.*

Proof. (a) $D\Gamma = \left(\begin{smallmatrix} 1 & 0 \\ \partial_t F & D_u F \end{smallmatrix}\right)$ is invertible, so Γ is a local diffeomorphism. It is injective on $\mathbb{R}\times\mathbb{B}(r_3)$, since $\Gamma(t,u) = \Gamma(t',u')$ forces $t = t'$ and then $u = u'$; hence it is a diffeomorphism of W onto its open image.

(b) Z is the union of $(-\infty,0]\times F_0(\mathbb{B}(r_2))$, $\Gamma([0,1]\times\mathbb{B}(r_2))$ and $[1,\infty)\times F_1(\mathbb{B}(r_2))$, each closed by compactness. $Z \subset O$ because $r_2 < r_3$. Injectivity gives $\mathrm{Push}(g) = g\circ(\Gamma|_W)^{-1}$ on O. Off Z, any $u\in\mathbb{B}(r_3)$ with $F_t(u) = y$ has $|u| > r_2$, so $g(t,u) = 0$.

(c) $\mathrm{Push}(g)$ is smooth on O and vanishes on the open set $(\mathbb{R}\times\mathbb{R}^{2n})\setminus Z$, and these two sets cover.

(d) Differentiate $\mathrm{Push}(g)\circ\Gamma = g$ in the direction $(0,\xi)$.

(e) The first statement follows from (b). For $t\notin[0,1]$, $F_t(\mathbb{B}(r_2))$ is $F_0(\mathbb{B}(r_2))$ or $F_1(\mathbb{B}(r_2))$, and K is compact as a continuous image of $[0,1]\times\mathbb{B}(r_2)$. The last statement is immediate. □

For ball i, take $r_\ell = r_{\ell,i}$, $F = F_i$ (Proposition 12.14(a)–(c)) and $g = \chi_i\,\mathfrak{h}_i$, where $\chi_i = \chi_{r_{1,i},r_{2,i}}$ is the cutoff of Definition M7.10; $\mathfrak{h}_i$ is C^∞ and vanishes for $t\notin(0,1)$ (Lemma M7.9), and χ_i is C^∞ and vanishes for $|u|\ge r_{2,i}$ (Lemma 2.7). Then $\mathrm{Push}(g)$ is H_i of Definition M7.10, and Lemma 12.16 with Proposition 12.14(d) gives Lemma M7.11.

13. A worked instance: ten balls in dimension four

We take $k = 10$, $\varepsilon = \frac{1}{10}$ in Theorem A and check the required inequalities stage by stage. The Hamiltonian, flows and maps f_i are not evaluated, nor are the extrema listed at the end. Tables give exact expressions, with seven-digit decimals for orientation only. The inequalities use the exact expressions, and the smallest gap in the displayed chains exceeds $4\cdot10^{-5}$.

13.1. **Input, normalisation and branch.** The input is $n = 2$, $k = 10$, $c_i = \pi$ and $s_i = (3i-3,0,0,0)$ for $1\le i\le 10$. Then $R^4 = \frac{k}{1-\varepsilon/4} = \frac{400}{39}$, $R\approx 1.789571$ and $C = \pi R^2 = \frac{20\pi}{\sqrt{39}} \approx 10.06115$. The unit balls $B(s_i,\pi)$ have centres at distance at least 3, so they are pairwise disjoint. By Definition M7.1, $\rho = \frac{\pi}{C} = \frac{\sqrt{39}}{20} \approx 0.3122499$, with $k\rho^2 = \frac{39}{40} = 1-\frac{\varepsilon}{4}$ and $\rho\sqrt{k} = \sqrt{39/40} \approx 0.9874209$.

Admissibility holds, since $k\rho^2 < 1$ and $3\rho \approx 0.9367497 < 1$. Since $n = 2$, $k\ge 10$, the capacities are equal and $\frac{3}{4} < \rho\sqrt{k} < 1$, Construction 11.1 applies: the equal-capacity maps of Remark 10.25, i.e. the equal case of Construction 10.22 with the linear comparison datum of Proposition M4.4. This branch is taken for every $k\ge 10$, $\varepsilon\in(0,1)$, as $\rho\sqrt{k} = \sqrt{1-\varepsilon/4} \in (\frac{\sqrt{3}}{2},1)$. The defect is $\eta_\star = 1-\sqrt{39/40} \approx 0.01257912$ (Remark 10.25).

13.2. **Model parameters.** Table 3 lists the parameters of Remarks 10.24, 10.25 and 8.18, §M4.2, Definitions 3.4, 8.2 and 8.4, Construction 8.15, and Lemmas 10.9 and 10.12.

Checks. Definition 8.1(a) and Proposition 8.17 require the chain

$$\rho \approx 0.3122499 < \rho^+ \approx 0.3127469 < \bar\rho \approx 0.3132444 < \hat\rho \approx 0.3134930$$
$$< a \approx 0.3142388 < h \approx 0.3312368 < d_n \approx 0.3322851 < \tfrac{1}{3}.$$

The condition $0 < g \le H_{\mathrm{b}}(h)$ holds with equality. In Proposition M4.4, the identity $F = K - P_{\mathrm{b}} = -g + c_s\Lambda$ uses $c_sA = k - 3\sqrt{k} \approx 0.5131670$, $c_sB = 3\sqrt{k} \approx 9.486833$ and $3\sqrt{k}\,a = 3-g$. Also $h + 2\delta_m \approx 0.3316142 < d_n < \frac{1}{3} - \frac{e_h}{9}$.

Quantity	Closed form	Value
$t_0 = \frac{1}{2}(1+\rho\sqrt{k})$	$1 - \frac{\eta_\star}{2}$	0.9937104
auxiliary capacity a	$t_0/\sqrt{10}$	0.3142388
height h	$t_0/3$	0.3312368
$d_n = \frac{1}{2}(h+\frac{1}{3})$	$(1+t_0)/6 = \frac{1}{3} - \frac{\eta_\star}{12}$	0.3322851
$a - \rho$	$\eta_\star/(2\sqrt{10})$	$1.988933 \cdot 10^{-3}$
$1 - 3h$	$\eta_\star/2$	$6.289559 \cdot 10^{-3}$
$d_n - h$, $1 - 3d_n$	$\eta_\star/12$, $\eta_\star/4$	$1.048260 \cdot 10^{-3}$, $3.144779 \cdot 10^{-3}$
profile $P_{\mathrm{b}}(p)$	$3(1-3p) - 10(a-p)_+$ on $[0,h]$	$P_{\mathrm{b}}(0) \approx -0.1423883$, $P_{\mathrm{b}}(h) = g$
$\int_0^h P_{\mathrm{b}}$	$t_0(1-t_0) = \frac{1}{4}(1-k\rho^2) = \frac{\varepsilon}{16}$	0.00625
comparison function $K(p) = c_s p$	$c_s = 3\sqrt{10} - 9$	$c_s \approx 0.4868330$
slopes of the tent Λ	$A = \sqrt{10}/3$, $B = 10 + 3\sqrt{10}$	1.054093, 19.48683
$g = 3(1-t_0) = H_{\mathrm{b}}(h)$	$3\eta_\star/2$	0.01886868
$e_q = \min(\frac{h}{100}, \frac{g}{128 c_s})$	$g/(128 c_s)$ (as $\frac{h}{100} \approx 3.312368 \cdot 10^{-3}$)	$3.027969 \cdot 10^{-4}$
planar cost $4 c_s e_q$	$g/32$	$5.896461 \cdot 10^{-4}$
comparison gap γ	$\frac{31}{32} g$	0.01827903
planar support P_{supp}	$h - \frac{31}{8} e_q < P_{\mathrm{supp}} < h$ (Lemmas 5.31, 4.12)	$\in$ (0.3300635, 0.3312368)
background e_h, cap scale μ_h	$g/16$, $g/160$	$1.179292 \cdot 10^{-3}$, $1.179292 \cdot 10^{-4}$
cap excess $k\mu_h/2$; residual	$g/32$; $\frac{7}{8} g$	$5.896461 \cdot 10^{-4}$; 0.01651009
$\delta_m = \min(\frac{d_n - h}{4}, \frac{g}{100})$	$g/100 = 3\eta_\star/200$ (as $\frac{d_n-h}{4} = \frac{\eta_\star}{48}$)	$1.886868 \cdot 10^{-4}$
moment cutoff $\mathfrak{c}$	1 for $p \le h + \frac{3}{2}\delta_m$, 0 for $p \ge h + \frac{7}{4}\delta_m$	0.3315198, 0.3315670
safe region bound $h + 2\delta_m$	$h + 3\eta_\star/100$	0.3316142
$\frac{15}{16} g - 18\delta_m$	lower bound for H^* on the safe region	0.01429302
nondegeneracy of Ω_λ	$p < \frac{1}{3} - \frac{e_h}{9}$	0.3332023
clock, layers	$u_0 = \frac{7}{48}$, $u_1 = \frac{3}{16}$, $N_\ell = 11$, $\ell_s = \frac{1}{264}$	$\ell_s \approx 3.787879 \cdot 10^{-3}$
layer i	$l_i = \frac{7}{48} + \frac{i-1}{264} + \frac{1}{1056}$, $r_i = l_i + \frac{1}{528}$	$l_1 \approx 0.1467803$, $r_{10} \approx 0.1827652$
strip slack $\delta = \frac{1}{2}(a-\rho)$	$\eta_\star/(4\sqrt{10})$	$9.944665 \cdot 10^{-4}$
Morse parameter of Ψ_δ	δ/π	$3.165485 \cdot 10^{-4}$
$\bar\rho = \frac{1}{2}(\rho + a)$	$\rho + \delta$	0.3132444
$\hat\rho = a - \frac{3}{4}\delta$	$\rho + \frac{5}{4}\delta$	0.3134930
$\varrho_4^2 = \bar\rho/\rho$, $\varrho_3 = \frac{1}{2}(1+\varrho_4)$	$\varrho_4^2 = \frac{4 - 3\eta_\star}{4(1-\eta_\star)}$	$\varrho_4 \approx 1.001591$, $\varrho_3 \approx 1.000796$
enlarged capacity $\rho^+ = \rho\varrho_3^2$	$\frac{\rho}{4}(1+\varrho_4)^2$	0.3127469
transport: $w_* = \frac{h}{d_n - h}$	$4t_0/\eta_\star$	315.9873
T_0, $c_* = \min(\frac{3}{2}, \frac{T_0^2}{d_n})$	$T_0 = (6w_*)^{-1/2}$, $c_* = \frac{\eta_\star}{4t_0(1+t_0)}$	0.02296622, $1.587334 \cdot 10^{-3}$
return: $1 - \frac{3}{2} d_n$	$(3-t_0)/4$	0.5015724

TABLE 3. Model parameters for $k = 10$, $\varepsilon = \frac{1}{10}$; $\eta_\star = 1 - \sqrt{39/40}$.

Of the unfilled volume $\frac{1}{2}(1-k\rho^2) = \frac{\varepsilon}{8}$ of $B^\circ(1)$, half, $\int_0^h P_{\mathrm{b}} = \frac{\varepsilon}{16}$, pays the gap; the other half, $\frac{1}{2}k(a^2-\rho^2) + \frac{1}{2}(1-3h)^2$, goes to capacity slack and height defect (Remark 10.25).

13.3. **The common isotopy and the Hamiltonian.** By Proposition M7.3, the physical maps are $f_i(u) = \sqrt{C}\, f_i^\natural(u/\sqrt{C})$, with $\sqrt{C} \approx 3.171931$, $c_i' = \pi\varrho_3^2$ and $\bar c_i = \pi\varrho_4^2$. Table 4 lists the radii of Definition M7.4; the source radius is $r_i = 1$ and the centre gap is $\min_{i \ne j}(|s_i - s_j| - 2) = 1$.

Quantity	Closed form	Value
c_i', $\bar{c}_i$	$\pi\varrho_3^2$, $\pi\varrho_4^2$	3.146593, 3.151598
r_i', $\bar{r}_i$	ϱ_3, ϱ_4	1.000796, 1.001591
step $\min(\frac{1}{4},\, r_i' - r_i)$	$\varrho_3 - 1$	$7.955775 \cdot 10^{-4}$
$r_{1,i}, r_{2,i}, r_{3,i}, r_{4,i}$	$1 + \frac{\ell}{5}(\varrho_3 - 1)$	1.000159, 1.000318, 1.000477, 1.000636
target radius $R_C = \sqrt{C/\pi}$	R	1.789571

TABLE 4. Radii of the common isotopy for $k = 10$, $\varepsilon = \frac{1}{10}$.

The Hamiltonian $H = H[2, 10, C, (\pi, \dots, \pi), s]$ of Definition M7.16 is assembled from the model embeddings $f_i^\natural = b_{\mathrm{ball}} \circ \psi_1^{-1} \circ \mathrm{pr} \circ \tau_1 \circ \iota \circ \mathsf{N} \circ \Phi^{-1} \circ \chi_{\mathcal{A}} \circ \mathcal{D}^{-1} \circ E_i$ on $B^\circ(\hat{\rho})$ with the parameters of Table 3 (E_i uses Ψ_δ, the shear and the layer root; $\mathcal{D}$ the planar isotopy with e_q; Φ the cut field of Construction 8.15 with δ_m; the transfer d_n and w_*; only E_i depends on i), the route translations, linear stage, shape stage and tail of Construction 12.12, and the cut-off radial primitive of Definition M7.10. By Theorem M7.14, $\phi_H^1(B(s_i, \pi)) = f_i(B(\pi))$. These images are pairwise disjoint and lie in $\bigcup_i f_i(B(c_i')) \subset B^\circ(\pi R^2)$, the Euclidean ball of radius R. Moreover $k/R^4 = \frac{39}{40} > 1 - \varepsilon$.

The proofs show only positivity or finiteness of C_b (Proposition 8.17), $m = \frac{1}{2}\min_{\mathcal{K}}(1 - \pi|z|^2)$ (Construction 10.22), y_i, A_i, and L, τ, δ_c, R_F (Definition 12.4, Proposition 12.14); these are not evaluated. The support area P_{supp} is a closed expression in the constants of Definitions 5.27 and 5.32; only its bracket is recorded in Table 3.

APPENDIX A. THE EXPLICITNESS CHECK

The Lean command `#check_explicit` (`SymplecticPacking/Explicit.lean`) audits the data dependencies of construction terms for the syntactic form of level L1 (Definition M2.5); it is used in no proof of this supplement.

A.1. **Data dependencies.** The command follows constants in data positions (including lambda bodies, local values, project and vendored definitions), skipping theorem bodies and proposition-typed arguments; choices supplying values are inspected. `sorryAx` and the recorded choice operators (`Classical.choose`, `Function.invFun`, …) are rejected outside §A.2. A source scan excludes new axioms and unchecked mechanisms; the main results use only `propext`, `Classical.choice`, `Quot.sound`.

A.2. **Definite operations and searches.** `Explicit.the` gives definite descriptions with uniqueness proofs (integral curves, inverse images, scalar roots, linear solves), the predicate and domain being part of the construction. Classical decisions are allowed as case distinctions at level L1 only. Searches over naturals, rationals or finite encoded data (also natural infima, filtered finite sets, descriptions over search types) must have genuinely decidable predicates, with requirements propagated to call sites; searches over function spaces are rejected. Mathlib's classical limits, integrals etc. are accepted only for the reviewed declarations in `Explicit/Whitelist.lean`, with dependency paths in the output.

A.3. **Definite descriptions in the construction.** Descriptions over continuous types are reviewed by hand (§M8). The report on `General.packingHamiltonian` lists 26 declarations with `Explicit.the`; each defining property is an explicit equation or inequality in the construction data, with uniqueness proved, and none refers to a packing specification.

- *Inverses* (fixed fallback off the image): `Inner.invR`, `Inner.TrotInv`, `Qhat.ordInv`, `Qhat.qhatInv`, `HighDim.ToricPackingStages.circleClockInv`, `layerChartInv`, `nativeInverse`, `BranchLarge3.CubicBridge.expInv`, `General.Ham.pushforward` ($u \in \mathbb{B}(r)$ with $F_t(u) = y$, Definition M7.10).
- *Roots, critical points*: `Strip.layerRoot`, `Strip.Cubic.layerRoot` ([Son26], §8.1), `Strip.vOfT`, `BranchLarge3.PrimExt.rootExt`, `Branch2.Planar.PlanarInput.pstar`, `HighDim.PlanarGen.Input.vstar`.
- *Linear solves*: `BranchLarge3.TargetMoser.solveL`, `BranchSmall.VT.transportField`, `returnField`, `General.Ham.Linear.posSqrt`.
- *Integral curves*: `Qhat.flowOf`, `Return.flowFrom`, `BranchSmall.VT.flowI`, `BranchLarge3.UMoser.curve`, `HighDim.ToricPackingStages.nativeAmbientFlow`.
- *Extrema, measures*: `HighDim.ToricPackingStages.finiteMinimumOrZero`, `Explicit.planeVolume`.

A.4. **Vendored declarations used.** The 3810 vendored declarations in the closure of the statement and algorithm theorems (§M8) are listed in `verification/Audit/VendoredUsage.lean`; none comes from `Compactness`, `Continuation`, `Degree`, `Cauchy`, `Campanato`.

A.5. **Formal statements that differ in form from the text.** Table 5 lists them (names relative to `SymplecticPacking`). The labels *needed consequence verified* and *full statement verified* record a human comparison, not a kernel check; unqualified tags claim no full coverage.

Table 5: Formal statements that differ in form from the text.

statement	formal form
S (Def. 2.3)	`Real.smoothTransition`, `expNegInvGlue` $= a_\flat$; the(h) is `Explicit.the`
real coordinates	ζ_l (Table 1) is coordinate $l-1$ of `Problem.E` n
Thm. 4.21	single ordinary path, no parameter y
Cor. 4.22	Theorem 5.34(e) with $g(I) = I+4e$, $\varepsilon = 0$, $\Upsilon \le h$ on $[0,h]$, $\Upsilon(\pi\lvert\mathfrak{e}_1(z)\rvert^2) < \pi\lvert z\rvert^2 + \frac{13}{24}e$ on $\bar{D}(h-4e)$
Def. 8.1	(q_t) only the whole-disc isotopy $\hat{q}$ of an ordinary path of area h
Prop. 8.16	(a)–(d), smoothness of $\Phi^{\pm1}$; $\Phi = \Phi_2 \circ \Phi_1 \circ \Phi_0$ not formalised
Prop. 8.17(c)	$\mathcal{T} f_i^N(X) = \mathcal{T} f_j^N(X') \Rightarrow f_i^N(X) = f_j^N(X')$, $\mathcal{T}$ of Thm. 10.21
Prop. 10.15, Lem. 10.20, Thm. 10.21	only along each $\mathcal{T} \circ f_i^N$: pullbacks along smooth unit-vector families and for $\mathcal{T}_{\mathrm{u}}$ on smooth families in $\mathcal{N}_h$, with Prop. 10.6; injectivity as above
Thm. 10.23	$u \mapsto \sqrt{C}\, f_i(u/\sqrt{C})$: smooth, symplectic on $B^\circ(C\rho_i^{\mathrm{dom}})$, injective on $B(C\rho_i^+)$, disjoint images in $B^\circ(C)$; m not stated, (iv) structural; injectivity, disjointness, $\pi\lvert f_i\rvert^2 < 1$ on $B^\circ(\hat{\rho}_i)$ in both cases
Rem. 10.25	`Params`, `Params.`$\varepsilon = 4\eta_\star$ (not ε of Theorem A), `Params.rho` $= (1-\eta_\star)/\sqrt{k} = \rho$; via `Branch2.CubicData.ofParams` to `Branch2.cubicEmbeddings_spec`; data of Theorem A from `General.epochHamiltonianQ`, `General.epochQ_equalCase` gives `Branch2.EqualCase`
support of H (§M7)	`General.Ham.isotopySupport` $= \bigcup_{i,t\in[0,1]} F_{i,t}(B(r_{2,i}))$, `isotopySupport_compact`, `ofIsotopy_support_precise`; for $H[n,k,C,c,s]$: `General.packingSupportSet_compact`, `packingHamiltonian_support_precise`, `packingSupportSet_subset_radius` (ball of radius `General.packingSupportRadius`), `main_support_radius`

A.6. **Reproduction.** The statement file runs the check on `General.packingHamiltonian`, the Epoch Hamiltonians, each embedding constructor, and the polynomial interpreter, rational verifier and output construction of the finite-output algorithm of [Son26]; any violation fails the

build. `test/ExplicitTest.lean` also contains admissible searches that must be accepted (besides the rejected ones of §M8).

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